UNITED STATES
SECURITIES AND EXCHANGE COMMISSION

Washington, DC 20549

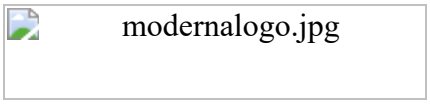
FORM 10-K

(Mark One)

☒ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**
For the fiscal year ended December 31, 2025

OR

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**
For the transition period from _ to _
Commission File Number: 001-38753



Moderna, Inc.

(Exact Name of Registrant as Specified in Its Charter)

Delaware	**81-3467528**
(State or Other Jurisdiction of Incorporation or Organization)	(IRS Employer Identification No.)
325 Binney Street Cambridge, Massachusetts	**02142**
(Address of Principal Executive Offices)	(Zip Code)

(617) 714-6500
(Registrant's Telephone Number, Including Area Code)

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Common stock, par value $0.0001 per share	**MRNA**	**The Nasdaq Stock Market LLC**

Securities registered pursuant to Section 12(g) of the Act: **None**

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. **Yes** ☑ **No** ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Act. **Yes** ☐ **No** ☑

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. **Yes** ☑ **No** ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§ 232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). **Yes** ☑ **No** ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer", "accelerated filer", "smaller reporting company", and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer ☑	Accelerated filer ☐	Non-accelerated filer ☐	Smaller reporting company ☐
			Emerging growth company ☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. **Yes ☑ No ☐**

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to §240.10D-1(b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Act). **Yes ☐ No ☑**

The aggregate market value of voting stock held by non-affiliates of the registrant, computed by reference to the closing price as of the last business day of the registrant's most recently completed second fiscal quarter, was approximately $9.9 billion. This excludes shares of common stock held by each executive officer and director and by each other person who may be deemed to be an affiliate of the registrant. The determination of affiliate status for this purpose is not necessarily a conclusive determination for other purposes. The registrant has no non-voting common stock.

As of February 13, 2026, there were 394,939,424 shares of the registrant's common stock, par value $0.0001 per share, outstanding.

DOCUMENTS INCORPORATED BY REFERENCE

Portions of the registrant's Definitive Proxy Statement relating to its 2026 Annual Meeting of Stockholders to be filed hereafter are incorporated by reference into Part III of this Annual Report on Form 10-K where indicated.

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SUMMARY OF THE MATERIAL RISKS ASSOCIATED WITH OUR BUSINESS

Our business is subject to numerous risks and uncertainties that you should be aware of before making an investment decision, including those highlighted in the section entitled "Risk Factors." These risks include, but are not limited to, the following:

- Regulatory and market uncertainty have and may continue to impact our business and the markets for our products;
- We may experience difficulties executing our near-term strategy and prioritized pipeline;
- The vaccine market, and pharmaceutical market more generally, is intensely competitive, and we may be unable to compete effectively in the market for existing or new products, treatment methods or technologies;
- We have experienced commercial challenges and may experience additional challenges in the future;
- The commercial success of our products depends on the degree of market acceptance by physicians, patients, third-party payors and others in the medical community;
- Sales of pharmaceutical products depend on the availability and extent of reimbursement from third-party payors, and we may be adversely impacted by changes to such reimbursement policies or rules;
- The market opportunities for our products and product candidates may be smaller than we believe, or we may be unable to successfully identify clinical trial participants;
- If we cannot obtain, or are delayed in obtaining, regulatory approvals and advisory committee recommendations, we will be unable to commercialize, or will be delayed in commercializing, our product candidates;
- Clinical development is lengthy and uncertain, and our clinical programs may be delayed or terminated, or may be more costly to conduct than we anticipate;
- Our products are, and any future products will be, subject to regulatory scrutiny;
- We or our third-party manufacturers may encounter difficulties in manufacturing, product release, shelf life, testing, storage, supply chain management or shipping for any of our products;
- As we grow as a commercial company and our drug development pipeline matures, the increased demand for clinical and commercial supplies from our facilities and third parties may impact our ability to operate;
- We are subject to operational risks associated with the physical and digital infrastructure at our manufacturing facilities and those of our external service providers;
- Our intismeran autogene product candidates are uniquely manufactured for each patient using a novel, complex manufacturing process and we may encounter difficulties in production;
- We are dependent on single-source suppliers for some of the components and materials used in, and the manufacturing processes required to develop and commercialize, our products and product candidates;
- We have entered into strategic alliances with third parties for product development and commercialization. If these alliances are unsuccessful, our business could be adversely affected;
- We may seek to establish additional strategic alliances and, if we are unable to do so on commercially reasonable terms, we may have to alter our development and commercialization plans. Certain of our strategic alliance agreements may restrict our ability to develop certain products;
- We may be unable to obtain and enforce patent protection for our discoveries and the intellectual property rights therein, or protect the confidentiality of our trade secrets;
- Uncertainty over intellectual property in the pharmaceutical and biotechnology industry has been the source of litigation and other disputes, which is inherently costly and unpredictable and can have adverse financial and freedom-to-operate consequences;
- We incurred net losses in 2025 and 2024, and expect to incur additional losses in the future; we have a limited history of recognizing revenue from product sales and may not achieve long-term sustainable profitability;
- Our quarterly and annual operating results may fluctuate. As a result, we may fail to meet or exceed the expectations of research analysts or investors, which could cause our stock price to decline;
- We may encounter difficulties in managing changes to the size, structure and scope of our company;
- We are subject to the risks of doing business outside of the United States;
- Our internal computer systems and physical premises, or those of third parties with which we share sensitive data or information, may fail or suffer security breaches, including from cybersecurity incidents, which could materially disrupt our product development programs and manufacturing operations; and
- The price of our common stock has been volatile, which could result in substantial losses for shareholders.

You should consider carefully the risks and uncertainties described below, in the section entitled "Risk Factors" and the other information contained in this Annual Report on Form 10-K, including our consolidated financial statements and the related notes, before you decide whether to purchase our common stock. The risks described above are not the only risks that we face. Additional risks and uncertainties not presently known to us or that we currently deem immaterial may also impair our business operations.

SPECIAL NOTE REGARDING FORWARD-LOOKING STATEMENTS

This Annual Report on Form 10-K, including the sections entitled "Business," "Risk Factors" and "Management's Discussion and Analysis of Financial Condition and Results of Operations," contains express or implied forward-looking statements within the meaning of the federal securities laws, Section 27A of the Securities Act of 1933, as amended (the Securities Act), and Section 21E of the Securities Exchange Act of 1934, as amended (the Exchange Act). All statements other than statements of historical facts contained in this Annual Report are forward-looking statements. Forward-looking statements in this Annual Report on Form 10-K include, but are not limited to, statements about:

- our ability to drive sales growth through geographic expansion and new product launches;
- our expectations regarding revenue growth in 2026;
- our ability to deliver cost efficiency across our business and to achieve targeted cash breakeven;
- our expectations regarding the size and durability of the respiratory vaccine market and future demand for and sales of our products;
- the potential and timing for future data readouts, regulatory filings, regulatory approvals and commercial launches;
- the timing of initiation, progress, completion, results (including interim data) and cost of our clinical trials, as well as those of our collaborators;
- our ability to successfully contract with third-party suppliers, distributors and manufacturers;
- our ability and the ability of third parties with whom we contract to successfully manufacture, supply and distribute our products, at scale, as well as drug substances, delivery vehicles and product candidates;
- the scope of protection we are able to establish and maintain for intellectual property rights covering our commercial products, product candidates and technology, including our ability to enter into license agreements, and our expectations regarding pending legal proceedings related to our intellectual property;
- participant enrollment in our clinical trials, including enrollment demographics and timing;
- potential advantages of mRNA as compared to traditional medicine;
- our ability to successfully commercialize our products, if approved, including in light of the size and growth potential of the markets for our products and the degree of market acceptance of our products;
- the pricing and reimbursement of our medicines, if approved;
- the buildout of our global manufacturing and commercial operations, and our ability to leverage our global manufacturing network;
- our financial performance and estimates of our future expenses, revenues and capital requirements;
- the potential benefits of strategic collaboration agreements and our ability to enter into strategic collaborations or other agreements with collaborators with development, regulatory and commercialization expertise;
- legal and regulatory developments in the United States and foreign countries;
- our ability to produce our products or product candidates with advantages in turnaround times and manufacturing cost;
- our ability to attract and retain key scientific, manufacturing, regulatory, commercial and management personnel; and

- developments relating to our competitors and our industry.

In some cases, forward-looking statements can be identified by terminology such as “may,” “should,” “expects,” “intends,” “plans,” “anticipates,” “believes,” “estimates,” “predicts,” “potential,” “continue,” or the negative of these terms or other comparable terminology, although not all forward-looking statements contain these identifying words. Forward-looking statements are based on our management’s belief and assumptions and on information currently available to our management. Although we believe that the expectations reflected in these forward-looking statements are reasonable, these statements relate to future events or our future operational or financial performance, and involve known and unknown risks, uncertainties, and other factors that may cause our actual results, performance, or achievements to be materially different from any future results, performance, or achievements expressed or implied by these forward-looking statements. We may not actually achieve the plans, intentions or expectations disclosed in our forward-looking statements, and you should not place undue reliance on forward-looking statements. Factors that may cause actual results or events to differ materially from current expectations include, among other things, those listed under the section entitled “Risk Factors” and elsewhere in this Annual Report on Form 10-K. If one or more of these risks or uncertainties occur, or if our underlying assumptions prove to be incorrect, actual events or results may vary significantly from those expressed or implied by the forward-looking statements. No forward-looking statement is a guarantee of future performance.

The forward-looking statements in this Annual Report on Form 10-K represent our views as of the date of this Annual Report on Form 10-K. We anticipate that subsequent events and developments will cause our views to change. However, while we may elect to update these forward-looking statements at some point in the future, we have no current intention of doing so except to the extent required by applicable law. You should therefore not rely on these forward-looking statements as representing our views as of any date subsequent to the date of this Annual Report on Form 10-K.

This Annual Report on Form 10-K includes statistical and other industry and market data that we obtained from industry publications and research, surveys, and studies conducted by third parties. Industry publications and third-party research, surveys, and studies generally indicate that their information has been obtained from sources believed to be reliable, although they do not guarantee the accuracy or completeness of such information. We have not independently verified the information contained in such sources.

NOTE REGARDING COMPANY REFERENCES

Unless the context otherwise requires, the terms “Moderna,” the “Company,” “we,” “us” and “our” in this Annual Report on Form 10-K refer to Moderna, Inc. and its consolidated subsidiaries.

TRADEMARKS

This Annual Report on Form 10-K contains references to our trademarks and to trademarks belonging to other entities. Solely for convenience, trademarks and trade names referred to, including logos, artwork and other visual displays, may appear without the ® or TM symbols, but such references are not intended to indicate, in any way, that their respective owners will not assert, to the fullest extent under applicable law, their rights thereto. We do not intend our use or display of other companies’ trade names or trademarks to imply a relationship with, or endorsement or sponsorship of us by, any other companies.

PART I

Item 1. Business

Moderna is a pioneer and leader in the field of mRNA medicine. Through the advancement of our technology platform, we are reimagining how medicines are made to transform how we treat and prevent diseases. Since our founding, our mRNA platform has enabled the development of vaccine and therapeutic candidates across infectious disease, oncology, rare disease and more.

With a global team and a unique culture, driven by our values and mindsets, our mission is to deliver the greatest possible impact to people through mRNA medicines.

We currently have three commercial products—Spikevax and mNEXSPIKE (our COVID vaccines) and mRESVIA (our vaccine against respiratory syncytial virus (RSV)). mNEXSPIKE, which we launched commercially in the third quarter of 2025, is now our leading product in the U.S. retail channel. In 2025, we achieved total revenue of $1.9 billion, largely from sales of our COVID vaccines.

Beyond our commercial products, we continue to demonstrate the potential of our platform technology and are advancing a pipeline of development candidates across oncology, rare disease and infectious disease. In January 2026, we and Merck announced five-year data from the Phase 2b study of intismeran autogene (mRNA-4157), our mRNA-based individualized neoantigen therapy, in combination with Merck's pembrolizumab (KEYTRUDA®), which demonstrated sustained improvement in recurrence-free survival in patients with high-risk melanoma (stage III/IV) following complete resection. We are advancing intismeran in collaboration with Merck, with eight Phase 2 and Phase 3 clinical trials underway across multiple tumor types. In oncology, we are also advancing mRNA-4359, a cancer antigen therapy designed to elicit T-cell immune responses against tumor and immunosuppressive cells.

In infectious disease, we have regulatory filings under review for our seasonal flu+COVID combination vaccine candidate (mRNA-1083) in Europe and Canada, and for our seasonal flu vaccine candidate (mRNA-1010) in the United States, Europe, Canada and Australia. For mRNA-1010, in response to a prior Refusal-to-File letter, we engaged with the U.S. Food and Drug Administration (FDA) in a Type A meeting and submitted an amended biologics license application (BLA) outlining a revised regulatory pathway based on age, seeking full approval for adults 50 to 64 years of age and accelerated approval for adults 65 and older, along with a post-marketing requirement to conduct an additional study in older adults. Following the meeting and submission of the amended application, the FDA accepted our BLA for review and assigned a Prescription Drug User Fee Act (PDUFA) goal date of August 5, 2026. In addition, we recently completed enrollment of a second Northern Hemisphere season (2025-2026) cohort in our ongoing Phase 3 study for our norovirus candidate (mRNA-1403).

In rare disease, our propionic acidemia therapeutic (mRNA-3927) has reached target enrollment in a registrational study. In January 2026, we entered into a strategic collaboration with Recordati, an international pharmaceutical group, to advance mRNA-3927 through the final stages of clinical development and, if approved, global commercialization. In addition, we expect the registrational study for our methylmalonic acidemia therapeutic (mRNA-3705) to begin in 2026.

Since 2022, we have streamlined our production sites into a global manufacturing network to support new product launches and deliver products for multi-year collaborations. In 2025, we announced new drug product capabilities in the U.S. and we have added three Moderna-built and managed facilities in the UK, Canada and Australia to enable local access to mRNA vaccines. Additionally, our Marlborough, Massachusetts facility was purpose-built for intismeran and began clinical batch supply in September 2025.

THE mRNA OPPORTUNITY

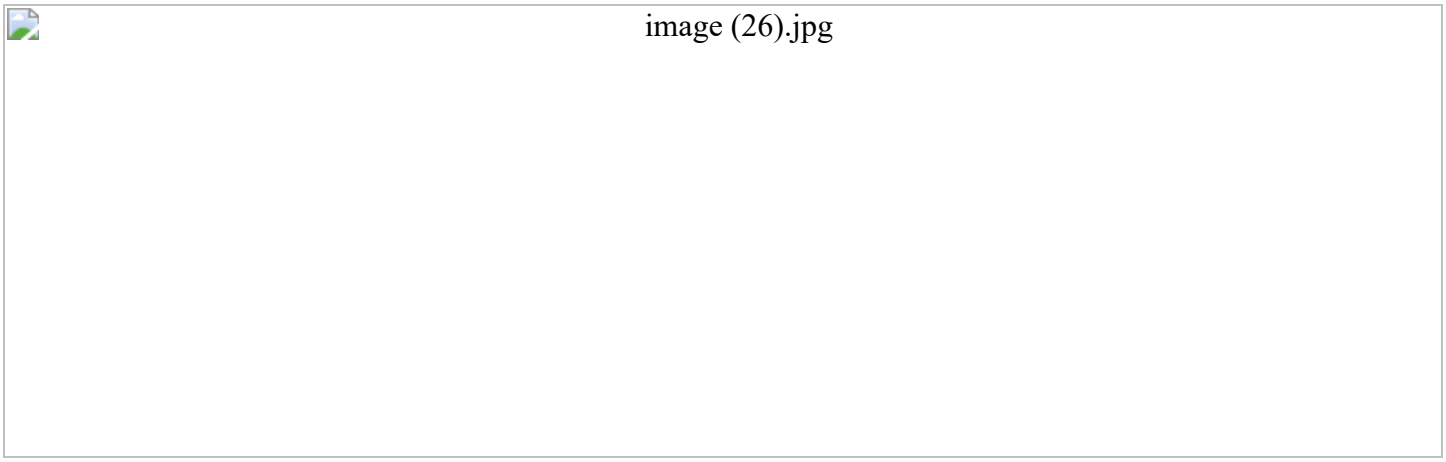
mRNA, the software of life

mRNA transfers the information stored in our genes to the cellular machinery that makes all the proteins required for life. Our genes are stored as sequences of DNA which contain the instructions to make specific proteins. DNA serves as a hard drive, safely storing these instructions in the cell's nucleus until they are needed by the cell.

When a cell needs to produce a protein, the instructions to make that protein are copied from the DNA to mRNA, which serves as the template for protein production. Each mRNA molecule contains the instructions to produce a specific protein with a distinct function in the body. mRNA transmits those instructions to cellular machinery, called ribosomes, that make copies of the required protein.

We see mRNA functioning as the "software of life." Every cell uses mRNA to provide real time instructions to make the proteins necessary to drive all aspects of biology, including in human health and disease. This was codified as the central dogma of molecular biology over 60 years ago, and is exemplified in the schematic below.

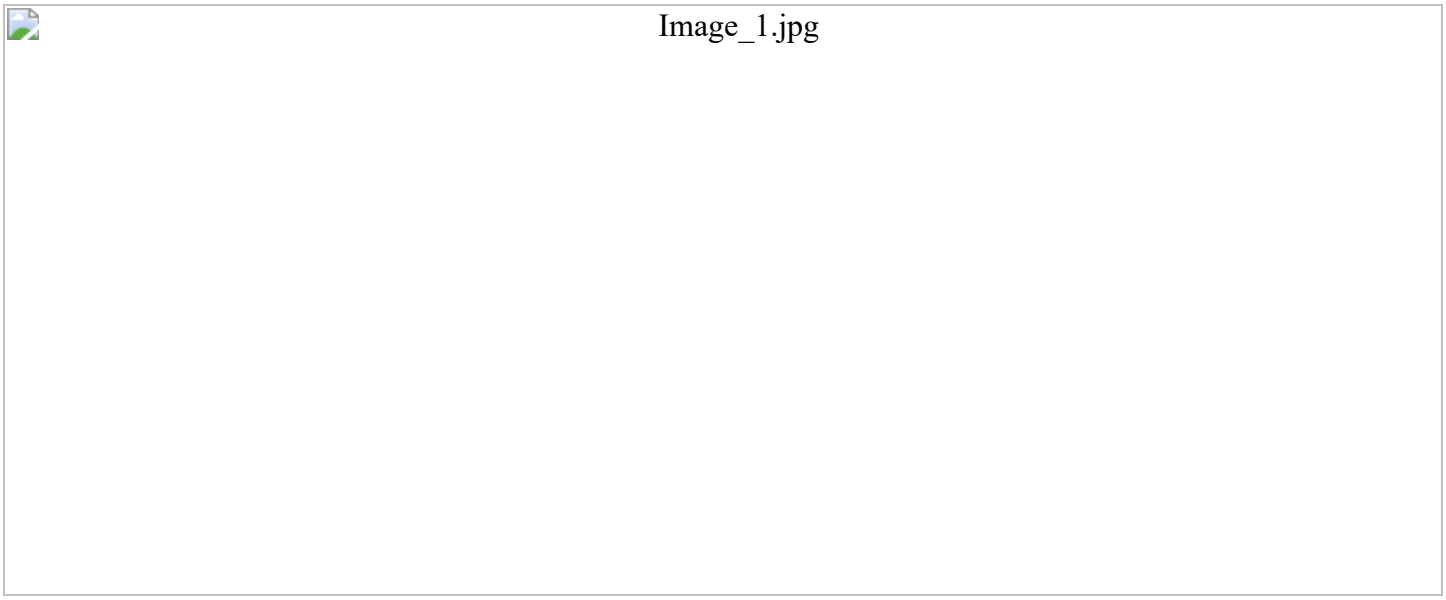
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The structure of mRNA

mRNA is a linear polymer comprising four monomers called nucleotides: adenosine (A), guanosine (G), cytosine (C) and uridine (U). Within the region of the molecule that codes for a protein (the coding region), the sequence of these four nucleotides forms a language made up of three-letter words called codons. The first codon, or start codon (AUG), signals where the ribosome should start protein synthesis. To know what protein to make, the ribosome then progresses along the mRNA one codon at a time, appending the appropriate amino acid to the growing protein. To end protein synthesis, three different codons (UAA, UAG, and UGA) serve as stop signals, telling the ribosome where to terminate protein synthesis. In total, there are 64 potential codons, but only 20 amino acids that are used to build proteins; therefore, multiple codons can encode for the same amino acid.

The process of protein production is called translation because the ribosome is reading in one language (a sequence of codons) and outputting in another language (a sequence of amino acids). The coding region is analogous to a sentence in English. Much like a start codon, a capitalized word can indicate the start of a sentence. Codons within the coding region resemble groups of letters representing words. The end of the sentence is signaled by a period in English, or a stop codon for mRNA.



In every cell, hundreds of thousands of mRNAs make hundreds of millions of proteins every day. A typical protein contains 200-600 amino acids; therefore, a typical mRNA coding region ranges from 600-1,800 nucleotides. In addition to the coding region, mRNAs contain four other key features: (1) the 5’ untranslated region (5’-UTR); (2) the 3’ untranslated region (3’-UTR); (3) the 5’ cap; and (4) a 3’ polyadenosine (poly-A) tail. The sequence of nucleotides in the 5’-UTR influences how efficiently the ribosome initiates protein synthesis, whereas the sequence of nucleotides in the 3’-UTR contains information about which cell types should translate that mRNA and how long the mRNA should last. The 5’ cap and 3’ poly-A tail enhance ribosome engagement and protect the mRNA from attack by intracellular enzymes that digest mRNA from its ends.

The intrinsic advantages of using mRNA as a medicine

mRNA possesses inherent characteristics that we believe position it to have a profound impact on human health:

- **mRNA is used by every cell to produce all proteins:** mRNA is used to make every type of protein, including secreted, membrane and intracellular proteins, in varying quantities over time, in different locations and in various combinations. Given the universal role of mRNA in protein production, we believe that mRNA medicines could have broad applicability across human disease.
- **Making proteins inside one's own cells mimics human biology:** Tailored mRNA can be sent into cells to instruct them to produce specific protein therapeutics or vaccine antigens and provides certain advantages over traditional approaches to medicine, where a protein or chemical is introduced to the body.
- **mRNA has a simple and flexible chemical structure:** Each mRNA molecule comprises four chemically similar nucleotides to encode proteins made from up to 20 chemically different amino acids. To make the full diversity of possible proteins, only simple sequence changes are required in mRNA, instead of starting from scratch for each new vaccine or therapy.
- **mRNA has classic pharmacologic features:** mRNA possesses many of the attractive pharmacologic features of most modern medicines, including reproducible activity, predictable potency and well-behaved dose dependency; mRNA also provides the ability to adjust dosing based on an individual patient's needs, including stopping or lowering the dose, to seek to promote safety and tolerability.

Our success in developing, manufacturing and commercializing mRNA medicines demonstrates the potential of our platform to help people and patients in far-reaching ways that could exceed the impact of traditional approaches to medicine.

We believe that the main advantages of mRNA as compared to traditional medicine are:

1. **mRNA could create an unprecedented abundance and diversity of medicines.** mRNA's breadth of applicability has the potential to create an extraordinary number of new mRNA medicines that are currently beyond the reach of recombinant protein technology.

2. **Advances in the development of our mRNA medicines reduce risks across our portfolio.** mRNA medicines share fundamental features that can be leveraged across our portfolio. We believe that once safety and proof of protein production has been established in one program, the technology and biology risks of related programs that use similar mRNA technologies, delivery technologies and manufacturing processes will decrease significantly.

3. **mRNA technology can accelerate discovery and development.** The software-like features of mRNA enable rapid *in silico* design and the use of automated high-throughput synthesis processes that permit discovery to proceed in parallel rather than sequentially. We believe these mRNA features can also accelerate drug development by allowing the use of shared manufacturing processes and infrastructure.

4. **The ability to leverage shared processes and infrastructure can drive significant capital efficiency over time.** We believe the manufacturing requirements of different mRNA medicines are similar and that at commercial scale, a portfolio of mRNA medicines will benefit from shared capital expenditures.

OUR STRATEGY

We believe that the development of mRNA medicines represents a significant breakthrough for patients, our industry and human health globally. We are currently focused on four strategic priorities:

1. **Deliver sales growth.** Our commercial growth drivers include geographic expansion and new product launches. In 2026, we expect to drive revenue growth from the annualized impact of our long-term partnerships in the UK, Canada and Australia, as well as continued strong uptake of mNEXSPIKE in the U.S. In addition, we expect multiple growth opportunities in 2027 and 2028.

2. **Deliver cost efficiency across the business.** Throughout 2025, we maintained disciplined cost management, improving productivity across manufacturing, R&D and SG&A. We expect to further reduce costs in 2026 and 2027. We plan to leverage our global production network, artificial intelligence (AI) and digital tools to improve cost efficiency.

3. **Execute on our prioritized pipeline.** We anticipate pivotal trial data readouts in 2026 across our oncology, rare disease and infectious disease portfolios. We expect to launch several new infectious disease products over the next few years (flu,

flu+COVID combination and Norovirus), which would expand our infectious disease vaccine franchise to as many as six approved products. We expect to invest the cash generated from these products into oncology and rare disease therapeutics.

4. **Continue to advance our early pipeline and platform technology.** We continue to advance our early-stage pipeline. This includes our early-stage oncology programs, which expand our oncology portfolio across cancer antigen therapies, T-cell engagers and cell-therapy enhancers, as well as multiple early-stage vaccine programs.

OUR PLATFORM

Overview of our platform

Our mRNA "platform" refers to our accumulated knowledge and capabilities in basic and applied sciences. Our platform incorporates advances across three key components—mRNA, delivery and the manufacturing process— to advance our medicines. We integrate these components and combine different versions of mRNA delivery and process into each of our medicines.

Our platform: mRNA science advancements

We continue to invest in both basic and applied research, seeking to advance both the state of our technology and the state of the scientific community's understanding of mRNA. Examples of advances in mRNA science that combine nucleotide chemistry, sequence engineering and targeting elements are described below.

mRNA chemistry: Modified nucleotides to mitigate immune system activation: The innate immune system has evolved to protect cells from foreign RNA, such as viral RNA, by inducing inflammation and suppressing mRNA translation once detected. Many cells surveil their environment through sensors called toll-like-receptors (TLRs). These include types that are activated by the presence of double-stranded RNA (TLR3) or uridine containing RNA fragments (TLR7, TLR8). Additionally, all cells have cytosolic double-stranded RNA, sensors, including retinoic acid inducible gene-I (RIG-I) that are sensitive to foreign RNA inside the cell.

The immune and cellular response to mRNA is complex, context specific, and often linked to the sensing of uridine. To minimize undesired immune responses to our potential mRNA medicines, our platform employs chemically-modified uridine nucleotides to minimize recognition by both immune cell sensors such as TLR3/7/8, and broadly-distributed cytosolic receptors such as RIG-I.

mRNA sequence engineering: Maximizing protein expression: mRNA exists transiently in the cytoplasm, during which time it can be translated into thousands of proteins before eventually being degraded. Our platform applies bioinformatic, biochemical, and biological screening capabilities, most of which have been invented internally that aim to optimize the amount of protein produced per mRNA. We have identified proprietary sequences for the 5'-UTR that have been observed to increase the likelihood that a ribosome bound to the 5'-end of the mRNA transcript will find the desired start codon and reliably initiate translation of the coding region. We additionally design the nucleotide sequence of the coding region to maximize its successful translation into protein.

Targeting elements: Enabling tissue-targeted translation: All nucleated cells in the body are capable of translating mRNA, resulting in pharmacologic activity in any cell in which mRNA is delivered and translated. To minimize or prevent potential off-target effects, our platform employs technologies that regulate mRNA translation in select cell types. Cells often contain short RNA sequences, called microRNAs or miRNAs, that bind to mRNA to regulate protein translation at the mRNA level. Different cell types have different concentrations of specific microRNAs, in effect giving cells a microRNA signature. microRNA binding directly to mRNA effectively silences or reduces mRNA translation and promotes mRNA degradation. We design microRNA binding sites into the 3'-UTR of our potential mRNA medicines so that if our mRNA is delivered to cells with such microRNAs, it will be minimally translated and rapidly degraded.

Our platform: Delivery science

Our mRNA can, in specific instances, be delivered by direct injection to a tissue in a simple saline formulation without lipid nanoparticles (LNPs) to locally produce small amounts of pharmacologically active protein. However, the blood and interstitial fluids in humans contain significant RNA degrading enzymes that rapidly degrade any extracellular mRNA and prevent broader distribution without LNPs. Additionally, cell membranes tend to act as a significant barrier to entry of large, negatively-charged molecules such as mRNA. We have therefore invested heavily in delivery science and have developed LNP technologies to enable delivery of larger quantities of mRNA to target tissues.

LNPs are generally composed of four components: an amino lipid, a phospholipid, cholesterol, and a pegylated-lipid (PEG-lipid). Each component, as well as the overall composition, or mix of components, contributes to the properties of each LNP system. LNPs

containing mRNA injected into the body rapidly bind proteins that can drive uptake of LNPs into cells. Once internalized in endosomes within cells, the LNPs are designed to escape the endosome and release their mRNA cargo into the cell cytoplasm, where the mRNA can be translated to make a protein and have the desired therapeutic effect. Any mRNA and LNP components that do not escape the endosome are typically delivered to lysosomes where they are degraded by the natural process of cellular digestion.

Examples of tools we developed by using our platform include proprietary LNP formulations that address the steps of mRNA delivery, including cell uptake, endosomal escape, and subsequent lipid metabolism, and for avoidance of counterproductive interactions with the immune system.

Chemistry: Novel lipid chemistry to potentially improve safety and tolerability: Our proprietary LNP systems are designed to be highly tolerated and minimize any LNP vehicle-related toxicities with repeat administration *in vivo*. To overcome limitations of previous LNP formulations, we have engineered amino lipids to avoid the immune system and to be rapidly biodegradable relative to prior lipids.

Composition: Proprietary LNPs enhance delivery efficiency: Our platform includes extensive in-house expertise in medicinal chemistry, which we have applied to design large libraries of novel lipids. Using these libraries in combination with our discovery biology capabilities, we have conducted high throughput screens for desired LNP properties and believe that we have made fundamental discoveries in preclinical studies about the relationships between structural motifs of lipids and LNP performance for protein expression.

Surface properties: Novel LNP design to avoid immune recognition: We have designed our proprietary LNP systems for sustained pharmacology upon repeat dosing by eliminating or altering features that activate the immune system. These are based on insights into the surface properties of LNPs. Upon repeated dosing, surface features on traditional LNPs such as amino lipids, phospholipids, and PEG-lipids, can be recognized by the immune system, leading to rapid clearance from the bloodstream, a decrease in potency upon repeat dosing, and an increase in inflammation. Based on our insights into these mechanisms, we have engineered our LNP systems to reduce or eliminate undesirable surface features. In clinical studies for our systemic therapeutic product candidates that use our novel LNP systems, we have been able to repeat dose with negligible or undetectable loss in potency, liver damage, and immune system activation.

Our platform: Manufacturing process science

We invest significantly in manufacturing process science to impart more potent features to our mRNA and LNPs, and to invent the technological capabilities necessary to manufacture our mRNA medicines at scales ranging from micrograms to kilograms, as well as achieve pharmaceutical properties such as solubility and shelf life. We view developing these goals of manufacturing and pharmaceutical properties as appropriate for each program, based on its stage of development.

mRNA manufacturing process: Improving pharmacology: Our platform creates mRNA using a cell-free approach called *in vitro* transcription in which an RNA polymerase enzyme binds to and transcribes a DNA template, adding the nucleotides encoded by the DNA to the growing RNA strand. Following transcription, we employ proprietary purification techniques to ensure that our mRNA is free from undesired synthesis components and impurities that could activate the immune system in an indiscriminate manner. Applying our understanding of the basic science underlying each step in the manufacturing process, we have designed proprietary manufacturing processes to impart desirable pharmacologic features, for example increasing potency in a vaccine.

LNP manufacturing process: Improving pharmacology: Our platform technology includes synthetic processes to produce LNPs. Traditionally LNPs are assembled by dissolving the four molecular components, amino lipid, phospholipid, cholesterol, and PEG-lipid, in ethanol and then mixing this with mRNA in an aqueous buffer. The resulting mixture is then purified to isolate LNPs from impurities. Such impurities include molecular components that have not been incorporated into particles, un-encapsulated mRNA that could activate the immune system, and particles outside of the desired size range. Going beyond optimization of traditional manufacturing processes, we have invested in understanding and measuring the various biochemical and physical interactions during LNP assembly and purification. We have additionally developed state-of-the-art analytical techniques necessary to characterize our LNPs and biological systems to analyze their *in vitro* and *in vivo* performance. With these insights, we have identified manufacturing process parameters that drive LNP performance, for example, the potency in a secreted therapeutic setting. These insights have allowed us to make significant improvements in the efficiency of our processing and the potency of our LNPs.

Harnessing the power of mRNA through modalities

Within our platform, we invest in science to invent novel ways to deliver mRNA into various cell types. Each novel delivery system is a new application, called a “modality.” While the programs within a modality may target diverse diseases, they share similar mRNA characteristics and manufacturing processes to achieve shared product features.

We believe that the high technological correlation within a modality allows us to rapidly accelerate the expansion of programs within that modality based on learnings from the earlier programs, while the lower technology correlation between modalities allows us to compartmentalize the technology risks. Additionally, because programs within a modality pursue diverse diseases, they often have

uncorrelated biology risk. New modalities and product candidates can create a network effect by helping us gain additional insight into the other programs in our pipeline.

Although developing a new modality is difficult, time-consuming and expensive, we believe our experience and technology provide us with unique advantages in the development of mRNA medicines. Over the last decade, we have developed a number of modalities, each with one or many product candidates in the clinic. We believe that our ongoing investments in our platform will lead to the identification of additional modalities and expand the utility of our existing modalities and the diversity of our pipeline.

OUR PIPELINE

Over the last decade, we have advanced in parallel a diverse development pipeline that currently consists of 35 therapeutic and vaccine programs, 6 of which are in late-stage development. The scope of our pipeline reflects the breadth of biology addressable using mRNA technology, and spans three franchises: infectious disease vaccines, oncology therapeutics, and rare disease therapeutics.

Our selection process for advancing new product candidates reflects both program-specific and portfolio-wide considerations. Program-specific criteria include, among other relevant factors, the severity of the unmet medical need, the biology risk of our chosen target or disease, the feasibility of clinical development, the costs of development and the commercial opportunity. Portfolio-wide considerations include the ability to demonstrate technical success for our platform components within a modality, thereby increasing the probability of success and learnings for subsequent programs.

Our full pipeline is shown in the figure below:



INFECTIOUS DISEASE FRANCHISE

Respiratory Vaccines

We have three commercial respiratory vaccines—Spikevax and mNEXSPIKE (our COVID vaccines) and mRESVIA (our RSV vaccine for older adults and high-risk adults ages 18-59). Additionally, we have regulatory filings under review for our seasonal flu+COVID vaccine (mRNA-1083) in Europe and Canada and for our seasonal flu vaccine (mRNA-1010) in the United States, Europe, Canada and Australia. Our current respiratory programs are summarized below.

COVID vaccines (Spikevax/mRNA-1273, mNEXSPIKE/mRNA-1283)

COVID-19 is caused by the SARS-CoV-2 virus that was first identified in humans in 2019, driving a global pandemic resulting in millions of deaths. The risk of mortality increases with age, and the risk of severe disease and mortality is higher among individuals with certain pre-existing conditions, including cardiovascular disease, diabetes, chronic lung disease and obesity. As the SARS-CoV-2 virus continues to evolve, our COVID vaccines continue to be key tools in fighting COVID-19.

We currently have two approved COVID vaccines: Spikevax (our original COVID vaccine) and mNEXSPIKE. mNEXSPIKE was approved by the FDA in May 2025 for individuals 65 years of age and older, and individuals 12 through 64 years of age who are at high risk for severe COVID-19. mNEXSPIKE focuses immune responses to the domains of the SARS-CoV-2 spike protein that are critical for neutralizing antibody and T cell responses. mNEXSPIKE's mRNA dose is one-fifth that of Spikevax. In April 2025, we shared Phase 3 data showing non-inferior vaccine efficacy relative to Spikevax, and a 13.5% relative vaccine efficacy (rVE) compared to Spikevax in participants 65 years of age and older.

As part of our strategy to continue to combat COVID-19, we develop and assess variant-specific versions of our COVID vaccines. We pursue updated vaccine compositions based on guidance from the FDA, and other regulatory bodies including European Medicines Agency (EMA) and the World Health Organization (WHO), with the goal of broadening vaccine-induced immunity and providing protection against circulating SARS-CoV-2 variants. Our 2025-2026 formulas for Spikevax and mNEXSPIKE target the LP.8.1 variant of SARS-CoV-2. The FDA has approved our 2025-2026 Spikevax formula for high-risk individuals aged six months through 64 years and all adults 65 years of age and older. We have received approval of our 2025-2026 Spikevax formula in 40 countries. The FDA has approved our 2025-2026 mNEXSPIKE formula for high-risk individuals 12 through 64 years of age and all adults 65 years of age and older. We have also received approvals of mNEXSPIKE in Europe, Canada and Australia.

RSV vaccine (mRESVIA/mRNA-1345)

RSV is one of the most common causes of lower respiratory disease in children under the age of five and in older adults. Populations that are especially vulnerable to developing severe RSV infections include infants, young children, children and adults with chronic medical conditions and older adults. Most children are infected at least once by age two. In the United States, it is estimated that up to 80,000 children are hospitalized due to RSV infection annually. RSV infection causes up to 160,000 hospitalizations and up to 10,000 deaths per year in adults aged 65 years or older in the United States. Adults 18 to 59 years of age with certain comorbidities also face a significant disease burden that is similar to that of older adults.

We have developed an RSV vaccine (mRNA-1345 or mRESVIA) for adults. mRESVIA encodes an engineered form of the RSV F protein stabilized in the prefusion conformation and is formulated in our proprietary LNP. In May 2024, we announced the FDA approved mRNA-1345, brand name mRESVIA, for the prevention of RSV-associated lower respiratory tract disease (RSV-LRTD) in adults 60 years or older. Subsequently, the Advisory Committee on Immunization Practices (ACIP) issued a recommendation for all unvaccinated people aged 75 years and older and unvaccinated people aged 60 to 74 who are at increased risk for RSV. In June 2025, we received FDA approval for use of mRESVIA in adults ages 18-59 who are at increased risk for RSV-LRTD. In April 2025, the ACIP expanded its RSV vaccination recommendation to include high-risk individuals ages 50-59. mRESVIA was approved in the EU in August 2024 and in Canada in November 2024. In Canada, mRESVIA is recommended for individuals 75 and over and high-risk individuals ages 50-74, while in Europe, the vaccine is recommended for individuals 60 and over as well as high-risk individuals ages 18-59. mRESVIA has been approved in 40 countries for all adults 60 years and older and also approved in 31 of those countries for high-risk adults aged 18-59.

We have completed additional Phase 3 studies in adults that have shown that mRNA-1345 restores immune response when revaccinated at 12 or 24 months, can be co-administered with a COVID mRNA vaccine, and is immunogenic in the solid organ transplant population.

Seasonal influenza vaccine (mRNA-1010)

The WHO estimates that seasonal influenza viruses cause three to five million cases of severe illness and 290,000 to 650,000 deaths each year, resulting in a severe challenge to public health. Currently licensed seasonal influenza vaccines rarely exceed 60% overall effectiveness and can provide low effectiveness during years when the circulating viruses do not match the strains selected for the vaccine antigens.

Our seasonal influenza vaccine candidate (mRNA-1010) encodes for the hemagglutinin (HA) proteins of the strains recommended by the WHO. We aim to work with the WHO, regulators and public health authorities to enable strain selection closer to the influenza season to provide a better match to the circulating viruses.

In February 2023, we announced interim results from the P301 study of mRNA-1010. The results indicated that mRNA-1010 achieved higher seroconversion rates for A/H3N2 and A/H1N1, as well as superiority on geometric mean titer ratios for A/H3N2 and non-inferiority on geometric mean titer rations for A/H1N1. Non-inferiority was not met for either endpoints for the influenza B/Victoria- or B/Yamagata-lineage strains. mRNA-1010 showed an acceptable safety and tolerability profile. In April 2023, we announced the P302 study of mRNA-1010 did not accrue sufficient cases at the interim efficacy analysis to declare early success in the Phase 3 Northern Hemisphere efficacy trial. In September 2023, we announced the P303 immunogenicity and safety study of mRNA-1010 met all eight co-primary endpoints with an updated composition that was able to generate an improved immune response to influenza B strains. mRNA-1010 also elicited higher titers than the licensed comparator against all strains in this study. In September 2024, we shared results in an older adult extension study of P303, where mRNA-1010 met all primary immunogenicity endpoints, including superiority for all strains, compared to a licensed enhanced flu vaccine and showed an acceptable reactogenicity profile. Consequently, we announced that we were planning to start a confirmatory vaccine efficacy study, P304, funded by project financing through Blackstone Life Sciences, a collaboration we announced in 2024. In October 2025, we presented the results of this study showing a 26.6% relative vaccine efficacy versus the standard dose comparator in adults 50 year of age and older.

mRNA-1010 has been filed for approval in the United States, Europe, Canada and Australia. In February 2026, in response to a prior Refusal-to-File letter, we engaged with the FDA in a Type A meeting and submitted an amended BLA outlining a revised regulatory pathway based on age, seeking full approval for adults 50 to 64 years of age and accelerated approval for adults 65 and older, along with a post-marketing requirement to conduct an additional study in older adults. Following the meeting and submission of the amended application, the FDA accepted our BLA for review and assigned a PDUFA goal date of August 5, 2026.

We have paused development of mRNA-1011, mRNA-1012, mRNA-1020, and mRNA-1030, which were flu vaccines that included additional antigens relative to what is included in mRNA-1010.

Combination vaccines (mRNA-1083 and mRNA-1365)

We are developing combination vaccine candidates to simplify and facilitate protection against a range of respiratory diseases.

Our COVID and seasonal influenza combination vaccine (mRNA-1083), encodes the same antigens as our seasonal influenza vaccine (mRNA-1010) and mNEXSPIKE. In June 2024, we announced that our Phase 3 clinical trial of mRNA-1083 met its primary endpoints, eliciting higher immune responses against influenza virus and SARS-CoV-2 than licensed flu and COVID vaccines in adults 50 years and older, including an enhanced influenza vaccine in adults 65 years and older. We filed for FDA approval for mRNA-1083 in November 2024 and also submitted regulatory applications in other geographies, including the EU, Canada, Australia and the UK. Demonstration of vaccine efficacy in our ongoing Phase 3 mRNA-1010 flu study was required by regulatory authorities in certain jurisdictions to support approval of mRNA-1083. As a result, we withdrew our application for mRNA-1083 in certain geographies until the mRNA-1010 Phase 3 study was completed. In October 2025, we reported full results of our mRNA-1010 vaccine efficacy study. mRNA-1083 is under regulatory review in Europe and Canada, and we are awaiting further guidance from the FDA for re-filing in the U.S. In addition, we plan to pursue regulatory submissions for mRNA-1083 in selected additional international markets.

We continue to evaluate the clinical development strategy for mRNA-1083, including the appropriate age populations and indications for future studies.

In addition, safety follow-up continues in our Phase 1 study for mRNA-1365, our RSV and human metapneumovirus (hMPV) combination vaccine, in children five to 23 months of age. We are also evaluating an updated version of mRNA-1365 in older adults.

Pandemic influenza vaccine (mRNA-1018)

An influenza pandemic is a global outbreak of a new influenza A virus that is very different from recently circulating human seasonal influenza A viruses. Historically, pandemic strains have arisen by antigenic shift, which is a major change in an influenza A virus caused by the exchange of genetic segments between a non-human influenza virus with another influenza virus; this can occur through a simultaneous infection of an animal (e.g., swine) or humans with multiple influenza viruses.

We are developing a pandemic influenza vaccine candidate that encodes for hemagglutinin (HA) glycoproteins. In 2023, we initiated a Phase 1/2 study to generate safety and immunogenicity data of our investigational pandemic influenza vaccine (mRNA-1018) in healthy adults 18 years of age and older. The study included vaccine candidates against H5 and H7 avian influenza viruses. In October 2025, we presented results of this study showing that mRNA-1018 demonstrated a rapid, potent and durable immune response. At Day 43, three weeks after the second vaccination, 97.8% of participants achieved hemagglutination inhibition (HAI) antibody titers ≥1:40, an HAI titer considered to correlate with protection.

In December 2025, we announced that the Coalition for Epidemic Preparedness Innovations (CEPI) will invest up to $54.3 million to support a pivotal Phase 3 clinical trial that aims to help advance our investigational mRNA-based H5 pandemic influenza vaccine candidate, mRNA-1018, to licensure. The Phase 3 trial is expected to begin in early 2026.

Latent and Other Vaccines

Our current latent and other vaccines programs are summarized below.

Vaccines against latent viruses

CMV vaccine (mRNA-1647)

Cytomegalovirus (CMV) is member of the herpes virus family and a common human pathogen that causes complications for people who are immunosuppressed or can be passed to babies during pregnancy and result in congenital malformation.

Our CMV vaccine candidate, mRNA-1647, combines six mRNAs in one vaccine. These six mRNAs encode proteins located on the surface of CMV: five mRNAs encode the subunits that form the membrane-bound pentamer complex and one mRNA encodes the full-length membrane-bound glycoprotein B (gB). Both pentamer and gB are essential for CMV to infect barrier epithelial surfaces and gain access to the body, which is the first step in CMV infection. mRNA-1647 is designed to produce an immune response against both pentamer and gB for the prevention of CMV infection.

Congenital CMV (cCMV) results when infected mothers transmit the virus to their unborn child and it is the leading infectious cause of birth defects in the United States. We conducted a cCMV Phase 3 study of mRNA-1647 in women of childbearing age and announced in October 2025 that the study failed to reach its primary clinical endpoint. The program in women of childbearing age was discontinued.

In addition to the health burden of cCMV infection, CMV is a major health risk in the transplant population. There are approximately 47,000 solid organ transplants and 23,000 hematopoietic stem cell transplants in the U.S. annually. Each recipient faces a significant risk of CMV infection post-transplant, and this could result in graft rejection or end-organ CMV disease. mRNA-1647 has the potential to prevent CMV viral replication and/or disease in transplant populations. A Phase 2 proof-of-concept study for mRNA-1647 in allogeneic hematopoietic cell transplant (HCT) patients is ongoing. In interim data from our Phase 2 study, we reported robust CMV-specific T-Cell responses, and we expect to present full Phase 2 data after the study’s conclusion.

EBV vaccines (mRNA-1189 and mRNA-1195)

Epstein-Barr virus (EBV) is a member of the herpesvirus family that infects approximately 90% of people in the U.S. by adulthood, with primary infection typically occurring during childhood or late adolescence (approximately 50% and 89% seropositivity, respectively). EBV is the major cause of infectious mononucleosis, accounting for over 90% of the cases in the U.S. each year. Infectious mononucleosis can debilitate patients for weeks to months and, in some cases, can lead to hospitalization due to complications such as splenic rupture. EBV infection is also associated with the development and progression of certain lymphoproliferative disorders, cancers and autoimmune diseases. In particular, EBV infection and infectious mononucleosis are associated with increased risk of developing multiple sclerosis (MS), an autoimmune disease of the central nervous system. MS is the most common neurodegenerative disorder of the central nervous system, affecting approximately 2.8 million individuals worldwide of which approximately 1 million live in the U.S. MS leads to progressive disability, with profound socioeconomic impact on the patients, caregivers and the healthcare system.

We are developing two mRNA candidates against EBV —a prophylactic vaccine to prevent infectious mononucleosis (mRNA-1189) and a therapeutic to treat multiple sclerosis (mRNA-1195). We believe that an effective EBV vaccine must generate an immune response against antigens that are required for viral entry and reactivation in susceptible cell types. mRNA-1189 is designed to elicit an immune response to EBV envelope glycoproteins, which are required for infection of both epithelial and B cells. mRNA-1195 encodes for entry glycoproteins and latent antigens aimed at inducing an antibody and T cell response, and will be investigated in the context of multiple sclerosis and post-transplant lymphoproliferative disorder.

We are currently conducting Phase 2, randomized, observer-blind, placebo-controlled studies of mRNA-1189 and mRNA-1195. The primary purpose of these studies is to assess the safety, tolerability and immunogenicity of these candidates, and to enable dose selection for further clinical development. In March 2024, we shared Phase 1 data for mRNA-1189, where the randomized, observer-blind, placebo-controlled study showed mRNA-1189 was immunogenic and generally well tolerated across all dose levels. We have since fully enrolled a dose finding Phase 2 study for mRNA-1189. Our Phase 1 study for mRNA-1195 is fully enrolled, and in November 2025 we shared data that showed mRNA-1195 was immunogenic and generally well tolerated across all dose levels. Our mRNA-1195 Phase 2 proof of concept trial in MS is ongoing, with an initial sentinel cohort fully enrolled as of December 2025.

HIV vaccine (mRNA-1645)

Human immunodeficiency virus (HIV) is responsible for acquired immunodeficiency syndrome (AIDS), a lifelong, progressive illness with no effective cure. Approximately 38 million people worldwide are currently living with HIV, with 1.2 million in the U.S. Approximately 1.5 million new infections of HIV are acquired worldwide every year and approximately 680,000 people die annually due to complications from HIV/AIDS. The primary routes of transmission are sexual intercourse and IV drug use, putting young adults at the highest risk of infection. From 2000 to 2015, a total of $562.6 billion globally was spent on care, treatment and prevention of HIV, representing a significant economic burden.

Our HIV vaccine candidate (mRNA-1645) is being studied in Phase 1 clinical trials, sponsored by IAVI, to determine the optimal design to induce broadly neutralizing antibodies capable of addressing the genetic diversity of HIV that is necessary in future proof of concept studies. In collaboration with the Gates Foundation, International AIDS Vaccine Initiative (IAVI), The Scripps Research Institute and the National Institutes of Health (NIH), Phase 1 studies of previous versions of our HIV vaccine have been completed.

Vaccines against enteric viruses

Norovirus vaccines (mRNA-1403 and mRNA-1405)

Norovirus is a leading cause of acute gastroenteritis (AGE), responsible for approximately 18% of all AGE cases worldwide and imposing a substantial global health care burden. Annually, norovirus causes an estimated 685 million illnesses and 200,000 deaths, including more than 50,000 deaths in children under the age of five. While the incidence of norovirus AGE is highest among children under five years of age, disease severity is most pronounced in infants, older adults, and individuals with underlying medical conditions. In high-income countries such as the United States, approximately 90% of norovirus-associated deaths occur among older adults.

Norovirus is highly infectious and difficult to control due to its environmental stability and persistence on contaminated surfaces, frequently leading to large, disruptive, and costly outbreaks in closed or semi-closed settings such as daycare centers, schools, cruise ships, long term care facilities and healthcare institutions. Globally, the economic impact of norovirus is estimated at $60 billion per year. In the United States alone, norovirus causes an estimated 20 million infections, 100,000 hospitalizations and 900 deaths each year,

with associated costs of approximately $2 billion. As populations age and reliance on institutionalized care increases, the burden of norovirus disease among older adults is expected to rise further.

Norovirus exhibits broad genetic diversity, with 10 genogroups and 49 genotypes identified to date, 30 of which infect humans. Vaccine development has been challenging to date for many reasons, including the lack of a robust cell culture system, no reliable immune markers of norovirus protection and the broad and shifting diversity of genotypes. A multivalent vaccine with broad genotype coverage is needed to maximize protection against the genotypes most frequently associated with AGE in older adults and young children.

We are currently developing pentavalent (mRNA-1405) and trivalent (mRNA-1403) vaccine candidates for norovirus. Both candidates were reviewed in a Phase 1 study to evaluate safety, reactogenicity and immunogenicity in healthy adult participants 18 to 49 years of age and 60 to 80 years of age. In March 2024, we presented data demonstrating that a single dose of trivalent vaccine candidate mRNA-1403 was well-tolerated across all dose levels evaluated and elicited robust antibody responses against vaccine-matched norovirus genogroup I & II selected genotypes, including functional histo-blood group antigen (HBGA) blocking antibodies. In September 2024, we presented additional data that mRNA-1403 elicited robust HBGA blocking antibody titers against a second GII genotype in older and younger adults.

In September 2024, we began dosing participants in a Phase 3 study of mRNA-1403. This Phase 3 trial is a randomized, observer-blind, placebo-controlled trial evaluating the efficacy, safety and immunogenicity of mRNA-1403. The trial has enrolled approximately 38,000 participants 18 years of age and older globally, across countries in the Northern Hemisphere (U.S., Canada, UK and Japan), and Panama and Australia. Approximately 33,000 participants 60 years of age and older and approximately 5,000 participants between 18 and 59 years of age have been enrolled to assess the ability of mRNA-1403 to protect against vaccine genotype-matched moderate to severe norovirus AGE, with a focus on the older adult age group that is at greatest risk of severe outcomes including hospitalization and death.

The Phase 3 study of mRNA-1403 completed its first seasons in the Northern and Southern Hemispheres. A second Northern Hemisphere season cohort is fully enrolled and the study is ongoing.

Bacterial vaccines

Lyme vaccines (mRNA-1975 and mRNA-1982)

Lyme disease is an illness caused by Borrelia bacteria transmitted by the bite of infected ticks. Lyme disease affects approximately 675,000 people annually in the U.S. and Europe, with incidence increasing due to expanding tick territories driven by rising temperatures. The disease burden follows a bimodal age distribution, predominantly impacting children under 15 and older adults. Symptoms include rash, fever, headaches, fatigue, joint pain, swelling, stiffness, and headaches. While no vaccines are currently approved, prior Phase 3 trials of outer surface protein A (OspA) antigen-based vaccines achieved efficacy up to 92%, with protection linked to high levels of antigen-specific antibodies.

We are conducting a randomized, observer-blind, placebo-controlled, dose-ranging Phase 1/2 trial in healthy participants aged 18 to 70. This trial evaluates the safety and immunogenicity of a heptavalent (mRNA-1975) and monovalent (mRNA-1982) approach in parallel. mRNA-1982 targets Borrelia burgdorferi, responsible for nearly all Lyme cases in North America, while mRNA-1975 targets the four major Borrelia species causing disease in North America and Europe.

No safety concerns have been identified across the evaluated dose levels for three injections of mRNA-1975/1982. Furthermore, three injections of mRNA-1975/1982 elicit robust anti-OspA IgG antibody responses, with titers up to ~1,300 times above baseline for OspA serotype 1 of Borrelia burgdorferi. The program will progress to a Phase 2 dose ranging study of mRNA-1982.

Public health vaccines

Nipah vaccine (mRNA-1215)

Nipah virus (NiV) is a zoonotic virus transmitted to humans from animals, contaminated food or through direct human-to-human transmission and causes a range of illnesses including fatal encephalitis. Severe respiratory and neurologic complications from NiV have no treatment other than intensive supportive care. The case fatality rate among those infected is estimated at 40-75%. NiV outbreaks cause significant economic burden to impacted regions due to loss of human life and interventions to prevent further spread, such as the slaughter of infected animals. NiV has been identified as the cause of isolated outbreaks in India, Bangladesh, Malaysia and Singapore since 2000 and is included on the WHO R&D Blueprint list of epidemic threats needing urgent research and development action.

In collaboration with the NIH-Vaccine Research Center (VRC), we conducted a Phase 1 clinical trial of mRNA-1215, our vaccine candidate against NiV, and testing was focused on pandemic preparedness. This Phase 1 dose-escalation, open-label clinical trial was the first study of mRNA-1215 in healthy adults to evaluate the safety, tolerability and immunogenicity of a NiV mRNA vaccine candidate. The trial was sponsored and funded by the National Institute of Allergy and Infectious Diseases (NIAID).

Mpox vaccine (mRNA-1769)

Mpox (formerly known as monkeypox) is an infectious disease caused by the mpox virus, a member of the Orthopoxvirus genus, which also includes Variola virus, the virus that caused smallpox. Although smallpox was eradicated in 1980, continued protection from smallpox is of great importance given the lethality of the infection and potential for use as an agent of bioterrorism. Other viruses associated with the Orthopoxvirus genus include cowpox, rabbitpox and camelpox.

There are two main types of the mpox virus, Clade I and Clade II, each with subtypes (a and b). Prior to 2022, mpox outbreaks were primarily sporadic, driven by zoonotic spillover in endemic regions. Mpox can infect humans and other animals and is spread mainly through close physical contact with an infected person or contaminated materials, as well as through contact with infected animals. Respiratory transmission via face-to-face interaction is also possible.

Clade II (subclade IIb) was responsible for the 2022 global mpox outbreak which led the WHO to declare a Public Health Emergency of International Concern (PHEIC). Transmission during the outbreak occurred primarily through sexual contact. In 2024, a second PHEIC was declared due to Clade I outbreak expansion in the Democratic Republic of the Congo and neighboring countries. Current epidemiology reports indicate that both Clade I and Clade II viruses continue to circulate globally, with recent expansions of Clade Ib transmission in some regions outside Africa, sometimes without direct travel links to endemic areas.

The incubation period for mpox typically ranges from 3 to 17 days. Initial symptoms include fever, headache, muscle aches, fatigue, and swollen lymph nodes, followed by a characteristic rash or skin lesions that may last 2-4 weeks. Most individuals recover fully, although severe disease can occur, particularly in people with weakened immune systems.

The current standard of care for mpox prevention is JYNNEOS, which is approved by the FDA for prevention of mpox and smallpox.

Our mpox vaccine candidate, mRNA-1769, is designed to express four antigens from the mpox virus. In preclinical studies using a stringent non-human primate model, vaccination with mRNA-1769 resulted in fewer lesions, reduced viral replication, and stronger neutralizing and functional antibody responses compared with existing treatments. The immune responses induced by mRNA-1769 also showed broad activity against several orthopoxviruses. These results were published in 2024.

We conducted a randomized, placebo-controlled, dose-ranging Phase 1/2 study to evaluate the safety, tolerability, and immune response of three dose levels of mRNA-1769 in healthy adult participants. Interim analysis showed that mRNA-1769 demonstrated a favorable safety profile and strong immunogenicity. These interim results were presented as an oral presentation at the 2025 ESCMID Global conference. The full study results are currently expected in 2026.

ONCOLOGY THERAPEUTICS FRANCHISE

Within our oncology therapeutics franchise, we are developing intismeran autogene (mRNA-4157) in collaboration with Merck. We and Merck are targeting a variety of tumor types, with eight Phase 2 and Phase 3 trials ongoing.

Separate from the collaboration with Merck, our Phase 1/2 study of mRNA-4359, an investigational wholly-owned cancer antigen therapy, is ongoing. We have also expanded our oncology portfolio with early-stage programs, including mRNA-4106 (a cancer antigen therapy), mRNA-2808 (a T-cell engager), and mRNA-4203 (a cell-therapy enhancer). mRNA-4203 is being developed in collaboration with Immatics.

Intismeran autogene (mRNA-4157)

As tumors grow, they acquire mutations which may lead to the creation of new protein segments (neoantigens) that can be presented on human leukocyte antigen (HLA) molecules in the tumor and recognized as non-self by T cells. While some neoantigens are shared across tumors, the majority are completely unique to an individual patient's tumor and their presentation is dependent on a patient's specific HLA type.

Intismeran autogene, mRNA-4157, uses next generation sequencing and a machine-learning based algorithm to design an mRNA that encodes up to 34 neoantigens against each individual patient's tumor mutations with specificity to their HLA type, and is predicted to elicit both class I (CD8+ T cells) and class II (CD4+ T cells) responses. The neoantigens are encoded in a single mRNA sequence and formulated in our proprietary LNPs designed for intramuscular injection. Intismeran autogene is manufactured using an automated workflow to enable a rapid turnaround time.

We are developing mRNA-4157 in collaboration with Merck. In September 2022, Merck exercised its option for personalized cancer vaccines (PCVs), including mRNA-4157, pursuant to the terms of our existing PCV Collaboration and License Agreement with Merck, which was amended and restated in 2018 (PCV Agreement, also referred to as the INT Agreement). Pursuant to the PCV Agreement, we and Merck will collaborate on further development and commercialization of mRNA-4157, and we will share costs and any profits and losses worldwide related to mRNA-4157 equally.

In December 2022, we announced that the randomized Phase 2 trial of mRNA-4157 had met its primary endpoint. The open-label Phase 2 study is investigating a 1 mg dose of mRNA-4157 in combination with Merck's KEYTRUDA, compared to pembrolizumab alone, for the adjuvant treatment of high-risk resected melanoma. The study showed that mRNA-4157 in combination with KEYTRUDA reduced the risk of recurrence or death by 44% (HR=0.56 [95% CI, 0.31-1.02]; one-sided p value=0.0266) compared with KEYTRUDA alone. The results were the first demonstration of efficacy for an investigational mRNA cancer treatment in a randomized clinical trial in melanoma. Adverse events observed were consistent with those previously reported in a Phase 1 clinical trial, which showed mRNA-4157 to be well-tolerated at all dose levels.

In February 2023, mRNA-4157 received a Breakthrough Therapy Designation from the FDA, and in April 2023, mRNA-4157 received PRIME Scheme Designation from the EMA.

In December 2023, we announced that at a planned median follow-up of approximately three years, mRNA-4157 in combination with KEYTRUDA showed sustained benefit, reducing the risk of recurrence or death by 49% (HR=0.510 [95% CI, 0.288-0.906]; one-sided nominal p=0.0095) and the risk of distant metastasis or death by 62% (HR=0.384 [95% CI, 0.172-0.858]; one-sided nominal p= 0.0077) compared to KEYTRUDA alone in stage III/IV melanoma patients with high risk of recurrence following complete resection. These data were presented at the American Society of Clinical Oncology (ASCO) in June 2024, as well as translational biomarker data that suggests mRNA-4157 may benefit a broad patient population, irrespective of the status of PD-L1, TMB, ctDNA, and HLA heterozygosity. Three-year exploratory endpoint data also showed an encouraging trend in overall survival (OS) with the combination versus pembrolizumab monotherapy.

In January 2026, we announced that at a planned median follow-up of approximately five years, mRNA-4157 in combination with KEYTRUDA showed continued sustained benefit, reducing the risk of recurrence or death by 49% (HR=0.510 [95% CI, 0.294-0.887]; one-sided nominal p=0.0075) compared to KEYTRUDA alone in stage III/IV melanoma patients with high risk of recurrence following complete resection.

We and Merck have expanded the intismeran autogene program, initiating Phase 3 studies in adjuvant melanoma, adjuvant non-small cell lung cancer (NSCLC), and adjuvant NSCLC post neoadjuvant treatment. We also have Phase 2 trials ongoing in adjuvant renal cell carcinoma (RCC), muscle-invasive bladder cancer (MIBC), non-muscle-invasive bladder cancer (NMIBC), metastatic melanoma, and first-line metastatic squamous NSCLC. In addition, we have an ongoing Phase 1 study in early and advanced solid tumors. Enrollment of the Phase 3 adjuvant melanoma study was completed in 2024 and enrollment of the Phase 2 RCC study was completed in 2025.

Cancer antigen therapy (mRNA-4359)

We are developing a cancer antigen therapy (mRNA-4359) that encodes Indoleamine 2,3-dioxygenase (IDO) and programmed death-ligand 1 (PD-L1) antigens. We designed mRNA-4359 with the goal of stimulating effector T cells that target and kill suppressive immune and tumor cells that express the target antigens. Our initial indications for mRNA-4359 are advanced or metastatic cutaneous melanoma and NSCLC.

We presented data from the Phase 1a study of mRNA-4359 at the European Society for Medical Oncology (ESMO) Congress in 2024, where in a population of patients with heavily pre-treated, advanced stage cancers, eight of 16 response-evaluable patients achieved a best overall response (BOR) of stable disease. Translational data showed antigen-specific T-cell responses were elicited by mRNA-4359 treatment; a proportion of activated, cytotoxic, and memory T cells were elevated and a proportion of regulatory T cells and myeloid-derived suppressor cells (MDSCs) were diminished on treatment. mRNA-4359 monotherapy was tolerable at all dose levels tested with most adverse events of low grade (grade 1–2) and manageable.

We presented data from the Phase 1b study at ESMO in 2025, where in a population of 29 checkpoint inhibitor resistant/refractory melanoma patients, across all evaluable patients, mRNA-4359 in combination with pembrolizumab, the objective response rate (ORR) was 24%, and the disease control rate, or the combination of patients achieving tumor response and stable disease, was 60%. mRNA also demonstrated a consistently manageable safety profile. Responses were enriched in patients with PD-L1 ≥1% tumor proportion score (TPS), with an ORR of 67%, indicating encouraging activity in this difficult to treat patient population. Translational analyses demonstrated biological activity of mRNA-4359, including induction of PD-L1– and IDO1-specific T-cell responses and novel clonal T-cell expansion in the periphery, consistent with the proposed mechanism of action.

Additional translational analyses presented at the Society for Immunotherapy of Cancer (SITC) 2025 Annual Meeting further demonstrated that the magnitude of immune activation, including T-cell clonal expansion, correlated with clinical outcomes and was accompanied by early and substantial reductions in circulating tumor DNA. Tumor-based analyses showed that responders had higher baseline gene expression of PD-L1 and IDO1 and exhibited on-treatment increases in immune-inflamed gene expression signatures.

The Phase 1/2 study of mRNA-4349 is ongoing. The Phase 2 portion of the study includes cohorts in first-line metastatic melanoma, second-line+ metastatic melanoma and first-line metastatic NSCLC

Cancer antigen therapy (mRNA-4106)

mRNA-4106 is a cancer antigen therapy offering broad coverage across tumor types. It encodes for multiple tumor targets and is designed to broaden coverage across and within patients. Our Phase 1 study commenced in 2025.

T-cell engager (mRNA-2808)

mRNA-2808 is a T-cell engager targeting surface antigens in multiple myeloma and is currently dosing in a Phase 1 / 2 clinical study.

Cell therapy enhancer (mRNA-4203 and Anzu-cel (IMA203))

mRNA-4203 is a cell therapy enhancer that encodes for the target of an ex-vivo cell therapy to enhance the persistence and efficacy of the cell therapy. mRNA-4203 is being developed in collaboration with Immatics and is dosed in combination with Immatics' IMA203. The Phase 1 study commenced in 2025.

RARE DISEASE FRANCHISE

Our current rare disease programs are summarized below.

Propionic acidemia (mRNA-3927)

Propionic acidemia (PA) is a rare, inherited metabolic disorder with significant morbidity and mortality, affecting one in 100,000-150,000 individuals worldwide. PA is caused by pathogenic variants in the propionyl-coenzyme A carboxylase (PCC) α or β subunits (PCCA and PCCB genes, respectively), leading to PCC deficiency and subsequent accumulation of toxic metabolites. PA is characterized by recurrent life-threatening metabolic decompensation events (MDEs) and multisystemic complications. Multisystemic complications include neurological manifestations, cardiomyopathy, arrythmias, growth retardation, recurrent pancreatitis, bone marrow suppression and predisposition to infection. Long-term, insults by toxic metabolites cause complications in various organs, and cognitive outcome is negatively correlated with the number of MDEs. Currently, there is no approved therapy for PA that targets the underlying root cause of the disease.

Our PA therapy candidate, mRNA-3927, is a novel, IV-administered, LNP-encapsulated dual mRNA therapy that encodes for PCCA and PCCB subunit proteins to restore functional PCC enzyme activity in the liver. By encoding for intracellular proteins, mRNA therapy has a potential role in preventing and treating acute metabolic decompensations.

We have received Rare Pediatric Disease Designation, Orphan Drug Designation and Fast Track Designation from the FDA and Orphan Designation from the European Commission for the PA program.

At the International Congress of Inborn Errors of Metabolism (ICIEM) in September 2025, we reported data from the ongoing global Phase 1 / 2 clinical trial for mRNA-3927 in patients one year and older with PA. In part one, 22 patients were dosed and mRNA-3927 was well tolerated in all doses administered. Treatment continued to demonstrate sustained reduction in MDEs, with benefit highest for patients receiving doses of 0.6 mg/kg or greater.

mRNA-3927 is in a registrational study and target enrollment has been reached. An infant dose-finding study is also ongoing.

Methylmalonic acidemia (mRNA-3705)

Methylmalonic acidemia (MMA) is a rare, inherited metabolic disorder with significant morbidity and mortality caused by a deficiency in an enzyme called methylmalonyl-CoA mutase (MUT). There are an estimated 500-2,000 people with MMA MUT deficiency in the United States based on estimated birth prevalence (0.3-1.2:100,000 newborns) and mortality rates. Mortality is significant, with rates of 50% for those with complete MUT deficiency (mut 0) (median age of death 2 years) and 40% for MMA patients with partial MUT deficiency (mut -) (median age of death 4.5 years) reported in a large European study. MMA mainly affects the pediatric population and usually presents in the first few days or weeks of life.

The occurrence of acute metabolic decompensations is the hallmark of the disorder, and decompensations are typically more frequent in the first few years of life. Each decompensation is life-threatening and often requires hospitalization and management at an intensive care unit. Survivors often suffer from numerous complications including chronic renal failure and neurologic complications such as movement disorders, developmental delays, and seizures. Consequently, the health-related quality of life for MMA patients and their families is significantly impaired. There are currently no approved therapies that address the underlying defect for MMA.

Our MMA therapy candidate, mRNA-3705, is an investigational, lipid nanoparticle-encapsulated therapy administered intravenously that codes for the human MUT (hMUT) enzyme and is hypothesized to restore normal MUT production.

In 2024, the FDA announced that it had selected our MMA therapy candidate, mRNA-3705, for the Support for Clinical trials Advancing Rare Disease Therapeutics (START) program.

In September 2025, we reported MMA data at the International Congress for Inborn Errors of Metabolism (ICIEM). In an ongoing Phase 1/2 study, eighteen participants have been dosed. All eligible participants opted to participate in the Open-Label Extension study. mRNA-3705 was generally well-tolerated. Initial data are encouraging, with reductions in methylmalonic acid and other pathway biomarkers. The registrational study is expected to begin in 2026.

Cystic Fibrosis (mRNA-3692/VX-522)

Cystic Fibrosis (CF) is a rare genetic disease, which is progressive from birth and leads to multi-organ damage and early death due to lung dysfunction. It is caused by the mutations in the cystic fibrosis transmembrane conductance regulator (CFTR) gene, which results in the loss of CFTR chloride ion channel function. This decreased function of CFTR at the cell surface leads to thick, sticky mucus in multiple organ systems but most pathologically the lungs. There are approximately 92,000 patients living with cystic fibrosis in the United States, Europe, Australia and Canada, with over 5,000 of these patients not being able to benefit from the approved CFTR modulators.

We are collaborating with Vertex on our CF candidate, mRNA-3692/VX-522, which is designed to treat the underlying cause of CF by enabling cells in the lungs to produce functional CFTR protein for the treatment of the over 5,000 patients who do not produce any modulator-responsive CFTR protein. This would be our first demonstration of a nebulized mRNA therapy. The FDA has granted VX-522 Fast Track designation.

The multiple ascending dose portion of the Phase 1/2 study of VX-522 is underway, with data expected in the second half of 2026.

PROGRAMS DISCONTINUED IN 2025

As a result of portfolio prioritization and emerging clinical data, we discontinued a number of programs in 2025 as we focus our research and development efforts. We do not have current plans for future development of the following programs:

- mRNA-1647: CMV in congenital CMV (we plan to continue to evaluate mRNA-1647 in an ongoing Phase 2 trial of bone marrow transplant patients)
- mRNA-1608: Herpes simplex virus (HSV) vaccine
- mRNA-1468: Varicella-Zoster virus (VZV) vaccine
- mRNA-3745: Glycogen Storage Disease Type 1a (GSD1a) therapeutic
- mRNA-3210: Phenylketonuria (PKU) therapeutic

MANUFACTURING

Our manufacturing capabilities play a critical role in our value chain and support every stage of the development of our products, from discovery to commercialization. During the research stage of product development, manufacturing provides mRNA drug substance and drug product for platform research and therapeutic area drug discovery. During early development of our product candidates, we manufacture drug substance and drug product for IND-enabling GLP toxicology studies and initial human clinical studies. For late clinical development, we produce mRNA and drug product for Phase 3 trials. At the commercial stage, we manufacture drug substance and drug product in collaboration with our contract manufacturing organizations (CMOs), both in the United States and internationally.

Overview of our manufacturing operating model

Our manufacturing activities generally focus on:

- **Commercial Production:** Our manufacturing capabilities include state-of-the-art technologies for mRNA and drug product manufacturing, as well as quality control testing to attain a robust and consistent supply that matches target product profiles. Our manufacturing technology is built to scale-up and support production of products for commercial approval. Our platform allows for efficient manufacturing at scale.

- **Research and Development Support:** The product supply enables platform research and drug discovery in our therapeutic and vaccine areas, in addition to activities related to clinical studies of our product candidates.

We have built a dedicated in-house, multi-building manufacturing campus in Norwood, Massachusetts, the Moderna Technology Center (MTC). In December 2024, we purchased the MTC campus to provide greater operational flexibility and long-term stability in supporting our manufacturing and development capabilities. The MTC provides supply for our preclinical research, IND-enabling GLP toxicology study supplies, our Phase 1 and Phase 2 pipeline activities, later-stage clinical development activities, as well as drug substance commercial production for vaccines. The MTC campus has been designed with a high level of automation and state-of-the-art digital integration to handle manufacturing execution, product testing and release, and regulatory filings. In 2025, we began construction at the MTC campus for new commercial drug product manufacturing and packaging capability to enable full end-to-end manufacturing for our mRNA medicines in the U.S.

In the second quarter of 2023, we acquired a biomanufacturing facility in Marlborough, Massachusetts. In 2025, we completed construction of the facility, which was purpose-built for intismeran autogene. Designed for speed and scalability with advanced automation and robotics, the site was approved for clinical supply and began shipping patient batches in September 2025. We are on track for potential commercial launch, pending regulatory approvals.

Internationally, we have built and manage mRNA manufacturing facilities in the United Kingdom, Canada and Australia. These facilities were fully licensed in 2025, and the government in each of these countries has entered into a multi-year commitment to purchase mRNA products from us. These local manufacturing facilities will provide direct access to rapid pandemic response capabilities and our respiratory vaccine candidates. We may seek to enter into future agreements with other governments to provide similar manufacturing capabilities in other geographies.

In addition to our internal manufacturing facilities, we also maintain relationships with CMOs in the United States and abroad, providing critical raw material production and fill-finish capacity for our vaccines.

Manufacturing technology development

To support our broad pipeline of products, which spans multiple therapeutic areas and routes of administration, our platform research and technical development teams closely collaborate to facilitate rapid and seamless clinical translation of scientific breakthroughs. This enables us to develop potential medicines to serve a broad patient population.

Technical development encompasses the design and optimization of robust and consistent manufacturing processes, product characterization, fit-for-purpose formulations and product presentations. For instance, our novel hardware platforms' automation and robotics, coupled with the flexibility of our in-house digital development systems, allows for thousands of experiments and process parameters across our projects, thus supporting our drug product pharmaceutical readiness. Moreover, our technical manufacturing advances have enabled internalization of new key capabilities, including DNA plasmids and small molecules.

In parallel, we continue to refine existing processes, resulting in increased manufacturing capabilities. These improvements may allow us better control over our supply chain, resulting in larger production yields and longer shelf life of our products.

Our substantial investments in recent years in technical development have enabled the breadth and depth of our pipeline, and laid the foundation to help meet the needs and requirements associated with late-stage development and the commercialization of our products.

Supply of mRNA for All Stages of Product Development and Commercialization

Supply for Research

High-throughput automation and custom engineered equipment allow us to produce and deliver high quality mRNA and formulated constructs in a short period of time: our proprietary platform is capable of producing up to 1,000 lots of mRNA sequences and formulations per month with a turnaround time of a few weeks from sequence to final product. The typical scale of mRNA manufactured by this team is 1-1,000 mg. This has been possible, in part, due to the ability of researchers in the Moderna ecosystem to order constructs through an integrated digital portal that tracks materials end-to-end in less than 45 days. In addition, multiple integrated algorithms that leverage artificial intelligence and machine learning optimize manufacturability, reduce failures and increase quality of mRNA sequences.

Supply for Clinical Development

We have established manufacturing capabilities that support the early development stage of product development in three key areas: GLP Tox, Clinical Studies and intismeran. We supply formulated product to conduct IND-enabling GLP toxicology studies. In addition, human clinical studies rely on supply to meet required cGMP standards. This is achieved via internal manufacturing at the MTC campus. Our MTC campus is also suited to enable rapid technology development and scale-up for future needs.

Our manufacturing also produces cGMP intismeran. Due to the specialized nature of personalized medicine (i.e., where a batch is specifically designed and manufactured for a single patient), the manufacturing process for intismeran has unique requirements. We digitally integrate patient-specific data from sequencing tumor samples to automatically design intismeran for patients. We have developed proprietary bioinformatics designed algorithms linked to an automated manufacturing process for rapid production of formulated mRNA, with a typical turnaround time of a few weeks. We have operationalized intismeran manufacturing to meet our Phase 1 and 2 pipeline supply needs by using single-use systems with fast "needle-to-needle" turnaround times. Unlike traditional process development, each intismeran batch is manufactured for a single patient and thus scaled-out (in parallel) with extensive use of automation and robotics to account for the larger number of patients involved in later phases of development and commercialization. We have shown consistent quality in our production of many patient batches, each with unique mRNA sequences. Our new Marlborough facility has state-of-the-art mRNA manufacturing areas, including a full manufacturing clean room, quality control laboratories, and a just-in-time satellite warehouse.

Our manufacturing capabilities have allowed us to build our broad pipeline of development programs, including the output required to supply related toxicological and human clinical studies. While the technology that underpins these programs is the same, each program typically requires customization based on target product profiles. These custom features range from varying molecular architecture to different routes of administration, often requiring multivalent products. All programs, except for intismeran, require that we progressively scale up supply to meet clinical demand requirements across development phases, in addition to the necessary preparation for regulatory approval and commercial production, which demand larger batch sizes. In contrast, the intismeran program seeks to develop a cancer therapeutic that is designed and manufactured for a specific patient, thus increasing the number of unique batches. As

we scale manufacturing output for each program, we plan to continuously improve yield, purity and the pharmaceutical properties of our product candidates.

Supply for Late-Stage Development and Commercialization

Our development pipeline continues to advance to later-stage development and towards commercialization. Our platform approach allows us to continue to evolve our manufacturing suites and other capabilities at our manufacturing facilities. mRNA manufacturing is flexible and one plant can manufacture multiple vaccines and therapeutics. Our manufacturing facilities also permit us to manufacture products in parallel.

Quality Unit

Quality is core to the way we operate. We seek to ensure quality at Moderna through a combination of a robust Quality Management System (QMS), our quality culture and our people. In accordance with applicable regulations, we have established, documented and implemented a QMS to assure continued compliance with the requirements therein. The QMS facilitates cGMP compliance by implementing practices that identify the various required processes, their application throughout the organization and the sequence of interaction of these processes.

The primary mode of documenting these key practices is through policies, standard operating procedures (SOPs), forms and other quality records, which include an overarching Quality Policy and Quality Manual. We have implemented tools and metrics to monitor, measure, and analyze these practices to support cGMP operations, achieve planned results, and support continuous improvement. We monitor these quality metrics through formal governance processes, including Quality Management Review (QMR), to enable continuous improvement. We have also established an independent Quality Unit that fulfills quality assurance and quality control responsibilities.

Environment, Health, and Safety

We have established a global Environment, Health, and Safety (EHS) organization to foster a safe and healthy work environment with a focus on sustainability and compliance. Our approach integrates environmentally responsible practices with health and safety measures focused on risk reduction to promote long-term workplace well-being and environmental stewardship. We achieve this through a combination of training, procedures, digital data collection and reporting tools, and corporate programs that drive towards continuous improvement.

Supply Chain Unit

We have established a global supply chain to enable supply of the raw materials and components used to produce our products, consistent with clinical and preclinical demands. We have worked with our external vendors to characterize critical raw materials and to understand their impact on the quality of drug substance and formulated drug product. We also assess the quality system and performance of our external vendors and work with them to comply with regulatory requirements. In addition, we have established an infrastructure to enable direct-to-customer shipments for our commercial products. We leverage third-party wholesalers and integrate with artificial intelligence-driven data analytics to ensure successful ordering and delivery.

Engineering

Our global engineering organization is structured to deliver exceptional facilities and services. We partner closely with external and internal service providers to incorporate robotics, automation and predictive analysis for equipment operation. Engineering plays a critical role in the design, build, operation and maintenance of our facilities.

DIGITAL AND AI STRATEGY

Since our founding, we have been a digital-first company, seeking to use the power of digital information to maximize our impact on patients. mRNA is an information molecule, and our company was built on the premise that the natural flow of information in life can be used to develop medicines. Leveraging over a decade of experience developing mRNA medicines, we have built a large library of data that, combined with our platform approach and cloud-native infrastructure, positions us well to scale a digital operating model using artificial intelligence (AI).

Led by our Chief People and Digital Technology Officer, our digital organization partners across all Moderna functions to perpetuate and grow Moderna's AI-native culture. To that end, we have implemented cross-organization AI training initiatives to support our employees in transforming the way we work. Through these initiatives, our employees learn how to leverage AI in their specific job functions to augment their capacity and capabilities, and to maximize our impact on patients.

AI helps optimize each aspect of our value chain, from drug design to commercial manufacturing and beyond. At the research stage, our digital and AI infrastructure allows our scientists to design novel mRNA constructs, use AI algorithms to optimize them and order them from our high throughput preclinical scale production line. Our capabilities allow us to design mRNA, protein and LNP components with desired properties, such as reduced toxicity or increased stability. At the product development stage, AI helps to improve the efficiency of our clinical trial operations by, for example, forecasting participant enrollment and automating clinical trial data processing.

Our manufacturing processes likewise utilize the power of AI. For example, we leverage a series of fully autonomous, integrated AI algorithms in connection with manufacturing intismeran autogene (mRNA-4157). Our machine-learning based algorithms design the specific therapy for each individual patient and optimize the timely manufacture and delivery of intismeran to each patient.

At the commercial stage, our digital and commercial organizations partner to drive performance and prepare for product launches. Digital and AI are key components of our commercialization strategy and are vital to our ability to increase our speed to market, enhance our commercial capabilities and continuously improve the quality of our products. We believe that our ability to move with both scale and speed positions us well to pursue our goals related to future product launches.

In early 2023, we began a collaboration with OpenAI to co-innovate with a shared vision of AI's transformative potential in the future of business and healthcare. In 2024, our AI culture led to the deployment across the company of ChatGPT Enterprise and its enhanced capabilities such as Advanced Analytics, Image Generation and GPTs. These GPTs are now embedded across our business functions, including legal, research, manufacturing and commercial, helping to drive productivity. For example, our Dose ID GPT uses ChatGPT Enterprise's Advanced Data Analytics feature to further evaluate the optimal vaccine dose selected by the clinical study team. By applying standard dose selection criteria and principles, Dose ID provides a rationale, references its sources, and generates informative charts illustrating the key findings. This allows for a detailed review, led by humans and augmented with AI input, while prioritizing safety and optimizing the vaccine dose profile prior to further development in late-stage clinical trials.

In 2025, we continued to scale our use of generative AI across the enterprise while further strengthening governance, security and responsible AI controls to support safe and compliant adoption. We also expanded the use of customized GPTs and advanced analytics in additional workflows, with a continued emphasis on human oversight, data integrity and alignment with our scientific, regulatory and ethical standards.

We believe that the integrated AI ecosystem we are building at Moderna will accelerate our mission to deliver the greatest possible impact to people through mRNA medicines.

COMMERCIAL

We continue to build our differentiated commercial model, with active commercial subsidiaries in key markets across North America, Europe and the Asia-Pacific region. Our commercial footprint provides us with local commercial teams in major markets where respiratory vaccines have high utilization rates and sales. To support the build out of our commercial activities in markets worldwide, we have hired talent with extensive pharmaceutical company experience. Our commercial teams also work with third-party distributors and other commercial alliances in countries where we do not have a direct presence. Our commercial activities are dependent on regulatory approvals and on agreements that we have made or may make in the future with strategic collaborators.

We currently have three commercial products—Spikevax and mNEXSPIKE (our COVID vaccines) and mRESVIA (our RSV vaccine). The commercial markets for these vaccines are seasonal and characterized, particularly in the U.S. (our largest market), by a fragmented end customer base, unpredictability in orders and seasonality of deliveries. The private vaccine market is also characterized by market practices regarding rebates, discounts and returns. The respiratory vaccine market, and markets for COVID and RSV vaccines in particular, depends on many factors such as medical need, viral evolution, public health authority recommendations and consumer motivation to vaccinate.

In addition, we have built and manage mRNA manufacturing facilities in the United Kingdom, Canada and Australia. These facilities were fully licensed in 2025, and the government in each of these countries has entered into a multi-year commitment to purchase mRNA products from us. See "—Manufacturing" above for further detail.

THIRD-PARTY STRATEGIC ALLIANCES

Strategic alliances

We are party to strategic alliances with a diverse group of collaborators, including pharmaceutical and biotechnology companies, government agencies, academic laboratories, foundations and research institutes with therapeutic area expertise and resources. Through our collaborations, we seek to advance our discovery and development programs, while leveraging our platform and our research and early development capabilities. From time to time, we also partner with and invest in companies developing other types of therapeutics where we believe we can leverage our core mRNA and LNP capabilities to expand the reach of our technology.

Through certain of our strategic alliances, we share the benefits and risks of developing a new mRNA modality or program, where we may have early research data and desire a strategic collaborator to join us in advancing early development candidates within such modality into the clinic. Representative relationships and associated programs include those with Merck, for our intismeran programs (mRNA-4157), and Vertex, for our CF program (mRNA-3692).

To maintain the integrity of our platform, our strategic collaboration agreements generally grant either us the rights to develop and commercialize potential mRNA medicines we design and manufacture, or grant our collaborators those rights. These agreements do not allow collaborators to use our platform to generate new mRNA technologies, and we generally retain ownership of intellectual property related to our platform arising from research conducted under the alliance. We may continue to identify potential strategic collaborators who can contribute meaningful technology and insights to our programs and allow us to expand our impact more rapidly to broader patient populations.

Below are brief descriptions of certain of our ongoing collaborations.

Merck—Strategic Alliance for Personalized mRNA Cancer Vaccines (Intismeran Autogene)

In June 2016, we entered into a Collaboration and License Agreement, which was subsequently amended in 2018, with Merck (the PCV Agreement) for the development and commercialization of personalized mRNA cancer vaccines (PCV), also known as Individualized Neoantigen Therapy (INT), which has been assigned the generic name intismeran autogene. Under the strategic alliance, we identify genetic mutations present in a particular patient's tumor cells, synthesize mRNA for these mutations, encapsulate the mRNA in one of our proprietary LNPs and administer to each patient a unique intismeran designed to specifically activate the patient's immune system against her or his own cancer cells.

Pursuant to the PCV Agreement, we received an upfront payment of $200 million from Merck and we were responsible for designing and researching intismeran, providing manufacturing capacity and manufacturing intismeran and conducting Phase 1 and Phase 2 clinical trials for intismeran, alone and in combination with KEYTRUDA (pembrolizumab), Merck's anti-PD-1 therapy, all in accordance with an agreed upon development plan and budget.

In September 2022, Merck exercised its option for intismeran, including mRNA-4157, pursuant to the terms of the PCV Agreement and in October 2022 paid us an option exercise fee of $250 million. Pursuant to the PCV Agreement, we and Merck have agreed to collaborate on further development and potential commercialization of intismeran, with costs and any profits or losses generally shared equally on a worldwide basis, subject to certain exceptions as outlined in the agreement.

Vertex—2016 Strategic Alliance in Cystic Fibrosis

In July 2016, we entered into a Strategic Collaboration and License Agreement (Vertex Agreement) with Vertex Pharmaceuticals Incorporated, and Vertex Pharmaceuticals (Europe) Limited (together, Vertex). The Vertex Agreement is aimed at the discovery and development of potential mRNA medicines for the treatment of CF by enabling cells in the lungs of people with CF to produce functional CFTR proteins.

Other Collaborations

We have entered into additional collaborations where we have agreed to provide funding in areas where we believe we can leverage our mRNA technology, such as our collaboration with Immatics N.V. to pioneer novel and transformative therapies for cancer patients with high unmet medical need. We have also entered into agreements with other parties under which we have received licenses of certain technology.

Further, in January 2026, we entered into a collaboration with Recordati, an international pharmaceutical group, to advance our investigational PA therapeutic (mRNA-3927) through the final stages of clinical development and, if approved, global commercialization. We will continue to lead the clinical development of mRNA-3927 through approval and Recordati will lead commercialization.

Strategic alliances with government organizations and foundations

Biomedical Advanced Research and Development Authority (BARDA)

In April 2020, we entered into an agreement with BARDA for an award of up to $483 million to accelerate development of mRNA-1273, our original COVID vaccine. The agreement has been subsequently amended to provide for additional commitments to support various late-stage clinical development efforts of mRNA-1273, including a 30,000 participant Phase 3 study, pediatric clinical trials, adolescent clinical trials and pharmacovigilance studies. The maximum award from BARDA, inclusive of all amendments, was approximately $1.8 billion. All contract options have been exercised. The BARDA contract concluded on June 15, 2025, upon completion of all contractual deliverables. As of December 31, 2025, the remaining available funding, net of revenue earned, was approximately $62 million, and we do not expect to utilize a material portion of this amount following the conclusion of the contract.

Coalition for Epidemic Preparedness Innovations (CEPI)

In December 2025, CEPI agreed to invest up to $54.3 million to support a pivotal Phase 3 clinical trial that aims to help advance our investigational mRNA-based H5 pandemic influenza vaccine candidate, mRNA-1018, to licensure.

Institute for Life Changing Medicines (ILCM)

In September 2021, we entered into a collaboration agreement with the ILCM to develop a new mRNA therapeutic (mRNA-3351) for type 1 Crigler-Najjar syndrome (CN-1). Under the terms of the agreement, we agreed to license mRNA-3351 to ILCM with no upfront fees, and without any downstream payments. ILCM will be responsible for the clinical development of mRNA-3351.

The Gates Foundation

In January 2016, we entered a global health project framework agreement with the Bill & Melinda Gates Foundation (n/k/a the Gates Foundation) to advance mRNA development projects for various infectious diseases. Under the framework agreement, the Gates Foundation committed funding to support certain preclinical and clinical development activities, including a program evaluating mRNA-based approaches for the prevention of HIV infection. The funded activities under the framework agreement concluded during 2025, following completion of the agreed-upon program scope. The collaboration supported multiple preclinical and early-stage clinical studies; results from these efforts were published in peer-reviewed scientific journals in 2022-2025 and additional publications are in preparation. As of December 31, 2025, there were no remaining funding commitments under the framework agreement. We continue to engage in scientific dialogue with the Gates Foundation regarding potential future areas of collaboration.

INTELLECTUAL PROPERTY

We rely on a combination of intellectual property laws, including patent, trademark, copyright and trade secret, as well as confidentiality and license agreements, to protect our intellectual property and proprietary rights.

Protecting our platform, modality and program investments: Building an expansive, multi-layered IP estate

We have built a substantial IP estate that includes numerous patents and patent applications related to the development and commercialization of mRNA vaccine and therapeutic development candidates, including related platform technologies. Our platform IP protects advances in mRNA design and engineering, proprietary LNP components, delivery systems, processes for the manufacture and purification of drug substances and products and analytical methods. A significant portion of our platform IP estate further provides multi-layered protection for our modalities and programs.

With respect to our IP estate, our solely-owned patent portfolio consists of more than 260 issued or allowed U.S. patents or patent applications and more than 140 granted or allowed patents in jurisdictions outside of the U.S. (including granted European patents that have been validated in numerous European countries) covering certain of our proprietary platform technology, inventions and improvements, and covering key aspects of our clinical and most advanced development candidates. We have 485 additional pending patent applications that, in many cases, are counterparts to the foregoing U.S. and foreign patents.

Most of the patents and applications (if issued) in our portfolio will not expire until 2033 at the earliest. Any patent that may issue from our most recently filed patent applications is projected to expire between 2044 and 2045, at the earliest. We file additional U.S. and foreign patent applications in key markets as necessary to protect our evolving intellectual property positions.

We also rely on trademarks, copyright, trade secrets and know-how relating to our proprietary technology and programs, continuing innovation, and in-licensing opportunities to develop, strengthen and maintain our proprietary position in the field of mRNA therapeutic and vaccine technologies. We take additional steps, such as entering into confidentiality and license agreements, to protect our intellectually property and proprietary rights. We additionally plan to rely on data exclusivity, market exclusivity and patent term extensions when and where available, and plan to seek and rely on regulatory protection afforded through orphan drug designations. We also possess substantial proprietary know-how associated with related manufacturing processes and expertise.

IP protecting our platform

We have a broad IP estate covering key aspects of our platform. This estate provides multiple layers of protection covering the making and use of the mRNA drug substance and delivery technologies.

With respect to our platform, we have a portfolio that includes U.S. and foreign patents or patent applications covering platform innovations that are related to the design, manufacturing and formulating of mRNA medicines. For example, these patents and patent applications include claims directed to:

- mRNA chemistry imparting improved properties for vaccine and therapeutic uses;
- methods for mRNA sequence optimization to enhance the levels and fidelity of proteins expressed from our mRNA medicines;
- methods for identifying epitopes having superior suitability in cancer vaccine contexts;
- engineering elements tailored to enhance stability and the *in vivo* performance of mRNA medicines;
- LNP delivery systems, including novel lipid components designed for optimal delivery and expression of both therapeutic and vaccine nucleic acids, in particular, prophylactic infectious disease and cancer therapy nucleic acids, intratumoral immuno-oncology therapeutics, local regenerative therapeutics, systemic therapeutics, and inhaled pulmonary therapeutics; and
- innovative processes for the manufacture and analysis of mRNA drug substance and formulated drug product.

IP protection

Our IP estate provides protection for the multiple programs both at the product-specific level and at various broader levels. For example, we have patent coverage for LNP-encapsulated mRNAs having specific chemical modification suited for vaccine and therapeutic mRNA use. Our estate also includes IP covering certain LNP-encapsulated mRNAs coding for infectious disease antigens for use in preventing or treating infectious diseases, including those caused by respiratory and latent viruses, as well as bacterial diseases known to threaten public health. Our mRNA chemistry, formulation and manufacturing patent applications and related know-how, along with trade secrets, may also provide us with additional IP protection relating to our development candidates.

Respiratory vaccines

For our respiratory vaccines programs, we have pursued patent protection featuring composition of matter and method of use claims. Where we may pursue patent protection may vary based on the unique geographic prevalence of various infectious diseases.

Approved products

We have filed several patent applications covering our betacoronavirus vaccine program. We are pursuing patent protection for both our existing and new betacoronavirus vaccines. We have also filed several patent applications covering our RSV vaccine. Our RSV patent portfolio includes multiple families of differing patent breadth.

The table below sets forth the year of projected expiration for the latest to expire, currently granted patents covering each of our approved products or a component thereof. Patent term extensions, supplementary protection certificates, and pediatric exclusivity periods (if any) are not reflected in the expiration dates listed in the table below. We also have additional patents covering each product.

Product	Projected Expiration of U.S. Patents	Projected Expiration of European Patents
Spikevax	2041	2036
mNEXSPIKE	2041	2036
mRESVIA	2041	2036

Influenza

We have multiple patent families spanning different levels of breadth, design and antigen valency pending in the U.S., Europe and around the world, including several granted patents.

hMPV

Human metapneumovirus (hMPV) is a single-stranded RNA virus that is used in a combination program. We have patent applications covering our hMPV vaccine pending in the U.S. and Europe, with a granted patent in the U.S.

Latent vaccines

We have vaccine programs and patent applications directed to diseases caused by various latent viruses, in some cases, using both preventative vaccines targeting the acute phase and therapeutic vaccines for treating the latent diseases in those who do become infected.

CMV

The patent coverage for our human CMV vaccine candidate is extensive and is based on a vaccine with six mRNAs encoding a pentamer surface glycoprotein complex and the gB surface glycoprotein. Both pentamer and gB facilitate entry of the virus into different cell types and therefore immune responses targeting these proteins can block virus entry, spread and reactivation. The current patent portfolio contains both compositions of matter and methods of treating subjects using the vaccine. In the U.S., our CMV vaccine is covered by multiple issued U.S. patents of differing breadth. Each family has counterparts consisting of pending applications and issued patents in non-U.S. jurisdictions, including, in some cases, Europe and Japan. A separate family of CMV patents, which includes mRNA-1647 for use in CMV vaccines for transplant indications, is also yielding patents and applications in foreign jurisdictions are pending.

EBV, HSV and VZV

Similar to CMV, we have filed patent applications, and in some cases multiple patent families, for each of EBV, HSV and VZV, e.g., covering prophylactic and/or therapeutic indications. In addition to patent applications filed in the United States, certain of these patent families have foreign counterparts, such as in Europe.

Public health vaccines

We maintain a multi-program effort at developing vaccines for potential future pandemics and for use in parts of the world with less well-established health care systems. This group of programs include infectious diseases such as flaviviruses such as Zika and dengue viruses, HIV, Nipah virus, and the Mpox virus. In addition, programs are ongoing in many bacterial diseases. While patent applications are filed on some potential public health targets, in some scenarios, platform patents rather than target specific patents may be used to provide patent protection for public health target vaccines.

Oncology

For our oncology programs, we are pursuing patent protection in numerous jurisdictions. The scope and geographic extent of patent protection may vary based on factors such as differences in regulatory regimes, competitive landscapes, and market considerations across these jurisdictions.

Intismeran

Composition of matter and method claims are being pursued to protect intismeran. Proprietary methods around the making and therapeutic use of our intismeran and resulting vaccine compositions are described and claimed in one granted and one allowed U.S. patent, eight pending U.S. patent applications, five pending European patent applications, two granted patent and five pending patent applications in Japan, three pending patent applications and one granted patent in China, and several pending patent applications in New Zealand, South Africa, Asian and South American countries, as well as two PCT applications. These applications also relate to various vaccine design formats, in particular, polyepitopic vaccine formats, and methods of treating cancer with intismeran. We also possess substantial know-how and trade secrets relating to the development and commercialization of our cancer therapy programs, including related manufacturing process and technology.

Cancer antigen therapy

For our checkpoint cancer antigen therapy program (mRNA-4359), we are pursuing patent protection in various jurisdictions. Any U.S. and foreign patents that may issue from these patent applications would be expected to expire in 2043, excluding any patent term adjustments, any patent term extensions and/or any terminal disclaimers.

Rare diseases

We have programs featuring expression of therapeutic proteins, e.g., intracellular enzymes for the treatment of rare diseases. For our rare disease programs, we generally pursue patent protection featuring composition of matter and method of use claims, for example, pharmaceutical composition and method of treatment claims. We have patent applications granted, pending and/or published for our most advanced rare disease development candidates targeting PA and MMA. In addition, we have patent applications granted, pending and/or published for our other rare disease candidates.

Any U.S. and foreign patents that may issue from these patent families would be expected to expire in 2036 for the earliest of the MMA patents and 2038 to 2042 for the remaining MMA and PA patents, excluding any patent term adjustments, any patent term extensions and any terminal disclaimers.

As further described below, as we continue the development of our intended products, we continue to identify additional means of protecting our assets that would potentially enhance commercial success, including possible patent protection for additional methods of use, formulation, or manufacture.

Cystic fibrosis

Our CF development candidate is covered by pending U.S., European and PCT patent applications.

Trademarks

Our trademark portfolio currently contains at least 1,400 trademark registrations, including at least 29 registrations in the United States and the remaining in Canada, the European Union, the United Kingdom, Israel, China, Japan, Australia, and elsewhere. In addition, we have at least 180 pending trademark applications in more than 55 jurisdictions, including in the aforementioned locations and additional countries throughout Africa, Asia, and South America.

In-licensed intellectual property

While we develop and manufacture our potential mRNA medicines using our internally created mRNA technology platform, we also seek out and evaluate third party technologies and IP that may be complementary to our platform.

Patent sublicense agreements with Cellscript and mRNA RiboTherapeutics

The Trustees of the University of Pennsylvania owns several issued U.S. patents, granted European patents and pending U.S. patent applications directed, in part, to nucleoside-modified mRNAs and their uses (the Penn Modified mRNA Patents). mRNA RiboTherapeutics, Inc. (MRT) obtained an exclusive license to the Penn Modified mRNA Patents and granted its affiliate, Cellscript, LLC (Cellscript), a sublicense to the Penn Modified mRNA Patents in certain fields of use.

In June 2017, we entered into two sublicense agreements, one with Cellscript, and one with MRT, which agreements we collectively refer to as the Cellscript-MRT Agreements. Together, the Cellscript-MRT Agreements grant us a worldwide, sublicensable sublicense to the Penn Modified mRNA Patents to research, develop, make, and commercialize products covered by the Penn Modified mRNA Patents (licensed products), for all *in vivo* uses in humans and animals, including therapeutic, prophylactic, and diagnostic applications. The Cellscript-MRT Agreements are non-exclusive, although Cellscript and MRT are subject to certain time restrictions on granting additional sublicenses for *in vivo* uses in humans under the Penn Modified mRNA Patents. The Cellscript-MRT Agreements require us to pay royalties based on annual net sales of licensed products at rates in the low single digits for therapeutic, prophylactic, and diagnostic uses, and royalties based on annual net sales of licensed products sold for research uses at rates in the mid-single digits, subject to certain reductions, with an aggregate minimum floor.

The Cellscript-MRT Agreements will terminate upon the expiration or abandonment of the last to expire or become abandoned of the Penn Modified mRNA Patents. Cellscript or MRT, as applicable, may terminate its respective Cellscript-MRT Agreement if we fail to make required payments or otherwise materially breach the applicable agreement, subject to specified notice and cure provisions. Cellscript or MRT, as applicable, may also terminate the applicable Cellscript-MRT Agreement upon written notice in the event of our bankruptcy or insolvency or if we challenge the validity or enforceability of the Penn Modified mRNA Patents. We have the right to terminate each Cellscript-MRT Agreement at will upon 60 days' prior notice to Cellscript or MRT, as applicable, provided that we cease all development and commercialization of licensed products upon such termination. If rights to MRT or Cellscript under the Penn Modified mRNA Patents are terminated (e.g., due to bankruptcy of MRT or Cellscript), the terminated party will assign its interest in the respective Cellscript-MRT Agreement to the licensor from which it received rights under the Penn Modified mRNA Patents and our rights will continue under the new licensor.

Patent license agreements with NIAID

In December 2022, we entered into a non-exclusive patent license agreement with the National Institute of Allergy and Infectious Diseases (NIAID), an Institute or Center of the National Institutes of Health (NIH), to license certain patent rights concerning stabilizing prefusion coronavirus spike proteins and the resulting stabilized proteins for use in COVID vaccine products. Pursuant to the agreement, we have agreed to pay low single-digit royalties on future net sales, a minimum annual royalty payment and certain contingent development, regulatory and commercial milestone payments on a licensed product-by-licensed product basis.

In January 2025, we entered into a non-exclusive patent license agreement with NIAID to license certain patent rights concerning prefusion RSV F proteins and their use. Pursuant to the agreement, we have agreed to pay tiered, low-to-mid single-digit royalties on net sales of our RSV vaccine, a minimum annual royalty payment, and certain contingent development, regulatory and commercial milestone payments on a licensed product-by-licensed product basis.

Formulation technology in-licenses

Our development candidates use internally developed formulation technology that we own. We do, however, have rights to use and exploit multiple issued and pending patents covering formulation technologies under licenses from other entities. If in the future we elect to use or to grant our strategic collaborators sublicenses to use these in-licensed formulation technologies, we or our strategic collaborators may be liable for milestone and royalty payment obligations arising from such use. We consider the commercial terms of these licenses and their provisions regarding diligence, insurance, indemnification and other similar matters, to be reasonable and customary for our industry.

HUMAN CAPITAL

We had approximately 4,700 full-time employees in 18 countries as of December 31, 2025. We operate in a highly competitive environment for talent, particularly as we seek to attract and retain talent with experience in the biotechnology and pharmaceutical sectors. Our workforce is highly educated, and as of December 31, 2025, 46% of our employees hold Ph.D., Doctorate, M.D., J.D. or Master's degrees. Among our employees, as of December 31, 2025, 49% are female. Among our leadership (which we define as employees at the vice president level and above), as of December 31, 2025, approximately 39% are female. 44% of our U.S. employees identify as racially or ethnically diverse as of December 31, 2025. In 2025, for the fourth year in a row, an outside statistical pay equity

analysis confirmed zero statistically significant differences in pay across gender globally and across gender, race and ethnicity in the United States.

Our approach to attracting and retaining talent

We are committed to ensuring that our employees find that their careers at Moderna are filled with purpose, growth and fulfillment. We believe that a career at Moderna provides opportunity for:

- **Impact**: Our people have the opportunity to do work that is unparalleled in terms of its innovation and scope of impact on people's lives.
- **Growth**: We provide incredible opportunities for growth and we obsess over learning (as demonstrated, in part, by our Mindsets (see below). We invest substantially in the development of our people.
- **Well-being**: We are deeply invested in the health and wellness of our employees and provide benefits and resources that support each person at work, at home and in their communities.
- **Belonging**: We believe that innovation happens through bringing together a broad set of perspectives and backgrounds, and creating an environment where differences are celebrated.
- **Compelling rewards**: To attract and retain the best talent, we provide competitive rewards that help to drive groundbreaking work and allow employees to share in the value we will create together, including through our equity programs.
- **Giving and volunteering:** Our people have the opportunity to give back to their communities and directly support causes that they are passionate about through volunteer and employee matching donation programs.

To help promote alignment between our employees and our shareholders, all employees participate in our corporate equity programs through the receipt of equity awards, and the percentage of equity as a component of overall pay mix increases with seniority. We also allow our employees to select how they want the value of their award to be split between stock options and restricted stock units (RSUs). We believe that in addition to incentivizing growth that leads to shareholder value, broad eligibility for our equity programs further embeds our "We behave like owners" mindset and helps promote employee retention as these awards generally vest over a four-year period.

None of our employees have entered into a collective bargaining agreement with us. A small number of employees in France, Italy and Spain are covered by statutory collective bargaining agreements governing certain benefits and working conditions. Employees in our Madrid work center are represented by a works council. None of our other employees are represented by a labor union or a works council. We consider our employee relations to be good.

We believe that our employees are highly engaged, and our company and team have been publicly recognized for our leadership, innovation and good corporate citizenship. *Science* magazine ranked us as a top employer for each of the last eleven years. Additionally, in 2025, *Biospace* ranked us a top large employer in its 2026 Best Places to Work in Biopharma report for the fifth consecutive year. We measure employee engagement through a vendor-supplied engagement software, using validated external benchmarks to track employee engagement factors.

We continually monitor employee turnover rates, as our success depends upon retaining our highly trained personnel. We believe that the competitive compensation we offer, along with the combination of the factors listed above, among other factors, have helped reduce voluntary turnover. In 2025, our voluntary turnover rate was approximately 11%.

Our approach to training our employees

To further invest in our teams, we have established a structured training curriculum for our employees so that every employee becomes deeply familiar with our core technology and technologies that might further enable our innovation. In addition, we are focused on creating strong leaders through various management and leadership trainings. We have also built an online library of videos of a variety of scientific material that our employees can access flexibly. This content includes presentations by external speakers at in-house scientific seminars, scientific courses at external universities and peer-to-peer video series in which in-house experts provide an introductory view of complex topics they tackle within their teams.

New employees participate in our Moderna ONE onboarding program, which is an interactive learning experience designed to immerse our people in our culture and Mindsets from day one. Following onboarding, our employees continue to learn throughout their careers at Moderna and we deploy a digital learning management system to track and administer training programs for each employee.

In December 2021, we launched our AI Academy. The AI Academy is intended to educate and empower our employees to identify and integrate AI and machine learning solutions into every Moderna system and process to bring mRNA medicines to patients. In 2025, we delivered a targeted learning portfolio focused on practical AI skills designed to build durable habits and fundamentally change how employees work.

Our culture

As an organization, we are bold, collaborative, curious and relentless. These values are underpinned by a core set of what we call "basecamp" values— integrity, quality, respect. Additionally, with the continued rapid growth of our company, we articulated the Moderna Mindsets, which define how we behave, lead and make decisions. We believe our Mindsets will be integral to our future success, and we integrate them into every facet of how we identify, onboard, grow and manage our talent.

To further develop and retain our workforce, we conduct periodic talent reviews that identify key talent within the organization. We use that data to inform specific development opportunities for key current and potential future leaders, and to support our periodic succession planning activities for key roles. These steps together ensure we have a robust understanding of our workforce and a talent pipeline to grow future leaders, and provide our employees an opportunity to continuously grow and advance in a way that meets their aspirations and talents.

CORPORATE SOCIAL RESPONSIBILITY

As we pursue our mission to deliver the greatest possible impact to people through mRNA medicines, we have developed a corporate social responsibility (CSR) program that demonstrates our commitment to patients, employees, the environment and local communities. Our CSR framework consists of five key focus areas: medicines for patients, community, governance and ethics, employees and environment. Please refer to our 2024 Impacting Human Health Report under the "Responsibility—Corporate policies & reporting" section of our website, which can be found at www.modernatx.com/responsibility/corporate-policies, as well as our proxy statement related to our 2026 Annual Meeting of Stockholders that we will file with the SEC, for a description of some of the measures we have taken to progress our commitment to corporate social responsibility.

COMPETITION

The biotechnology and pharmaceutical industries utilize rapidly advancing technologies and are characterized by intense competition. There is also a strong emphasis on defense of intellectual property and proprietary products.

We believe that mRNA as a medicine coupled with our capabilities across mRNA technology, drug discovery, development and manufacturing provide us with a competitive advantage. However, we face competition from others developing mRNA medicines, as well as other medicines that compete or could compete with our products. We face competition from various sources, including large pharmaceutical companies, biotechnology companies, academic institutions, government agencies and public and private research institutions. The continued growth of the mRNA field is leading to increased competitive pressure, including from large and more established pharmaceutical companies. We also face competition when entering into strategic alliances to advance and grow our pipeline.

Seasonal vaccines

We largely compete against Pfizer and BioNTech for sales of our COVID vaccines, whose vaccine is also based on mRNA technology. We also compete against other vaccines, including Sanofi and Novavax's. Additionally, some competitors have developed COVID treatments, including Pfizer's antiviral pill, which may reduce demand for vaccines. With respect to our RSV vaccine, we compete against Pfizer and GlaxoSmithKline, who entered the U.S. market prior to us, and our RSV sales have been minimal to date. We will also face competition for other seasonal vaccines we expect to launch in the future; for example, we are developing a seasonal flu vaccine for which there is a well-developed market.

Competition for the sale of our products is impacted by many factors, including, among others, actual and perceived vaccine efficacy, safety and tolerability, perceptions of mRNA technology, storage and handling conditions and the relative ease of distribution and administration, the timing and scope of regulatory approvals, reimbursement coverage and production and distribution costs.

In markets that we enter after competitors have already introduced a competing product, we may have difficulty achieving market share, as was the case in connection with the launch of our RSV vaccine. See "Risk Factors—The vaccine market, and pharmaceutical market more generally, is intensely competitive, and we may be unable compete effectively in the market for existing or new products, treatment methods or technologies."

Oncology

We face intense competition in the oncology space, including from large pharmaceutical and biotechnology companies developing a broad range of oncology products and platforms. Oncology markets evolve rapidly and are subject to significant clinical and regulatory uncertainty; competitors may generate superior data, advance more quickly, and establish new standards of care. As a result, competition could reduce the commercial opportunity and adversely affect development timelines.

Collaborations

We and our strategic collaborators also face competition in therapeutic areas beyond seasonal vaccines and oncology. These competitors include a number of pharmaceutical and biotechnology companies and academic and research institutions developing a broad range of therapeutic approaches and platforms that may compete with our current or future programs.

GOVERNMENT REGULATION

Government authorities in the United States and in other countries and regions, such as in the EU, regulate the research, development, manufacture and marketing of our products. Before a new medicine is marketed, data demonstrating its quality, safety and efficacy must be generated, submitted for review, and approved by the competent regulatory authority.

U.S. drug and biologic development and approval

In the United States, the FDA regulates drugs and biologics under the Federal Food, Drug, and Cosmetic Act and the Public Health Service Act, together with their implementing regulations and other applicable federal, state, and local laws. Failure to comply with these requirements at any stage of development, approval, or post-approval may result in administrative or judicial sanctions, including the FDA's refusal to approve pending applications, license revocation, clinical holds, untitled or warning letters, product recalls, market withdrawals, product seizures, suspension of production or distribution, injunctions, fines, refusals of government contracts, restitution, disgorgement, and civil or criminal penalties.

Our product candidates must be approved through a biologics license application (BLA) or new drug application (NDA), or a supplement, before they may be marketed in the United States.

Preclinical studies

Before testing in humans, our development candidates undergo preclinical studies that include laboratory evaluation of product chemistry and formulation, as well as *in vitro* and animal studies designed to assess safety and, in some cases, establish a rationale for therapeutic use. Preclinical safety and toxicology studies are subject to GLP requirements.

To begin human clinical trials, a sponsor must submit an IND to the FDA containing, among other things, the results of preclinical studies, manufacturing and analytical information, any available clinical data or literature, and proposed clinical trial protocols. An IND must become effective before human clinical trials may begin. Unless the FDA raises concerns, an IND generally becomes effective 30 days after receipt. If the FDA identifies deficiencies, the sponsor and the FDA must resolve those issues before clinical trials may proceed.

Clinical trials

Clinical trials involve the administration of an investigational product to volunteers or patients under the supervision of qualified investigators and in accordance with Good Clinical Practice (GCP) requirements. Clinical trials are conducted pursuant to protocols that specify, among other things, trial objectives, dosing, subject eligibility criteria, and safety and efficacy assessments. Each protocol and any amendments must be submitted to the FDA under the IND.

Each clinical trial must also be reviewed and approved by an Institutional Review Board (IRB) at each participating institution to ensure that risks to subjects are minimized and reasonable in relation to anticipated benefits. IRBs approve informed consent documents and provide ongoing oversight of the trial. Sponsors must submit periodic reports to the FDA, including safety reports and annual progress reports, and must report serious adverse events on an expedited basis. Information regarding certain clinical trials must be submitted for posting on the clinicaltrials.gov website within required timeframes.

Certain gene-based clinical trials are subject to oversight under the NIH Guidelines for Research Involving Recombinant or Synthetic Nucleic Acid Molecules, including review by an institutional biosafety committee. Although the NIH Guidelines are mandatory only for institutions receiving NIH funding, many sponsors voluntarily follow these guidelines.

Foreign clinical studies conducted under an IND must comply with the same requirements applicable to U.S. studies. Data from foreign studies not conducted under an IND may be submitted in support of a BLA if the studies were conducted in accordance with GCP and the FDA is able to validate the data.

Clinical development generally proceeds in three sequential phases, which may overlap. Phase 1 clinical trials typically assess safety, tolerability, and pharmacology of the investigational product in a small number of healthy volunteers or disease-affected patients. Phase 2 clinical trials generally evaluate dosing and preliminary efficacy of an investigational product in disease-affected patients while continuing to assess safety. Phase 3 clinical trials are designed to generate the pivotal evidence of safety and efficacy of an investigational product for its intended use in a large number of disease-affected patients at multiple sites to support regulatory approval and labeling.

The FDA may also require post-approval Phase 4 non-registrational studies to further characterize safety, efficacy, or other scientific questions during commercial use.

The FDA, an IRB, or a clinical trial site may suspend or terminate a clinical trial at any time for safety, protocol compliance, or other reasons. In addition, certain trials are overseen by an independent data safety monitoring board or committee that periodically reviews accumulating data and may recommend continuation, modification, or termination of the trial.

FDA review process

Following completion of the clinical trials, data are analyzed to determine whether the investigational product is safe and effective for its proposed indication. The results of preclinical and clinical studies, together with proposed labeling, chemistry, manufacturing, and controls information, and other required data, are submitted to the FDA as part of a BLA or NDA.

A BLA seeks approval to market a biologic and must demonstrate the product's safety, purity, and potency. An NDA seeks approval to market a drug and must demonstrate safety and efficacy. In all cases, the submission must contain sufficient evidence to satisfy the FDA's standards for approval. FDA approval must be obtained before a product may be marketed in the United States.

As part of its review, the FDA conducts pre-approval inspections of manufacturing facilities to assess compliance with cGMP requirements and may audit clinical trial data to confirm compliance with GCP. The FDA may also convene an advisory committee to review applications that present novel scientific or regulatory issues. Advisory committee recommendations are not binding but are often considered by the FDA in its decision-making.

Following review, the FDA may approve the application, request additional information, or issue a complete response letter identifying deficiencies that must be addressed before approval can be granted. A complete response letter may require additional preclinical or clinical studies or other data, and submission of such information does not guarantee eventual approval. If approved, the FDA issues an approval letter authorizing commercial marketing for specified indications and with approved labeling.

Expedited development and review programs

The FDA has several programs to facilitate and expedite the development and review of medicines intended to treat serious or life-threatening conditions, including fast track designation, breakthrough therapy designation, accelerated approval, and priority review. Fast track designation is designed to facilitate the development and review of medicines that address unmet medical needs. Breakthrough therapy designation is designed to expedite the development and review of medicines for which preliminary clinical evidence demonstrates substantial improvement over available therapies. Priority review designation establishes a goal of FDA action on an application within six months of filing and may be granted for medicines that offer significant improvements in the safety or effectiveness of the treatment, diagnosis, or prevention of serious conditions.

A product may be eligible for accelerated approval if it treats a serious or life-threatening condition and generally provides a meaningful advantage over available therapies based on an effect on a surrogate endpoint or an intermediate clinical endpoint reasonably likely to predict clinical benefit. As a condition of approval, the FDA may require post-marketing confirmatory clinical trials and may impose additional post-approval requirements or restrictions on distribution or use to assure safe use of the product. Pursuant to the Food and Drug Omnibus Reform Act of 2022, the FDA may require, as appropriate, that confirmatory trials be underway prior to approval or completed within specified timelines and has enhanced authority to expedite withdrawal of accelerated approval if clinical benefit is not verified or trials are not conducted with due diligence.

Even if a product qualifies for one or more of these programs, the FDA may later decide that the product no longer meets the applicable criteria or that review timelines will not be shortened. Participation in these programs does not change the statutory standards for approval.

Emergency Use Authorization and PREP Act

The Secretary of Health and Human Services (HHS) may authorize unapproved medical products , or unapproved uses of approved medical products, during a declared public health emergency. Following such a declaration, the FDA may issue Emergency Use Authorizations (EUA) for specific products if statutory criteria are met, including a determination that the product may be effective in diagnosing, treating, or preventing serious or life-threatening disease and that no adequate, approved, and available alternatives exist. An Emergency Use Authorization terminates when the underlying emergency declaration ends and is not a long-term substitute for FDA approval or licensure. The FDA may revoke an EUA at any time, including if the public health emergency no longer warrants such authorization. In August 2025, the FDA revoked the EUA for emergency use of our COVID vaccine.

In the United States, the Public Readiness and Emergency Preparedness Act (PREP Act) provides liability immunity for manufacturers and other covered persons against certain claims arising from the administration or use of covered countermeasures during a declared public health emergency, subject to a limited exception for willful misconduct. Covered countermeasures include qualified pandemic or epidemic products, such as vaccines intended to diagnose, prevent, or treat pandemic or epidemic diseases. To qualify for immunity, the Secretary of HHS must issue a declaration identifying the applicable emergency or credible risk of a future emergency. The Secretary issued a PREP Act declaration in March 2020 relating to COVID-19 and has issued subsequent amendments. While we believe our products sold to the U.S. Government continue to fall within the scope of applicable PREP Act declarations, this cannot be assured.

Pediatric information

Under the Pediatric Research Equity Act of 2003, all marketing applications for new active ingredients, indications, dosage forms, dosing regimens or routes of administration must contain an assessment of the safety and effectiveness of the product for the claimed indication in pediatric patients unless this requirement is waived, deferred or inapplicable.

Under the Best Pharmaceuticals for Children Act, a product may be eligible for pediatric exclusivity, which adds six months to existing exclusivity periods and patent terms. This exclusivity may be granted based on the voluntary completion of a pediatric study in accordance with an FDA-issued written request for such a study.

Rare Pediatric Disease Designation and Priority Review Vouchers

Under the Federal Food, Drug, and Cosmetic Act, as amended, the FDA incentivizes the development of product candidates that meet the definition of a “rare pediatric disease,” defined to mean a serious or life-threatening disease in which the serious of life-threatening manifestations primarily affect individuals aged from birth to 18 years and the disease affects fewer than 200,000 individuals in the United States or affects 200,000 or more in the United States and for which there is no reasonable expectation that the cost of developing and making in the United States a drug or biologic for such disease or condition will be received from sales in the United States of such drug or biologic. The sponsor of a product candidate for a rare pediatric disease may be eligible for a voucher that can be used to obtain a priority review for a subsequent marketing application after the date of approval of the rare pediatric disease drug product. A sponsor may request rare pediatric disease designation from the FDA prior to the submission of its BLA. A rare pediatric disease designation does not guarantee that a sponsor will receive a priority review voucher (“PRV”) upon approval of its BLA. Moreover, a sponsor who chooses not to submit a rare pediatric disease designation request may nonetheless receive a PRV upon approval of their marketing application if they request such a voucher in their original marketing application and meet all of the eligibility criteria. If a PRV is received, it may be sold or transferred an unlimited number of times. Under current law, after September 30, 2029, the FDA may not award any rare pediatric disease priority review vouchers, although the FDA’s authority to do so could be extended by Congress in the future.

Post-approval requirements

Following approval, manufacturers and approved products remain subject to ongoing FDA regulation. These requirements include, among other things, adverse event reporting, recordkeeping, pharmacovigilance, compliance with promotional and advertising restrictions, and limitations on industry-sponsored scientific and educational activities. Although physicians may prescribe approved products for off-label uses, manufacturers are prohibited from marketing or promoting products for unapproved indications. Certain prescription drug and biologic promotional materials must be submitted to the FDA in connection with their initial use.

Changes to an approved product, including changes to indications, labeling, manufacturing processes, or manufacturing facilities, may require the submission and approval of a new application or a supplemental BLA or NDA and may necessitate the generation of additional preclinical or clinical data.

The FDA may impose post-approval conditions to ensure that a product's benefits outweigh its risks, including the requirement to implement a Risk Evaluation and Mitigation Strategy (REMS). A REMS may include medication guides, communication plans, restricted distribution systems, patient registries, or other measures designed to mitigate specific risks. Newly identified safety or effectiveness information may require labeling updates, additional risk management measures, or post-marketing studies. Failure to comply with regulatory requirements or the emergence of post-marketing safety issues may result in enforcement actions, including withdrawal of approval.

Approved drugs and biologics must be manufactured in registered facilities in compliance with current good manufacturing practice requirements. Manufacturers and third-party contractors involved in manufacturing or distribution must maintain quality systems, documentation, and procedures to investigate and correct deviations. Facilities are subject to periodic, unannounced FDA inspections. Identified violations or post-approval product issues may result in enforcement actions, including manufacturing restrictions or product recalls.

U.S. patent term restoration and regulatory data exclusivity

In certain circumstances, some U.S. patents may be eligible for limited patent term extension under the Drug Price Competition and Patent Term Restoration Act of 1984, commonly referred to as the Hatch Waxman Amendments. Patent term restoration is intended to compensate for patent life lost during product development and the FDA review process and may extend a patent term by up to five years, subject to a maximum remaining patent term of 14 years from the product's approval date. Only one patent per approved product may be eligible for such restoration, and any application for patent term restoration must be submitted prior to patent expiration. The U.S. Patent and Trademark Office, in consultation with the FDA, determines eligibility for patent term restoration.

Certain drug products may also qualify for periods of non-patent regulatory data exclusivity. A drug product containing an active ingredient not previously approved by the FDA is generally entitled to five years of regulatory data exclusivity. Products approved based on the FDA's reliance on new clinical investigations essential to approval may receive three years of regulatory data exclusivity. If pediatric studies are conducted in response to an FDA request, pediatric exclusivity may be granted, which for drugs extends existing patent and regulatory exclusivities by six months and for biologics extends existing regulatory exclusivities by six months.

The Biologics Price Competition and Innovation Act of 2009 established an abbreviated licensure pathway for biological products demonstrated to be biosimilar to, or interchangeable with, an FDA-licensed reference biological product. A biosimilar product must be shown to be highly similar to the reference product and to have no clinically meaningful differences in safety, purity, or potency. An interchangeable product must additionally be shown to produce the same clinical result in any given patient and, for products administered more than once, to be capable of alternating or switching with the reference product without increased risk. A reference biological product is entitled to 12 years of regulatory data exclusivity from the date of first licensure, and the FDA may not accept an application for a biosimilar or interchangeable product until four years after that date.

Pursuant to the Orphan Drug Act, certain product candidates may also receive orphan drug designation for the treatment of rare diseases or conditions. In the United States, a rare disease or condition is statutorily defined as a condition that affects fewer than 200,000 individuals in the United States or that affects more than 200,000 individuals in the United States and for which there is no reasonable expectation that the cost of developing and making available the biologic for the disease or condition will be recovered from sales of the product in the United States.

Orphan drug designation qualifies a company for tax credits and market exclusivity for seven years following the date of the product's marketing approval if granted by the FDA. A sponsor may apply for orphan designation at any time prior to submitting a marketing application. Designation is granted by the FDA's Office of Orphan Products Development (OOPD) based on a confidential request, after which the product must undergo the standard review and approval process for commercial distribution.

A sponsor may seek orphan designation for an unapproved product, a new orphan indication for an approved product, or, in limited circumstances, for a product that is the same as an already approved orphan drug if they can demonstrate a plausible hypothesis of clinical superiority. Multiple sponsors may receive orphan designation for the same product and indication, provided each submits a complete request.

Orphan drug exclusivity begins on the date of FDA approval and applies only to the designated indication. During the exclusivity period, the FDA generally may not approve the same product by another manufacturer for the same indication unless the original sponsor consents, cannot supply sufficient quantities, or the subsequent product is clinically superior.

Drug development in the European Economic Area

Medicinal products can be marketed in the European Economic Area (EEA), which is comprised of the 27 Member States of the EU and Norway, Iceland and Liechtenstein, only if a marketing authorization from the competent regulatory authority has been obtained. Similar to the United States, the various phases of preclinical and clinical research in the EU/EEA are subject to significant regulatory controls. Effective since January 2022, the Clinical Trials Regulation (EC) No. 536/2014 aims to streamline and harmonize the procedures for assessment and governance of clinical trials throughout the EU and to require that information on the authorization, conduct and results of each clinical trial conducted in the EU be publicly available.

Pediatric investigation plan

An application for marketing authorization of a medicinal product for human use that is not yet authorized in the EU must include a Pediatric Investigational Plan (PIP) pursuant to the Regulation No. 1901/2006 on medicinal products for pediatric use (known as the Paediatric Regulation), unless a waiver applies. A scientific committee established at the European Medicines Agency (EMA), the Paediatric Committee (PDCO) assesses the content of any PIP, waivers, and deferrals for a medicinal product submitted to it and formulates an opinion thereon.

Review and approval process

In the EU/EEA, in order to obtain a marketing authorization from the applicable regulatory authority, a company may submit marketing authorization applications either under a centralized or national procedure. The centralized procedure is mandatory for certain categories of medicinal products, including those developed using specified biotechnological processes, advanced therapy medicinal products, orphan medicinal products, and certain innovative medicinal products, and is optional for other medicinal products that are considered highly innovative or contain a new active substance. The centralized procedure results in a single marketing authorization that is valid throughout the EU and EEA.

In addition, a marketing authorization may be obtained through a national procedure, which requires separate applications to and approvals by individual Member States, through a decentralized procedure under which applications are submitted simultaneously to multiple Member States, or through a mutual recognition procedure, under which a marketing authorization granted in one Member State may be recognized by others.

A conditional marketing authorization may be granted in the EU when comprehensive clinical data for the safety and efficacy of the medicinal product have not been supplied but all the following requirements are met: (i) the risk-benefit balance of the medicine is positive; (ii) it is likely that the applicant will be in a position to provide the comprehensive clinical data post-authorization; (iii) the medicine fulfills an unmet medical need; and (iv) the benefit to public health of the immediate availability on the market of the medicine outweighs the risk that additional data is still required. Conditional marketing authorizations are valid for one year, on a renewable basis. The marketing authorization holder will be required to fulfil specific obligations within certain timeframes, which may include completing ongoing trials or conducting new trials to confirm that the benefit-risk balance is positive. Once such obligations are fulfilled, provided the benefit-risk balance is still positive, a conditional marketing authorization can be converted into a standard marketing authorization.

European regulatory data protection

In the EU, innovative medicinal products authorized for marketing qualify for regulatory data protection consisting of eight years of data exclusivity and an additional two years of market protection upon the grant of a marketing authorization. Data exclusivity prevents generic or biosimilar applicants from referencing the innovator’s data when applying for a generic or biosimilar marketing authorization. During the additional two-year period of market protection, a generic or biosimilar marketing authorization application can be submitted, and the innovator’s data may be referenced, but no generic or biosimilar product can be marketed until the expiration of the market protection period. There is no guarantee that a product will be considered by the EU’s regulatory authorities to be an innovative medicinal product, and products may not qualify for regulatory data protection.

European orphan designation and exclusivity

Orphan medicinal product designation is available in the EU to promote the development of products that are intended for the diagnosis, prevention, or treatment of life threatening or chronically debilitating conditions affecting not more than five in 10,000 persons in the EU community, or where it is unlikely that the development of the medicine would generate sufficient return to justify the necessary investment in its development, and in each case for which no satisfactory method of diagnosis, prevention, or treatment has been authorized (or, if a method exists, the product would be a significant benefit to those affected). Medicinal products that receive and maintain orphan drug designation following approval are entitled to 10 years of market exclusivity, which protects against applications for and the grant of marketing authorizations for similar medicinal products in the same therapeutic indication. This period may be reduced to six years if, at the end of the fifth year, it is established that the orphan drug designation criteria are no longer met. During the period of market exclusivity, a marketing authorization for a similar medicinal product for the same therapeutic indication may only be granted if the subsequent product is demonstrated to be clinically superior, the marketing authorization holder consents, or the holder of the authorized orphan medicinal product is unable to supply sufficient quantities of the product to meet patient needs.

The aforementioned EU rules are generally applicable in the EEA.

Reform of the Regulatory Framework in the European Union

The European Commission introduced legislative proposals in April 2023 that, if implemented, will replace the current regulatory framework in the EU for all medicines (including those for rare diseases and for children). In April 2024, the European Parliament adopted its position on the legislative proposals and, in June 2025, the Council of the European Union adopted its position. A common position on the text has been agreed upon on December 11, 2025, in the context of subsequent inter-institutional trilogue negotiations. The proposed revisions remain to be adopted, and are not expected to become applicable before 2028.

European data protection regulations

The EU General Data Protection Regulation (EU GDPR) governs the collection and use of personal data in the EU. The EU GDPR imposes strict requirements relating to the consent of the individuals to whom the personal data relates, the collection and processing of sensitive data (such as health data), the information provided to the individuals, the security and confidentiality of the personal data, data breach notification and the use of third-party processors in connection with the processing of the personal data. The EU GDPR also imposes strict rules on the transfer of personal data out of the EU, provides an enforcement authority and imposes large penalties for noncompliance, including the potential for fines of up to €20.0 million or 4% of the annual global revenues of the infringer, whichever is greater. The GDPR also confers a private right of action on data subjects and consumer associations to lodge complaints with competent national data protection authorities, seek judicial remedies and obtain compensation for damages resulting from violations of the GDPR. Non-compliance could also result in the imposition of orders to stop data processing activities.

The UK has incorporated the GDPR into UK law (the UK GDPR). The UK GDPR and the UK Data Protection Act 2018 set out the UK's data protection regime, which is independent from but aligned to the EU's data protection regime. Although the UK is regarded as a third country under the EU's GDPR, the European Commission has issued a decision recognizing the UK as providing adequate protection under the EU GDPR and, therefore, transfers of personal data originating in the EU to the UK remain unrestricted. Like the GDPR, the UK GDPR restricts personal data transfers outside the UK to countries not regarded by the UK as providing adequate protection. The UK government has confirmed that personal data transfers from the UK to the EEA remain free flowing. Non-compliance with the UK GDPR may result in monetary penalties of up to £17.5 million or 4% of worldwide revenue, whichever is higher.

Marketing of medicines in the EU

Similar to the Anti-Kickback Statute prohibition in the United States discussed below, the provision of benefits or advantages to physicians to induce or encourage the prescription, recommendation, endorsement, purchase, supply, order, or use of medicinal products is prohibited in the EU. Infringement of relevant EU laws could result in substantial fines and imprisonment. Payments may be made to physicians in limited circumstances, and in certain EU Member States such payments must be publicly disclosed. Moreover, agreements with physicians for the provision of services often must be the subject of prior notification and approval by the physician's employer, his or her competent professional organization, and/or the regulatory authorities of the individual EU Member States. These requirements are provided in the national laws, industry codes, or professional codes of conduct, applicable in the EU Member States. Failure to comply with these requirements could result in reputational risk, public reprimands, administrative penalties, fines, or imprisonment.

Rest of the world regulation

Outside of the United States and the EU, the requirements governing the conduct of clinical trials, product licensing, pricing, and reimbursement vary from country to country. If we fail to comply with such requirements, we may be subject to, among other things, fines, suspension or withdrawal of regulatory approvals, product recalls, seizure of products, operating restrictions, or criminal prosecution.

Coverage and reimbursement

Patients who are provided medical treatment for their conditions generally rely on third-party payors to reimburse all or part of the costs associated with their treatment. Coverage and adequate reimbursement from governmental healthcare programs, such as Medicare and Medicaid, and commercial payors is critical to new product acceptance. Government authorities and other third-party payors, such as private health insurers and health maintenance organizations, decide which drugs and treatments they will cover and the amount of reimbursement.

In the United States, there is no uniform policy for coverage and reimbursement among third-party payors. As a result, obtaining coverage and reimbursement approvals can be time-consuming and costly and may require submission of scientific, clinical, and economic data to individual payors on a case-by-case basis, with no assurance of favorable outcomes. Coverage and reimbursement decisions for new medicines are often influenced by determinations made by the Centers for Medicare and Medicaid Services, and private payors frequently follow CMS policies. Even if coverage is obtained, reimbursement rates may be inadequate to achieve or sustain profitability or may require patient cost sharing that limits access or utilization. In addition, third-party payors may limit or exclude coverage for follow-up care or monitoring associated with the use of approved products. For vaccines, coverage decisions may be influenced by recommendations of the Advisory Committee on Immunization Practices, although such recommendations do not guarantee coverage or reimbursement. It is therefore difficult to predict at this time what third-party payors will decide with respect to the coverage and reimbursement for our product candidates.

Beginning in 2026, the Medicare Drug Price Negotiation Program established under the Inflation Reduction Act of 2022 requires CMS to implement negotiated prices for certain high-expenditure Medicare Part D drugs. CMS has issued guidance for multiple cycles of the negotiation program, and additional negotiated pricing rounds are expected in future years. The scope, timing, and impact of this program, including its potential application to additional drugs or coverage categories, remain subject to change and could affect pricing benchmarks, reimbursement levels, and net realized prices for our products.

Net prices for drugs may also be reduced by mandatory discounts or rebates required by government healthcare programs or negotiated by private payors. Manufacturers are subject to complex price reporting obligations, including reporting of metrics such as average sales price and best price, and may be subject to penalties for failure to report accurately or timely. Changes in laws or policies affecting drug importation or pricing methodologies could further pressure pricing in the United States.

Current and future healthcare reform initiatives may result in more restrictive coverage criteria, additional pricing pressure, expanded post-approval requirements, or further limitations on sales and promotional activities for pharmaceutical products. The impact of these measures on coverage, reimbursement, and pricing for our products remains uncertain.

In addition, in certain foreign jurisdictions, proposed pricing for a drug must be approved before the product may be marketed. Pricing and reimbursement requirements vary widely by country. In the EU, Member States may limit the medicinal products eligible for reimbursement under national health insurance systems and may impose direct or indirect controls on pricing or profitability. To obtain reimbursement or pricing approval, some Member States may require clinical or health economic data, including comparative cost effectiveness analyses, relative to existing therapies. A Member State may approve a specific price for a medicinal product or instead apply other pricing or profitability controls.

There can be no assurance that any country with price controls or reimbursement limitations will permit favorable pricing or reimbursement for any of our products or product candidates. Prices for pharmaceutical products in the EU are generally lower than prices in the United States.

For further information on risks associated with pricing and reimbursement, see “Risk Factors—Sales of pharmaceutical products depend on the availability and extent of reimbursement from third-party payors, and we may be adversely impacted by changes to such reimbursement policies or rules.”

Other healthcare laws

Healthcare providers, physicians, and third-party payors, including governmental payors such as Medicare and Medicaid in the United States, play a significant role in the recommendation, prescription, and reimbursement of pharmaceutical products. Our arrangements with these parties are subject to extensive federal, state, and foreign healthcare laws and regulations. In the United States, these laws include, among others:

- The federal Anti-Kickback Statute, which prohibits knowingly and willfully offering, paying, soliciting, or receiving remuneration to induce or reward referrals or the purchase, recommendation, or prescription of products reimbursable under federal healthcare programs.
- The federal False Claims Act, which imposes civil liability, including through whistleblower actions, for knowingly submitting or causing the submission of false or fraudulent claims for payment to federal healthcare programs or for avoiding or decreasing an obligation to pay the federal government.
- Federal criminal healthcare fraud provisions, including those under the Health Insurance Portability and Accountability Act (HIPAA), which impose criminal and civil penalties for schemes to defraud healthcare benefit programs or for making false statements in connection with healthcare benefits, items, or services.
- HIPAA, as amended by the Health Information Technology for Economic and Clinical Health Act, and their implementing regulations, which impose obligations relating to the privacy and security of individually identifiable health information.
- The Physician Payments Sunshine Act, enacted as part of the Patient Protection and Affordable Care Act (ACA), which requires certain manufacturers to disclose annually payments and other transfers of value made to physicians, teaching hospitals, and certain other healthcare professionals.
- Federal government price reporting laws, which require manufacturers to calculate and report pricing metrics accurately and timely to government programs.
- Federal consumer protection and unfair competition laws that regulate marketing and business practices.
- Analogous state fraud and abuse laws, including state anti-kickback and false claims laws, which may be broader in scope and apply regardless of payor.

Additionally, certain state and foreign data privacy and security laws govern the processing and protection of personal information and may impose requirements that differ from, and are not preempted by, HIPAA. For example, the California Consumer Privacy Act and the California Privacy Rights Act establish comprehensive privacy obligations and enforcement mechanisms, including expanded consumer rights and potential statutory damages. While certain clinical trial data governed by HIPAA are exempt, other personal information we process may be subject to these laws. Similar privacy laws have been enacted or proposed in other U.S. states and foreign jurisdictions. The scope and enforcement of these laws continue to evolve, increasing compliance complexity and risk.

The scope and enforcement of each of these laws is uncertain and subject to rapid change in the current environment of healthcare reform.

Current and future healthcare reform legislation

In the United States and internationally, legislative and regulatory changes affecting the healthcare system may prevent or delay marketing approval, restrict post-approval activities, or affect our ability to commercialize products profitably. Existing healthcare reform laws and future reforms may result in more restrictive coverage criteria, increased pricing pressure, or expanded compliance obligations.

In the United States, the ACA introduced measures intended to increase competition and reduce healthcare costs, including provisions affecting biosimilar competition, Medicaid rebate obligations, Medicare discounts, and industry fees. Other federal laws have further affected reimbursement dynamics. The Budget Control Act of 2011 resulted in reductions to Medicare payments that are currently scheduled to remain in effect through 2031 absent further legislative action. The American Rescue Plan Act eliminated the statutory Medicaid drug rebate cap beginning in 2024, which may increase rebate liabilities for certain drugs.

In August 2022, the Inflation Reduction Act was enacted and includes provisions that are being implemented over time and are expected to affect pharmaceutical pricing and reimbursement. These provisions include a cap on Medicare Part D beneficiary out-of-pocket spending, increased manufacturer financial liability under Medicare Part D, inflation-based rebates, and a program allowing the federal government to negotiate prices for certain high-expenditure drugs and biologics without generic or biosimilar competition, with negotiated prices taking effect beginning in 2026. The scope and impact of these measures, including future negotiation cycles, remain subject to change and uncertainty.

In addition, pharmaceutical pricing practices continue to be subject to heightened scrutiny by federal and state governments, including legislative proposals and investigations focused on pricing transparency, patient assistance programs, and reimbursement methodologies. Governments in the United States and abroad have also implemented or proposed cost-containment measures, such as price controls, reference pricing, reimbursement restrictions, upper-payment limits, drug price transparency reporting and substitution requirements, which could further affect pricing, coverage, and utilization of pharmaceutical products.

Environment

We are subject to state and federal laws regarding environmental protection and hazardous substances, including the Occupational Safety and Health Act, the Resource Conservation and Recovery Act and the Toxic Substances Control Act. These and other laws govern the use, handling and disposal of various biologic, chemical and radioactive substances used in, and wastes generated by, operations. If our operations result in contamination of the environment, breach of our regulatory obligations or expose individuals to harm, we could be liable for damages and governmental fines. Equivalent laws have been adopted in foreign countries that impose similar obligations.

CORPORATE INFORMATION

We were incorporated under the laws of the State of Delaware on July 22, 2016. We are the successor in interest to Moderna LLC, a limited liability company formed under the laws of the State of Delaware in 2013. Moderna LLC was the successor in interest to Moderna Therapeutics, Inc., a Delaware corporation incorporated in 2009 as Newco LS18, Inc. by Flagship Pioneering. In August 2018, we changed our name from Moderna Therapeutics, Inc. to Moderna, Inc. Our principal corporate office is located at 325 Binney Street, Cambridge, MA 02142, and our telephone number is (617) 714-6500.

Our website, www.modernatx.com, including the Investor Relations section, www.investors.modernatx.com; corporate blog www.modernatx.com/moderna-blog, and our Statements and Perspectives webpage, https://investors.modernatx.com/Statements--Perspectives/default.aspx; as well as our social media channels: Facebook, www.facebook.com/modernatx; X, www.x.com/moderna_tx; and LinkedIn, www.linkedin.com/company/modernatx; contain a significant amount of information about us, including financial and other information for investors. We encourage investors to visit these websites and social media channels as information is frequently updated and new information is shared. The information on our website and that we disclose through social media channels is not incorporated by reference in this Annual Report on Form 10-K or in any other filings we make with the Securities and Exchange Commission (the SEC).

We make available free of charge on or through our website certain reports and amendments to those reports that we file with or furnish to the SEC pursuant to Section 13(a) or 15(d) of the Exchange Act as soon as reasonably practicable after we electronically file such material with, or furnish it to, the SEC. These include our Annual Reports on Form 10-K, our Quarterly Reports on Form 10-Q and our Current Reports on Form 8-K, and amendments to those reports.

The SEC also maintains an Internet site (http://www.sec.gov) that contains reports, proxy and information statements, and other information regarding us and other issuers that file electronically with the SEC.

Item 1A. Risk Factors

You should carefully consider the following risks and uncertainties, together with all other information in this Annual Report on Form 10-K. Any of the risk factors we describe below could adversely affect our business, financial condition or results of operations and the market price of our common stock.

Risks related to commercialization and our products

Regulatory and market uncertainty have and may continue to impact our business and the markets for our products.

Our ability to successfully commercialize our approved vaccines and any future products depends in part on evolving regulatory requirements, public health recommendations and market acceptance. In 2025, changes at U.S. regulatory agencies impacted policies and priorities related to our industry. For example, changes in FDA regulatory policies, post-marketing safety monitoring, and evidentiary expectations, as well as CDC and Advisory Committee on Immunization Practices (ACIP) recommendations regarding eligibility, target populations, and vaccination practices, may have affected and may continue to affect demand for our vaccines. Some of these changes, including new post-marketing commitments, have impacted and may in the future impact our costs. Ongoing regulatory uncertainty and evolving regulatory and public health guidance could impact future approvals, advisory committee recommendations, acceptance and demand for our existing or future products.

Vaccination rates in the future may also be lower than our expectations due to other factors, including viral evolution, medical need and consumer motivation to vaccinate. These and other factors may make it difficult to accurately forecast vaccination rates for our products. If demand for COVID vaccines continues to decline, we lose significant market share, or our products are subject to significant competitive pricing pressure, our product sales may not materialize consistent with our projections. Beyond COVID, we have experienced challenges in the RSV market, in part due to advisory committee recommendations that were more limited than anticipated.

Furthermore, our decisions regarding seasonal vaccine development, including for COVID-19, will be informed by annual guidance from the FDA and foreign regulators, which may impact the timing of development for our COVID vaccines. Different regulators have in the past and may in the future issue varying guidance regarding vaccine composition or populations who should receive a vaccine. Additionally, if our efforts to develop variant-specific vaccines are delayed or not as successful as our competitors', we could suffer reputational harm, loss of market share and adverse financial results. We may expend significant resources adapting our vaccines or conducting clinical trials to protect against variants, but a market for our adapted vaccines may fail to develop or demand may not align with our projections or cost expenditures.

We may experience difficulties executing our near-term strategy and prioritized pipeline.

While we expect to launch multiple new products over the next several years, our ability to commercialize our pipeline is subject to many risks, including those described elsewhere in these *Risk Factors* under "Risks related to our pipeline, product development and regulatory review" and "Risks related to the manufacturing of our commercial products and product candidates." If we do not successfully execute our near-term plan for growing our infectious disease vaccine franchise, our ability to invest in our pipeline, including our oncology and rare disease programs, may suffer.

Furthermore, our broad clinical success and post-pandemic commercial challenges have necessitated a more selective and paced approach to our research and development investment. If we do not successfully implement our cost efficiency and prioritization programs, we may fail to meet our cash breakeven goal. Additionally, as we pursue development of our prioritized programs, we may forego or delay pursuit of other opportunities that could prove to have greater commercial potential. If our prioritized programs are unsuccessful, or not as successful as others could have been, we may be unable to realize a sustainable return on our investments and or achieve long-term growth.

The vaccine market, and pharmaceutical market more generally, is intensely competitive, and we may be unable to compete effectively in the market for existing or new products, treatment methods or technologies.

We compete with larger, well-established and experienced pharmaceutical companies for sales of our products, including against Pfizer and Sanofi for sales of our COVID vaccines and Pfizer and GSK for our RSV vaccine. Additionally, we have been excluded from selling our COVID vaccines in many European markets due to a competitor's contract with the European Commission, which does not

lapse until year-end 2026. For RSV vaccines, we entered a market already occupied by two larger competitors and may continue to face difficulties achieving market share. These competitors have exploited and may in the future exploit their greater size, infrastructure, resources and experience to gain advantages in contracting with customers by, among other things, bundling their products, leveraging larger supply chains and greater purchasing power and utilizing their global networks.

In addition, certain U.S. private vaccine market practices, including regarding discounts, rebates and returns, may cause us to realize significantly lower revenues than list prices. We may also be adversely affected by similar market practices outside of the United States. In some instances, our competitors have been able to offer more attractive terms than we can, and they may continue to do so in the future.

More generally, the pharmaceutical market is highly competitive and certain of the disease areas we are targeting present particularly high competitive risk. For example, the oncology market is intensely competitive, innovative and fast-moving. Other companies, academic institutions, governmental agencies and public and private research organizations are developing products that may compete with programs in our development pipeline. These competitors may have products that have already been approved and accepted by the medical community or are in later stages of development. For example, we are developing a seasonal flu vaccine, for which there is a well-developed market, and we may need to offer more favorable terms to gain market share (which we may be unable to do), which may negatively impact our profitability. Additionally, competitive products may make any products we develop obsolete or noncompetitive before we can recover the expenses of developing and commercializing our products. In oncology, standards of care often evolve rapidly, and changes during the development or commercialization of our product candidates could reduce their clinical relevance, complicate clinical trial design, increase development costs or timelines, or limit commercial adoption, even for programs that show promising early results.

Additionally, our products may struggle to compete against competitors' products for a variety of reasons, including relative safety and effectiveness, degree of any side effects, shelf-life, ease of administration and the extent to which patients accept relatively new routes of administration, the timing and scope of regulatory approvals, the availability and cost of manufacturing, distribution, marketing and sales capabilities, price, reimbursement coverage and patent protection. Even if our products demonstrate superiority to competitors' products, consumers and retailers may fail to appreciate that benefit, or existing purchase commitments with competitors may discourage them from purchasing from us. These factors, or the perception of these factors, could lead to a competitor's product being more successfully commercialized. Further, the mRNA medicines field is growing rapidly, with increased competitive pressure from large and more established pharmaceutical companies. The actual or perceived success or failure of others may adversely impact our ability to commercialize our products. See "Business—Government Regulation—Coverage and reimbursement."

We have experienced commercial challenges and may experience additional challenges in the future.

We face risks and uncertainties related to successfully commercializing our products. In the past, commercial challenges have led to lower-than-expected sales and required us to adapt our business strategy. Moving forward, we may need to dedicate greater resources to our commercial efforts than we anticipate, and we may not realize a return on this investment. Additionally, we may be unsuccessful in accurately anticipating future rates of return for our products, which may adversely impact our accounting estimates.

Furthermore, we may seek to enter into agreements with others to utilize their marketing and distribution capabilities, but may be unable to enter into agreements on favorable terms, if at all. If we rely on others to commercialize our products, our revenues may be lower than if we commercialized these products ourselves. In addition, we may have little or no control over such third parties' sales efforts. If our partners commit insufficient resources to commercialize our products, and we cannot independently develop necessary marketing capabilities, we may be unable to generate sufficient product revenue to sustain our business.

The commercial success of our products depends on the degree of market acceptance by physicians, patients, third-party payors and others in the medical community.

The degree of market acceptance of our products will depend on numerous factors, including:

- efficacy and potential advantages over alternative treatments;
- the duration of protection provided by our products compared to those of our competitors;
- acceptance of mRNA products generally and the availability of competing non-mRNA medicines that may be preferred by the medical community or the public;
- safety and the prevalence and severity of any side effects, including any limitations, restrictions (including for use together with other medicines) or warnings contained in a product's approved labeling;
- the prevalence and severity of any side effects resulting from checkpoint inhibitors or other products or therapies with which our products are co-administered;
- relative convenience and ease of administration;
- the willingness of the target patient population to try, and physicians to prescribe, new therapies;
- the strength of marketing and distribution support and timing of market introduction of competitive products;

- whether our product presentation meets customer demand;
- publicity and health authority communications concerning our products or competing products and treatments; and

- product cost and sufficient third-party insurance coverage or reimbursement, and patients' willingness to pay out-of-pocket in the absence of third-party coverage or adequate reimbursement.

Even if a product candidate shows a favorable efficacy and safety profile in clinical trials, market acceptance will be unknown until after it is launched. Our efforts to educate the medical community and third-party payors on the benefits of our products may require significant resources and may never be successful.

Sales of pharmaceutical products depend on the availability and extent of reimbursement from third-party payors, and we may be adversely impacted by changes to such reimbursement policies or rules.

Sales of pharmaceutical products depend on adequate coverage, pricing and reimbursement from third-party payors. When a new product is approved, the availability and extent of government and private reimbursement, and the pricing, for that product may be uncertain. Additionally, pricing and reimbursement for any product we develop may be adversely affected by various factors, including:

- changes in, and implementation of, federal, state or foreign government regulations or private third-party payors' reimbursement policies;
- pressure by employers on private health insurance plans to reduce costs; and
- consolidation and increasing assertiveness of payors seeking price discounts or rebates in connection with the placement of our products on their formularies and, in some cases, the imposition of restrictions on access or coverage of particular drugs or pricing determined based on perceived value.

Our ability to set the price for any product we develop will vary significantly by country. Our inability to obtain and maintain adequate prices in a particular country may limit the revenues from our products within that country and adversely affect our ability to secure acceptable prices in existing and potential new markets, which may limit market growth. This may create the opportunity for third-party cross-border trade or influence our decision whether to sell a product, thus adversely affecting our geographic expansion plans and revenues.

Many payors continue to adopt benefit plan changes that shift a greater portion of prescription costs to patients, including more limited benefit plan designs, higher patient co-pay or co-insurance obligations and limitations on patients' use of commercial manufacturer co-pay payment assistance programs. Significant consolidation in the health insurance industry has resulted in a few large insurers and pharmacy benefit managers exerting greater pressure in pricing and usage negotiations with drug manufacturers, significantly increasing discounts and rebates required of manufacturers and limiting patient access and usage. Further consolidation among insurers, pharmacy benefit managers and other payors would increase the negotiating leverage such entities have over us and other drug manufacturers. Additional discounts, rebates, coverage or plan changes, restrictions or exclusions as described above could have a material adverse effect on sales of our affected products, particularly our therapeutic products or those that are individualized for a particular patient. Coverage and reimbursement by a third-party payor may depend on various factors, including the third-party payor's determination that use of a product is a covered benefit under its health plan, safe, effective and medically necessary, appropriate for the specific patient, cost-effective and neither experimental nor investigational.

Additionally, target patient populations for some of our product candidates may be small (e.g., for rare diseases) or require individual customization (e.g., for intismeran autogene). The pricing and reimbursement of our medicines, if approved, must be adequate to support commercial infrastructure. If we cannot obtain adequate levels of reimbursement, we may be unable to successfully market and sell our products. Further, inadequate reimbursement for services related to our products (e.g., for administration to patients) may discourage physicians from prescribing or recommending our products, adversely affecting our ability to market or sell those products.

Federal legislative and regulatory efforts to implement reference pricing or most-favored-nation pricing models and other, similar regulatory actions could impact our product revenues and materially harm our business.

On May 12, 2025, President Trump issued an executive order calling on pharmaceutical manufacturers to voluntarily reduce the prices of medicines in the U.S. and directing the Secretary of Health and Human Services (HHS) to communicate price targets to pharmaceutical manufacturers to align prices with those in comparably developed nations, an approach commonly referred to as "most-favored-nation" (MFN) pricing and, in the event significant progress towards MFN pricing is not delivered, to propose rulemaking to impose MFN pricing.

MFN pricing models in the U.S. could also affect our international pricing strategy and future decisions on reimbursement and commercialization in certain jurisdictions. If our U.S. pricing becomes tied to international reference prices, we may face decisions

regarding pricing in foreign markets that could result in reduced patient access internationally, affect our relationships with foreign regulatory authorities and payers, or impact our ability to obtain or maintain reimbursement approvals in ex-U.S. markets.

These reforms remain subject to change, potential legal challenges, or expansion through additional rulemaking or sub-regulatory guidance, creating uncertainty for our overall pricing strategy. It remains to be seen whether and how these drug pricing initiatives will apply to our products, how they will affect the broader pharmaceutical industry, and whether similar reform measures may be adopted in the future.

The market opportunities for our products and product candidates may be smaller than we believe, or we may be unable to successfully identify clinical trial participants.

We focus certain of our research and development activities on treatments for severe rare genetic diseases, where patient populations are small and difficult to ascertain. Additionally, our products may only be approved for certain populations or lines of treatment.

Our estimates of addressable patient populations are based on our beliefs and are derived from a variety of sources, including scientific literature, surveys of clinics, patient foundations or market research, and may prove to be incorrect. Further, new studies may change the estimated incidence or prevalence of these diseases. The number of trial participants may be lower than expected and potential clinical trial participants or patients may not be otherwise amenable to treatment with our product candidates or products, or new clinical trial participants or patients may become increasingly difficult to identify or gain access to. Even if we obtain significant market share for our products, if approved, because the potential target populations are small, these medicines may never be profitable. Furthermore, the size of markets that we target may be impacted by health authority recommendations regarding who should receive our products, which may be impacted by the new U.S. federal government administration.

Risks related to our pipeline, product development and regulatory review

If we cannot obtain, or are delayed in obtaining, regulatory approvals and advisory committee recommendations, we will be unable to effectively commercialize, or will be delayed in commercializing, our product candidates.

Any products we may develop are subject to comprehensive regulation by the FDA and comparable foreign regulators. To obtain required regulatory approvals to commercialize any product candidate, we must demonstrate through extensive preclinical studies and clinical trials that our products are safe and effective for their intended use. In addition, regulators conduct pre-approval inspections and negative findings may lead to delay in approval and failure to commercialize a product candidate. Even once approved, products may be subject to recommendations from advisory committees, such as the ACIP, before or after they can be brought to market.

To date, we have only received regulatory approvals for our COVID and RSV vaccines, and our current or future product candidates may never obtain regulatory approval or advisory committee recommendations. We may need to rely on third parties to assist us in making applications for marketing approvals. Preclinical studies and clinical trials conducted in one country may not be accepted by regulators elsewhere, and regulatory approval in one country does not guarantee approval in another. Regulators may require that we conduct additional clinical trials to support our applications for approval beyond our clinical development plans, which may result in increased expense and delays in bringing our products to market.

Additionally, the process of obtaining marketing approvals is expensive, time-consuming and uncertain, and can vary substantially based on many factors, including the type, complexity and novelty of the product candidates involved. Changes in marketing approval policies during the development period, changes in law or changes in regulatory review may delay the review of a submitted product application.

For example, recent FDA policy updates regarding COVID vaccine approvals and the evidence expected for certain populations could require additional clinical data or trials to support broader indications, delay approvals or supplemental applications, or result in more limited labeled populations, which could adversely affect the timing and scope of commercialization for our COVID vaccines, including updated or variant-adapted formulations.

The FDA and comparable foreign regulators have substantial discretion in the approval process and may refuse to accept any marketing approval application or may decide that our data are insufficient for marketing approval and require additional preclinical, clinical or other studies. In addition, varying interpretations of the data obtained from preclinical and clinical testing could delay, limit or prevent marketing approval of a product candidate. Additional delays or non-approval may result if an FDA advisory committee or other regulator recommends non-approval or restrictions on approval.

Clinical development is lengthy and uncertain, and our clinical programs may be delayed or terminated, or may be more costly to conduct than we anticipate.

Clinical testing is expensive, complex and lengthy, and its outcome is inherently uncertain. Most product candidates that commence clinical trials are never approved as products. Although we have demonstrated historical success with our platform technology, we may not continue to realize the same levels of success with clinical trials in the future. We and our strategic collaborators also may experience unforeseen events during, or as a result of, clinical trials that could delay or prevent us or them from successfully developing our product candidates and gaining approval from regulators. Such events could include:

- regulators, Institutional Review Boards (IRBs) or ethics committees may not authorize us or our investigators to commence a clinical trial or conduct a clinical trial at a prospective trial site;
- we may experience delays in reaching, or fail to reach, agreement on favorable terms with prospective trial sites and prospective contract research organizations (CROs);
- changes to the scale or site of our manufacturing could cause significant delays or changes in our clinical trial designs;
- the outcome of our preclinical studies and our early clinical trials may not be predictive of the success of later clinical trials, and interim results of a clinical trial do not necessarily predict final results;
- we may be unable to establish or achieve clinically meaningful endpoints for our studies;
- if we make changes to our product candidates after clinical trials have commenced (which we have done in the past), we may be required to repeat earlier stages or delay later stages of clinical testing;
- clinical trials of any product candidates may fail to show safety or efficacy, or may produce negative or inconclusive results, and we may decide, or regulators may require us, to conduct additional nonclinical studies or clinical trials, or we may decide to abandon product development programs;
- our product candidates, or other medicines in the same class as ours, may cause significant adverse events or other undesirable side effects, such as the immunogenicity of the LNPs or their components, the immunogenicity of the protein made by the mRNA or degradation products, any of which could lead to serious adverse events, or other effects;
- our third-party contractors may fail to comply with regulatory requirements or meet their contractual obligations to us in a timely manner, or at all, or may deviate from the applicable clinical trial protocol or withdraw from the applicable clinical trial, which may require that we add new clinical trial sites;
- regulators may impose a complete or partial clinical hold on a clinical trial (or a trial of another company working on mRNA medicines), or we or our investigators, IRBs or ethics committees may suspend or terminate clinical research or trials for various reasons, including quality events, noncompliance with regulatory requirements or a finding that participants are being exposed to an unacceptable benefit-risk ratio;
- the supply or quality of our product candidates or other materials necessary to conduct clinical trials may be insufficient or inadequate;
- safety and efficacy concerns regarding our product candidates will be considered by us and by the FDA and other regulators as we pursue clinical trials of new product candidates, develop effective informed consent documentation and work with IRBs and scientific review committees (SRCs);
- safety or efficacy concerns could arise from nonclinical or clinical testing of other therapies targeting a similar disease state or other therapies, such as gene therapy, that are perceived as similar to ours;
- adverse side effects could be observed in future clinical trials where our product candidates are administered in combination with other therapies (such as the co-administration of intismeran autogene);
- delays in developing assays acceptable to the FDA or other regulators; and
- a lack of adequate funding to continue a particular clinical trial, including due to higher-than-anticipated costs.

Additionally, we have conducted and may conduct in the future “open-label” clinical trials, where both the patient and investigator know whether the patient is receiving the investigational product candidate, an approved drug or a placebo. The results from an open-label clinical trial may not be predictive of future clinical trial results from a controlled environment with a placebo or active control. Further, the FDA or other regulators may change the requirements for approval even after they have reviewed and commented on the design for our clinical trials. Significant preclinical or nonclinical testing and studies or clinical trial delays for our product candidates could allow our competitors to bring products to market before we do and could harm our business.

There are risks unique to each of our programs and modalities and risks applicable across programs and modalities, which may delay or prevent our ability to advance one or more of our programs in clinical development, obtain regulatory approval or commercialize our products.

Certain features in our product candidates, including those related to mRNA, chemical modifications, surface chemistries, LNPs and their components, may result in risks that apply to some or all of our programs and modalities. As our product candidates progress, we or others may determine that certain of our risk allocation decisions were incorrect or insufficient, we made platform-level technology mistakes, individual programs or our mRNA science in general has technology or biology risks that were unknown or under-appreciated, our choices on how to develop our infrastructure to support our scale will result in an inability to manufacture our product candidates for clinical trials or otherwise impair our manufacturing or we have allocated resources in such a way that we cannot recover large investments or rapidly re-direct capital.

We utilize earlier programs in a modality to understand the technology risks within the modality, including the program's manufacturing and pharmaceutical properties. Even if our earlier programs in a modality are successful in any phase of development, other programs may fail at that stage, and any program may fail at a later phase of development. This may be a result of technical challenges or biology risk unique to that program. The biology risk across much of our pipeline represents targets and pathways not clinically validated by one or more approved drugs, and the risk that the targets or pathways that we have selected may not be effective will continue to apply across our current and future programs.

As we progress our programs through clinical development, new technical challenges may arise that cause an entire modality to fail. Additionally, any portfolio-spanning risks, whether known or unknown, such as an increased risk of a particular type of side effect, if realized in any one of our programs, would have a material and adverse effect on our other programs and on our overall business.

There are also specific additional risks to certain of our modalities and programs. For example, prophylactic vaccines typically require clinical testing in up to tens of thousands of healthy volunteers to define an approvable benefit-risk profile. The need to show a high degree of safety and tolerability when dosing healthy individuals could result in rare and even spurious safety findings, negatively impacting a program prior to or after commercial launch. Even if we observe positive safety, tolerability and levels of immunogenicity in early clinical trials, we may not observe acceptable safety or efficacy profiles in later-stage trials required for approval of these programs.

There are many clinical and manufacturing challenges specific to intismeran autogene and any other neoantigen therapies we may develop. These include a rapid production turn-around time measured in weeks in order to supply patients before further progression and mutation of their tumors, the significant costs incurred in making individualized medicines and potential lack of immune responses due to the biology of the tumor or patient's immune status. These risks apply to intismeran and other neoepitope investigational medicine programs.

In addition, the translatability of target selection from preclinical animal models to successful clinical trial results is uncertain and may fail, particularly for systemic therapies and cancer antigen therapies. In general, several biological steps are required for delivery of mRNA to translate into therapeutically active medicines. These processing steps may differ between individuals or tissues, potentially leading to variable levels of therapeutic protein, activity, immunogenicity or distribution to tissues for a therapeutic effect. mRNA medicines may elicit innate or adaptive immune responses to the mRNA, delivery vehicle (such as an LNP) or the encoded protein, which could result in adverse events or reduced tolerability. In certain settings, immune responses to the encoded protein limit the effectiveness or durability of treatment, including upon repeat administration, or require changes to dosing, formulation or clinical development plans.

We may experience delays in enrolling participants in our clinical trials.

Enrolling participants in our clinical trials is critical to our success. Difficulties or delays in enrolling a sufficient number of participants could increase costs or impact the timing or outcome of our planned clinical trials, which could prevent trial completion and adversely affect our ability to advance the development of and obtain regulatory approval for our product candidates. Participant enrollment is affected by many factors, including:

- severity of the disease under investigation;
- complexity and design of the clinical trial protocol;
- size of the patient population and eligibility criteria for the clinical trial in question, including age-based eligibility criteria limiting subject enrollment to adolescent or pediatric populations;
- proximity and availability of clinical trial sites for prospective trial participants;

- availability of competing therapies and clinical trials, and clinicians' and participants' perceptions of the relative advantages, risks and side effects of our product candidates compared to other available therapies, including any new drugs or treatments;
- patient referral practices of physicians;

- ability to monitor trial participants adequately during and after treatment;
- ability to recruit qualified clinical trial investigators;
- adverse results or other safety signals in our trials or related to other product candidates, and any resulting negative publicity, which could discourage potential participants and their physicians from participating in our trials; and
- our ability to obtain and maintain participant informed consent.

Additionally, we may have limited or no ability to influence enrollment in clinical trials where our collaborators control clinical development. Even if we or our collaborators successfully enroll participants, some may not be dosed or complete the trial.

mRNA drug development involves substantial clinical and regulatory risks, and negative perceptions of our platform, products and product candidates could adversely affect our business and ability to obtain regulatory approvals.

Few mRNA medicines have been approved to date by the FDA or other regulators, and efficacy, safety, tolerability and immunogenicity data and real-world evidence with respect to mRNA medicines continue to accumulate. We may observe new, more frequent or more severe adverse events in clinical trial participants or among vaccinated individuals. For example, some studies have suggested that Spikevax may be associated with higher rates of myocarditis and pericarditis in young males compared to other COVID vaccines. Myocarditis and pericarditis have also been reported following vaccination with other COVID vaccines, including mNEXSPIKE. If similar observations are made in recipients of our other products or product candidates, or if other unexpected safety issues arise, we could suffer significant damage to our reputation and that of our mRNA platform. Such events could lead to other issues, including delays in our other programs, the need to re-design our clinical trials and the need for significant additional financial resources. In addition, the FDA and other regulators may interpret data from our clinical trials differently than we do and such agencies may require us to conduct additional studies or analyses, which could delay or prevent us from obtaining full regulatory approvals in certain jurisdictions or for certain demographics. Further, local legislatures may attempt to regulate or restrict the use of mRNA medicines in their jurisdictions.

Successful discovery and development of mRNA medicines by us or our strategic collaborators is highly uncertain and depends on many factors beyond our or their control, and our business decisions and development efforts relating to mRNA technology, delivery systems and manufacturing processes may ultimately be unsuccessful. Promising early-phase product candidates may fail to advance, be delayed in the clinic, be subject to clinical holds or fail to reach the market for many reasons, including:

- nonclinical, preclinical or clinical trial results may show potential mRNA medicines to be less effective than desired or to have harmful or problematic side effects or toxicities;
- adverse results in our clinical trials, or in those of others developing similar products, or adverse effects relating to mRNA, or our LNPs, may lead to negative publicity or delays in or termination of our programs;
- the efficacy or safety of a combination vaccine product candidate could be less than that seen with the administration of the vaccines separately, which could prevent the combination product from obtaining regulatory approval;
- adverse events related to products that are perceived to be similar to mRNA medicines, such as those related to gene therapy or gene editing, could result in a decrease in the perceived benefit of one or more of our programs, increased regulatory scrutiny, decreased confidence by patients and clinical trial collaborators in our product candidates and less demand for any product that we may develop;
- the insufficient ability of our translational models to reduce risk or predict outcomes in humans, particularly given that each component of our product candidates may have a dependent or independent effect on safety, tolerability and efficacy, which may be species-dependent;
- manufacturing failures or insufficient supply of current Good Manufacturing Practice (cGMP) materials for clinical trials, or higher than expected cost could delay or set back clinical trials, or make mRNA medicines commercially unattractive;
- changes that we make to optimize our manufacturing, testing or formulating of cGMP materials could impact the safety, tolerability and efficacy profile of our product candidates;
- pricing or reimbursement issues or other factors that delay clinical trials or make any mRNA medicine uneconomical or noncompetitive with other therapies;
- our large pipeline of product candidates could result in a greater quantity of reportable adverse events, including suspected unexpected serious adverse reactions, other reportable negative clinical outcomes, manufacturing reportable events or material clinical events that could lead to clinical delay or hold by the FDA or applicable regulatory authority or other clinical delays, any of which could negatively impact the perception of one or more of our programs, as well as our business as a whole;
- failure to timely advance our programs or a failure or delay in receiving necessary regulatory approvals due to, among other factors, slow or failure to complete enrollment in clinical trials, withdrawal by trial participants from trials, failure to achieve trial endpoints, additional time requirements for data analysis, data integrity issues, preparation of a BLA or the equivalent

application, discussions with the FDA or EMA, a regulatory request for additional nonclinical or clinical data or safety formulation or manufacturing issues may lead to our inability to obtain sufficient funding;

- new legislation or regulations passed by U.S., state or foreign governments in response to negative public perception of mRNA medicines; and
- the proprietary rights of others and their competing products and technologies that may prevent our mRNA medicines from being commercialized.

Because we are developing some of our product candidates for the treatment of diseases in which there is little clinical experience and, in some cases, using new endpoints or methodologies, the FDA or other regulators may not consider the endpoints of our clinical trials to provide clinically meaningful results.

There are no pharmacologic therapies approved to treat the underlying causes of many diseases that we are currently attempting to address or may address in the future. For many rare diseases, few clinical trials have been attempted. As a result, the design and conduct of clinical trials of product candidates for the treatment of these disorders may take longer, be more costly or be less effective.

Even if the FDA finds our success criteria to be sufficiently validated and clinically meaningful, we may not achieve the pre-specified endpoint to a degree of statistical significance in our pivotal or other clinical trials, or our results may be inconsistent or unpredictable compared to traditional efficacy endpoints. Further, the FDA may give overriding weight to other efficacy endpoints over a primary endpoint, even if we achieve statistically significant results on that endpoint, if we do not do so on our secondary efficacy endpoints. The FDA also weighs the benefits of a product against its risks and may view the efficacy results in the context of safety as not being supportive of licensure. Regulators in other countries may make similar findings with respect to these endpoints.

Certain mRNA therapies are classified as gene therapies by the FDA and the EMA. The association of our products with gene therapies could result in increased regulatory burdens, impair the reputation of our products or negatively impact our platform or our business.

There are only a few approved gene therapy products in the United States and other jurisdictions, and significant adverse events associated with their testing and use have been reported. Regulatory requirements governing these products have evolved and may continue to change in the future, and the implications for mRNA therapies are unknown. For example, the FDA has established the Office of Therapeutics Products within its Center for Biologics Evaluation and Research (CBER) to consolidate review of gene therapy and related products, and convenes the Cellular, Tissue and Gene Therapies Advisory Committee to advise CBER. In the EU, certain mRNA therapies have been characterized as gene therapy medicinal products, which are subject to additional regulatory requirements. In certain other countries, such as Japan, mRNA therapies have not yet been classified. Notwithstanding the differences between mRNA medicines and gene therapies, the classification of some of our mRNA product candidates as gene therapies could adversely impact our ability to develop our product candidates, including if a clinical hold applicable to gene therapy products were applied to our mRNA product candidates.

Adverse events reported with respect to gene therapies could adversely impact one or more of our programs. Although our mRNA product candidates are designed not to make any permanent changes to cell DNA, regulatory agencies or others could believe that adverse effects of gene therapies caused by introducing new DNA and irreversibly changing the DNA in a cell could also be a risk for our mRNA investigational therapies, and as a result may delay one or more of our clinical trials or impose additional testing for long-term side effects. Any new requirements and guidelines promulgated by regulatory agencies may negatively affect our business by lengthening the regulatory review process, requiring us to perform additional or larger studies or increasing our development costs, any of which could lead to changes in regulatory positions and interpretations, delay or prevent advancement or approval and commercialization of our product candidates, or lead to significant post-approval studies, limitations or restrictions. As we advance our product candidates, we will be required to consult with these regulatory agencies and advisory committees and comply with applicable requirements and guidelines. If we fail to do so, we may be required to delay or discontinue development of some or all of our product candidates.

Our work in genomic editing is subject to all risks associated with gene therapies. Genome editing technologies are relatively new and their therapeutic utility is largely unproven. Public perception and related media coverage of potential therapy-related efficacy or safety issues, as well as ethical concerns related specifically to genome editing, may adversely influence the willingness of subjects to participate in clinical trials. In addition, any review conducted by an institutional biosafety committee may result in delay or prevent initiation of a gene therapy clinical trial.

Additionally, if any such therapeutic is approved, physicians, health care providers, third-party payors and patients may be slow or fail to accept these novel and personalized treatments. Physicians may be unwilling to undergo necessary training, view a particular therapy as too complex or potentially risky to adopt without appropriate training and choose not to administer the therapy. Further, certain patients may not be candidates for the therapies, for many reasons, including due to health conditions or genetic profile. In addition, responses by federal and state agencies, congressional committees and foreign governments to negative public perception, ethical concerns or financial considerations may result in new legislation, regulations or medical standards that could limit our ability to develop or commercialize any product candidates, obtain or maintain regulatory approval or otherwise achieve profitability. Based on these and other factors, health care providers and payors may decide that the benefits of these new therapies do not or will not outweigh their costs.

Our products are, and any future products will be, subject to regulatory scrutiny.

Even if we obtain regulatory approval in a jurisdiction, we may remain subject to significant restrictions on the indicated uses or marketing of our product and ongoing requirements for potentially costly post-approval studies, such as those required under an accelerated approval by the FDA or other similar type of approval, or post-market surveillance. For example, a BLA holder must monitor and report adverse events and any failure of a product to meet applicable specifications, as well as submit and obtain FDA approval for certain new or supplemental applications relating to the approved product, labeling or manufacturing process. Additionally, pharmacovigilance obligations under the regulatory regimes of the jurisdictions where our products are distributed require us to collect, process, analyze and monitor safety data and to identify and evaluate adverse reactions to our products as they are administered in those jurisdictions. If we or any of our partners assisting us in meeting these obligations cannot comply with relevant regulations, we may be subject to sanctions, increased costs and reputational harm, or our regulatory authorizations to distribute our vaccines in the relevant jurisdiction may be revoked or curtailed. In addition, regulators may increasingly rely on post-approval studies or real-world evidence to determine whether to maintain, modify or expand approved indications or population eligibility, which could increase costs, delay commercialization or limit the scope of approved use of our products.

Furthermore, advertising and promotional materials must comply with FDA rules and regulations and are subject to FDA review, in addition to other applicable federal and state laws. In addition, regulatory agencies may not approve the labeling claims that are necessary or desirable for the successful commercialization of our product candidates.

We are required by the FDA to conduct post-marketing studies for mRNA-1273, including to assess the risk of myocarditis and pericarditis and to evaluate outcomes in pregnant women and infants post-vaccination. We or others could identify previously unknown side effects, or known side effects could be observed as being more frequent or severe than in clinical trials or earlier post-marketing periods. If we, our contract manufacturers or other collaborators fail to comply with any post-approval regulatory requirements, a regulator may issue a warning letter asserting that we are in violation of the law, seek an injunction or impose civil or criminal penalties or monetary fines, suspend or withdraw regulatory approval or revoke a license, suspend any ongoing clinical trials, refuse to approve a pending BLA or supplements to a BLA submitted by us, seize or recall products or product candidates, or require field alerts to physicians, pharmacists and hospitals or refuse to allow us to enter into supply contracts. We could also be required to conduct additional nonclinical studies or clinical trials or implement changes in labeling or to our manufacturing processes, specifications or facilities. Initiation of any government investigation or lawsuit, including class-action lawsuits, would require us to expend significant time and resources in response and would likely generate negative publicity. The occurrence of any event or penalty described above may inhibit our ability to commercialize our COVID vaccine, or any future approved products, and generate revenues.

Additionally, the FDA or other regulators could require us to adopt Risk Evaluation and Mitigation Strategies (REMS) for any product to ensure that the benefits of treatment outweigh the risks for each potential patient, which may include, among other things, a medication guide outlining the risks of the product for distribution to patients, a communication plan to health care practitioners, extensive patient monitoring or distribution systems and processes that are highly controlled, restrictive and more costly than what is typical for the industry. Furthermore, if we or others later identify undesirable side effects caused by any product that we develop, several potentially significant negative consequences could result, including the suspension or withdrawal of approvals and licenses, the addition of warning labels, changes to the way a product is administered, the requirement to conduct further clinical trials, lawsuits or increased liability for harm to patients and their children and reputational harm to us. Any of these events could prevent us from achieving or maintaining market acceptance of any products we develop.

Additionally, the U.S. Supreme Court's June 2024 decision in *Loper Bright Enterprises v. Raimondo* overturned the longstanding *Chevron* doctrine, under which courts were required to give deference to regulatory agencies' reasonable interpretations of ambiguous federal statutes. The *Loper* decision could result in additional legal challenges to regulations and guidance issued by federal agencies, including the FDA, on which we rely. Additionally, the *Loper* decision may result in increased regulatory uncertainty, inconsistent judicial interpretations and other impacts to the agency rule-making process. We cannot predict the likelihood, nature or extent of government regulation that may arise from future legislation or administrative action or as a result of legal challenges, either in the United States or abroad. If we are slow or unable to adapt to changes in existing requirements or the adoption of new requirements or policies, or if we are not able to maintain regulatory compliance, our business could be materially harmed.

Preclinical development is lengthy and uncertain, especially for mRNA medicines.

Much of our pipeline is in preclinical development, and these programs could be delayed or not advanced into the clinic. Before initiating clinical trials for a product candidate, we must complete extensive preclinical studies, including IND-enabling good laboratory practice (GLP) toxicology testing, as well as significant Chemistry, Manufacturing, and Controls (CMC) activities for inclusion in an IND submission. CMC activities for mRNA medicines involve complex manufacturing processes and analytical development, which are uncertain and lengthy. We have had and may in the future have difficulty identifying appropriate buffers and storage conditions to enable sufficient shelf life of batches of our product candidates. If we must produce new batches, our preclinical studies could be delayed. We cannot be certain of the timely completion of our preclinical testing and studies, whether the FDA or other regulators will accept the results or if the outcome of our preclinical testing, studies and CMC activities will ultimately support further development of our programs. As a result, we may be unable to submit INDs or similar applications for our preclinical programs on the timelines we expect, if at all, and such applications may not result in the FDA or other regulators allowing clinical trials to begin.

Although we have obtained rare pediatric disease designation for mRNA-3927, we may not be eligible to receive a priority review voucher in the event the FDA determines we no longer meet the criteria for designation, revokes the designation or FDA approval does not occur by September 30, 2029.

The sponsor of an application for a rare pediatric disease drug product may be eligible for a voucher that can be used or sold to obtain a priority review for a subsequent application submitted under section 505(b)(1) of the Federal Food, Drug, and Cosmetic Act or section 351 of the Public Health Service Act. We received rare pediatric disease designation from the FDA for mRNA-3927. We may, in the future, apply for rare pediatric disease designation from the FDA for future product candidates that may qualify for designation. Vouchers for rare pediatric disease drugs are awarded for qualifying applications when the drug receives approval. Under current law, after September 30, 2029, the FDA may not award any rare pediatric disease priority review vouchers, although the FDA's authority to do so could be extended by Congress in the future.

Risks related to the manufacturing of our commercial products and product candidates

We or our third-party manufacturers may encounter difficulties in manufacturing, product release, shelf life, testing, storage, supply chain management or shipping for any of our products.

The manufacturing processes for our mRNA medicines are novel and complex. We and our collaborators have experienced and may continue to encounter difficulties in manufacturing, product release, shelf life, testing, storage, supply chain management or shipping, including delays as our supply chain expands and grows more complex. We may experience difficulties effectively managing and executing the complexities of a larger and multi-product supply plan, including those resulting from complexities of producing batches at larger scale, equipment failure, human error, choice and quality of raw materials and excipients, analytical testing technology and product instability. Further, mRNA medicines encapsulated in LNPs must be developed and manufactured under well-controlled conditions, or pharmacological activity can be adversely impacted.

In an effort to optimize product features, we have in the past and may in the future make changes to our product candidates in their manufacturing and stability formulation and conditions. This has in the past and may in the future result in our having to resupply batches for preclinical or clinical activities when there is insufficient product stability during storage and insufficient supply. Insufficient stability or shelf life of our product candidates could materially delay our ability to continue clinical trials or require us to begin a new clinical trial with a newly formulated drug product, due to the need to manufacture additional preclinical or clinical supply.

Additionally, our high rate of innovation may result in a high degree of technology change that could negatively impact product comparability during and after clinical development. These technology changes may necessitate changes to, or the sourcing of, new manufacturing infrastructure or may adversely affect third-party relationships.

In many cases, we may need to utilize multiple batches of drug substance and drug product to meet the supply requirement of a single preclinical study or clinical trial. Failure in our ability to scale up batch size or failure in any batch, which we have experienced in the past, may lead to a substantial delay in our clinical trials or in the commercialization of any approved product. For example, the changes we make as we continue developing new manufacturing processes for our drug substance and drug product may impact specification and stability of the drug product, and may lead to failure of batches, resulting in a substantial delay in delivery of commercial product or conduct of our clinical trials. Our mRNA product candidates may prove to have a stability profile that leads to a lower than desired shelf life of the final approved medicine. This poses risk in supply requirements, wasted stock and higher cost of goods. We may also experience wasted stock from other factors such as volatile demand forecasts or our inability to scale down for products that have low demand.

We are dependent on a number of equipment providers who are also implementing novel technology. Further, we have developed our own custom manufacturing equipment for certain of our medicines. If we encounter unexpected performance issues with such equipment, we could encounter delays or interruptions to clinical and commercial supply. Due to the number of different programs, we may have cross contamination of product candidates inside of our factories, CROs, suppliers or in the clinic that affect the integrity of our product candidates.

As we scale the manufacturing output for commercial production and particular programs, we plan to continuously improve yield, purity and the pharmaceutical properties of our products and product candidates from IND-enabling studies through commercial launch, including shelf-life stability, and solubility properties of drug product and drug substance. Because of continuous improvement in manufacturing processes, we may switch processes for a particular program during development. However, after a change in process, more time will be required for pharmaceutical property testing, such as 6 or 12 month stability testing. That may require resupplying clinical material or making additional cGMP batches to keep up with clinical trial demand before such pharmaceutical property testing is completed.

We utilize a number of raw materials and excipients that may have a single source of supply, are new to the pharmaceutical industry and are being employed in a novel manner. Some of these raw materials and excipients have not been scaled to a level to support commercial supply and could experience unexpected manufacturing or testing failures or supply shortages. Such issues with raw materials and excipients could cause delays or interruptions to clinical and commercial supply of our products or product candidates.

We have established several analytical assays and may have to establish several more to assess the quality of our mRNA product candidates. We may identify gaps in our analytical testing strategy that might prevent release of product or could require product withdrawal or recall. For example, we may discover new impurities that have an impact on product safety, efficacy or stability. This may lead to an inability to release mRNA product candidates until the manufacturing or testing process is rectified.

As we grow as a commercial company and our drug development pipeline matures, the increased demand for clinical and commercial supplies from our facilities and third parties may impact our ability to operate.

Completion of our trials and commercialization of our products depend on the availability of facilities capable of manufacturing our vaccines at sufficient yields and at commercial scale. We expect to continue to make significant investments in our manufacturing network as we expand our commercial launch efforts. Our new manufacturing plants in Australia, Canada and the United Kingdom are subject to risks associated with operating these facilities, and we may encounter difficulties in optimizing our global manufacturing network.

To supplement our internal manufacturing infrastructure, we may rely on third parties for product manufacturing, packaging, testing, warehousing, distribution and other logistical services, and the manufacture and testing of raw materials, components, parts and consumables, and we may need to engage additional third parties in the future to meet capacity needs. If we cannot enter into or maintain such arrangements on favorable terms, or at all, our ability to develop, manufacture and distribute our products could be adversely affected. Further, efforts to establish and scale these capabilities may not meet expectations regarding scheduling, scale-up, reproducibility, yield, purity, cost, potency or quality. If we are unable to institute necessary controls related to product development, manufacturing and quality, we may encounter difficulties producing our products on the timelines and in the quantities required to meet demand or supply obligations.

We have in the past, and may in the future, seek to resize our manufacturing infrastructure to reflect anticipated demand for our products. This resizing may result in our incurring costs associated with exiting commitments, such as with suppliers for raw materials and contract manufacturing organizations (CMOs). Additionally, as a result of lower demand for our products, we have experienced, and may in the future experience, increased costs with respect to raw material suppliers as we seek to exit or modify our purchase

commitments. Further, we have entered, and may in the future enter, into non-cancellable or take-or-pay purchase commitments for raw materials that require a long lead time to procure, increasing our commitment exposure.

Demand for third-party manufacturing, raw materials or testing facilities may grow faster than available capacity, which could disrupt our ability to secure sufficient quantities of materials and services. The use of service providers and suppliers could expose us to risks, including:

- termination or non-renewal of supply and service agreements in a manner or at a time that is costly or damaging to us;
- disruptions to the operations of these suppliers and service providers caused by conditions unrelated to our business or operations, including the bankruptcy of the supplier or service provider; and
- inspections of third-party facilities by regulators that could have a negative outcome and result in delays to or termination of their ability to supply our requirements.

Our reliance on third-party manufacturers may also result in unforeseen delays or operational problems beyond our control. Because of contractual restrictions and the limited number of manufacturers with the expertise, required regulatory approvals and facilities to manufacture our bulk vaccines on a commercial scale, replacing a third-party manufacturer may be expensive, time-consuming or disruptive. Regulators may require us to register our facilities or those of another supplier if we terminate an existing third-party manufacturer relationship, which could delay or limit our ability to supply a particular market. In addition, third-party manufacturers may experience production interruptions due to factors such as production cost pressures, scale-up challenges, shortages of qualified personnel, availability of raw materials and supplies, quality control failures, compliance with regulatory requirements or lack of capital funding.

We are subject to operational risks associated with the physical and digital infrastructure at our manufacturing facilities and those of our external service providers.

Our manufacturing facilities incorporate a significant level of automation of equipment with integration of several digital systems, including those that may utilize artificial intelligence (AI), to improve efficiency of operations. The digitization of our facilities exposes us to the risk of process equipment malfunctions. These risks include potential system failures or shutdowns due to internal or external factors including design issues, system compatibility or potential cybersecurity compromises, incidents or breaches. Upgrades or changes to our systems, infrastructure or the software that we implement, use, or upon which our business relies, may result in the introduction of new cybersecurity vulnerabilities and risks.

Our facilities and infrastructure or those of our contract manufacturers or other third-parties may also be subject to attacks or acts of sabotage by outside actors, contractors or employees. Any disruption in these manufacturing capabilities could delay scaling up production capacity for our drug substances or products or shut down facilities, add costs, cause us to fail to meet certain product volume or delivery timing obligations, or may require us to identify, qualify and establish an alternative manufacturing site, which could adversely affect our business.

As we expand our development and commercial capacities, we have and expect that we will continue to establish additional manufacturing capabilities in our global network. This expansion may lead to regulatory delays or prove more costly than anticipated. If we fail to select suitable locations, complete construction in an efficient manner, engage effectively with local regulators, recruit the required personnel or manage our growth effectively, the development and production of products or our product candidates could be delayed or curtailed.

Our products and product candidates are sensitive to shipping and storage conditions, which, in some cases, requires cold-chain logistics and subjects them to risk of loss or damage.

Our products and product candidates are sensitive to temperature, storage and handling conditions, and we could lose medicines if the product or product intermediates are not stored or handled properly. Shelf life for our products and product candidates is variable, and they may expire prior to use. Further, if we or third-party distributors do not maintain effective cold-chain supply logistics required for certain of our products and product candidates, products may be returned, become out-of-date or be rendered unusable. This has led and could lead to added manufacturing costs and delays in our ability to supply required quantities for clinical trials or commercial sale. In addition, the cost associated with such transportation services and the limited pool of vendors could cause supply disruptions.

Our manufacturing facilities or those of third-party manufacturers or suppliers may fail to meet regulatory requirements, which could delay approval of or increase production costs for our products.

The manufacturing of medicines for clinical trials or commercial sale is subject to extensive regulation, and components of such products must be manufactured in accordance with cGMP requirements. These regulations govern manufacturing processes and procedures, including record keeping and the implementation and operation of quality systems to control and assure the quality of

products and materials used in clinical trials and for commercial supply. Poor control of cGMP manufacturing processes can lead to product quality failures that may impact our ability to supply product, resulting in cost overruns and cause significant delays to clinical development timelines. Such manufacturing process issues may include:

- critical deviations in the manufacturing process;
- ineffective corrective or preventative actions taken to address such deviations as processes scale;
- facility and equipment failures;
- product or facility contamination, including due to ineffective quality control strategies or failures identified through environmental monitoring programs;
- failures of raw materials, components or consumables, including due to ineffective supplier qualification or regulatory compliance issues at critical suppliers; and
- ineffective product stability;

Regulators typically require representative manufacturing site inspections to assess compliance with cGMP and manufacturing controls. Such inspections may occur at any time during product development or commercialization, may be unannounced, and may be product- or facility-specific or conducted in response to issues identified by regulators. If one of our, or a third-party manufacturer's, sites fails to maintain sufficient quality assurance or control, or receives a deficient inspection outcome, a product may not receive or maintain regulatory approval, and our manufacturers or suppliers may be unable to meet their supply obligations, impacting or delaying supply or programs.

In addition, the manufacturing processes for our products are subject to regulatory approval, and if we or our third-party manufacturers are unable to reliably produce products or product candidates to specifications acceptable to regulators, we or our strategic collaborators may not obtain or maintain the approvals needed to commercialize such products. Even after regulatory approval is obtained, there is no assurance that we or our CMOs will be able to manufacture the approved medicine to specifications acceptable to the FDA or other regulators. Any of these challenges could delay or prevent completion of clinical trials, require bridging or repeat clinical trials, increase trial costs, delay regulatory approval, impair commercialization efforts or increase our cost of goods, any of which could have an adverse effect on our business.

In addition, we may not have direct control over the ability of our contract manufacturers to maintain adequate quality control, quality assurance and qualified personnel. Our contract manufacturers supply or manufacture materials or products for other companies and their failure to meet applicable regulatory requirements may affect the regulatory status of their facilities. In addition, to the extent that we rely on foreign contract manufacturers, we are subject to additional risks, including the need to comply with import and export regulations.

The FDA, the EMA and other regulators may require us to submit product samples of any lot of any approved product, together with the protocols showing the results of applicable tests, at any time. In some cases, regulators may prohibit us from distributing a lot until it authorizes release. Deviations in the manufacturing process, including those affecting quality attributes and stability, may cause unacceptable changes in the product, resulting in lot failures or product recalls. In the past, our third-party contract manufacturers experienced lot failures resulting in product recalls of our COVID vaccine. Lot failures have caused, and lot failures or product recalls in the future with respect to product produced by either our own or our third-party manufacturers' facilities could cause, us and strategic collaborators to delay clinical trials or product launches.

We and our manufacturing partners also may encounter problems hiring and retaining the experienced scientific, quality control and manufacturing personnel, which could result in delays in production or difficulties in maintaining compliance with applicable regulatory requirements. Additionally, we may not be able to control for or detect intentional sabotage or negligence by any employee or contractor.

Our intismeran autogene product candidates are uniquely manufactured for each patient using a novel, complex manufacturing process and we may encounter difficulties in production.

We custom design and manufacture intismeran tailored for each patient. As a result, manufacturing is susceptible to product loss or failure due to issues with:

- logistics associated with the collection and shipping of a patient's tumor, blood or other tissue sample;
- next-generation sequencing of the tumor mRNA and identification of appropriate tumor-specific mutations;
- the use of a software program, including proprietary and open source components, which is hosted in the cloud and a part of our product candidate, to assist with the design of the patient-specific mRNA, which software must be maintained and secured;
- effective design of the patient-specific mRNA that encodes for the required neoantigens;
- batch specific manufacturing failures or issues that arise due to the uniqueness of each patient-specific batch;

- quality control testing failures;
- unexpected failures of batches placed on stability;

- shortages or quality control issues with single-use assemblies, consumables or critical parts sourced from third-party vendors that must be changed out for each patient-specific batch;
- significant costs associated with individualized manufacturing that may adversely affect our ability to continue development;
- successful and timely manufacture and release of the patient-specific batch;
- shipment issues encountered during transport of the batch to the patient site of care; and
- the ability to define a consistent safety profile at a given dose when each participant receives a unique therapy.

We have purpose-built our facility in Marlborough, Massachusetts for intismeran, and it may encounter difficulties or it take longer than anticipated to scale up the facility to meet future commercial demand, if intismeran is approved. Manufacturing demand for intismeran may also increase as we continue our current late-stage trials and expand to additional tumor types. Further, we may encounter difficulties executing our plans to improve turnaround time and reduce costs. The addition of new facilities could also lead to product comparability issues, which could further delay introduction of new capacity. In addition, if our equipment does not function as designed, we could experience deviations in the drug product produced, which could lead to increased batch failure and the inability to supply patients enrolled in a clinical trial.

Because intismeran is manufactured for each individual patient, we are required to maintain a chain of identity with respect to each patient's tissue sample, sequence data derived from such tissue sample, results of analysis of such patient's genomic analysis and the custom manufactured product for each patient. Maintaining such a chain of identity is difficult and complex, and failure to do so has resulted and may in the future result in product mix up, adverse patient outcomes, loss of product or regulatory action, including withdrawal of any approved products from the market. Further, as intismeran is developed through clinical trials towards approval and commercialization, multiple aspects of the complicated collection, analysis, manufacture and delivery process may be modified to optimize processes and results. These changes may not achieve the intended objectives, and any of these changes could cause intismeran to perform differently than we expect, potentially affecting the results of clinical trials.

Risks related to our reliance on third parties

We are dependent on single-source suppliers for some of the components and materials used in, and the manufacturing processes required to develop and commercialize, our products and product candidates.

We cannot ensure that the suppliers we rely on will remain in business, have sufficient capacity or supply to meet our needs or that they will not be purchased by one of our competitors or another company that will cease working with us. Our use of single-source suppliers exposes us to several risks, including disruptions in supply, price increases, late deliveries or business interruptions.

There are, in general, few alternative sources of supply for substitute components. If we must switch suppliers, the manufacture and delivery of our products or product candidates could be interrupted for an extended period. Establishing additional or replacement suppliers for any of the components or processes used in our products or product candidates may not be accomplished quickly, if at all. Any replacement supplier (or us, if we produced directly) would need to be qualified and may require additional regulatory approval, resulting in further delay. Any such interruption or delay, or our inability to obtain components or materials from alternate sources on acceptable terms, could impair our ability to meet demand for our products or product candidates. Additionally, as part of the FDA's approval of our product candidates, the FDA will review the manufacturing processes and facilities of our single-source suppliers.

We have entered into strategic alliances with third parties for product development and commercialization. If these alliances are unsuccessful, our business could be adversely affected.

We have entered into strategic alliances with collaborators that have provided funding, intellectual property licenses and other resources for developing, manufacturing and commercializing our product candidates. Additionally, we have entered into strategic alliances where we agree to provide funding, intellectual property licenses and other resources to third parties. Our strategic alliances may pose a number of risks, including:

- strategic collaborators may not perform their obligations as expected;
- strategic collaborators may cease or deprioritize development or commercialization of our products due to unfavorable clinical trial results, changes in their focus or funding, or other factors that divert their resources or create competing priorities, such as potential competition with their own products or candidates;
- strategic collaborators may delay, stop or provide insufficient funding or resources for clinical trials (whether as a result of a business decision or necessitated by financial difficulties of such collaborator);
- a strategic collaborator with marketing and distribution rights to one or more of our products may commit insufficient resources to the marketing and distribution of any such product;

- disagreements with strategic collaborators, including over proprietary rights, contract interpretation or the course of development of any product candidates, may cause delays or termination of the research, development or commercialization of such product candidates, lead to additional responsibilities for us with respect to such product candidates or result in litigation or arbitration, any of which would be time-consuming and expensive;
- strategic collaborators may use our proprietary information in such a way as to invite litigation that could jeopardize or invalidate our IP or proprietary information;
- disputes may arise with respect to the ownership of IP developed pursuant to our strategic alliances;
- strategic collaborators may infringe the IP rights of third parties, exposing us to potential litigation and liability;
- future relationships may require us to incur non-recurring and other charges, increase our near- and long-term expenditures, issue securities that dilute our existing shareholders or disrupt our management and business;
- any equity investments we make in collaborators could decrease in value or become worthless; and
- our international operations through any future collaborations, acquisitions or joint ventures may expose us to certain operating, legal and other risks not encountered in the United States.

Our strategic collaborators generally may materially amend or terminate their agreements with us in accordance with the terms of those agreements, which has happened in the past. If any collaboration agreement is terminated, we may not receive anticipated future research or development funding or milestone, earn-out royalty, profit share or other contingent payments and the research or development of our product candidates may be delayed or discontinued. It may also be difficult to attract new strategic collaborators to continue development or commercialization of the applicable product candidate, and our reputation could be adversely affected. All the risks relating to product development, regulatory approval and commercialization described elsewhere in these *Risk Factors* apply to our strategic collaboration activities.

We may seek to establish additional strategic alliances and, if we are unable to do so on commercially reasonable terms, we may have to alter our development and commercialization plans. Certain of our strategic alliance agreements may restrict our ability to develop certain products.

Our development programs and the potential commercialization of our product candidates will require substantial cash to fund expenses. We may collaborate with other companies to develop and potentially commercialize some of our product candidates, and we face significant competition in seeking collaborators. Our ability to establish additional strategic alliances will depend, among other things, on our assessment of the collaborator's resources and expertise, proposed terms and conditions and the collaborator's evaluation of a number of factors. Those factors may include the design or results of clinical trials, the likelihood of regulatory approval, potential market sizes and competition, the costs and complexities of manufacturing and delivering a product candidate to trial participants, uncertainty with respect to our or the proposed collaborator's ownership of technology (regardless of the merits of the challenge), and industry and market conditions generally.

We are also restricted under some of our existing strategic alliance agreements from entering into agreements on certain terms with potential strategic collaborators to pursue other targets on our own. These restrictions on working with targets, polypeptides, routes of administration and fields could limit our ability to enter into strategic collaborations with other collaborators or to pursue certain potentially valuable product candidates.

Strategic alliances are complex and time-consuming to negotiate and document. If we cannot enter alliances on a timely basis, on favorable terms or at all, we may need to curtail the development of the product candidate for which we are seeking to collaborate, reduce or delay its development program or one or more of our other development programs, delay its potential commercialization or reduce the scope of any sales or marketing activities, or increase our expenditures and undertake development or commercialization activities at our own expense.

We expect to continue to rely on third parties to conduct aspects of our research, preclinical studies, protocol development and clinical trials. If these third parties do not perform satisfactorily or comply with regulatory requirements, we may be unable to obtain regulatory approval for or commercialize our product candidates.

We rely on third parties such as CROs to help manage certain preclinical work, and on medical institutions, clinical investigators and CROs to assist in the design, review and conduct of our clinical trials, including enrolling qualified patients. In addition, we engage third-party contractors and collaborators to support other research, commercial and administrative activities, which reduces our control over these activities but does not relieve us of our responsibilities, such as ensuring that each clinical trial is conducted in accordance with its general investigational plan and protocols. Moreover, the FDA requires us to comply with GLPs and good clinical practices for conducting, recording and reporting the results of preclinical studies and clinical trials to assure that data and reported results are

credible and accurate and that in the case of clinical trials the rights, integrity and confidentiality of trial participants are protected. Such standards will evolve and subject us and third parties to new or changing requirements.

If third parties do not successfully carry out their contractual duties or meet expected deadlines, we may need to replace them, which could delay clinical trials, drug development or other activity. If clinical trials are not conducted in accordance with our contractual expectations or regulatory requirements, action by regulators may significantly and adversely affect the conduct or progress of such trials or require a clinical trial to be redone. Accordingly, our efforts to obtain regulatory approvals and commercialize our drug candidates could be delayed. In addition, failure of any third-party contractor to timely provide access to our data in a format acceptable to us may result in delays or impediments to our regulatory submissions or other development activities.

Risks related to our intellectual property

We may be unable to obtain and enforce patent protection for our discoveries and the intellectual property rights therein, or protect the confidentiality of our trade secrets.

Our success depends, in part, on our ability to protect proprietary methods and technologies that we develop under the patent and other IP laws of the United States and other countries, so that we can prevent others from unlawfully using our inventions and proprietary information. Because certain U.S. patent applications are confidential until the patents issue, third parties may have filed patent applications for technology covered by our pending patent applications without our being aware of those applications, and our patent applications may not have priority over those applications. In addition, publications of discoveries in the scientific literature often lag behind the actual discoveries, and patent applications in the United States and other jurisdictions are typically not published until 18 months after filing, if at all. Therefore, we cannot be certain that we were the first to make the inventions claimed in our patents or pending patent applications or that we were the first to file for patent protection of such inventions. We therefore may be unable to secure desired patent rights, thereby losing exclusivity. In the past, we have obtained licenses under third-party patents to market our products or conduct our research and development or other activities, and we may do so in the future if necessary. If licenses are not available to us on favorable terms, we may not be able to market the affected products or conduct the desired activities.

The process of obtaining and enforcing patent protection is expensive and time-consuming and our pending patent applications may not result in issued patents. Obtaining and maintaining our patent protection depends on compliance with various procedural, document submission, fee payment and other requirements imposed by governmental patent agencies, and our patent applications may fail to result in valid enforceable patents, or our patent protection could be reduced or eliminated, for non-compliance with these requirements. If we or our strategic collaborators fail to file and prosecute all necessary and desirable patent applications at a reasonable cost and in a timely manner, our business may be adversely affected.

Despite our and our strategic collaborators' efforts to protect our proprietary rights, unauthorized parties may obtain and use information that we regard as proprietary. While issued patents are presumed valid, they may not survive a validity challenge and could be held unenforceable. Any patents we have obtained, or obtain in the future, may be challenged, invalidated, adjudged unenforceable or circumvented by parties seeking to design around our IP. Also, third parties through the USPTO or European Patent Office (EPO) may commence patent office proceedings involving our patents or patent applications. Any challenge to, finding of unenforceability or invalidation, or circumvention of, our patents or patent applications, would be costly, would require significant time and attention of our management, could reduce or eliminate royalty payments to us from third-party licensors and could have a material adverse impact on our business.

The standards that the USPTO and its foreign counterparts use to grant patents are not always applied predictably or uniformly and can change. Similarly, the ultimate degree of protection that will be afforded to biotechnology inventions, including ours, in the United States and foreign countries, remains uncertain and is dependent upon the scope of the protection decided upon by patent offices, courts and lawmakers. One major provision of the America Invents Act changed U.S. patent practice from a first-to-invent to a first-to-file system. If we fail to file an invention before a competitor files on the same invention, we no longer can provide proof that we were in possession of the invention prior to the competitor's filing date, and thus would not be able to obtain patent protection for our invention. There is also no uniform, worldwide policy regarding the subject matter and scope of claims granted or allowable in pharmaceutical or biotechnology patents. In certain countries, for example, methods for the medical treatment of humans are not patentable.

Accordingly, we do not know the degree of future protection for our proprietary rights or the breadth of claims that will be allowed in any patents issued to us or to others. We also rely to a certain extent on trade secrets, know-how and technology, which are not protected by patents, to maintain our competitive position. We also rely on non-disclosure agreements and invention assignment agreements with our employees, consultants and third parties. If any trade secret, know-how or other technology not protected by a patent were to be disclosed to or independently developed by a competitor, our business and financial condition could be materially adversely affected.

Failure to obtain and maintain all available regulatory exclusivities and broad patent scope and to maximize patent term restoration or extension on patents covering our products may lead to loss of exclusivity and early biosimilar entry resulting in a loss of market share or revenue.

In addition, we may choose not to enforce our IP rights in certain circumstances or for certain periods of time. For example, in March 2022, we announced that we will not enforce our patents for COVID vaccines against companies manufacturing in or for the Gavi COVAX Advance Market Commitment (AMC) countries, provided that the manufactured vaccines are solely for use in the AMC 92 countries. In addition, we are willing to license our IP for COVID vaccines to manufacturers, but licensing of our IP may be limited, and our business may be otherwise adversely impacted if we are unable to enforce our IP.

Uncertainty over IP in the pharmaceutical and biotechnology industry has been the source of litigation and other disputes, which is inherently costly and unpredictable and can have adverse financial and freedom-to-operate consequences.

mRNA medicines are a relatively new scientific field and, as the field continues to mature, patent applications are being processed by national patent offices globally. There is uncertainty about which patents will issue, and, if they do, as to when, to whom and with what claims. Litigation is ongoing over the underlying technology to mRNA medicines between many mRNA market participants. It is likely that there will continue to be significant litigation and patent office proceedings in various patent offices relating to patent rights in the mRNA field.

We have issued patents and pending patent applications in the United States and in key markets globally that claim many different methods, compositions and processes relating to the discovery, development, manufacture and commercialization of mRNA medicines and our delivery technology, including LNPs. Oppositions and inter partes review petitions have been filed against some of our patents, and we expect that further proceedings will be filed in the EPO, USPTO and elsewhere relating to patents and patent applications in our portfolio. In many cases, the possibility of appeal exists for any party, and it may be years before final, unappealable rulings are made with respect to these patents in certain jurisdictions. The timing and outcome of these and other proceedings is uncertain and may adversely affect our business if we are not successful in defending the patentability and scope of our pending and issued patent claims. We cannot be certain that any patent will survive or that the claims will remain in the current form. Even if our rights are not directly challenged, disputes could lead to the weakening of our IP rights.

In certain instances, we have instituted and may in the future institute inter partes review proceedings against issued U.S. patents and opposition proceedings against European patents owned by third parties in the field of mRNA medicines. We have a number of these proceedings ongoing against third-party patents. If we are unsuccessful in narrowing or invalidating such third-party patents, those third parties may attempt to assert those patents against our products. As the biotechnology and pharmaceutical industries expand and more patents are issued, the risk increases that our products or product candidates may be subject to claims of infringement of the patent rights of third parties.

We have invested billions of dollars in creating our patented mRNA platform, which is integral to the development of our mRNA medicines, and we are involved in various intellectual property litigation as described under Part I, Item 3, “Legal Proceedings.” We expect to expend substantial financial and managerial resources in connection with these legal proceedings, and the ultimate outcome of each proceeding is uncertain.

We are, and may in the future become, involved in patent litigation or other proceedings related to a determination of rights, we could incur substantial costs and expenses, substantial liability for damages or be required to stop our product development and commercialization efforts.

There may be third-party patents or patent applications with claims to materials, formulations, methods of manufacture or methods for treatment related to the use or manufacture of our products and product candidates, and third parties may assert that we are employing their proprietary technology without authorization. In addition, third parties may obtain patents in the future and claim that our technologies infringe upon these patents. If any third-party patents were held by a court of competent jurisdiction to cover the manufacturing process of any of our products or product candidates, any molecules formed during the manufacturing process or any final product itself, the holders of any such patents may obtain injunctive or other equitable relief, which could effectively block our ability to commercialize such product unless we obtained a license under the applicable patents, or until such patents expire. Similarly, if any third-party patents were held by a court of competent jurisdiction to cover aspects of our formulations, processes for manufacture or methods of use, including combination therapy, the holders of any such patents may be able to block our ability to develop and commercialize the applicable product or product candidate unless we obtained a license or until such patent expires.

Defense of infringement and other claims, regardless of their merit, involves substantial litigation expense and diverts employee resources from our business. In the event of a successful infringement claim against us, we may have to pay substantial damages, including treble damages and attorneys’ fees for willful infringement, pay royalties, redesign our infringing products or obtain one or

more licenses from third parties, which may not be made available on commercially favorable terms, if at all, or may require substantial time and expense.

In addition, any such licenses are likely to be non-exclusive and, therefore, our competitors may have access to the same technology licensed to us. If we fail to obtain a required license and are unable to design around a patent, we may be unable to effectively market some of our technology and products, which could limit our ability to generate revenues or achieve profitability, which could jeopardize our ability to sustain our operations. Moreover, we expect that a number of our collaborations will provide that royalties payable to us for licenses to our IP may be offset by amounts paid by our collaborators to third parties who have competing or superior IP positions in the relevant fields, which could result in significant reductions in our revenues from products developed through collaborations.

In addition, in connection with certain license and strategic alliance agreements, we have agreed to indemnify certain third parties for certain costs incurred in connection with litigation relating to IP rights or the subject matter of the agreements. The cost to us of any litigation or other proceeding relating to IP rights, even if resolved in our favor, could be substantial, and litigation would divert our management's efforts. Some of our competitors may be able to sustain the costs of complex patent litigation more effectively than we can because they have substantially greater resources. Uncertainties resulting from the initiation and continuation of any litigation could delay our research, development and commercialization efforts and limit our ability to continue our operations.

If third-party owners of any patent rights that we license do not properly or successfully obtain, maintain or enforce the patents underlying such licenses, our competitive position and business prospects may be harmed.

We may become party to licenses that give us rights to third-party IP necessary or useful for our business and our success may depend in part on the ability of our licensors to obtain, maintain and enforce patent protection for our licensed IP. Our licensors may fail to successfully prosecute the patent applications we license, fail to maintain these patents, determine not to pursue litigation against other companies that are infringing these patents or pursue such litigation less aggressively than we would. Without protection for the IP we license, other companies might be able to offer substantially identical products for sale, which could adversely affect our competitive business position and harm our business prospects. In addition, we sublicense our rights under various third-party licenses to our strategic collaborators. Any impairment of these sublicensed rights could result in reduced revenues under our strategic alliance agreements or result in termination of an agreement by one or more of our strategic collaborators.

If we fail to comply with our obligations in the agreements under which we license IP rights from third parties or otherwise experience disruptions to our business relationships with our licensors, we could lose license rights that are important to our business.

We license IP, which involves complex legal, business and scientific issues and is complicated by the rapid pace of scientific discovery in our industry. We are a party to certain IP license agreements and expect to enter into additional license agreements in the future. Our existing license agreements impose, and we expect that future license agreements will impose, various diligence, milestone payment, royalty and other obligations on us. If we fail to comply with our obligations under these agreements, or we are subject to a bankruptcy, the licensor may have the right to terminate the license, in which event we would not be able to market products covered by the license and may be subject to additional liabilities.

In certain cases, we control the prosecution of patents resulting from licensed technology. In the event we breach any of our obligations related to such prosecution, we may incur significant liability to our strategic collaborators. Disputes may arise regarding IP subject to a licensing agreement, including:

- the scope of rights granted under the license agreement and other interpretation-related issues;
- whether our technology and processes that are not subject to the licensing agreement infringe on IP of the licensor;
- the sublicensing of patent and other rights under our collaborative development relationships;
- our diligence obligations under the license agreement and what activities satisfy those diligence obligations;
- the ownership of inventions and know-how resulting from the joint creation or use of IP by our licensors and us and our strategic collaborators; and
- the priority of invention of patented technology.

If disputes over IP that we have licensed prevent or impair our ability to maintain our current licensing arrangements on favorable terms, we may be unable to successfully develop and commercialize the affected product candidates. We are generally also subject to the same risks with respect to protection of IP that we license as we are for IP that we own. If we or our licensors fail to adequately protect this IP, our ability to commercialize products could suffer.

We may be subject to claims that our employees, consultants or independent contractors have wrongfully used or disclosed confidential information of third parties or that our employees have wrongfully used or disclosed alleged trade secrets of their former employers.

We employ individuals who were previously employed at universities or other biotechnology or pharmaceutical companies, including our competitors or potential competitors. From time to time, we are subject to claims that we or our employees, consultants or independent contractors have inadvertently or otherwise used or disclosed IP, including trade secrets or other proprietary information, of third parties, including our employees' former employers. Litigation may be necessary to defend against these claims. If we fail to defend such claims, in addition to paying monetary damages, we may lose valuable IP rights or personnel, which could adversely impact our business. Even if we are successful in defending against such claims, litigation could result in substantial costs and be a distraction to management and other employees.

We may be subject to claims challenging the inventorship or ownership of our patents and other IP.

We may be and have been subject to claims that former employees, collaborators or other third parties have an ownership interest in our patents or other IP. Ownership disputes may arise, for example, from conflicting obligations of consultants or others who are involved in developing our product candidates. Litigation may be necessary to defend against these and other claims challenging inventorship or ownership. If we fail in defending any such claims, in addition to paying monetary damages, we may lose valuable IP rights, including exclusive ownership of, or right to use, valuable IP. Such an outcome could have a material adverse impact on our business. Even if we are successful in defending against such claims, litigation could result in substantial costs and distract management and other employees, and could impact or patenting strategy.

Changes in U.S. patent and regulatory law could impair our ability to protect our products.

U.S. patent law continues to evolve, and certain U.S. Supreme Court rulings have narrowed the scope of patent protection available in certain circumstances and weakened the rights of patent owners in certain situations. These rulings have increased uncertainty regarding our ability to obtain patents in the future, as well as with respect to the value of patents, once obtained. Depending on actions or decisions by the U.S. Congress, the federal courts and the USPTO, the laws and regulations governing patents could change in unpredictable ways that would weaken our ability to obtain new patents or to enforce our existing patents and patents that we might obtain in the future.

We may be unable to protect our IP rights throughout the world.

Filing, prosecuting and defending patents in every country would be prohibitively expensive, and our foreign IP rights can be less extensive than those in the United States. In addition, the laws of some foreign countries do not protect IP rights to the same extent as U.S. federal and state laws. Consequently, we may be unable to prevent third parties from practicing our inventions in all countries outside the United States, or from selling or importing products made using our inventions in and into the United States or other jurisdictions. Competitors may use our technologies to develop their own products in jurisdictions where we have not obtained patent protection or may export infringing products to territories where we have patent protection, but enforcement is not as strong as in the United States.

Many companies have encountered significant problems in protecting and defending IP rights in foreign jurisdictions. The legal systems of certain countries, particularly certain developing countries, do not favor the enforcement of patents, trade secrets and other IP protection, particularly those relating to biotechnology products, which could make it difficult for us to stop the infringement of our patents or marketing of competing products in violation of our proprietary rights generally. Proceedings to enforce our patent rights in foreign jurisdictions could result in substantial costs and divert our efforts and attention from other aspects of our business, could put our patents at risk of being invalidated or interpreted narrowly and our patent applications at risk of not issuing and could provoke third parties to assert claims against us. We may not prevail in any lawsuits that we initiate and the damages or other remedies awarded, if any, may not be commercially meaningful. Accordingly, our efforts to enforce our IP rights around the world may be inadequate to obtain a significant commercial advantage from the IP that we develop or license.

Additionally, many countries have compulsory licensing laws under which a patent owner may be compelled to grant licenses to third parties. In addition, many countries limit the enforceability of patents against government agencies or government contractors. In these countries, the patent owner may have limited remedies, which could materially diminish the value of the relevant patent rights. If we or any of our licensors is forced to grant a license to third parties with respect to any patents relevant to our business, our competitive position may be impaired.

Our reliance on government funding and collaboration from governmental and quasi-governmental entities for certain of our programs adds uncertainty to our research and development efforts with respect to those programs and may impose requirements related to IP rights and requirements that increase the costs of development, commercialization and production of any programs developed under those government-funded programs.

Contracts and grants funded by the U.S. government and its agencies, including our agreements funded by BARDA and DARPA and our collaboration with NIAID, and with foreign governments and agencies, including in the United Kingdom, Canada and Australia, may include provisions that reflect the government's substantial rights and remedies, many of which are not typically found in commercial contracts, including powers of the government to:

- terminate agreements, in whole or in part, for any or no reason;
- reduce or modify the government's obligations under such agreements without our consent;
- claim rights, including IP rights, in products and data developed under such agreements;
- audit contract-related costs and fees, including allocated indirect costs;
- suspend the contractor or grantee from receiving new contracts pending resolution of alleged violations of procurement laws or regulations;
- impose manufacturing requirements for products that embody inventions conceived or first reduced to practice under such agreements;
- suspend or debar the contractor or grantee from doing future business with the government;
- control and potentially prohibit the export of products;
- pursue criminal or civil remedies under the False Claims Act, False Statements Act and similar remedy provisions specific to government agreements; and
- limit the government's financial liability to amounts appropriated by the U.S. Congress on a fiscal-year basis, thereby leaving some uncertainty about the future availability of funding for a program even after it has been funded for an initial period.

We may not have the right to prohibit the U.S. government from using certain technologies developed by us and we may be unable to prohibit other companies, including our competitors, from using those technologies in providing products and services to the U.S. government. The U.S. government generally takes the position that it has the right to royalty-free use of technologies that are developed under U.S. government contracts.

In addition, government contracts and grants, and subcontracts and subawards awarded in the performance of those contracts and grants, normally contain additional requirements that may increase our costs of doing business, reduce our profits and expose us to liability for failure to comply with these terms and conditions. These requirements include, for example:

- specialized accounting systems unique to government contracts and grants;
- mandatory financial audits and potential liability for price adjustments or recoupment of government funds after such funds have been spent;
- public disclosures of certain contract and grant information, which may enable competitors to gain insights into our research program; and
- mandatory socioeconomic compliance requirements, including labor standards, non-discrimination, and affirmative action programs and environmental compliance requirements.

Further, under these agreements we are subject to the obligations to and the rights of the U.S. government set forth in the Bayh-Dole Act of 1980 (Bayh-Dole Act). As a result, the U.S. government may have rights in certain inventions developed under these government-funded programs, including a non-exclusive, non-transferable, irrevocable worldwide license to use inventions for any governmental purpose. In addition, the U.S. government has the right to require us to grant exclusive, partially exclusive or nonexclusive licenses to any of these inventions to a third party if it determines that: (i) adequate steps have not been taken to commercialize the invention; (ii) government action is necessary to meet public health or safety needs; or (iii) government action is necessary to meet requirements for public use under federal regulations, also referred to as "march-in rights."

Any exercise of the march-in rights by the U.S. government could harm our competitive position, business, financial condition, results of operations and prospects. In December 2023, the Biden administration released a proposed framework specifying for the first time that price can be a factor in considering whether an invention is sufficiently available to the public. The proposed framework could potentially enable march-in rights to be used as a tool to regulate drug pricing. The potential inclusion of price as a factor in a march-in determination is expected to draw extensive criticism and challenge, and the ultimate impact is currently unknown. If the U.S. government exercises such march-in rights, we may receive compensation deemed reasonable by the U.S. government, which may be less than what we might be able to obtain in the open market. IP generated under a government-funded program is also subject to certain reporting requirements, compliance with which may require us to expend substantial resources.

We have significant contracts with the U.S. and other governments, and as such are subject to related regulatory compliance obligations. If we fail to maintain compliance with those obligations, we may be subject to potential liability and to termination of our contracts.

Risks related to our financial condition and results of operations

We incurred net losses in 2025 and 2024, and expect to incur additional losses in the future; we may not achieve long-term sustainable profitability.

We incurred net losses of $2.8 billion and $3.6 billion in 2025 and 2024, respectively, and other than in 2022 and 2021, we have incurred net losses in each year since our inception. Currently, our COVID and RSV vaccines are our only commercial products and our RSV sales have been minimal. Although we anticipate near-term product launches, their occurrence and timing is uncertain. Our ability to achieve profitability depends on our ability to successfully develop and obtain the regulatory approvals necessary to commercialize our products.

We have incurred, and expect to continue to incur, significant costs associated with commercializing our products and our clinical development activities. Although we continue to implement cost reduction and prioritization measures, these efforts may fail to reduce our costs in line with expectations, which could require us to seek additional funding to continue operations. Our expenses could increase for many reasons, including if and as we:

- initiate additional clinical trials, particularly large pivotal trials;
- continue to build out our manufacturing capabilities;
- build out a sales, marketing and distribution infrastructure to commercialize any products;
- acquire or in-license other product candidates and technologies; and
- experience any delays or encounter issues with any of the above.

Our quarterly and annual operating results may fluctuate. As a result, we may fail to meet or exceed the expectations of research analysts or investors, which could cause our stock price to decline and negatively impact our financing or funding ability, as well as our ability to exist as a standalone company.

Our financial condition and operating results may fluctuate from quarter-to-quarter and year-to-year due to many factors, many of which are beyond our control. As such, a period-to-period comparison of our operating results may not be predictive of our future performance. In any quarter, our operating results could be below market expectations, which could cause our stock price to decline. Our stock price could be affected by events not necessarily tied to our actual operating results, including recommendations by securities analysts, the timing of certain public disclosures by us, our collaborators or our competitors and our ability to accurately report our financial results in a timely manner. Other factors relating to our business that may contribute to these fluctuations include those described in these *Risk Factors* and elsewhere in this Annual Report on Form 10-K.

The investment of our cash, cash equivalents and investments is subject to risks which may cause losses and affect the liquidity of these investments.

As of December 31, 2025, we had approximately $8.1 billion in cash, cash equivalents and investments, which includes a $600 million draw down from our term loan facility, which is subject to general credit, liquidity, market, inflation and interest rate risks. We may realize losses in the fair value of these investments. In addition, if our investments cease paying or reduce the amount of interest paid to us, our interest income would suffer. These and other market risks associated with our investment portfolio may adversely affect our results of operations, liquidity and financial condition.

Failure to comply with the covenants in our credit agreement could adversely affect our business

A failure to comply with the covenants in our credit agreement with lenders led by Ares Capital Corporation, as administrative agent, could result in an event of default, which, if not cured or waived, could have a material, adverse effect on our business, financial condition, and results of operations. In the event of any such default, the holders of our indebtedness (i) would not be required to lend any additional amounts to us; and (ii) could elect to declare all then outstanding indebtedness, including accrued interest and unpaid fees, to be due and payable and terminate all commitments to extend further credit, if applicable.

If we are unable to repay those amounts, the holders of our secured indebtedness could proceed against their secured collateral to seek repayment from the proceeds of the sale or liquidation of our assets. If our indebtedness were to be accelerated, there can be no assurance that our assets would be sufficient to repay such indebtedness in full.

Risks related to our business and operations

We may encounter difficulties in managing changes to the size, structure and scope of our company.

We have experienced significant growth in our operations and workforce in recent years and, more recently, have undertaken initiatives to reduce costs, prioritize our portfolio and adjust the size and structure of our organization. As a result, we must continue to adapt and improve our managerial, operational, and financial systems to effectively support our evolving business, including potential future growth and expansion initiatives. These efforts may require our management to devote substantial time and attention away from day-to-day operations and strategic execution.

Successfully developing, manufacturing and commercializing products across the many therapeutic areas and diseases we seek to address requires significant depth of talent, resources and robust corporate processes to support simultaneous execution. Organizational changes, including workforce reductions, restructuring, and shifts in strategic priorities, may make it more difficult to retain personnel, recruit and train qualified employees, maintain institutional knowledge, and preserve employee morale and productivity. If we are unable to effectively manage these challenges, we may experience weaknesses in our infrastructure, operational mistakes or inefficiencies, loss of business opportunities and reduced productivity.

In addition, we continue to operate and expand a global manufacturing and operational footprint, including our Norwood and Marlborough campuses in Massachusetts and manufacturing facilities outside the United States. Managing the costs, operational complexity, regulatory requirements, and execution risks associated with these facilities may require significant financial and managerial resources and could divert capital from other priorities, including the development of our product candidates. If we are unable to effectively manage organizational change, operational scale, and resource allocation, our financial performance and ability to commercialize our products may be affected negatively.

We are subject to the risks of doing business outside of the United States.

Our business is subject to risks associated with doing business globally. We are not permitted to market or promote any of our product candidates before receiving regulatory approval or other authorization from applicable authorities, which we may never obtain. To obtain regulatory approval in various jurisdictions, we must comply with numerous regulatory requirements regarding safety and efficacy and governing, among other things, clinical trials, manufacturing, commercial sales, pricing and distribution of our product candidates, and we may fail to obtain approval. We have rapidly expanded our global operations, establishing commercial subsidiaries and entering into arrangements to support the worldwide manufacture and distribution of our products, which is a complex task. For example, we are building regional manufacturing facilities and investing in research and development in several countries. Our business may be adversely affected by many factors associated with our expanding global business, including:

- efforts to develop an international commercial sales, marketing and supply chain and distribution organization, including efforts to mitigate longer accounts receivable collection times, longer lead times for shipping and potential language barriers;
- our customers' ability to obtain reimbursement for our products in foreign markets;
- our inability to directly control commercial activities because we rely on third parties;
- different medical practices and customs in foreign countries affecting acceptance in the marketplace;
- changes in a specific country's or region's political and cultural climate or economic condition;
- an increased legal and compliance burden to establish, maintain and operate legal entities in foreign countries;
- the burden of complying with complex and changing foreign regulatory, tax, accounting, reporting and legal requirements, including the European General Data Protection Regulation 2016/679 (GDPR);
- the interpretation of contractual provisions governed by foreign laws in the event of a contract dispute, and the difficulty of effective enforcement of contractual provisions in local jurisdictions;
- inadequate IP protection in foreign countries and the existence of potentially relevant third-party IP rights;
- trade-protection measures including trade restrictions, import or export licensing requirements such as Export Administration Regulations promulgated by the U.S. Department of Commerce and fines, penalties, or suspension or revocation of export privileges, the imposition of government controls and changes in tariffs;
- the effects of applicable foreign tax structures and potentially adverse tax consequences; and
- significant adverse changes in foreign currency exchange rates.

We are also subject to extensive federal, state and foreign anti-bribery regulations, including the U.S. Foreign Corrupt Practices Act (FCPA), the UK Bribery Act and similar laws in other countries. Compliance with the FCPA is expensive and difficult, particularly in countries where corruption is a recognized problem. Additionally, the FCPA presents particular challenges to the pharmaceutical industry because, in many countries, hospitals are operated by the government, and doctors and other hospital employees are considered foreign officials. Certain payments to hospitals in connection with clinical trials and other work have been deemed to be improper payments to government officials and have led to FCPA enforcement actions. Various laws, regulations and executive orders also restrict the use and dissemination outside the United States, or the sharing with certain non-U.S. nationals, of information classified for national security purposes, as well as certain products and technical data relating to those products. As we expand our presence outside

the United States, we will need to dedicate additional resources to comply with these laws, and these laws may preclude us from developing, manufacturing or selling certain products and product candidates outside the United States, which could limit our growth potential and increase our development costs.

Additionally, in many jurisdictions outside the United States, we have or may become subject to additional industry-specific codes of conduct that impose additional compliance responsibilities on us. Failure to comply with these codes particularly as we enter new markets has, and could in the future, result in findings against us, which could result in financial sanctions or reputational harm.

We cannot guarantee that we, or our employees, consultants or third-party contractors are or will be in compliance with all federal, state and foreign regulations regarding bribery and corruption. Moreover, our strategic collaborators and third-party contractors outside the United States may have inadequate compliance programs or fail to respect the laws and guidance of the territories where they operate, which may result in substantial civil and criminal penalties and suspension or debarment from government contracting. The SEC also may suspend or bar issuers from trading securities on U.S. exchanges for violations of the FCPA's accounting provisions. Even if we are not determined to have violated these laws, government investigations typically require the expenditure of significant resources and generate negative publicity.

Our failure to maintain our enterprise resource planning (ERP) system could adversely impact our business and results of operations.

Any disruptions or difficulties using our global ERP system could adversely affect our controls, resulting in harm to our business, including our ability to forecast or make sales and collect our receivables. Significant delays in documenting, reviewing and testing our internal controls could cause our non-compliance with our SEC reporting obligations related to our management's assessment of our internal control over financial reporting, resulting in unanticipated costs and the diversion of management's attention.

Our success depends on our ability to retain key employees, consultants and advisors and to attract, retain and motivate qualified personnel.

Our ability to compete in the highly competitive biotechnology and pharmaceutical industries depends on our ability to attract and retain highly qualified managerial, scientific, technical, quality-control, manufacturing, medical, regulatory and commercial personnel. The turnover rate in our industry is high and we compete with other companies, as well as academic institutions, for individuals with certain skill sets. Failure to attract and retain personnel, or to maintain relationships with key consultants, contractors and advisors, could result in delays in production or difficulties in maintaining compliance with regulatory requirements. In addition, negative business developments and changes in immigration laws, policies or enforcement patterns, or announcements thereof, that make it more difficult or less desirable for immigrants to work in the U.S. or introduce greater uncertainty into the immigration system may make it difficult to recruit and retain qualified personnel.

We are highly dependent on members of our management and scientific teams. Each of our executive officers and key employees are employed "at will," and the loss of any of their services may adversely impact the achievement of our business objectives. We do not have "key person" insurance on any of our employees. Several of our executives have valuable, fully vested stock options or other long-term equity awards. Additionally, significant declines in our stock price have in the past reduced, and may in the future reduce, the retentive power of equity awards for our executives or other key personnel. We may not be able to retain these employees due to the competitive environment in the biotechnology industry, particularly in Cambridge, Massachusetts.

If we cannot maintain our corporate culture, we could lose the innovation, teamwork and passion that we believe contribute to our success.

It may be increasingly difficult to maintain our culture, which we have invested substantial resources in building. The evolution of our workforce and shifts in workplace and workstyle increase this risk. Failure to preserve our culture could negatively affect our future success.

Our internal computer systems and physical premises, or those of third parties with which we share sensitive data or information, may fail or suffer security breaches, including from cybersecurity incidents, which could materially disrupt our product development programs and manufacturing operations.

Our internal computer systems and infrastructure and those of our strategic collaborators, vendors, contractors, consultants or regulators with whom we share confidential, protected or sensitive data or information, or upon which our business relies, are vulnerable to damage from computer viruses, unauthorized access, misuse, natural disasters, terrorism, cybersecurity threats, war and telecommunication and electrical failures, as well as security compromises or data breaches, which may compromise our systems, infrastructure, data or that of those with whom we share such data or information or upon which our business relies, or lead to data compromise, misuse, misappropriation or leakage. We have experienced, and may experience additional, cyber-attacks on our

information technology systems and infrastructure by threat actors of all types (including nation states, criminal enterprises, individual actors or advanced persistent threat groups). Bad actors around the world use increasingly sophisticated methods, including the use of artificial intelligence, to engage in illegal activities involving the theft and misuse of personal information, confidential information

and intellectual property. In addition, the use of generative AI models in our internal or third-party systems may create new attack surfaces or methods for adversaries, which could impact us and our vendors. Any of these effects could damage our reputation, result in the loss of valuable property and information, cause us to breach applicable laws and regulations, and adversely impact our business. In addition, we may experience intrusions on our physical premises by these threat actors. In addition to extracting sensitive information, such attacks could include the deployment of harmful malware, ransomware, digital extortion, business email compromises and denial-of-service attacks, social engineering (including phishing attacks) and other means to affect server reliability and threaten the confidentiality, integrity and availability of information, systems or infrastructure. If any such cyber-attack or physical intrusion against us or those with whom we share confidential, protected or sensitive data or information, or upon which our business relies, were to result in a loss of or damage to our data, systems or infrastructure, or interrupt our operations, such as a material disruption of our development programs or our manufacturing operations, or due to a loss of any of our proprietary or confidential information, it would have a material adverse effect on us. For example, the loss of clinical trial data could delay our regulatory approval efforts and increase our costs to recover or reproduce the data. In addition, because we run multiple clinical trials in parallel, any breach or compromise of our computer systems or infrastructure or physical premises may result in a loss of data or compromised data integrity across multiple programs in many stages of development. While we seek to take steps to address cybersecurity risks, our efforts may not wholly mitigate such risks. Further, our contracts may not contain limitations of liability, and even where they do, there can be no assurance that limitations of liability in our contracts are sufficient to protect us from liabilities, damages or claims related to our privacy and data security obligations. Further, although we maintain cyber liability insurance, this insurance may not provide adequate coverage against potential liabilities related to any experienced cybersecurity incident or breach.

Any data breach, security incident or compromise of confidential, protected or personal information, including any clinical trial participant personal data, may also result in notification requirements or other disclosure obligations and may subject us to civil fines and penalties, litigation, regulatory investigations or enforcement actions, or claims for damages under the GDPR and UK GDPR and relevant member state law in the EU, other foreign laws, and the federal Health Insurance Portability and Accountability Act of 1996 (HIPAA), and other relevant state and federal privacy laws in the United States including the California Consumer Privacy Act, as amended by the California Privacy Rights Act (the CCPA). We have from time to time received information that companies working on vaccine research and development may be a particular focus for those planning cyberattacks, including by nation states and affiliated cyber actors. To the extent that any disruption or security compromise incident or breach were to result in a loss of, or damage to, our data, systems, infrastructure or applications, or inappropriate use or disclosure of confidential or proprietary information, including information related to the research and manufacturing of our products, we could incur liability, our competitive and reputational position could be harmed and the further development and commercialization of our product candidates could be delayed.

We may use our financial and human capital to pursue a particular research program or product candidate and fail to capitalize on programs that may be more profitable or for which there is a greater likelihood of success.

We pursue and fund the development of selected research programs or product candidates and may choose to forego or delay pursuit of other opportunities that could later prove to have greater commercial potential. For example, we are currently focusing our efforts on certain prioritized programs that we expect to file for approval over the next several years. Our resource allocation decisions, or our contractual commitments to provide resources to our strategic collaborators, may cause us to fail to capitalize on other commercial products or market opportunities. Additionally, spending on research and development programs may not yield commercially viable products. We may also seek to enter into strategic collaborations or financing arrangements pursuant to which we may relinquish valuable rights to product candidates, including the rights to a portion of future revenues, through a strategic alliance, licensing or other royalty arrangements in cases where it would have been more advantageous for us to retain sole development and commercialization rights. Additionally, we may allocate internal resources to a product candidate in a therapeutic area in which it would have been more advantageous to enter into a strategic alliance.

If we are not successful in discovering, developing and commercializing additional products, our ability to expand our business and achieve our strategic objectives would be impaired.

A key element of our longer-term strategy is to discover, develop and commercialize products beyond our current portfolio to treat various conditions and in a variety of therapeutic areas. We intend to do so by investing in our drug discovery efforts, exploring potential strategic alliances for the development of new products and in-licensing technologies. Identifying new product candidates requires substantial technical, financial and human resources. We may fail to identify promising candidates or to successfully develop and commercialize products for many reasons, which would impair our potential for growth.

Our business could be harmed if we suffer damage to our reputation, including as a result of a product recall.

The FDA or foreign regulators could require the recall of our products. The FDA has authority to recall a biologic product if it finds that a batch, lot or other quantity of the biologic product presents an imminent or substantial hazard to the public health. In addition, foreign governmental bodies may require the recall of any products in the event of material deficiencies or defects in design or manufacture. Manufacturers may independently recall a product if a material deficiency in a product is found. A government-mandated or voluntary recall by us or our strategic collaborators could occur because of manufacturing errors, design or labeling defects or other deficiencies and issues, as occurred with the recall in 2021 of certain batches of our COVID vaccine shipped to Japan that were found to contain foreign particulate. Recalls of any of our products would divert managerial and financial resources and adversely affect our financial condition and results of operations. A recall announcement could harm our reputation and negatively affect our sales. Our reputation could be further impacted by public discourse regarding our business and perception of our business strategy.

Product liability lawsuits against us could cause us to incur substantial liabilities and limit commercialization of our products.

We are exposed to product liability risk related to the development, testing, manufacturing and marketing of our products and product candidates in clinical trials. Product liability claims and related cross-claims and claims for indemnification may be brought against us by patients, healthcare providers or others using, prescribing, selling or otherwise coming into contact with our products or product candidates. For example, we may be sued if any product or product candidate allegedly causes injury or is found to be otherwise unsuitable during clinical trials, manufacturing or, if approved, marketing, sale or commercial use. If we cannot successfully defend ourselves against such claims, we could incur substantial liabilities.

We could also face product liability claims relating to the worsening of a patient's condition, injury or death alleged to have been caused by our products or product candidates. Any such claims may include allegations of defects in manufacturing or design, a failure to warn of dangers inherent in the product, including due to interactions with alcohol or other drugs, knowledge of risks, negligence, strict liability and a breach of warranties. Claims could also be asserted under state consumer protection acts. Such claims might not be fully covered by product liability insurance. For any marketed products, product liability claims could result in an FDA investigation of the safety and effectiveness of our products, our manufacturing processes and facilities or our marketing programs, and potentially a recall of our products or more serious enforcement action, limitations on the approved indications for which they may be used, suspension or withdrawal of approvals or license revocation. Regardless of the merits or eventual outcome, liability claims may cause decreased demand for our products, injury to our reputation and significant negative media attention, costs to defend the related litigation, withdrawal of clinical trial participants, loss of revenue, a diversion of management's time and our resources, substantial monetary awards to trial participants, patients or their family members, payments to indemnify clinical trial sites and other clinical trial partners and a decline in our stock price. On occasion, large judgments have been awarded in individual, mass tort and class-action lawsuits based on drugs or medical treatments that had unanticipated adverse effects.

With respect to our COVID vaccine, although the U.S. and certain foreign governments contractually agreed to indemnify us or make statutory immunity available to us for sales during the pandemic public health emergency, which has since ended, such indemnification or statutory immunity may be unavailable to cover potential claims or liabilities resulting from the research, development, manufacture, distribution or commercialization of the vaccine. In the United States, our COVID vaccine currently benefits from liability protections under the Public Readiness and Emergency Preparedness (PREP) Act. However, if the Secretary of the U.S. Department of Health and Human Services were to revoke or materially modify the declaration underlying such PREP Act coverage, or if our COVID vaccine is not successfully transitioned to the National Vaccine Injury Compensation Program (VICP), our COVID vaccine could become subject to significantly greater product liability exposure in the United States. In addition, any significant changes to, or overhaul of, the VICP could increase our potential liability exposure. Additionally, other foreign governments that we contract with have not provided us with similar contractual indemnity or statutory immunity, and we generally no longer have the benefit of such indemnities or immunities for future sales of our commercial products. Substantial claims arising from the vaccine outside the scope of or in excess of U.S. or foreign government indemnity or statutory immunity could harm our financial condition and operating results. Moreover, any adverse event or injury for which we are liable, even if fully covered under an indemnity or immunity, could negatively affect our reputation.

We may be unable to maintain our product liability insurance coverage at a reasonable cost or in sufficient amounts to protect us against losses due to liability. If the costs of maintaining adequate insurance coverage increase significantly in the future, our operating results could be materially adversely affected. Likewise, if insurance coverage became unavailable to us or economically impractical, we would be required to operate our business without indemnity from commercial insurance providers. Additionally, even if we maintain insurance coverage for a type of liability, a particular claim may not be covered if it is subject to a coverage exclusion or we do not otherwise meet the conditions for coverage. If we operate our business with inadequate insurance, we could be responsible for paying claims or judgments against us, which could adversely affect our results of operations or financial condition.

Federal legislation and actions by federal, state and local governments may permit reimportation into the United States of drugs from foreign countries where the drugs are sold at lower prices.

We may face competition in the United States from medicines sourced from foreign countries with price controls on pharmaceutical products. For example, in October 2020, the FDA published a final rule that would allow for the importation of certain prescription drugs from Canada, where there are government price controls. In January 2024, the FDA approved Florida's request to import certain lower-priced medications from Canada. While the full implications of the final rule are unknown, legislation or regulations allowing the reimportation of drugs could decrease the price we receive for our products and adversely affect our future revenues and potential profitability.

Healthcare legislative reform discourse and potential or enacted measures may have a material adverse impact on our business and results of operations and legislative or political discussions surrounding the desire for and implementation of pricing reforms may adversely impact our business.

In the United States, federal and state legislatures, health agencies and third-party payors continue to focus on containing the cost of health care. Legislative and regulatory proposals, enactments to reform health care insurance programs and increasing pressure from social sources could significantly influence how our products are prescribed and purchased. For example, provisions of the ACA resulted in changes in the way health care is paid for by governmental and private insurers, including increased rebates owed by manufacturers under the Medicaid Drug Rebate Program, annual fees and taxes on manufacturers of certain branded prescription drugs, the requirement that manufacturers participate in a discount program for certain outpatient drugs under Medicare Part D and the expansion of the number of hospitals eligible for discounts under Section 340B of the PHSA. Additionally, the Inflation Reduction Act of 2022 includes provisions, such as drug pricing controls and Medicare redesign, that have begun to be implemented and are likely to impact our business, but its ultimate effect on our business and the healthcare industry in general is not yet known. See "Business—Government Regulation-Current and future healthcare reform legislation."

We may face uncertainties due to efforts to repeal, substantially modify or invalidate some or all of the provisions of the ACA. There is no assurance that the ACA, as currently enacted or as amended in the future, will not adversely affect our business and financial results, and we cannot predict how future federal or state legislative or administrative changes relating to healthcare reform will affect our business.

There is increasing public attention on the costs of prescription drugs and there have been, and are expected to continue to be, legislative proposals to address prescription drug pricing, which could have significant effects on our business. These actions and the uncertainty about the future of the ACA and healthcare laws may put downward pressure on pharmaceutical pricing and increase our regulatory burdens and operating costs.

In addition, the U.S. government is pursuing measures intended to reduce the cost of drugs in the United States. For example, in May 2025, the President issued an executive order directing the U.S. Department of Health and Human Services and other federal agencies to take certain steps intended to, among other things, reduce the prices of drugs sold in the United States to match the lowest price available for the same drugs in comparably developed nations. It is currently unclear how and to what extent these measures may be implemented and what impact any such implementation would have on us.

There is also significant economic pressure on state budgets that may result in states increasingly seeking to achieve budget savings through mechanisms that limit coverage or payment for drugs or that would allow for importation of pharmaceutical products from lower cost jurisdictions outside the United States. State Medicaid programs are increasingly requesting manufacturers to pay supplemental rebates and requiring prior authorization by the state program for use of any drug for which supplemental rebates are not being paid. In addition, changes to federal Medicaid rebate requirements may increase manufacturer rebate liabilities. Government efforts to reduce Medicaid expenses may lead to increased use of managed care organizations by Medicaid programs. This may result in managed care organizations influencing prescription decisions for a larger segment of the population and a corresponding limitation on prices and reimbursement for our products, if approved.

In the EU and some other international markets, the government provides health care at low cost to consumers and regulates pharmaceutical prices, patient eligibility or reimbursement levels to control costs for the government-sponsored health care system. Many countries have announced or implemented measures, and may in the future implement new or additional measures, to reduce health care costs to limit the overall level of government expenditures. These measures vary by country and may include, among other things, patient access restrictions, suspensions on price increases, prospective and possible retroactive price reductions and other recoupments and increased mandatory discounts or rebates, recoveries of past price increases and greater importation of drugs from lower-cost countries. These measures may adversely affect our revenues and results of operations.

We are subject, directly or indirectly, to federal and state healthcare fraud and abuse laws and false claims laws. If we cannot comply, or have not fully complied, with such laws, we could face substantial penalties.

Healthcare providers, physicians and third-party payors in the United States and elsewhere play a primary role in the recommendation and prescription of pharmaceutical products. Arrangements with third-party payors and customers can expose pharmaceutical manufacturers to broadly applicable fraud and abuse and other healthcare laws and regulations, which may constrain the business or financial arrangements and relationships through which such companies sell, market and distribute pharmaceutical products. In particular, the promotion, sales and marketing of healthcare items and services, as well as a wide range of pricing, discounting, structuring and commissions, certain customer incentive programs and other business arrangements, are subject to extensive laws designed to prevent fraud, kickbacks, self-dealing and other abusive practices, which may restrict or prohibit such activities and arrangements. Activities subject to these laws also involve the improper use of information obtained during patient recruitment for clinical trials. See "Business—Government Regulation—Other healthcare laws."

The scope and enforcement of each of these laws is uncertain and subject to rapid change in the current environment of healthcare reform. Ensuring business arrangements comply with applicable healthcare laws, as well as responding to possible government investigations, is time- and resource-consuming and can divert a company's attention from the business. If our operations are found to violate any of these laws or any other regulations, we may be subject to significant sanctions, including civil, criminal and administrative penalties, damages, fines, disgorgement, imprisonment, reputational harm, exclusion from participation in federal and state funded healthcare programs, contractual damages and the curtailment or restricting of our operations, as well as additional reporting obligations and oversight if we become subject to a corporate integrity agreement or other agreement to resolve allegations of non-compliance. Furthermore, if any physician or other healthcare provider or entity with whom we do business is found to be not in compliance with applicable laws, they may be subject to similar penalties. Any action for violation of these laws, even if successfully defended, could cause us to incur significant legal expenses and divert management's attention from the operation of the business. In addition, the approval and commercialization of any product candidate we develop outside the United States will subject us to foreign healthcare laws.

The provision of benefits or advantages to physicians to induce or encourage the prescription, recommendation, endorsement, purchase, supply, order or use of medicinal products is prohibited in the EU and the UK. The provision of benefits or advantages to induce or reward improper performance generally is also governed by the national anti-bribery laws of EU Member States, and the Bribery Act 2010 in the UK. Infringement of these laws could result in substantial fines and imprisonment. The EU Directive (2001/83/EC, as amended) governing medicinal products for human use provides that, where medicinal products are being promoted to persons qualified to prescribe or supply them, no gifts, pecuniary advantages or benefits in kind may be supplied, offered or promised to such persons unless they are inexpensive and relevant to the practice of medicine or pharmacy. This provision has been retained in UK law despite the UK's departure from the EU.

Payments made to physicians in certain EU Member States must be publicly disclosed, and agreements with physicians often are the subject of prior notification and approval by the physician's employer, competent professional organization or applicable regulatory authorities. Failure to comply with these requirements could result in reputational harm, public reprimands, administrative penalties, fines or imprisonment.

We are subject to various and evolving laws and regulations governing the privacy and security of personal data, and our failure to comply could result in fines or criminal penalties and damage our reputation.

We are subject to data privacy and security laws and regulations in the United States, Europe and other jurisdictions where we operate or collect personal information. These laws and regulations apply to the collection, storage, use, sharing and security of personal data, including health information, and impose significant compliance obligations. In addition, numerous other federal and state laws, including state security breach notification laws, state health information privacy laws and federal and state consumer protection and privacy laws, impose additional requirements with respect to personal information.

The EU GDPR and UK GDPR are wide-ranging in scope and impose numerous requirements on companies that process personal data, including imposing special requirements in respect of the processing of health and other sensitive data, requiring that consent of individuals to whom the personal data relates is obtained in certain circumstances, requiring additional disclosures to individuals regarding data processing activities, requiring that safeguards are implemented to protect the security and confidentiality of personal data, limiting retention periods for personal data, creating mandatory data breach notification requirements in certain circumstances, and requiring that certain measures (including contractual requirements) are put in place when engaging third-party service providers. Different EEA Member States have interpreted the EU GDPR differently and many have imposed additional requirements, adding to the complexity of processing personal data in the EEA. The EU GDPR and UK GDPR also impose strict rules on the transfer of personal data to countries outside the EEA that are not considered to provide "adequate" protection to personal data, including the United States, and permits data protection authorities to impose large penalties for violations. Compliance with the EU GDPR and UK GDPR is a rigorous and time-intensive process that may increase our cost of doing business or require us to change our business

practices. We could be subject to fines and penalties, litigation and reputational harm in connection with any activities falling within the scope of the EU GDPR or UK GDPR.

In the United States, at the federal level, failing to take appropriate steps to keep consumers' personal information secure may constitute unfair acts or practices in or affecting commerce in violation of Section 5(a) of the Federal Trade Commission Act. In addition, numerous states have introduced comprehensive laws which govern the privacy and security of personal information. For example, California has passed the CCPA and more than a dozen other states have passed their own similar comprehensive consumer privacy laws. Other states proposed or passed laws that target particular aspects of privacy, including health, genetic and biometric information, which could have particular impact for our business. For example, Washington passed a law that protects the privacy of consumer health-related information that is not covered by HIPAA and includes a private right of action. In addition, state laws such as the Texas Genomic Privacy Act and others focus specifically on genetic information and could materially impact our business.

Regulators and legislators are also increasingly scrutinizing and restricting certain personal data transfers and transactions involving foreign countries. For example, the Department of Justice's January 8, 2025, rule on "Preventing Access to U.S. Sensitive Personal Data and Government-Related Data by Countries of Concern or Covered Persons," prohibits data brokerage transactions involving certain sensitive personal data categories, including health data, genetic data, and biospecimens, to countries of concern, including China. The regulations also restrict certain investment agreements, employment agreements and vendor agreements involving such data and countries of concern, absent specified cybersecurity controls. Actual or alleged violations of these rules may be punishable by criminal and/or civil sanctions, result in exclusion from participation in federal and state programs or restrict our ability to use certain vendors, sites, investigators, or service providers in global clinical trials.

Additionally, many foreign jurisdictions have passed or are considering various proposals for data privacy and protection laws, and data privacy remains an evolving landscape at the domestic and international levels, subject to continued legal challenges. We must devote significant resources to understanding and complying with the changing landscape in this area. Each law is also subject to various interpretations by courts and regulatory agencies, creating additional uncertainty, and any failure or perceived failure to comply with the evolving data protection laws could expose us to risk of enforcement actions taken by authorities, private rights of action in some jurisdictions and potential significant penalties, including criminal sanctions, if we are found to be non-compliant. For example, we could be subject to penalties, including criminal penalties, if we knowingly obtain or disclose individually identifiable health information from a HIPAA-covered health care provider or research institution that has not complied with HIPAA's requirements for disclosing such information. Furthermore, the number of government investigations related to data security incidents and privacy violations continues to increase and government investigations typically require significant resources and generate negative publicity, which could harm our business and our reputation.

The Clinical Trials Regulation (EU) No. 536/2014 (the Clinical Trials Regulation) and the EMA policy on publication of clinical data for medicinal products for human use provide for the publication of certain clinical information submitted in marketing authorization applications (MAAs). The ability of third parties to review or analyze data from our clinical trials may increase the risk of commercial confidentiality breaches and result in enhanced scrutiny of our clinical trial results. Such scrutiny could result in public misconceptions regarding our drugs and drug candidates. These publications could also result in the disclosure of information to our competitors that we might otherwise deem confidential, which could harm our business.

Our use of generative AI ("GenAI") and other AI technologies presents certain risks and challenges given the emerging nature of AI technologies.

The development and use of GenAI and other AI technologies (collectively, AI Technologies), together with an uncertain and rapidly evolving global regulatory landscape, pose risks that could harm our reputation, expose us to liability, increase compliance costs, or otherwise adversely affect our business. In particular, new and emerging regulations, such as the European Union Artificial Intelligence Act (the EU AI Act), impose or are expected to impose enhanced obligations related to the development, deployment, transparency, governance, risk management, and use of certain AI systems, including those classified as high-risk, and may subject us to significant fines, penalties, operational restrictions, or changes to our business practices if we fail to comply. These requirements may apply extraterritorially, including to AI systems developed or deployed outside the European Union if their outputs are used in the EU.

In the United States, over the past year, states have advanced, and in some cases passed, dozens of laws focusing on AI and GenAI governance and regulation, including on deployment of AI Technologies in healthcare settings. At the same time, the Trump Administration has endorsed a federal moratorium on the enforcement of state AI laws, including through an executive order on "Ensuring a National Policy Framework for Artificial Intelligence." So far, these efforts have not been successful at curtailing state action on AI regulation, contributing to a complicated legislative patchwork, which may be litigated in state and federal courts. Moreover, the FDA has advanced guidance and proposed frameworks for regulating AI Technologies in drug discovery, marketing submissions, and medical device development.

The integration of AI Technologies into our and our vendors' systems (potentially without the vendor disclosing such use to us) subjects us to the risk that the providers of AI Technologies may not meet existing or evolving legal, regulatory or industry standards relating to privacy, data protection, cybersecurity, intellectual property, product safety, human oversight, or algorithmic accountability. Compliance with these requirements may require substantial investments in governance, documentation, monitoring, testing, or operational controls, and may limit our ability to deploy or scale certain AI-enabled tools or features.

In addition, bad actors globally use increasingly sophisticated methods, including AI Technologies, to engage in illegal activities involving the theft and misuse of personal information, confidential information and intellectual property. The use of GenAI models in our internal or third-party systems may create new attack surfaces or methods for adversaries, which could impact us and our vendors. These factors may lead to breaches of security or privacy, reduced system performance or availability, loss of intellectual property, unauthorized disclosure of confidential or proprietary information. Sophisticated cyber-attacks, including those leveraging AI, could exacerbate these risks.

Further, GenAI's potential for producing false, misleading or biased outputs, or to generate content that may not be subject to IP protection or that may infringe the proprietary rights of third parties, poses additional risks to our business. Regulatory changes or reinterpretations could introduce new compliance risks, including potential government enforcement actions or civil lawsuits. Our competitors' faster or more effective adoption of AI could also disadvantage us.

We could be adversely affected by outbreaks of epidemic, pandemic or other contagious diseases.

In the event of a future epidemic or pandemic, our clinical trials could be paused or delayed due to restrictions (such as quarantines or travel limitations) or reprioritization of resources. Travel limitations could also create challenges and potential delays in our development and production activities, increasing the expense and timelines for producing our products and product candidates.

We utilize third parties to, among other things, manufacture raw materials, components, parts and consumables, perform quality testing and ship our products. If these third parties were to experience delays or disruptions in providing their services in response to an epidemic or pandemic, our supply chain could be disrupted, limiting our ability to manufacture and sell our products and manufacture product candidates for our clinical trials, as well as negatively impacting our research and development operations. Such delays or disruptions could adversely impact our strategic collaborators' ability to fulfill their obligations, which could affect the clinical development or regulatory approvals of product candidates under joint development.

In addition, during a global health crisis, one or more government entities could take actions (such as via the Defense Production Act in the U.S.) that diminish our rights or economic opportunities with respect to our products. Our third-party service providers could be impacted by government-imposed restrictions on services they might otherwise offer. Any such action could cause us to experience delays in the development, production, distribution or export of our products and product candidates and increased expenses.

Engaging in acquisitions, joint ventures or strategic collaborations may increase our capital requirements, dilute our shareholders and cause us to incur debt or assume contingent liabilities.

We may engage in acquisitions, joint ventures and collaborations, including licensing or acquiring complementary products, IP rights, technologies or businesses. Such transactions and relationships may entail numerous risks, including:

- increased operating expenses and cash requirements;
- assimilation of operations, IP and products, including difficulties associated with integrating new personnel;
- the diversion of management's attention from our existing product programs and initiatives;
- the loss of key personnel and uncertainties in our ability to maintain key business relationships;
- risks and uncertainties associated with the other party to such a transaction, including the prospects of that party and their existing products or product candidates and regulatory approvals; and
- our inability to generate revenue from acquired technology or products sufficient to meet our objectives in undertaking the acquisition or even to offset the associated acquisition and maintenance costs.

If we undertake acquisitions in the future, we may utilize cash, issue dilutive securities, assume or incur debt obligations, incur large one-time expenses and acquire intangible assets that could result in significant future amortization expense. Moreover, if we cannot locate suitable acquisition or strategic collaboration opportunities, our ability to grow or obtain access to technology or products that may be important to the development of our business may be impaired.

The illegal distribution and sale by third parties of counterfeit or stolen versions of mRNA products could negatively impact our financial performance or reputation.

Third parties could illegally distribute and sell, especially online, counterfeit versions of mRNA products that do not meet the rigorous cGMP manufacturing and testing standards. Counterfeit medicines may contain harmful substances or the wrong dose, are frequently unsafe or ineffective and could be life-threatening. However, to distributors and users, counterfeit products may be visually indistinguishable from the authentic version.

Reports of adverse reactions to counterfeit products, increased levels of counterfeiting or unsafe mRNA products could materially affect patient confidence in our mRNA products. Adverse events caused by unsafe counterfeit or other non-mRNA products could mistakenly be attributed to our mRNA products. In addition, thefts of inventory at warehouses, plants or while in-transit, which are not properly stored and which are sold through unauthorized channels, could adversely impact patient safety, our reputation and our business. The public could lose confidence in the integrity of mRNA products because of counterfeiting, theft or improper manufacturing processes.

Climate change or legal, regulatory or market measures to address climate change may negatively affect our business, results of operations and financial condition.

We are exposed to physical risks (such as rising temperatures, flooding and severe storms), risks in transitioning to a low-carbon economy (such as additional legal or regulatory requirements, changes in customer behavior and cost and availability of raw materials) and social and human effects (such as population dislocations and harm to health and well-being) associated with climate change. These risks can be either acute (short-term) or chronic (long-term). Climate-related physical risks to our facilities and those of our suppliers could disrupt our operations and supply chain, which may result in increased costs.

New legal or regulatory requirements may be enacted to prevent, mitigate or adapt to the implications of climate change and its effects on the environment. These regulations, which may differ across jurisdictions, could subject us to new or expanded carbon pricing or taxes, increased compliance costs, restrictions on greenhouse gas emissions, investment in new technologies, increased carbon disclosure and transparency, investments in data gathering and reporting systems, upgrades of facilities to meet new building codes and the redesign of utility systems, which could increase our operating costs, including the cost of electricity and energy. Our supply chain would likely be subject to these same transitional risks and would likely pass along any increased costs to us, which may affect our procurement of raw materials or other supplies required to operate our business at the quantities and levels we require.

Evolving environmental, social and governance (ESG) standards and expectations may expose us to numerous risks.

Many investors use ESG screening criteria to determine whether we qualify for inclusion in their investment portfolios, and such investors and other stakeholders expect us to set ESG goals and provide robust disclosure regarding such goals, progress toward goals and other ESG matters of interest. Potential changes in voting practices and the fragmented influence of proxy advisors may impact investor expectations. In addition, we also face ESG-related expectations from certain customers outside the U.S. that can impact our competitiveness in commercial tenders. In response, we have established ESG goals and adapted our tracking and reporting to various ESG frameworks; however, these goals reflect our current plans and aspirations and may not be achieved.

ESG reporting standards are relatively new, have not been harmonized and continue to evolve, and differing interpretations, changing frameworks and evolving regulatory expectations could make compliance more complex and result in inconsistent or revised disclosures over time. If our ESG practices, disclosures or progress do not meet evolving stakeholder or regulatory expectations, or if we fail or are perceived to fail to achieve or accurately report progress toward our goals, our reputation, ability to attract and retain employees and attractiveness as an investment or business partner could be adversely affected, and we could be subject to enforcement actions or litigation. Additionally, the increased focus on ESG matters by policymakers, regulators and investors has in some instances resulted in diverging expectations and standards, which could make it more difficult to comply with ESG-related regulations across our global business.

If we fail to comply with environmental, health and safety laws and regulations, we could become subject to fines or penalties or incur costs that could harm our business.

We are subject to numerous environmental, health and safety laws and regulations, including those governing laboratory procedures and the handling, use, storage, treatment and disposal of hazardous and flammable materials and wastes, including chemicals and biological materials. We generally contract with third parties for the disposal of these materials and waste products, and we cannot eliminate the risk of contamination or injury from these materials. In the event of contamination or injury resulting from any use by us of hazardous materials, we could be held liable for any resulting damages, and any liability could exceed our resources. We also could incur significant costs associated with civil or criminal fines, penalties or other sanctions for failure to comply with such laws and regulations. We may also incur substantial costs to comply with such laws and regulations, and these laws and regulations could impair our research, development or production efforts.

Our workers' compensation insurance may not provide adequate coverage against potential liabilities due to injuries to our employees resulting from the use of hazardous materials. We do not maintain insurance for environmental liability or toxic tort claims that may be asserted against us in connection with our storage or disposal of biological or hazardous materials.

Risks related to ownership of our common stock

The price of our common stock has been volatile, which could result in substantial losses for shareholders.

Our stock price has been, and is expected to continue to be, subject to substantial volatility. Since our IPO in December 2018, our stock price has ranged from a high of $497.49 to a low of $11.54 per share. Over the last several years, our stock has experienced pronounced and extended periods of volatility, which could cause our shareholders to incur substantial losses. Public statements by us, government agencies, the media, competitors, financial analysts or others have resulted, and may result, in significant fluctuations in our stock price. Public information, regardless of accuracy, has had and will likely continue to have an outsized impact on our stock price.

The stock market generally, and the biopharmaceutical sector in particular, has experienced significant volatility that may be unrelated to company operating performance, and you may be unable to sell your shares at or above your initial purchase price. Our stock price may be influenced by many factors, including:

- our product sales and anticipated product revenue;
- our ability to effectively reduce costs;
- the commercial launch of any additional products;
- timing and results of clinical trials or progress of our product candidates or those of our competitors;
- the exclusion or removal of our stock from market indices;
- the success of competitive products or technologies;
- the emergence or decline of new or existing variants of the SARS-CoV-2 virus;
- developments regarding our manufacturing, regulatory and commercialization efforts, or information regarding such efforts by competitors;
- regulatory or legal developments in the United States and other countries;
- developments or disputes concerning patent applications, issued patents or other proprietary rights;
- the recruitment or departure of key personnel;
- expenses related to any of our products or clinical development programs;
- the results of our efforts to discover, develop, acquire or in-license additional product candidates;
- actual or anticipated changes in estimates of financial results, development timelines or recommendations by securities analysts;
- variations in our financial results or those of companies that are perceived to be similar to us;
- changes in the structure of healthcare payment systems;
- economic, industry and market conditions generally, and in the biopharmaceutical sector specifically; and
- announcement by us or our competitors of the commencement or termination of significant acquisitions, strategic collaborations, joint ventures or capital commitments.

Securities class-action litigation often has been instituted against companies following periods of volatility in their stock price. In 2024, class action and derivative litigations were filed against us related to statements made about our RSV vaccine. See Item 3. "Legal Proceedings." We could incur substantial costs in defense of such litigation or any future lawsuits that may be filed against us, and management's attention and resources could be diverted.

Our principal shareholders and management own a significant percentage of our stock and will be able to exert significant control over matters subject to shareholder approval.

As of February 13, 2026, our executive officers, directors and affiliated shareholders owned, directly or indirectly, approximately 8% of our outstanding common stock. In addition, non-affiliated five percent or greater shareholders owned approximately 32% of our outstanding common stock. These shareholders have the ability to influence us through their ownership positions. For example, if they were to act together, they could exert significant influence over matters such as elections of directors, amendments of our organizational documents or approval of any merger, sale of assets or other major corporate transaction. This may prevent or discourage unsolicited acquisition proposals or offers for our common stock that our shareholders may believe to be in their best interests.

Provisions in our organizational documents and Delaware law could make it more difficult or costly for a third party to acquire us or remove our current management, even if doing so would benefit our shareholders.

Our restated certificate of incorporation (charter), second amended and restated by-laws (by-laws) and Delaware law contain provisions that could delay or prevent a hostile takeover or change in control of us or changes in our management. Our charter and by-laws include provisions that:

- authorize "blank check" preferred stock, which could be authorized for issuance by our board of directors without shareholder approval and may contain voting, liquidation, dividend and other rights superior to our common stock;
- create a classified board of directors whose members serve staggered three-year terms;
- limit shareholders' ability to call a special meeting of shareholders to shareholders representing at least 20% of our outstanding shares;
- prohibit shareholder action by written consent;
- establish an advance notice procedure for shareholder approvals to be brought before an annual meeting of our shareholders, including proposed nominations of persons for election to our board of directors;
- provide that our directors may be removed only for cause;
- provide that vacancies on our board of directors may be filled only by a majority of directors then in office, even though less than a quorum;
- specify that no shareholder is permitted to cumulate votes at any election of directors;
- expressly authorize our board of directors to modify, alter or repeal our by-laws; and
- require supermajority votes of the holders of our common stock to amend specified provisions of our charter and by-laws.

In addition, because we are incorporated in Delaware, we are governed by the provisions of Section 203 of the Delaware General Corporation Law, which limits the ability of shareholders owning in excess of 15% of our outstanding voting stock to merge or combine with us. Any provision of our charter, by-laws or Delaware law that has the effect of delaying or deterring a change in control could limit the opportunity for our shareholders to receive a premium for their shares, and could also affect the price that some investors are willing to pay for our common stock.

We do not expect to pay cash dividends for the foreseeable future.

We do not currently intend to declare or pay cash dividends on our capital stock and instead intend to retain any future earnings to finance the growth and development of our business or to return cash to shareholders through share repurchases. In addition, our existing credit agreement contains restrictions on our ability to pay dividends, and any future debt agreements may preclude us from paying dividends. As a result, capital appreciation, if any, of our common stock will be your sole source of gain for the foreseeable future.

Our by-laws designate the Court of Chancery of the State of Delaware or the U.S. District Court for the District of Massachusetts as the exclusive forum for certain litigation that may be initiated by our shareholders, which could limit our shareholders' ability to obtain a favorable judicial forum for disputes with us.

Pursuant to by-laws, unless we consent in writing to the selection of an alternative forum, the Court of Chancery of the State of Delaware is the sole and exclusive forum for state law claims for (1) any derivative action or proceeding brought on our behalf, (2) any action asserting a claim of or based on a breach of a fiduciary duty owed by any of our current or former directors, officers, or other employees to us or our shareholders, (3) any action asserting a claim against us or any of our current or former directors, officers, employees or shareholders arising pursuant to any provision of the Delaware General Corporation Law or our by-laws or (4) any action asserting a claim governed by the internal affairs doctrine (the Delaware Forum Provision). The Delaware Forum Provision will not apply to any causes of action arising under the Securities Act or the Exchange Act. Our by-laws further provide that the U.S. District Court for the District of Massachusetts is the exclusive forum for resolving any complaint asserting a cause of action arising under the Securities Act (the Federal Forum Provision). Our by-laws provide that any person or entity purchasing or otherwise acquiring any interest in shares of our common stock is deemed to have notice of and consented to the Delaware Forum Provision and the Federal Forum Provision.

The Delaware Forum Provision and the Federal Forum Provision may impose additional litigation costs on shareholders in pursuing any such claims, particularly if the shareholders do not reside in or near the State of Delaware or the Commonwealth of Massachusetts, as applicable. Additionally, the forum selection clauses in our by-laws may limit our shareholders' ability to obtain a favorable judicial forum for disputes with us or our directors, officers or employees, which may discourage the filing of lawsuits against us and our directors, officers and employees, even though an action, if successful, might benefit our shareholders. The Federal Forum Provision may also impose additional litigation costs on shareholders who assert that the provision is unenforceable or invalid, and if the Federal Forum Provision is found to be unenforceable, we may incur additional costs in resolving such matters. The Court of Chancery of the State of Delaware and the U.S. District Court for the District of Massachusetts may also reach different judgments or results than would

other courts, including courts where a shareholder considering an action may be located or would otherwise choose to bring the action, and such judgments may be more or less favorable to us than our shareholders.

General risk factors

Our employees, principal investigators and consultants may engage in misconduct or other improper activities, including non-compliance with regulatory standards and requirements and insider trading.

We are exposed to the risk of fraud or other misconduct by our employees, principal investigators leading our clinical trials and consultants. Such misconduct could include failures to comply with FDA regulations or similar regulations in other jurisdictions, provide accurate information to the FDA, the EMA and other regulatory authorities, comply with healthcare fraud and abuse laws and regulations in the United States and abroad, comply with industry codes of conduct, report financial information or data accurately, or disclose unauthorized activities to us. Such misconduct also could involve improper use of information obtained during clinical trials or interactions with the FDA or other regulatory authorities, which could result in regulatory sanctions and serious harm to our reputation. It is not always possible to identify and deter employee misconduct, and we may fail to control unknown or unmanaged risks or losses or take steps that protect us from government investigations or other actions or lawsuits stemming from a failure to comply with these laws or regulations. If any such actions are instituted against us, those actions could have a significant impact on our business, financial condition, results of operations and prospects, including through the imposition of significant fines or other sanctions.

We could face unfavorable U.S. or global economic conditions, including as a result of disease outbreak, war, conflict or other political instability, or geopolitical risks.

Our results of operations could be adversely affected by general conditions in the global economy and financial markets, including disruptions caused by pandemic, war, conflict or other political instability. Adverse macroeconomic conditions, and perceptions or expectations about current or future conditions, such as inflation, slowing growth, rising interest rates, the imposition of tariffs (either on imports or exports, which may impact us directly or indirectly through increased costs in our supply chain), rising unemployment and recession, could negatively affect our business and financial condition. Additionally, global events, including war, conflict, political instability or other adverse economic conditions have and may in the future cause governments to divert spending away from healthcare, negatively impacting the market for our products.

Any severe or prolonged economic downturn could create a variety of risks to our business, including weakened demand for our medicines, and negatively impact our ability to raise additional capital or financing if needed on favorable terms, if at all. A weak or declining economy could strain our suppliers, possibly resulting in supply disruption, or cause delays in payments for our services by third-party payors or our collaborators. Any of the foregoing could harm our business and we cannot anticipate all the ways in which the current economic climate and financial market conditions could adversely impact our business.

Additionally, geopolitical tensions could adversely impact our business and our commercial plans. For instance, restrictions by the U.S. government on the export of mRNA technology or our products to certain markets, or restrictions on the establishment of manufacturing in certain foreign jurisdictions, may prevent us from seeking commercial growth opportunities in a manner that harms our competitive position.

Changes in tariffs and other governmental trade policies could negatively affect our business and results of operations.

Recent governmental actions and proposals relating to tariffs and other trade policies have created uncertainty about future trading arrangements and the possibility of imposing or increasing tariffs on certain goods. For example, certain governments (including the U.S. and other countries where we do business) have imposed or may impose tariffs on a wide range of products, raw materials, and intermediate goods. Additional tariffs, or retaliatory measures by other countries in response, may be implemented at any time. Should these or similar tariffs remain in place (or be re-imposed or increased), or if additional tariffs or trade restrictions are enacted in the future, they could cause us or our customers to face higher supply costs, delays or a need to switch suppliers. These actions could adversely affect our margins, profitability, financial condition, and results of operations. While we have not experienced adverse impacts of tariffs to date, we cannot predict future changes in trade policy or the terms of any renegotiated trade agreements, nor can we determine the impact they may have on our business. Any such changes could have a material adverse effect on our business, financial condition, and results of operations.

Employee litigation and unfavorable publicity could negatively affect our future business.

Our employees may, from time to time, bring lawsuits against us regarding injury, creating a hostile workplace, discrimination, wage and hour disputes, sexual harassment or other employment issues. Recently, there has been an increase in the number of discrimination and harassment claims generally. Coupled with the expansion of social media platforms and similar devices that allow individuals access to a broad audience, these claims have had a significant negative impact on some businesses. Certain companies that have faced employment- or harassment-related lawsuits have had to terminate management or other key personnel, and have suffered reputational harm.

Ineffective internal controls could adversely impact our business and operating results.

Our internal control over financial reporting may not prevent or detect misstatements because of its inherent limitations, including the possibility of human error, failure or interruption of information technology systems, the circumvention or overriding of controls, or fraud. Even effective internal controls can provide only reasonable assurance with respect to the preparation and fair presentation of financial statements. If we fail to maintain the adequacy of our internal controls, including any failure to implement required new or improved controls, or if we experience difficulties in their implementation, our business and operating results could be harmed and we could fail to meet our financial reporting obligations.

Inadequate funding for the FDA, CDC and other government agencies, including from government shut downs, or other disruptions to these agencies' operations, could hinder their ability to hire and retain key leadership and other personnel, prevent new products and services from being developed or commercialized in a timely manner or otherwise prevent those agencies from performing normal business functions on which the operation of our business may rely.

The FDA's ability to review and approve new products can be affected by a variety of factors, including government priorities, budget and funding levels, the ability to hire and retain key personnel and accept the payment of user fees, and statutory, regulatory and policy changes. Average FDA review times have fluctuated in recent years as a result. In addition, funding of other government agencies on which our operations may rely, including those that fund research and development activities, is subject to the political process, which is inherently fluid and unpredictable. Disruptions at the FDA, CDC and other agencies may delay the review or approval of new product candidates, or the development of usage recommendations. If a prolonged government shutdown occurred, it could significantly impact the ability of the FDA to timely review and process our regulatory submissions.

Changes in tax law could adversely affect our business and financial condition.

We are subject to evolving and complex tax laws in the jurisdictions in which we operate. The rules dealing with U.S. federal, state, local and non-U.S. income taxation are constantly under review by legislative and tax authorities. Changes to tax laws (which could apply retroactively) could adversely affect us and our shareholders. In recent years, such changes have been made and changes are likely to occur in the future, which could have a material adverse effect on our business, cash flow, financial condition and results of operations.

The increasing use of social media platforms presents risks and challenges.

Social media is increasingly being used to communicate about our research, product candidates, commercial products and the diseases our products and product candidates are designed to treat. Social media practices in the biopharmaceutical industry continue to evolve and regulations relating to such use are not always clear. This uncertainty creates risk of noncompliance with regulations applicable to our business, resulting in potential regulatory actions against us. For example, subjects may use social media channels to comment on their experience in an ongoing blinded clinical trial or to report an alleged adverse event. When such disclosures occur, we may fail to monitor and comply with applicable adverse event reporting obligations or we may be unable to defend our business in the face of political and market pressures generated by social media due to restrictions on what we may say about our product candidates. There is also a risk of inappropriate disclosure of sensitive information or negative or inaccurate posts or comments about us on any social networking website. If any of these events were to occur or we otherwise fail to comply with applicable regulations, we could incur liability, face regulatory actions or incur other harm to our business.

Item 1B. Unresolved Staff Comments

None.

Item 1C. Cybersecurity

Cyber Risk Management and Strategy

Our cybersecurity organization's mission is to provide a targeted set of services, support and capabilities to reduce the risk of cyberattacks, rapidly detect and contain threats, and mitigate risks to critical data.

Recognizing the threat of security breaches and cyberattacks globally, we have developed a cybersecurity program, overseen by our Chief Technology and Information Security Officer (CISO), that is designed to protect patient trust, defend the Moderna brand, and reduce the risk and impact of cyber-attacks. Our cybersecurity program is informed by industry standards and includes periodic risk assessments and security testing supported by cybersecurity technologies, including third-party security solutions, vulnerability management, and monitoring tools, designed to monitor, identify, and manage risks from cyber threats. In addition, we have implemented employee security and awareness training.

Management has established a cyber incident response plan (CIRP) designed to assess, identify and manage risks from cybersecurity threats and enable prompt response in the event that a cybersecurity incident is detected. We have a process in place for notification to our leadership response team in the event of a significant cyber incident, and for escalation of these events to our Audit Committee and Board, as appropriate. To date, we have not experienced a cybersecurity incident that has had a material impact on our business strategy, results of operations, or financial condition.

We undergo several annual internal compliance audits and external reviews to evaluate our controls, including cybersecurity controls. In an effort to minimize third-party risk, we have established a process to assess the security practices of third-party suppliers and related risks, including through review of relevant supplier certifications and security and responses to standardized information gathering (SIG) questionnaires, as applicable and appropriate.

Governance Related to Cybersecurity Risks

Our Board of Directors oversees Moderna's overall risk management strategy. The Board exercises oversight of risks from cybersecurity threats primarily through its Audit Committee, which oversees our risk management processes for information security and technology risks. Our cybersecurity risk management processes are integrated into our overall risk management strategy, which is overseen by the Audit Committee. At least annually, the Audit Committee discusses our risk management program, including information security and technology risks and findings from any audits, with our internal audit staff.

The Audit Committee receives cyber-related updates from management, including our CISO at committee meetings. During meetings, our CISO updates the committee on Moderna's cybersecurity posture, potential threats and risk mitigation strategies, and the progress of the Company's cybersecurity initiatives, as appropriate. The Chair of the Audit Committee and management provide regular briefings on such matters to the full Board of Directors, as appropriate.

At the management level, our CISO is primarily responsible for leading our cybersecurity strategy for assessing and managing material risks from cybersecurity threats. Our current CISO has over 25 years of cybersecurity experience across a wide array of industries, most recently serving in leadership positions at two different public companies and previous roles of increasing responsibility at multinational technology companies. Our CISO reports directly to our Chief People and Digital Technology Officer, who is a member of our Executive Committee and reports to our Chief Executive Officer.

We have built a cybersecurity leadership team designed to align with key services, with a separate lead overseeing each service offering, all reporting to the CISO. We also maintain relationships with law enforcement and industry groups to support our cybersecurity intelligence and risk management efforts.

Item 2. Properties

We have two main campuses in Massachusetts. During the third quarter of 2023, we commenced a lease for a property in Cambridge, Massachusetts. This building, spanning approximately 462,000 square feet, is designated as our Moderna Science Center (MSC). The MSC accommodates a combination of scientific and office spaces, including our principal executive offices, and is the location of our corporate headquarters and platform, drug discovery, and clinical development activities. The lease has a term of 15 years, with options for us to extend the lease for up to two additional seven-year terms.

The Moderna Technology Center (MTC) is located in Norwood, Massachusetts and is primarily comprised of three buildings (MTC South, MTC North and MTC East). The MTC campus is approximately 722,000 square feet and includes lab and office space, directly supporting our manufacturing capabilities and commercial and clinical activities. In December 2024, we completed the acquisition of the MTC campus, including the underlying land and buildings. This acquisition transitioned the facilities from leased to owned properties, providing greater operational flexibility and long-term stability in supporting our manufacturing and development capabilities.

In the second quarter of 2023, we acquired a newly constructed biomanufacturing facility, encompassing 140,000 square feet, in Marlborough, Massachusetts. This facility underwent enhancements, including the addition of 60,000 square feet to the existing structure. The facility includes state-of-the-art mRNA manufacturing areas, including a full manufacturing clean room, quality control laboratories, and a just-in-time satellite warehouse. The Marlborough facility became operational in August 2025 and is intended to support our intismeran autogene program.

In September 2024, we completed construction of our mRNA manufacturing facility in Laval, Quebec, Canada, which we own, and the facility received a Drug Establishment License from Health Canada authorizing the production of mRNA vaccine drug substance. The facility encompasses approximately 100,000 square feet of manufacturing space and has the capacity to produce up to 100 million doses annually. In 2025, the facility began manufacturing drug substance for our Spikevax, supporting domestic mRNA vaccine production for Canada. In December 2025, Health Canada approved the addition of the Laval drug substance site for mRESVIA production, representing the second product approval for the Canadian supply chain.

We own a large-scale mRNA manufacturing facility, the Moderna Technology Centre – Melbourne (MTC-M), located in Clayton, Victoria, Australia, within Monash University's Technology Precinct. The facility encompasses approximately 130,000 square feet of manufacturing space and is designed to support end-to-end mRNA vaccine manufacturing, with the capacity to produce up to 100 million doses annually. In August 2025, the facility received full Good Manufacturing Practice (GMP) licensing, authorizing the manufacture of mRNA vaccine drug substance and finished drug product in compliance with applicable regulatory standards.

We own the Moderna Innovation and Technology Centre (MITC), a manufacturing and research facility located at the Harwell Science and Innovation Campus in Oxfordshire, United Kingdom. The facility, which officially opened in September 2025, encompasses approximately 100,000 square feet and is designed to support mRNA vaccine manufacturing and related research activities. The MITC has the capacity to produce up to 100 million doses annually and up to approximately 250 million doses in the event of a pandemic. The facility is intended to produce British-made mRNA respiratory vaccines to support the National Health Service (NHS) seasonal vaccination programs and also supports research into the application of mRNA technology in additional disease areas.

We also own and lease various parcels of land, office and lab spaces across the globe for our business operations.

Item 3. Legal Proceedings

We are involved in various claims and legal proceedings of a nature considered ordinary course in our business, including the intellectual property litigation described below. Most of the issues raised by these claims are highly complex and subject to substantial uncertainties. For a description of risks relating to these and other legal proceedings we face, see Part I, Item 1A, "Risk Factors," including the discussion under the headings entitled "Risks related to our intellectual property" and "Risks related to the manufacturing of our commercial products and product candidates."

The outcome of any such proceedings, regardless of the merits, is inherently uncertain; therefore, assessing the likelihood of loss and any estimated damages is difficult and subject to considerable judgment. We describe below those legal matters for which a material loss is either (i) possible but not probable, and/or (ii) not reasonably estimable at this time.

Pfizer/BioNTech Patent Litigation

U.S. Proceedings

In August 2022, we filed a lawsuit in the U.S. District Court for the District of Massachusetts against Pfizer Inc. (Pfizer) and BioNTech SE, BioNTech Manufacturing GmbH and BioNTech US Inc. (collectively, BioNTech), asserting infringement of certain U.S. patents concerning our mRNA platform technology and disease-specific vaccine designs in Pfizer and BioNTech's manufacture and sale of their mRNA COVID vaccines. The complaint seeks a judgment of infringement of the asserted patents and monetary damages. The case has been stayed pending the outcome of two Inter Partes Review proceedings (IPRs) before the U.S. Patent and Trademark Office's Patent Trial and Appeal Board (PTAB) regarding the validity of two of the three asserted patents. In March 2025, the PTAB found the patents invalid and we appealed to the U.S. Court of Appeals for the Federal Circuit (Federal Circuit).

European Proceedings

Also in August 2022, we initiated patent infringement proceedings in the UK (in the High Court of Justice of England & Wales), the Netherlands (in the District Court of The Hague) and Germany (in the Dusseldorf Regional Court), against Pfizer, BioNTech and related entities with respect to certain European patents that also concern our mRNA platform technology and disease-specific vaccine designs, including coronaviruses. In May 2023, we initiated patent infringement proceedings in Ireland (in the High Court) and Belgium (in the Brussels Business Court) against Pfizer, BioNTech and related entities with respect to the same European patents. As in the U.S. action, we seek a judgment of infringement of the asserted patents and monetary damages.

In July 2024, the High Court of Justice of England & Wales ruled that EP3590949 (the '949 patent), which relates to chemically-modified mRNA, was valid and infringed by Pfizer and BioNTech. The court further ruled that EP3718565 (the '565 patent), which relates to coronavirus mRNA vaccines, was invalid. In August 2025, the UK Court of Appeal upheld the High Court's decision that the '949 patent is valid and infringed by Pfizer and BioNTech. In December 2025, the UK Supreme Court denied permission for appeal by Pfizer and BioNTech.

In December 2023, the District Court of The Hague ruled that the '949 patent was invalid. We appealed to the Court of Appeal of The Hague, which held a hearing on September 22, 2025, and is expected to issue its decision on March 31, 2026.

In March 2025, the Dusseldorf Regional Court in Germany ruled that Pfizer and BioNTech infringed the '949 patent. Pfizer and BioNTech have appealed to the Higher Regional Court of Dusseldorf, which will hold a hearing November 26, 2026.

Pfizer Inc. and BioNTech SE have filed oppositions seeking revocation of the '949 and '565 patents at the European Patent Office. In May 2024, the Opposition Division (OD) ruled that the '949 patent was valid, and Pfizer and BioNTech appealed to the Technical Appeal Board (TBA), which will hold a hearing on September 8-10, 2026. In November 2023, the OD ruled that the '565 patent was invalid and we appealed to the TBA, which confirmed the invalidity on January 27, 2026.

Arbutus Patent Litigation

In February 2022, Arbutus Biopharma Corporation (Arbutus) and Genevant Sciences GmbH (Genevant) filed a complaint against us in the U.S. District Court for the District of Delaware asserting that our manufacture and sale of our COVID vaccine willfully infringes certain U.S. patents concerning lipid nanoparticles but does not seek to prevent or stop the marketing or sales of our COVID vaccines. The complaint seeks a judgment of infringement of the asserted patents and monetary damages. The Court has set trial to begin March 9, 2026.

In March 2025, Arbutus and Genevant filed five international lawsuits asserting that our manufacture and sale of Spikevax and/or mRESVIA infringes certain patents concerning lipid nanoparticles. The companies filed lawsuits in the Canadian Federal Court, the Tokyo District Court in Japan and the Swiss Federal Patent Court and submitted two lawsuits to the Unified Patent Court (UPC). The UPC actions seek relief in all UPC member states and in ten additional European Patent Convention contracting states. Each proceeding identifies Spikevax and mRESVIA as accused products except for the Canadian proceeding, which only identifies Spikevax. The complaints seek monetary relief and injunctive relief after a trial decision and before any appeals are decided. Hearings for the UPC actions are set for May 19 and 20, 2026. The Canadian lawsuit is scheduled for trial beginning in September 2027.

GSK Patent Litigation

In October 2024, GlaxoSmithKline Biologicals SA (GSK) filed two separate complaints against us in the U.S. District Court for the District of Delaware asserting that our manufacture and sale of our COVID and RSV vaccines willfully infringe certain U.S. patents directed to lipid-mRNA vaccine formulation technology. Each complaint seeks a judgment of infringement of the asserted patents and unspecified damages, and injunctive relief related to RSV after any appeals are decided. The Court has set trial to begin on July 19, 2027 with respect to our COVID vaccines, and on August 23, 2027 with respect to our RSV vaccine.

In July 2025, GSK submitted two lawsuits to the UPC asserting that our manufacture and sale of our COVID and RSV vaccines infringe three European patents related to liposomes with specified characteristics for RNA delivery. The complaints seek monetary and injunctive relief after appeal. Hearings are set for September 2026.

In December 2025, GSK served us with two separate complaints in the Barcelona Commercial Court and the Madrid Commercial Court (the Spanish Courts) asserting that our manufacture and sale of our COVID and RSV vaccines infringe the same patents asserted in the UPC actions. The complaints seek monetary and injunctive relief after an appeal decision.

Northwestern University Patent Litigation

In October 2024, Northwestern University filed a complaint against us in the U.S. District Court for the District of Delaware asserting that our COVID and RSV vaccines infringe several U.S. patents concerning lipid nanoparticle technology. The complaint seeks a judgment of infringement of the asserted patents and unspecified damages.

Alnylam Patent Litigation

In March 2022 and July 2022, Alnylam Pharmaceuticals, Inc. (Alnylam) filed two complaints against us in the U.S. District Court for the District of Delaware asserting that our manufacture and sale of our COVID vaccine infringed certain U.S. patents concerning cationic lipids. On August 25, 2023, the Court entered a Final Judgment of non-infringement of all asserted patents in these lawsuits. In the first case, Alnylam appealed and the Federal Circuit decided the appeal in our favor in June 2025.

Relatedly, in May 2023, Alnylam filed a third complaint against us in the U.S. District Court for the District of Delaware asserting three additional U.S. patents concerning cationic lipids. The complaint sought judgments of infringement of the asserted patents and monetary damages. In November 2023, the District Court entered an order of partial dismissal with respect to two of the patents at issue. Subsequently, in October 2024, the District Court entered a final judgment of non-infringement in our favor with respect to the third patent at issue.

Following the June 2025 Federal Circuit decision in our favor, we and Alnylam entered into an agreement in September 2025 to settle all disputes between the parties, including filing a stipulated dismissal of Alnylam's claims with prejudice. The settlement did not include payments of any kind by us.

Bayer CropSciences Patent Litigation

In January 2026, Bayer CropScience LLC, Monsanto Company and Monsanto Technology, LLC (collectively, Bayer) filed a complaint against us in the U.S. District Court for the District of Delaware asserting that our manufacture and sale of our COVID vaccines infringe a U.S. patent directed to methods of modifying gene sequences. The complaint seeks a judgment of infringement of the asserted patent and unspecified damages.

mNG Bio Patent Litigation

In January 2026, mNG Bio, LLC (mNG) filed a complaint against us in the U.S. District Court for the District of Massachusetts asserting that certain tests of our COVID vaccines willfully infringe a U.S. patent directed to a yellow-green fluorescent protein. The complaint seeks a judgment of infringement of the asserted patent and unspecified damages.

BioNTech Patent Litigation

In February 2026, BioNTech SE filed a complaint against us in the U.S. District Court for the District of Delaware asserting that our manufacture and sale of mNEXSPIKE willfully infringes a U.S. patent directed to modified mRNA compositions encoding a spike

protein fragment. The complaint seeks a judgment of infringement of the asserted patent and unspecified damages.

Securities Class Action Litigation

In August 2024, a putative shareholder class action complaint was filed against us and certain officers in the U.S. District Court for the District of Massachusetts. The action was purportedly brought on behalf of a class of shareholders who purchased Moderna common stock between January 18, 2023 and June 25, 2024. The complaint asserts claims under the Securities Exchange Act of 1934 regarding statements about our RSV vaccine (mRNA-1345) and seeks unspecified damages. On August 25, 2025, plaintiff filed an amended complaint, which, among other things, narrowed the asserted class period to February 15, 2024 to May 31, 2024. On October 24, 2025, the Company filed a motion to dismiss, which remains pending.

Derivative Litigation

Between September and November 2024, purported shareholder derivative complaints were filed in the U.S. District Court for the District of Massachusetts against certain of our officers and directors and against us as a nominal defendant. The complaints allege breaches of fiduciary duty and claims under the Securities Exchange Act of 1934 regarding statements about mRNA-1345 and seek declaratory and injunctive relief and unspecified damages payable to us. The cases were stayed pending a decision on defendants' motion to dismiss the putative securities class action described above.

Item 4. Mine Safety Disclosures

Not applicable.

PART II

Item 5. Market for Registrant's Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities

Market for Our Common Stock

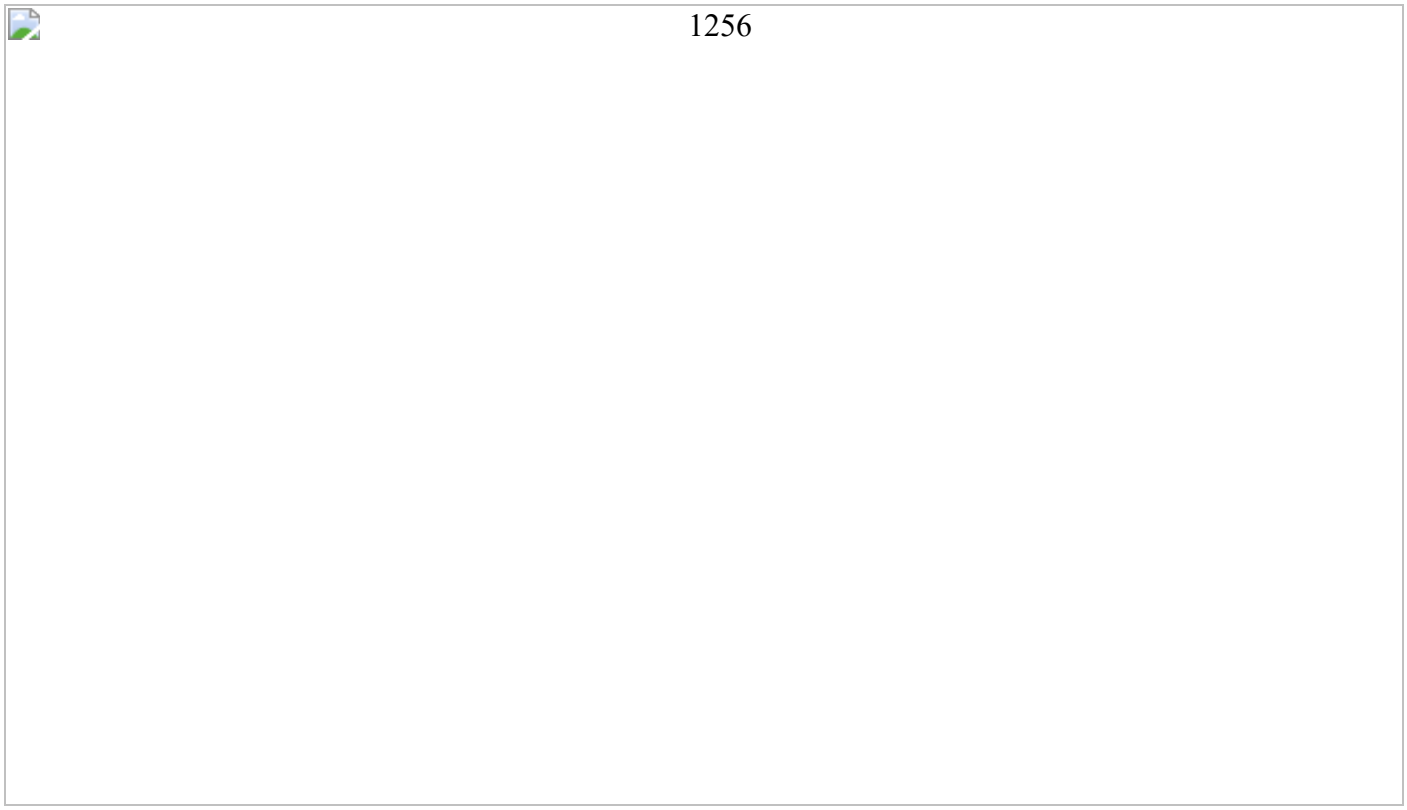
Our common stock trades on the Nasdaq Global Select Market under the symbol “MRNA”.

Stock Performance Graph

The following performance graph shall not be deemed “soliciting material” or to be “filed” with the SEC for purposes of the Exchange Act or otherwise subject to the liabilities under that section, and shall not be deemed to be incorporated by reference into any filing of Moderna, Inc. under the Securities Act or the Exchange Act.

The following graph illustrates a comparison for the five years ended December 31, 2025 of the cumulative total return for our common stock, the Nasdaq Biotechnology Index, and the Standard & Poor’s 500 Stock Index (the S&P 500) each of which assumes an initial investment of $100 and reinvestment of all dividends. Such returns are based on historical results and are not intended to suggest future performance.

The comparisons shown in the graph below are based upon historical data. We caution that the stock price performance shown in the graph below is not necessarily indicative of, nor is it intended to forecast, the potential future performance of our common stock.



Stockholders

We had approximately 62 stockholders of record as of February 13, 2026. Because many of our outstanding shares are held in accounts with brokers and other institutions, the number of beneficial owners is significantly greater than the number of record holders. This number of holders of record also does not include stockholders whose shares may be held in trust by other entities.

Dividend Policy

We have never declared or paid cash dividends on our common stock and do not expect to pay dividends on our common stock for the foreseeable future. Any future determination to pay dividends will be made at the discretion of our board of directors and will depend on various factors, including applicable laws, our results of operations, financial condition, future prospects, then applicable contractual restrictions and any other factors deemed relevant by our board of directors. In addition, our Credit Agreement with lenders led by Ares Capital Corporation includes covenants that, subject to certain exceptions, restrict our ability and the ability of our subsidiaries to pay dividends or distributions. Investors should not purchase our common stock with the expectation of receiving cash dividends.

Securities Authorized for Issuance Under Equity Compensation Plans

Information about our equity compensation plans in Item 12 of Part III of this Annual Report on Form 10-K is incorporated herein by reference.

Recent Sales of Unregistered Securities

None.

Issuer Purchases of Equity Securities

On August 1, 2022, our Board of Directors authorized a share repurchase program for our common stock of up to $3.0 billion, with no expiration date. During the three months ended December 31, 2025, there were no shares repurchased. As of December 31, 2025, $1.7 billion of our Board of Directors' authorization for repurchases of our common stock remained outstanding, with no expiration date.

See Note 13 to consolidated financial statements for information regarding our share repurchase programs.

Item 6. [Reserved]

Item 7. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

You should read the following discussion and analysis of our financial condition and results of operations together with our consolidated financial statements and related notes and other financial information appearing elsewhere in this Annual Report on Form 10-K. Some of the information contained in this discussion and analysis or set forth elsewhere in this Annual Report on Form 10-K, including information with respect to our plans and strategy for our business, includes forward-looking statements that involve risks and uncertainties. As a result of many factors, including those factors set forth in "Part I, Item 1A - Risk Factors" section of this Annual Report on Form 10-K, our actual results could differ materially from the results described in or implied by the forward-looking statements contained in the following discussion and analysis.

Overview

We are a biotechnology company advancing a new class of medicines made of messenger RNA (mRNA). mRNA medicines are designed to direct the body's cells to produce intracellular, membrane or secreted proteins that have a therapeutic or preventive benefit with the potential to address a broad spectrum of diseases. Our platform builds on continuous advances in basic and applied mRNA science, delivery technology and manufacturing, providing us the capability to pursue in parallel a robust pipeline of new development candidates. We are developing medicines across infectious disease vaccines, oncology therapeutics and rare disease therapeutics.

Since our founding in 2010, we have transformed from a research-stage company advancing programs in the field of mRNA to a commercial enterprise with a diverse clinical portfolio of vaccines and therapeutics across several modalities, a broad intellectual property portfolio and integrated manufacturing capabilities that allow for rapid clinical and commercial production at scale. We currently have three commercial products—Spikevax® and mNEXSPIKE®, our COVID vaccines, and mRESVIA®, our vaccine against respiratory syncytial virus (RSV). We also have a diverse development pipeline of 25 development candidates across our 35 development programs currently in clinical studies.

2025 Business Highlights

U.S. Manufacturing Capabilities Expansion

In November 2025, we announced the expansion of our U.S. manufacturing capabilities through the onshoring of drug product manufacturing to our existing Moderna Technology Center in Norwood, Massachusetts. Upon completion, this expansion will enable end-to-end mRNA manufacturing in the United States, supporting both commercial and clinical supply needs. This expansion is part of our continued investment in U.S.-based manufacturing infrastructure to support our mRNA vaccine and therapeutic pipeline. Construction for the new drug product manufacturing capability has commenced, with completion targeted in the first half of 2027.

$1.5 Billion Five-Year Credit Facility

In November 2025, we entered into a five-year term loan facility providing for up to $1.5 billion of capital to enhance our balance sheet and provide increased financial flexibility. The facility includes a $600 million initial term loan funded at closing, a $400 million delayed draw term loan facility available through November 2027, and an additional $500 million delayed draw term loan facility available through November 2028, subject to the achievement of specified regulatory milestones.

Total Revenue and Loss Per Share

For the year ended December 31, 2025, we recognized total revenue of $1.9 billion compared to $3.2 billion and $6.8 billion for the years ended December 31, 2024 and 2023, respectively. Loss per share was $(7.26) for the year ended December 31, 2025, compared to loss per share of $(9.28) and $(12.33) for the years ended December 31, 2024 and 2023, respectively.

Program Developments

Infectious Disease Vaccines

- *COVID vaccines:* We have received approval in 40 countries for our 2025-2026 formula for Spikevax. We have also received U.S. Food and Drug Administration (FDA) approval of our 2025-2026 formula for mNEXSPIKE, our second COVID vaccine, in all adults aged 65 and older, as well as individuals aged 12 to 64 years with at least one underlying risk factor.

mNEXSPIKE is also approved in Europe, Canada and Australia and we have filed and are targeting 2026 approvals in Japan and Taiwan.

- *mRESVIA*: Our RSV vaccine has been approved for adults aged 60 years and older in 40 countries. It is also approved in 31 of those countries for individuals 18 to 59 years of age who are at increased risk for disease.

- *Seasonal flu vaccine:* Our mRNA-1010 regulatory filings are under review in the United States, Europe, Canada and Australia. In February 2026, in response to a prior Refusal-to-File letter, we engaged with the FDA in a Type A meeting and submitted an amended biologics license application (BLA) outlining a revised regulatory pathway based on age, seeking full approval for adults 50 to 64 years of age and accelerated approval for adults 65 and older, along with a post-marketing requirement to conduct an additional study in older adults. Following the meeting and submission of the amended application, the FDA accepted our BLA for review and assigned a Prescription Drug User Fee Act (PDUFA) goal date of August 5, 2026.

- *Seasonal flu + COVID vaccine:* Our mRNA-1083 regulatory filing is under review in Europe and Canada. We are awaiting further guidance from the FDA on refiling.

- *Norovirus vaccine:* We recently completed enrollment of a second Northern Hemisphere season (2025-2026) cohort in our ongoing Phase 3 study for our norovirus vaccine candidate (mRNA-1403).

Oncology Therapeutics

- *Intismeran autogene:* We are advancing intismeran (mRNA-4157), our mRNA-based individualized neoantigen therapy, in collaboration with Merck, with eight total Phase 2 and Phase 3 clinical trials underway across multiple tumor types including melanoma, non-small cell lung cancer (NSCLC), bladder cancer and renal cell carcinoma. In January 2026, we and Merck announced five-year data from the Phase 2b study of intismeran in combination with KEYTRUDA, which demonstrated sustained improvement in recurrence-free survival in patients with high-risk melanoma (stage III/IV) following complete resection.

- *mRNA-4359:* Our Phase 1/2 study of mRNA-4359, an investigational wholly-owned cancer antigen therapy, is ongoing. The Phase 2 portion of the study includes cohorts in first-line metastatic melanoma, second-line+ metastatic melanoma and first-line metastatic NSCLC, and we expect a potential Phase 2 data readout in 2026.

Rare Disease Therapeutics

- *Propionic acidemia (PA) therapeutic:* mRNA-3927 is in a registrational study and target enrollment has been reached. In January 2026, we entered into a strategic collaboration with Recordati, an international pharmaceutical group, to advance mRNA-3927 through the final stages of clinical development and, if approved, global commercialization.

- *Methylmalonic acidemia (MMA) therapeutic:* mRNA-3705 has been selected by the FDA for the Support for Clinical Trials Advancing Rare Disease Therapeutics (START) pilot program, with a registrational study expected to begin in 2026.

Other Business Updates

During the third quarter of 2025, our mRNA manufacturing facilities in Australia and the United Kingdom were licensed and became operational. These facilities were established under long-term strategic agreements with the respective governments to support domestic mRNA manufacturing and pandemic preparedness. The facilities will enable local supply of our mRNA vaccines and provide rapid response capabilities in the event of future public health emergencies.

In September 2025, we delivered the first mRNA vaccines fully manufactured in Canada from our facility in Laval, Quebec, marking a significant milestone in our collaboration with the Government of Canada. The site, which received its Drug Establishment License from Health Canada in 2024, now produces the drug substance for our updated Spikevax targeting the LP.8.1 variant. Fill-and-finish of the vaccine, including the new single-use pre-filled syringes, is completed by Novocol Pharma in Cambridge, Ontario. This milestone demonstrates the execution of our end-to-end manufacturing strategy and our ability to provide timely access to locally manufactured mRNA vaccines through collaboration with government and industry.

In December 2025, we entered into an agreement with the Coalition for Epidemic Preparedness Innovations (CEPI) under which CEPI will provide up to $54 million in funding to support a pivotal Phase 3 clinical trial for our investigational mRNA-based H5 pandemic influenza vaccine candidate, mRNA-1018, which is expected to begin in early 2026. If licensure is granted and in the event of an influenza pandemic, Moderna is committed to working to provide people around the world with rapid, equitable access to the resulting

H5 vaccine, including, as part of this agreement, allocating 20% of its H5 pandemic vaccine manufacturing capacity for timely supply to low- and middle-income countries at affordable pricing.

Financial Operations Overview

Revenue

Net product sales

Net product sales by customer geographic location were as follows for the periods presented (in millions):

	Years Ended December 31,		
	2025	2024	2023
United States	$ 1,165	$ 1,726	$ 1,720
Europe	50	573	1,353
Rest of world	603	810	3,598
Total	$ 1,818	$ 3,109	$ 6,671

Net product sales by product were as follows (in millions):

	Years Ended December 31,		
	2025	2024	2023
COVID [(1)]	$ 1,810	$ 3,084	$ 6,671
RSV	8	25	—
Total	$ 1,818	$ 3,109	$ 6,671

[(1)]Includes sales of Spikevax and mNEXSPIKE.

As of December 31, 2025, we have three commercial products, our COVID vaccines, Spikevax and mNEXSPIKE, and our RSV vaccine, mRESVIA. mRESVIA was approved by the FDA in May 2024 for adults aged 60 years and older, and in June 2025, the approved use was expanded to include adults aged 18 through 59 years who are at increased risk for lower respiratory tract disease (LRTD) caused by RSV. We launched commercial sales of mRESVIA in the third quarter of 2024. In May 2025, mNEXSPIKE was approved for use in adults aged 65 years and older, as well as individuals aged 12 through 64 years with at least one underlying risk factor. We launched commercial sales of mNEXSPIKE in the third quarter of 2025.

We sell our COVID and RSV vaccines to the commercial market as well as to foreign governments and international organizations. In the U.S., our COVID and RSV vaccines are sold primarily to wholesalers and distributors, and to a lesser extent, directly to retailers and healthcare providers. Net product sales are recognized net of estimated wholesaler chargebacks, invoice discounts for prompt payments and pre-orders, provisions for sales returns and government rebates, and other related deductions.

The following table summarizes product sales provision for the periods presented (in millions):

	Years Ended December 31,		
	2025	2024	2023
Gross product sales	$ 3,304	$ 4,517	$ 8,203
Product sales provision:			
Wholesaler chargebacks, discounts and fees	(1,037)	(1,141)	(976)
Returns, rebates and other fees	(449)	(267)	(556)
Total product sales provision[(1)]	$ (1,486)	$ (1,408)	$ (1,532)
Net product sales	$ 1,818	$ 3,109	$ 6,671

[(1)]Includes an adjustment of approximately $216 million in 2024, reflecting a reduction in prior year provision estimates, primarily related to returns and chargebacks for the previous COVID vaccine season. Adjustments recorded in 2025 related to prior year provision estimates were not material.

Certain agreements may include upfront payments for our vaccine supply, initially recorded as deferred revenue. As of December 31, 2025, we had deferred revenue of $107 million related to product sales, of which $42 million is expected to be realized in less than one year.

Other revenue

Other revenue comprises grant revenue, collaboration revenue, licensing and royalty revenue and stand-ready manufacturing revenue. Grant and collaboration revenues have been primarily derived from government-sponsored and private organizations including the Biomedical Advanced Research and Development Authority (BARDA), the Defense Advanced Research Projects Agency (DARPA) and the Gates Foundation and from strategic alliances with Merck & Co., Inc (Merck), Vertex Pharmaceuticals Incorporated and Vertex Pharmaceuticals (Europe) Limited (together, Vertex) and others to discover, develop, and commercialize potential mRNA medicines. Licensing and royalty revenue is related to our out-licensing agreement for mRNA COVID related intellectual property in Japan, executed in 2024. Stand-ready manufacturing revenue relates to long-term strategic agreements with certain government entities to maintain manufacturing readiness, while the remaining consideration under these agreements is related to product sales.

The following table summarizes other revenue for the periods presented (in millions):

	Years Ended December 31,		
	2025	2024	2023
Grant revenue	$ 22	$ 37	$ 94
Collaboration revenue	13	48	83
Licensing and royalty revenue	11	42	—
Stand-ready manufacturing revenue	80	$ —	$ —
Total other revenue	$ 126	$ 127	$ 177

Cost of sales

Cost of sales includes raw materials, personnel, facility and other indirect overhead costs associated with manufacturing our commercial products. These costs include production materials, production costs at our manufacturing facilities, third-party manufacturing costs, and final formulation and packaging costs. Cost of sales also includes shipping costs, third-party royalties on net sales of our products, and charges for inventory valuation, excess and obsolete inventory and losses on firm purchase commitments.

Research and development expenses

The nature of our business and primary focus of our activities generate a significant amount of research and development costs.

Research and development expenses represent costs incurred by us for the following:

- cost to develop our platform;
- discovery efforts leading to development candidates;
- preclinical, nonclinical, and clinical development costs for our programs;
- cost to develop our manufacturing technology and infrastructure; and
- digital infrastructure costs related to our drug discovery efforts and clinical trials.

The costs above comprise the following categories:

- personnel-related expenses, including salaries, benefits, and stock-based compensation expense;
- expenses incurred under agreements with third parties, such as consultants, investigative sites, contract research organizations (CROs), that conduct our preclinical studies and clinical trials, and in-licensing arrangements;
- expenses associated with developing manufacturing, modification of formulation or design of a product or process, advancing the design to meet specific functional and economic requirements for manufacture and obtaining materials for preclinical studies, clinical trials and pre-launch inventory from internal and third-party contract manufacturing organizations (CMOs);
- expenses incurred for the procurement of materials, laboratory supplies, and non-capital equipment used in the research and development process;
- upfront fees and milestones paid to third-parties for licenses and technologies that had not reached technological feasibility and did not have an alternative future use; and
- facilities, depreciation, and amortization, and other direct and allocated expenses incurred as a result of research and development activities.

We use our employee and infrastructure resources for the advancement of our platform, and for discovering and developing programs. Due to the number of ongoing programs and our ability to use resources across several projects, indirect or shared operating costs incurred for our research and development programs are generally not recorded or maintained on a program- or therapeutic area-specific basis.

The following table reflects our research and development expenses, including direct program specific expenses summarized by therapeutic area and indirect or shared operating costs summarized under other research and development expenses during the years ended December 31, 2025, 2024, and 2023 (in millions):

	Years Ended December 31,		
	2025	2024	2023
Program-specific expenses by therapeutic area:			
Infectious disease vaccines	$ 652	$ 1,297	$ 1,903
Oncology	201	154	62
Rare disease therapeutics	43	82	67
Total program-specific expenses by therapeutic area[(1)]	$ 896	$ 1,533	$ 2,032
Other research and development expenses:			
Discovery and platform research	$ 356	$ 520	$ 751
Technical development and manufacturing expenses	792	1,081	1,116
Shared discovery and development expenses	797	1,145	789
Stock-based compensation	291	264	157
Total research and development expenses	$ 3,132	$ 4,543	$ 4,845

(1) Includes a total of 25 development candidates at December 31, 2025, 34 development candidates at December 31, 2024, and 42 development candidates at December 31, 2023. Program-specific expenses are reflected as of the beginning of the period in which the program was internally advanced to development.

The program-specific expenses by therapeutic area summarized in the table above include external development costs, such as fees paid to outside consultants, central laboratories, investigative sites, and CROs in connection with our preclinical studies and clinical trials. We generally do not allocate personnel-related costs, including stock-based compensation, costs associated with our general platform research, technical development, and other shared costs on a program-specific basis. These costs were therefore excluded from the summary of program-specific expenses by therapeutic area.

Discovery and platform research expenses include costs associated with early-stage research activities for preclinical programs and the development of technical advances in mRNA science, delivery science, and manufacturing process design. These costs include external CRO and lab services, personnel-related costs, technology and digital, facilities, and other directly attributable research support costs.

Technology development and manufacturing expenses are primarily related to manufacturing process development and manufacturing costs, including expenses for acquiring and manufacturing pre-launch inventory, mRNA supply for preclinical studies and clinical trials.

Shared discovery and development expenses are research and development costs such as personnel-related costs and other costs, such as the purchase of priority review vouchers. These expenses are not otherwise included in development programs, discovery and platform research, technical development, manufacturing expenses, or stock-based compensation.

Historically, we have made substantial investments in research and development activities, including development of our platform, mRNA technologies, and manufacturing technologies. We expense research and development costs as incurred and cannot reasonably estimate the nature, timing, and estimated costs required to complete the development of the development candidates and investigational medicines we are currently developing or may develop in the future.

Changes in expectations or outcomes of any of the known or unknown risks and uncertainties may materially impact our expected research and development expenditures. Continued research and development is central to the ongoing activities of our business. Investigational medicines in later stages of clinical development, such as our intismeran autogene, norovirus, and flu+COVID combination vaccine candidates, and development of any new COVID vaccines against variants of SARS-CoV-2, generally have higher development costs than those in earlier stages of clinical development, primarily due to the increased size and duration of later-

stage clinical trials. In addition, we may incur research and development costs related to postmarketing commitments, though the timing and scope of such commitments remain uncertain. We expect our research and development costs to be substantial in the near term as our investigational medicines advance through development phases and as we identify and develop additional programs.

There are numerous factors associated with the successful commercialization of any of our investigational medicines, including future trial design and various regulatory requirements, many of which cannot be determined with accuracy at this time due to the early stage of development of many of our investigational medicines. Moreover, future commercial and regulatory factors beyond our control will impact our clinical development programs and plans.

Selling, general and administrative expenses

Selling, general and administrative expenses consist primarily of personnel-related costs, including stock-based compensation, for executives, finance, legal, human resources, business development, commercial, marketing, and other administrative and operational functions, professional fees, accounting and legal services, technology and digital and facility-related costs, and expenses associated with obtaining, maintaining, and defending intellectual property (IP). These costs relate to the operation of the business, unrelated to the research and development function, or any individual program.

Prior to 2024, selling, general and administrative expenses increased annually as we expanded our commercial infrastructure to support the global commercialization of our COVID vaccine. These increases were driven by investments in building regulatory, sales, and marketing teams, establishing subsidiaries in multiple countries, and supporting operational growth. In 2025 and 2024, selling, general and administrative expenses decreased significantly year over year by $156 million and $375 million, or 13% and 24%, respectively. This decline reflects our focus on productivity improvements, cost-saving initiatives, and targeted investments that strengthen our overall efficiency.

Interest income

Interest income consists of interest generated from our investments in cash and cash equivalents, money market funds, and high-quality fixed income securities.

Other expense, net

Other expense, net consists of interest expense, gains or losses related to the sale of investments in marketable securities, changes in fair value of investments in equity securities, foreign currency transactions, remeasurements and hedges, and other income and expense unrelated to our core operations. Interest expense is primarily derived from our finance leases related to our Moderna Technology Center prior to its acquisition in December 2024, certain contract manufacturing service agreements, and, beginning in November 2025, interest associated with our long-term debt (see Note 10 and Note 11 to the consolidated financial statements).

Critical accounting policies and significant judgments and estimates

Our management's discussion and analysis of our financial condition and results of operations is based on our consolidated financial statements, which have been prepared in accordance with U.S. generally accepted accounting principles. The preparation of these consolidated financial statements requires us to make judgments and estimates that affect the reported amounts of assets, liabilities, revenues, and expenses and the disclosure of contingent assets and liabilities in our consolidated financial statements. We base our estimates on historical experience, known trends and events, and various other factors that we believe to be reasonable under the circumstances, the results of which form the basis for making judgments about the carrying values of assets and liabilities that are not readily apparent from other sources. Actual results may differ from these estimates under different assumptions or conditions. On an ongoing basis, we evaluate our judgments and estimates in light of changes in circumstances, facts, and experience. The effects of material revisions in estimates, if any, are reflected in the consolidated financial statements prospectively from the date of change in estimates.

While our significant accounting policies are described in more detail in the notes to our consolidated financial statements appearing elsewhere in this Annual Report on Form 10-K, we believe the following accounting policy used in the preparation of our consolidated financial statements requires the most significant judgments and estimates.

Net product sales

Prior to the third quarter of 2023, we sold our COVID vaccine to the U.S. Government, foreign governments and organizations. The agreements and related amendments with these entities generally do not include variable consideration, such as discounts, rebates or returns. In the third quarter of 2023, we commenced sales of Spikevax to the U.S. commercial market, in addition to continuing sales to foreign governments and organizations, and subsequently expanded our commercial COVID vaccine portfolio with the launch of mNEXSPIKE in the third quarter of 2025. We also commenced sales of our RSV vaccine in the third quarter of 2024. In the U.S., our COVID and RSV vaccines are sold primarily to wholesalers and distributors, and to a lesser extent, directly to retailers and healthcare providers.

We recognize net product sales when control of the product transfers to the customer, typically upon delivery. Net product sales are recognized net of estimated wholesaler chargebacks, invoice discounts for prompt payments and pre-orders, provisions for sales returns, government rebates, and other related deductions. These provisions are recorded based on contractual terms, our estimate of returns for product sold, and other relevant considerations, during the period, using the expected value method or the most likely amount method. Estimates are assessed each period and adjusted as required to revise information or actual experience. See Note 2 to our consolidated financial statements for further discussion and analysis of each significant category of product sales provisions and the accounting policy.

The application of our critical accounting policies necessitates substantial management judgment and estimation, particularly when determining the amount of variable consideration to recognize. The subjectivity of this process is heightened when assessing factors outside our direct control such as the limited historical data, constrained third-party information, and evolving market dynamics. Among all variables, estimating returns presents the most significant judgment due to the broad range of potential outcomes and the current lack of established return trends. While we now have two years of data on our product returns, this remains limited and subject significant variability given current market conditions. We will continue to refine our projections as additional information becomes available. The actual results could differ from our estimates, and such differences could have a material impact to our financial statements.

Recently issued accounting pronouncements

See Note 2 - Summary of Significant Accounting Policies, to the consolidated financial statements, under the caption “Recently Issued Accounting Standards”.

Results of operations

A discussion regarding our results of operations for the year ended December 31, 2025 compared to 2024 is presented below. A discussion regarding our results of operations for the year ended December 31, 2024 compared to 2023 can be found under Part II -Item 7 of our Annual Report on Form 10-K for the year ended December 31, 2024, which was filed with the Securities and Exchange Commission (SEC) on February 21, 2025.

The following table summarizes our consolidated statements of operations for the periods presented (in millions):

	Years Ended December 31,		Change 2025 vs. 2024	
	2025	2024	Change	%
Revenue:				
Net product sales	$ 1,818	$ 3,109	$ (1,291)	(42)%
Other revenue	126	127	(1)	(1)%
Total revenue	1,944	3,236	(1,292)	(40)%
Operating expenses:				
Cost of sales	868	1,464	(596)	(41)%
Research and development	3,132	4,543	(1,411)	(31)%
Selling, general and administrative	1,018	1,174	(156)	(13)%
Total operating expenses	5,018	7,181	(2,163)	(30)%
Loss from operations	(3,074)	(3,945)	871	(22)%
Interest income	314	425	(111)	(26)%
Other expense, net	(8)	(87)	79	(91)%
Loss before income taxes	(2,768)	(3,607)	839	(23)%
Provision for (benefit from) tax provision	54	(46)	100	(217)%
Net loss	$ (2,822)	$ (3,561)	$ 739	(21)%

Revenue

Total revenue decreased by $1.3 billion, or 40%, in 2025, primarily driven by a decline in net product sales. Net product sales decreased by $1.3 billion, or 42%, in 2025, largely due to lower sales volumes of our COVID vaccine across all regions. In the U.S., net product sales declined compared to 2024, reflecting a year-over-year decline in overall COVID vaccine demand. This decrease was partially offset by increased market share in the retail channel, supported by the successful launch and performance of mNEXSPIKE during the 2025/2026 respiratory vaccination season. In Europe and the rest of the world, net product sales declined compared to 2024, driven largely by the completion of advance purchase agreements related to COVID vaccines with government customers that contributed to revenue in the prior year, as well as lower overall demand in these markets. These impacts were partially offset by product sales generated under long-term strategic partnerships with government entities through our in-country manufacturing arrangements. Other revenue was relatively consistent compared to 2024, as lower grant revenue, collaboration revenue, and licensing and royalty revenue were largely offset by higher stand-ready manufacturing revenue.

Product sales are expected to return to growth in 2026, supported by the full-year impact of long-term strategic partnerships with government entities, as well as continued uptake of mNEXSPIKE in the U.S.

Operating expenses

Cost of sales

Our cost of sales was $868 million, or 48% of our net product sales, in 2025, including third-party royalties of $88 million, inventory write-downs of $291 million, primarily for raw materials and our finished and semi-finished COVID vaccine inventory, and unutilized manufacturing capacity and wind-down costs of $93 million. Our cost of sales was $1.5 billion, or 47% of our net product sales, in 2024, including third-party royalties of $155 million, inventory write-downs of $495 million, unutilized manufacturing capacity and wind down costs of $368 million, and losses on firm purchase commitments and cancellation fees of $60 million. These charges in both 2025 and 2024, other than royalties, were largely driven by customer demand forecast adjustments related to the seasonal nature of the COVID vaccine market, termination of third-party contract manufacturing service agreements, and commitments for

manufacturing capacity and raw material purchase agreements. Forecasting demand for COVID vaccines requires advance production planning ahead of each season, including production of multiple COVID vaccine products, which resulted in excess inventory and capacity as actual demand differed from forecasted demand. See Note 7 to our consolidated financial statements for further details on inventory related charges and this initiative.

Cost of sales for 2025 decreased by $596 million, or 41%, compared to 2024. As a percentage of net product sales, cost of sales was 48% in 2025 compared to 47% in 2024, remaining relatively consistent year over year. The decrease in cost of sales was primarily driven by reduced inventory write-downs, reduced unutilized manufacturing capacity and wind-down costs, lower losses on purchase commitments, and lower sales volume. In 2024, cost of sales included $238 million of wind-down costs related to the termination of a contract manufacturing agreement, whereas wind-down costs incurred in 2025 were significantly lower. As a result, the cost of sales as a percentage of net product sales remained relatively consistent year over year, as the benefit of a more efficient manufacturing cost structure was largely offset by lower product sales.

In 2026, we anticipate that cost of sales will remain at a relatively consistent level compared to 2025, reflecting continued manufacturing productivity improvements and operational efficiencies. To the extent net product sales increase, cost of sales as a percentage of net product sales may decrease modestly.

Research and development expenses

Research and development expenses decreased by $1.4 billion, or 31%, in 2025. The decrease was primarily attributable to lower clinical trial expenses of $578 million and lower clinical manufacturing expenses of $239 million, reflecting the wind-down of large Phase 3 respiratory programs and the timing of certain clinical trial activities shifting into 2026, partially offset by increased investment in our norovirus vaccine and oncology program. In addition, preclinical research expenses decreased by $107 million due to continued portfolio prioritization across the organization, and consulting and outside service costs declined by $175 million, consistent with the lower level of clinical and manufacturing activities. Separately, the prior-year period included an expense related to the purchase of two priority review vouchers, which did not recur in 2025.

We anticipate a modest reduction in research and development expenses in 2026 compared to 2025, driven by continued portfolio prioritization, disciplined cost management, and a focused approach to pipeline execution. These reductions are expected to be partially offset by certain clinical trial activities shifting into 2026 and additional costs related to post-marketing commitments. We remain committed to advancing our pipeline and late-stage programs, including intismeran autogene and norovirus vaccine programs, while continuing to manage research and development investment levels in line with our long-term objectives.

Selling, general and administrative expenses

Selling, general and administrative expenses decreased by $156 million, or 13%, in 2025. The decrease was mainly due to a $57 million broad-based reduction in consulting and outside services, along with a $59 million reduction in commercial, marketing and distribution costs. These decreases were attributable to continued cost discipline and ongoing efforts to streamline operations.

We expect selling, general and administrative expenses in 2026 to remain at a level relatively consistent with 2025, reflecting an efficient and scalable operating structure. While we will continue to make selective investments to support our key priorities, including our global commercial and regulatory activities, we expect these investments to be largely offset by ongoing efficiency initiatives and disciplined resource allocation.

Interest income

Interest income generated from our investments in marketable securities decreased by $111 million in 2025, mainly due to lower average investment balances.

Other expense, net

The following table summarizes other expense, net for the periods presented (in millions):

	Years Ended December 31,		Change 2025 vs. 2024	
	2025	2024	Change	%
Loss on investments	$ (5)	$ (56)	$ 51	(91)%
Interest expense	(10)	(24)	14	(58)%
Other income (expense), net	7	(7)	14	(200)%
Total other expense, net	$ (8)	$ (87)	$ 79	(91)%

Total other expense, net decreased by $79 million, or 91%, in 2025. The decrease was primarily driven by lower equity investment losses and lower interest expense. Interest expense is primarily related to our finance leases related to our Moderna Technology Center prior to its acquisition in December 2024, certain contract manufacturing service agreements, and, beginning in November 2025, interest associated with our long-term debt. See Note 10 and Note 11 to our consolidated financial statements for additional information.

Provision for income taxes

Provision for income taxes for 2025 remained immaterial and consistent with 2024 and 2023, reflecting the valuation allowance established on deferred tax assets in 2023, which has been applied consistently since its initial recognition. Provision for income taxes increased by $100 million, or 217%, in 2025, mainly driven by the absence of a discrete tax benefit related to research and development credits that was recognized in 2024, as well as taxable income in certain of our foreign subsidiaries, while we incurred a consolidated pre-tax loss. See Note 14 to our consolidated financial statements for additional details.

Liquidity and capital resources

The following table summarizes our cash, cash equivalents, investments and working capital for each period presented (in millions):

	December 31,	
	2025	2024
Financial assets:		
Cash and cash equivalents	$ 2,595	$ 1,927
Investments	3,204	5,098
Investments, non-current	2,336	2,494
Total	$ 8,135	$ 9,519
Working capital:		
Current assets	$ 6,544	$ 8,099
Current liabilities	1,987	2,206
Total	$ 4,557	$ 5,893

Our cash, cash equivalents and investments are invested in accordance with our investment policy, primarily with a view to liquidity and capital preservation. Our investments, consisting primarily of government and corporate debt securities, are stated at fair value. Cash, cash equivalents and investments as of December 31, 2025 decreased by $1.4 billion, or 15%, compared to December 31, 2024. For the year ended December 31, 2025, we had a net cash outflow from operations of $1.9 billion, and purchases of property, plant and equipment of $192 million, partially offset by $578 million of net proceeds from our credit facility.

Working capital, defined as current assets less current liabilities, as of December 31, 2025 decreased by $1.3 billion, or 23%, compared to December 31, 2024. This was primarily due to a $1.2 billion decrease in cash, cash equivalents and short-term investments, mainly used to fund our operating activities, a $191 million decrease in prepaid expenses and other current assets driven by reduced manufacturing and capital spend, and a $174 million decrease in accounts receivable, primarily driven by lower revenue and invoice volumes as well as timing of collections. These decreases were partially offset by a $129 million reduction in accounts payable and accrued liabilities, reflecting an overall reduction in operating expenses, and a $54 million reduction in short-term deferred revenue, mainly resulting from revenue recognized from deferred revenue in excess of customer advance payments received.

Cash flow

The following table summarizes the primary sources and uses of cash for the periods presented (in millions):

	Years Ended December 31,	
	2025	2024
Net cash (used in) provided by:		
Operating activities	$ (1,873)	$ (3,004)
Investing activities	1,946	1,949
Financing activities	593	56

Operating activities

We derive cash flows from operations primarily from cash collected from customer advance payments and accounts receivable related to our product sales, as well as other revenue and funding arrangements. Our cash flows from operating activities are significantly affected by our use of cash for operating expenses and working capital to support the business. We sell our COVID and RSV vaccines to the commercial market as well as to foreign governments and international organizations. Certain supply agreements include upfront payments, which are initially recorded as deferred revenue. In the U.S., our COVID and RSV vaccines are sold primarily to wholesalers and distributors, and to a lesser extent, directly to retailers and healthcare providers, which typically do not make upfront payments. In addition, we receive customer advance payments related to certain other revenue arrangements. As of December 31, 2025, we had $252 million in deferred revenue related to customer advance payments received or billable.

Net cash used by operating activities in 2025 was $1.9 billion and consisted of net loss of $2.8 billion and non-cash adjustments of $716 million, plus a net change in assets and liabilities of $233 million. Non-cash items primarily included stock-based compensation of $483 million and depreciation and amortization of $215 million. The net change in assets and liabilities was primarily due to a decrease in accounts receivable of $156 million, mainly due to lower invoice volumes and timing of collections, a $153 million decrease in prepaid expenses and other current assets, primarily due to reduced manufacturing and capital spend, and a $41 million decrease in deferred revenue due to revenue recognized from deferred revenue in excess of customer advance payments received. This was partially offset by a decrease in accounts payable and accrued liabilities of $94 million resulting from overall lower operating expenses in the period.

Net cash used by operating activities decreased by $1.1 billion, or 38%, in 2025 compared to 2024. This decrease was primarily driven by a decrease in net loss of $739 million, a change in deferred revenue of $480 million, reflecting less revenue recognized from deferred revenue in excess of customer advance payments received in 2025, and a change in accounts payable and accrued liabilities of $360 million, driven by decreased clinical and manufacturing spend. This was partially offset by a change in accounts receivable of $378 million, driven by lower invoice volumes and timing of collections, and a change in inventory of $117 million, driven by a smaller reduction in inventory write-downs.

Investing activities

Our primary investing activities consist of purchases, sales, and maturities of our investments, capital expenditures for land, building, leasehold improvements, manufacturing, laboratory, computer equipment and software, and business development.

Net cash provided by investing activities in 2025 was $1.9 billion, which primarily included proceeds from maturities of marketable securities of $5.6 billion and proceeds from sales of marketable securities of $2.4 billion, partially offset by purchases of marketable securities of $5.8 billion, and purchases of property, plant and equipment of $192 million.

Net cash flows provided by investing activities were relatively consistent in 2025 compared to 2024, primarily due to a decrease in proceeds from sale of marketable securities of $1.6 billion, which was partially offset by a decrease of $859 million in purchases of property, plant, and equipment, and a $761 million decrease in purchases of marketable securities.

Financing activities

Our primary financing activities consist of issuance of common stock related to our equity plans, repurchases of common stock, borrowings under our credit facility, and finance leases.

Net cash provided by financing activities in 2025 was $593 million, consisting of $578 million of net proceeds from our credit facility and net proceeds from the issuance of common stock in connection with the exercise of stock options and employee stock purchases under our equity plans of $35 million.

Net cash provided by financing activities increased by $537 million, or 959%, in 2025 compared to 2024, mainly due to the net proceeds from our credit facility.

Operation and funding requirements

Our principal sources of funding as of December 31, 2025 consisted of cash and cash equivalents, investments, and cash we may generate from operations. We reported a net loss of $2.8 billion, $3.6 billion and $4.7 billion for the years ended December 31, 2025, 2024 and 2023, respectively. In contrast, we generated net income of $8.4 billion and $12.2 billion for the years ended December 31, 2022 and 2021, respectively, following the authorization of our first commercial product in December 2020. From our inception to the end of 2020, we incurred significant losses from operations due to our significant research and development expenses. We have retained earnings of $7.2 billion as of December 31, 2025.

We have significant future capital requirements including expected operating expenses to conduct research and development activities, operate our organization, and meet capital expenditure needs. We anticipate maintaining substantial expenses across all areas of our ongoing activities, particularly as we continue research and development of our development candidates and clinical activities for our investigational medicines. This also extends to our manufacturing costs, including our arrangements with our manufacturers and suppliers. Our ongoing work on our intismeran autogene, norovirus, and flu+COVID combination vaccine candidates, development of any new COVID vaccines against variants of SARS-CoV-2, late-stage clinical development, investments in digital capabilities and artificial intelligence technologies, and buildout of global commercial, regulatory, sales and marketing infrastructure and manufacturing facilities will require significant cash outflows in future periods, most of which may not be reimbursed or otherwise paid for by our collaborators or alliances. We may also incur additional costs related to postmarketing commitments, though the timing and scope of such commitments remain uncertain. In addition, we have substantial facility, lease and purchase obligations. We have also entered into various collaboration and licensing agreements, as well as a research and development funding arrangement with a third party. These arrangements collectively encompass the funding of specific research and development activities, with the distinction that under the research and development funding arrangements, we receive funding. However, for all these arrangements, we may be obligated to make potential future milestone and royalty payments.

In November 2025, we entered into a Credit Agreement providing for a term loan facility with aggregate commitments of $1.5 billion. As of December 31, 2025, we had drawn $600 million under the initial term loan. The Credit Agreement also provides for $900 million of delayed draw term loan commitments, consisting of $400 million available, subject to applicable conditions, through November 2027 and $500 million available, subject to applicable conditions and specified regulatory approval milestones, through November 2028. Borrowings under the Credit Agreement bear interest at a variable rate based on Term SOFR or a base rate, at our option, plus an applicable margin, with interest payable periodically and the principal amount due in full at maturity in November 2030. The Credit Agreement includes customary affirmative and negative covenants, with which we were in compliance as of December 31, 2025. See Note 11 to our consolidated financial statements for additional information.

We believe that our cash, cash equivalents, and investments as of December 31, 2025, together with cash expected to be generated from product sales and available borrowings under our credit facility, will be sufficient to enable us to fund our projected operations and capital expenditures through at least the next 12 months from the issuance of the financial statements included in this Annual Report on Form 10-K. We are subject to all the risks related to the development and commercialization of novel medicines, and we may encounter unforeseen expenses, difficulties, complications, delays, and other unknown factors, which may adversely affect our business. For example, we experienced a decline in customer demand for our COVID vaccine in 2023 and 2024, and this trend has continued into 2025 as the market transitions to a more competitive and commercially driven environment. We foresee that our commitment to investing in our business for future product launches may lead to continued negative cash flows from operations in upcoming periods. Our forecast of the period of time through which our financial resources will be adequate to support our operations is a forward-looking statement and involves risks and uncertainties, and actual results could vary as a result of a number of factors. We have based this estimate on assumptions that may prove to be wrong, and we could utilize our available capital resources sooner than we currently expect.

Contractual obligations and commitments

The following table summarizes our contractual obligations as of December 31, 2025, and the effects that such obligations are expected to have on our liquidity and cash flows in future periods (in millions):

	Payments Due by Period				
	Total	Less than 1 year	1 - 3 years	3 - 5 years	More than 5 years
Operating leases	$ 1,066	$ 65	$ 159	$ 162	$ 680
Financing leases	47	27	20	—	—
Purchase obligations[(1)]	1,438	481	441	343	173
Long-term debt[(2)]	900	62	130	708	—
Total contractual cash obligations	$ 3,451	$ 635	$ 750	$ 1,213	$ 853

(1) The amounts represent non-cancelable fixed payment obligations related to purchases of raw materials, contract manufacturing services, research and development and other goods or services in the normal course of business. As of December 31, 2025, $5 million of the purchase commitments related to raw materials was recorded as an accrued liability for loss on future firm purchase commitments and is included in the purchase obligations line above.

(2) In November 2025, we entered into the Credit Agreement providing for a senior secured term loan facility with aggregate commitments of $1.5 billion, consisting of a $600 million initial term loan and $900 million of delayed draw term loan commitments. As of December 31, 2025, the $600 million initial term loan was outstanding and matures on November 24, 2030. Borrowings under the Credit Agreement bear interest at a variable rate. Interest amounts preset in the table are based on the applicable interest rate in effect as of December 31, 2025. The principal amount of $600 million is due in full at maturity, and no principal payments are required prior to that date. See Note 11 to the consolidated financial statements for additional details.

We have agreements with certain vendors for various services, including services related to clinical operations and support and contract manufacturing, which we are not contractually able to terminate for convenience. Certain agreements provide for termination rights subject to termination fees or wind down costs. Under such agreements, we are contractually obligated to make certain payments to vendors, mainly to reimburse them for their unrecoverable outlays incurred prior to cancellation. The exact amounts of such obligations are dependent on the timing of termination, and the exact terms of the relevant agreement and cannot be reasonably estimated. At December 31, 2025, we had cancelable open purchase orders of approximately $1.8 billion, reflecting amounts associated with our significant vendor arrangements, under such agreements for our clinical operations and support and contract manufacturing. These amounts represent only our estimate of those items for which we had a contractual commitment to pay at December 31, 2025, assuming we would not cancel these agreements. The actual amounts we pay in the future to the vendors under such agreements may differ from purchase order amounts.

In addition to the above obligations, we enter into a variety of agreements and financial commitments in the normal course of business. The terms generally allow us the option to cancel, reschedule, and adjust our requirements based on our business needs, prior to the delivery of goods or performance of services. It is not possible to predict the maximum potential amount of future payments under these agreements due to the conditional nature of our obligations and the unique facts and circumstances involved in each particular agreement.

As of December 31, 2025, we did not have any off-balance sheet arrangements, other than those obligations and commitments disclosed herein, that were material or reasonably likely to become material to our financial condition or results of operations.

Item 7A. Quantitative and Qualitative Disclosures about Market Risk

Interest Rate Risk

As of December 31, 2025 and 2024, we had cash, cash equivalents, restricted cash, and investments in marketable securities of $8.1 billion and $9.5 billion, respectively. Our investment portfolio comprises money market funds and marketable debt securities (including U.S. Treasury securities, debt securities of U.S. government agencies and corporate entities, and commercial paper), which are classified as available-for-sale securities. Our primary investment objectives are the preservation of capital and the maintenance of liquidity, and our investment policy defines allowable investments based on quality of the institutions and financial instruments designed to minimize risk exposure. Our exposure to interest rate sensitivity is affected by changes in the general level of U.S. interest rates. Our marketable securities are subject to interest rate risk and will fall in value if market interest rates increase. Due to the short-term maturities and low risk profiles of our investments, we do not anticipate a significant exposure to interest rate risk. If market interest rates were to increase immediately and uniformly by one percentage point from levels at December 31, 2025, the net fair value of our marketable securities would decrease by approximately $47 million.

In November 2025, we entered into a variable-rate Credit Agreement providing for borrowings of up to $1.5 billion, under which outstanding amounts bear interest at rates based on Term SOFR plus a margin of 5.50% or a base rate plus a margin of 4.50%, at our option. As a result, our interest expense is exposed to changes in market interest rates. As of December 31, 2025, we had $600 million outstanding under this facility. See Note 11 to the consolidated financial statements for additional details.

Foreign Currency Risk

We transact business in various foreign currencies and have international sales and expenses denominated in foreign currencies. As a result, we are exposed to risks arising from changes in foreign currency exchange rates driven by both our business operations and economic conditions. For the year ended December 31, 2025, approximately 27% of our revenue was denominated in foreign currencies, although our revenue generating activities and operations continued to be primarily denominated in U.S. dollars. Our foreign currency revenue exposure in 2025 was primarily associated with the Canadian dollar, Euro, Korean won, and Mexican peso markets.

As our business evolves, we have transitioned to receiving payments in local currencies rather than U.S. dollars. In addition, during 2025 we established and expanded manufacturing operations in the United Kingdom, Canada and Australia, where we also transact in foreign currencies. These developments are expected to result in higher levels of foreign currency denominated revenues and expenses on an annualized basis in the future, which may increase our exposure to fluctuations in foreign currency exchange rates and could impact our results of operations and cash flows. To manage this exposure, we have focused on balance sheet hedging activities as part of our strategy to address foreign currency fluctuations. While cash flow hedging programs are in place, there were no foreign currency cash flow hedging activities during 2025.

Balance Sheet Hedging Activities

We enter into foreign currency forward contracts to hedge fluctuations associated with foreign currency denominated monetary assets and liabilities, primarily cash, receivables, payables and lease liabilities in the Australian dollar, British pound, Canadian dollar, and Euro that are not designated for hedge accounting treatment. Therefore, these forward contracts are accounted for as derivatives whereby the fair value of the contracts are reported as prepaid expenses and other current assets or other current liabilities in our consolidated balance sheets, and gains and losses resulting from changes in the fair value are recorded as a component of other income (expense), net, in our consolidated statements of operations. The gains and losses on these foreign currency forward contracts generally offset the gains and losses in the underlying foreign currency denominated assets and liabilities, which are also recorded to other income (expense), net, in our consolidated statements of operations. As of December 31, 2025, our outstanding balance sheet hedging derivatives, carried at fair value, had maturities of less than three months.

We enter into these foreign exchange contracts to hedge our monetary assets and liabilities denominated in foreign currency in the normal course of business and accordingly, they are not speculative in nature. We believe the counterparties to our foreign currency forward contracts are creditworthy multinational commercial banks. While we believe the risk of counterparty nonperformance is not material, a sustained decline in the financial stability of financial institutions as a result of disruption in the financial markets could affect our ability to secure creditworthy counterparties for our foreign currency hedging programs.

Notwithstanding our efforts to mitigate some foreign currency exchange risks, there can be no assurance that our hedging activities will adequately protect us against the risks associated with foreign currency fluctuations. As of December 31, 2025, a hypothetical adverse movement of 10 percent in foreign currency exchange rates compared to the U.S. dollars across all maturities would have resulted in potential declines in the fair value on our foreign currency forward contracts used in balance sheet hedging of approximately $58 million. We expect that any increase or decrease in the fair value of the portfolio would be substantially offset by increases or decreases in the underlying exposures being hedged.

Item 8. Financial Statements and Supplementary Data

MODERNA, INC.
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Report of Independent Registered Public Accounting Firm

To the Stockholders and the Board of Directors of Moderna, Inc.

Opinion on the Financial Statements

We have audited the accompanying consolidated balance sheets of Moderna, Inc. (the Company) as of December 31, 2025 and 2024, the related consolidated statements of operations, comprehensive income (loss), stockholders' equity, and cash flows for each of the three years in the period ended December 31, 2025, and the related notes (collectively referred to as the "consolidated financial statements"). In our opinion, the consolidated financial statements present fairly, in all material respects, the financial position of the Company at December 31, 2025 and 2024, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2025, in conformity with U.S. generally accepted accounting principles.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the Company's internal control over financial reporting as of December 31, 2025, based on criteria established in Internal Control—Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework) and our report dated February 20, 2026 expressed an unqualified opinion thereon.

Basis for Opinion

These financial statements are the responsibility of the Company's management. Our responsibility is to express an opinion on the Company's financial statements based on our audits. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the financial statements are free of material misstatement, whether due to error or fraud. Our audits included performing procedures to assess the risks of material misstatement of the financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the financial statements. We believe that our audits provide a reasonable basis for our opinion.

Critical Audit Matter

The critical audit matter communicated below is a matter arising from the current period audit of the financial statements that was communicated or required to be communicated to the audit committee and that: (1) relates to accounts or disclosures that are material to the financial statements and (2) involved our especially challenging, subjective or complex judgments. The communication of the critical audit matter does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matter below, providing a separate opinion on the critical audit matter or on the accounts or disclosures to which it relates.

Provisions for returns on product sales

Description of the Matter

During the year ended December 31, 2025, the Company's net product sales were $1.8 billion. As explained in Note 2 of the consolidated financial statements, revenue from product sales includes estimates of variable consideration for which provisions are established, including provisions for product sales returns.

Auditing the Company's measurement of provisions for returns on product sales under its contracts with customers was especially challenging because (1) the estimate relies on management's assumptions regarding inventory remaining in the distribution channel as of the balance sheet date that may be subject to return in future periods, as well as projected market demand, and (2) the Company has limited historical returns on which to base its assumptions.

How We Addressed the Matter in Our Audit

We obtained an understanding, evaluated the design and tested the operating effectiveness of internal controls over the Company's process to determine provisions for returns on product sales. For example, we tested controls over management's review of the completeness and accuracy of the data used in the process and the assumptions about the amount of inventory in the distribution channel that could be subject to return in future periods.

To test the Company's provisions for returns on product sales, our audit procedures included, among other procedures, testing the accuracy and completeness of the underlying data used in the calculations and evaluating the assumptions applied by management in estimating the provisions. To test management's assumptions, we inspected agreements with significant customers to validate the rights of return, made inquiries of members of the commercial function regarding any changes to the terms and conditions of commercial contracts, and assessed the historical accuracy of management's estimates. We also examined credit memos issued during and after year end to identify any unusual items or trends inconsistent with the Company's analysis of product returns and performed revenue cutoff testing at period end to evaluate whether any unusual trends should have been considered in the Company's analysis of product returns. In addition, we reviewed inventory on hand-reporting from significant customers around the balance sheet date and subsequent to the balance sheet date and inspected vaccination data from third-party sources after the balance sheet date. We also performed sensitivity analyses on the Company's return rate to assess the impact of changes in assumptions.

/s/ Ernst & Young LLP
We have served as the Company's auditor since 2014.
Boston, Massachusetts
February 20, 2026

MODERNA, INC.
CONSOLIDATED BALANCE SHEETS
(In millions, except per share data)

	December 31,	
	2025	**2024**
Assets		
Current assets:		
Cash and cash equivalents	$ 2,595	$ 1,927
Investments	3,204	5,098
Accounts receivable, net	184	358
Inventory	153	117
Prepaid expenses and other current assets	408	599
Total current assets	6,544	8,099
Investments, non-current	2,336	2,494
Property, plant and equipment, net	2,134	2,196
Right-of-use assets, operating leases	719	759
Other non-current assets	605	594
Total assets	$ 12,338	$ 14,142
Liabilities and Stockholders' Equity		
Current liabilities:		
Accounts payable	$ 317	$ 405
Accrued liabilities	1,386	1,427
Deferred revenue	99	153
Other current liabilities	185	221
Total current liabilities	1,987	2,206
Deferred revenue, non-current	153	58
Operating lease liabilities, non-current	653	671
Financing lease liabilities, non-current	20	39
Long-term debt	590	—
Other non-current liabilities	285	267
Total liabilities	3,688	3,241
Commitments and contingencies (Note 12)		
Stockholders' equity:		
Preferred stock, $0.0001; 162 shares authorized as of December 31, 2025 and 2024; no shares issued or outstanding at December 31, 2025 and 2024	—	—
Common stock, par value $0.0001; 1,600 shares authorized as of December 31, 2025 and 2024; 394 and 386 shares issued and outstanding as of December 31, 2025 and 2024, respectively	—	—
Additional paid-in capital	1,382	866
Accumulated other comprehensive income (loss)	45	(10)
Retained earnings	7,223	10,045
Total stockholders' equity	8,650	10,901
Total liabilities and stockholders' equity	$ 12,338	$ 14,142

The accompanying notes are an integral part of these consolidated financial statements.

MODERNA, INC.
CONSOLIDATED STATEMENTS OF OPERATIONS
(In millions, except per share data)

	Years Ended December 31,		
	2025	**2024**	**2023**
Revenue:			
Net product sales	$ 1,818	$ 3,109	$ 6,671
Other revenue	126	127	177
Total revenue	1,944	3,236	6,848
Operating expenses:			
Cost of sales	868	1,464	4,693
Research and development	3,132	4,543	4,845
Selling, general and administrative	1,018	1,174	1,549
Total operating expenses	5,018	7,181	11,087
Loss from operations	(3,074)	(3,945)	(4,239)
Interest income	314	425	421
Other expense, net	(8)	(87)	(124)
Loss before income taxes	(2,768)	(3,607)	(3,942)
Provision for (benefit from) tax provision	54	(46)	772
Net loss	$ (2,822)	$ (3,561)	$ (4,714)
Net loss per share			
Basic and diluted	$ (7.26)	$ (9.28)	$ (12.33)
Weighted average common shares used in calculation of loss per share:			
Basic and diluted	389	384	382

The accompanying notes are an integral part of these consolidated financial statements.

MODERNA, INC.
CONSOLIDATED STATEMENTS OF COMPREHENSIVE INCOME (LOSS)
(In millions)

	Years Ended December 31,		
	2025	**2024**	**2023**
Net loss	$ (2,822)	$ (3,561)	$ (4,714)
Other comprehensive income, net of tax:			
Available-for-sale securities:			
Unrealized gains on available-for-sale securities	31	120	210
Less: net realized (gains) losses on available-for-sale securities reclassified in net loss	(3)	4	38
Net increase from available-for-sale securities	28	124	248
Cash flow hedges:			
Net realized losses on derivative instruments reclassified in net loss	—	—	8
Net increase from derivatives designated as hedging instruments	—	—	8
Gains (losses) on foreign currency translation	11	(8)	—
Pension and postretirement obligation adjustments	16	(3)	(9)
Total other comprehensive income	55	113	247
Comprehensive loss	$ (2,767)	$ (3,448)	$ (4,467)

The accompanying notes are an integral part of these consolidated financial statements.

MODERNA, INC.
CONSOLIDATED STATEMENTS OF STOCKHOLDERS' EQUITY
(In millions)

	Common Stock		Additional Paid-In Capital	Accumulated Other Comprehensive Loss	Retained Earnings	Total Stockholders' Equity
	Shares	Amount				
Balance at December 31, 2022	385	$ —	$ 1,173	$ (370)	$ 18,320	$ 19,123
Vesting of restricted common stock	1	—	—	—	—	—
Exercise of options to purchase common stock	4	—	25	—	—	25
Issuance of common stock under employee stock purchase plan	—	—	21	—	—	21
Stock-based compensation	—	—	305	—	—	305
Other comprehensive income, net of tax	—	—	—	247	—	247
Repurchase of common stock, including excise tax	(8)	—	(1,153)	—	—	(1,153)
Net loss	—	—	—	—	(4,714)	(4,714)
Balance at December 31, 2023	382	$ —	$ 371	$ (123)	$ 13,606	$ 13,854

	Common Stock		Additional Paid-In Capital	Accumulated Other Comprehensive Loss	Retained Earnings	Total Stockholders' Equity
	Shares	Amount				
Balance at December 31, 2023	382	$ —	$ 371	$ (123)	$ 13,606	$ 13,854
Vesting of restricted common stock	2	—	—	—	—	—
Exercise of options to purchase common stock	1	—	42	—	—	42
Issuance of common stock under employee stock purchase plan	1	—	24	—	—	24
Stock-based compensation	—	—	429	—	—	429
Other comprehensive income, net of tax	—	—	—	113	—	113
Net loss	—	—	—	—	(3,561)	(3,561)
Balance at December 31, 2024	386	$ —	$ 866	$ (10)	$ 10,045	$ 10,901

	Common Stock		Additional Paid-In Capital	Accumulated Other Comprehensive (Loss) Income	Retained Earnings	Total Stockholders' Equity
	Shares	Amount				
Balance at December 31, 2024	386	$ —	$ 866	$ (10)	$ 10,045	$ 10,901
Vesting of restricted common stock	7	—	—	—	—	—
Exercise of options to purchase common stock	1	—	17	—	—	17
Tax payments related to net share settlements on equity awards	—	—	(2)	—	—	(2)
Issuance of common stock under employee stock purchase plan	—	—	18	—	—	18
Stock-based compensation	—	—	483	—	—	483
Other comprehensive income, net of tax	—	—	—	55	—	55
Net loss	—	—	—	—	(2,822)	(2,822)
Balance at December 31, 2025	394	$ —	$ 1,382	$ 45	$ 7,223	$ 8,650

The accompanying notes are an integral part of these consolidated financial statements.

MODERNA, INC.
CONSOLIDATED STATEMENTS OF CASH FLOWS
(In millions)

	Years Ended December 31,		
	2025	2024	2023
Operating activities			
Net loss	$ (2,822)	$ (3,561)	$ (4,714)
Adjustments to reconcile net loss to net cash used in operating activities:			
Stock-based compensation	483	429	305
Depreciation and amortization	215	189	621
Amortization/accretion of investments	(67)	(95)	(61)
Loss on equity investments, net	8	52	35
Other non-cash items	77	60	7
Changes in assets and liabilities:			
Accounts receivable, net	156	534	493
Prepaid expenses and other assets	153	145	1,802
Inventory	(34)	83	747
Right-of-use assets, operating leases	38	(53)	(605)
Accounts payable	(92)	(69)	13
Accrued liabilities	(2)	(385)	(340)
Deferred revenue	41	(439)	(2,060)
Operating lease liabilities	(21)	28	551
Other liabilities	(6)	78	88
Net cash used in operating activities	(1,873)	(3,004)	(3,118)
Investing activities			
Purchases of marketable securities	(5,768)	(6,529)	(3,760)
Proceeds from maturities of marketable securities	5,563	5,562	5,575
Proceeds from sales of marketable securities	2,353	3,967	3,206
Purchases of property, plant and equipment	(192)	(1,051)	(707)
Acquisition of business, net of cash acquired	—	—	(85)
Purchase of intangible asset	(10)	—	—
Investment in convertible notes and equity securities	—	—	(23)
Net cash provided by investing activities	1,946	1,949	4,206
Financing activities			
Proceeds from credit facility	600	—	—
Payment of credit facility issuance costs	(22)	—	—
Proceeds from issuance of common stock through equity plans	35	66	46
Tax payments related to net share settlements on equity awards	(2)	—	(1,153)
Changes in financing lease liabilities	(18)	(10)	(270)
Net cash provided by (used in) financing activities	593	56	(1,377)
Effect of changes in exchange rates on cash and cash equivalents	2	—	—
Net increase (decrease) in cash, cash equivalents and restricted cash	668	(999)	(289)
Cash, cash equivalents and restricted cash, beginning of year	1,929	2,928	3,217
Cash, cash equivalents and restricted cash, end of year	$ 2,597	$ 1,929	$ 2,928
Supplemental cash flow information			
Cash paid (received) for income taxes	$ 26	$ 197	$ (357)
Cash paid for interest	$ 10	$ 24	$ 39
Non-cash investing and financing activities			
Purchases of property, plant and equipment included in accounts payable and accrued liabilities	$ 83	$ 97	$ 130

The accompanying notes are an integral part of these consolidated financial statements.

MODERNA, INC.
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS

1. Description of the Business

Moderna, Inc. (collectively, with its consolidated subsidiaries, any of Moderna, we, us, our or the Company) is a biotechnology company advancing a new class of medicines made of messenger RNA (mRNA). mRNA medicines are designed to direct the body's cells to produce intracellular, membrane or secreted proteins that have a therapeutic or preventive benefit with the potential to address a broad spectrum of diseases. Our platform builds on continuous advances in basic and applied mRNA science, delivery technology and manufacturing, providing us the capability to pursue in parallel a robust pipeline of new development candidates. We are developing medicines across infectious disease vaccines, oncology therapeutics and rare disease therapeutics.

Since our founding in 2010, we have transformed from a research-stage company advancing programs in the field of mRNA to a commercial enterprise with a diverse clinical portfolio of vaccines and therapeutics across several modalities, a broad intellectual property portfolio and integrated manufacturing capabilities that allow for rapid clinical and commercial production at scale. We currently have three commercial products—Spikevax® and mNEXSPIKE®, our COVID vaccines, and mRESVIA®, our vaccine against respiratory syncytial virus (RSV). Additionally, we also have a diverse development pipeline of 25 development candidates across our 35 development programs currently in clinical studies.

2. Summary of Significant Accounting Policies

Basis of Presentation and Principles of Consolidation

Our consolidated financial statements are prepared in accordance with U.S. generally accepted accounting principles (GAAP). Any reference in these notes to applicable guidance is meant to refer to the authoritative accounting principles generally accepted in the United States as found in the Accounting Standards Codification (ASC) and Accounting Standards Updates (ASU) of the Financial Accounting Standards Board (FASB).

The consolidated financial statements include the Company and its subsidiaries. All intercompany transactions and balances have been eliminated in consolidation.

Use of Estimates

We have made estimates and judgments affecting the amounts reported in our consolidated financial statements and the accompanying notes. We base our estimates on historical experience and various relevant assumptions that we believe to be reasonable under the circumstances, the results of which form the basis for making judgments about the carrying values of assets and liabilities at the date of the financial statements and the reported amounts of revenue and expenses during the reporting periods that are not readily apparent from other sources. Changes in our estimates are recorded in the financial results of the period in which the new information becomes available. The actual results that we experience may differ materially from our estimates.

Segment Information

The Company operates as a single operating and reportable segment, reflecting the integrated nature of our business focused on the research, development, and commercialization of mRNA-based medicines. Our Chief Executive Officer serves as the Chief Operating Decision Maker (CODM), responsible for assessing the Company's performance and making resource allocation decisions. The CODM evaluates financial information on a consolidated basis, focusing on key metrics such as total revenue, operating expenses, and net income or loss, which is our reported measure of segment profit or loss. The CODM allocates resources based on the Company's available cash resources, forecasted cash flow, and expenditures on a consolidated basis, as well as an assessment of the probability of success of its research and development activities. Resource allocation decisions are informed by budgeted and forecasted expense information, along with actual expenses incurred to date. Disaggregated profit or loss information at the program or functional level is not regularly provided to or relied upon by the CODM, as our integrated operating model emphasizes shared resources and centralized decision-making. The interdependent nature of our research, development, and commercialization activities, supported by common infrastructure such as our mRNA platform, makes further disaggregation of expenses less meaningful for assessing performance. Additionally, there are no segment managers accountable for operations, operating results, and planning for levels or components below the consolidated unit level.

Revenue Recognition

To determine the appropriate amount of revenue to be recognized for arrangements that we determine are within the scope of ASC 606, we perform the following five steps (the five-step model): (i) identify the contract(s) with our customer; (ii) identify the performance obligations in the contract; (iii) determine the transaction price; (iv) allocate the transaction price to the performance obligations in the contract; and (v) recognize revenue when or as each performance obligation is satisfied.

Net Product Sales

We sell our COVID and RSV vaccines to the commercial market as well as to foreign governments and international organizations. In the U.S., our COVID vaccine and RSV vaccine are sold primarily to wholesalers and distributors, and to a lesser extent, directly to retailers and healthcare providers.

We recognize net product sales when control of the product transfers to the customer, typically upon delivery. Payment terms generally range from 30 to 60 days, in line with customary practices in each country. Net product sales are recognized net of estimated wholesaler chargebacks, invoice discounts for prompt payments and pre-orders, provisions for sales returns, government rebates, and other related deductions. These provisions are recorded based on contractual terms, our estimate of returns for product sold during the period, and other relevant considerations, using the expected value method or the most likely amount method. We update our estimates quarterly and record necessary adjustments in the period when we identify the adjustments. Product sales, net of provisions, are recorded only to the extent a significant reversal in the amount of cumulative revenue recognized is not probable when the uncertainty associated with the provisions is subsequently resolved. Shipping and handling activities are considered fulfillment activities and not a separate performance obligation. Taxes assessed by governmental authorities that are imposed on and collected from our product sales are excluded from net product sales.

Wholesaler chargebacks, discounts and fees

We contract with retailers, healthcare providers, and group purchasing organizations (GPO) to broaden our customer reach and offer contractual discounts. The chargeback represents the difference between the invoice price billed to the wholesaler and the negotiated price charged to the retailers, healthcare providers and GPO members. For distribution and related services, such as stocking and cold chain storage, we provide compensation to our wholesalers and distributors. We typically offer our customers invoice discounts on product sales for prompt payments and pre-orders. The estimation of these discounts and fees is based on contractual terms and our expectations regarding future customer payment behaviors. Wholesaler fees and invoice discounts are deducted from our gross product sales and accounts receivable at the time such product sales are recognized.

Product returns

We typically offer customers in the U.S. the right to return products, up to a certain limit as stipulated in our contracts. Estimated returns for our vaccines are determined considering available return rates for similar products, estimated levels of inventory in the distribution channel, projected market demand, and our historical experience with returns. The estimated amount for product returns is presented within accrued liabilities on our consolidated balance sheets and is deducted from our gross product sales in the period the related product sales are recognized.

Government rebates and other fees

Fees payable to third party payers and healthcare providers, along with fees to our direct customers that are settled via cash payments, including certain patient assistance programs, are recorded as accrued liabilities on our consolidated balance sheets. In 2024, we began recognizing Medicare rebates associated with our RSV product. The estimation of Medicare rebates requires judgment and is based on historical utilization trends, and the mix of customers and payers. The estimated liability for unpaid or unbilled rebates is presented as accrued liabilities on our consolidated balance sheets. For 2024 and 2025, the product sales subject to Medicare rebates were immaterial.

Determining the amount of variable consideration to recognize necessitates substantial judgment, especially when assessing factors outside our direct control, such as the limited historical data, constrained third-party information, and evolving market dynamics. Among all variables, estimating returns continues to present the most significant judgment due to the broad range of potential outcomes and the lack of established return trends. While we now have two years of data on our product returns, this remains limited and subject to significant variability given current market conditions. We will continue to refine our projections as additional information becomes

available. The actual results could differ from our estimates, and such differences could have a material impact to our financial statements.

Other Revenue

Other revenue consists primarily of grant revenue, collaboration revenue, licensing and royalty revenue, and stand-ready manufacturing revenue.

Grant revenue

We have contracts with government-sponsored and private organizations for research and development related activities that provide for payments for reimbursed costs, which may include overhead and general and administrative costs as well as a related profit margin. We recognize grant revenue from these contracts as we perform services under these arrangements when the funding is committed. Associated expenses are recognized when incurred as research and development expense. Grant revenue and related expenses are presented gross in the consolidated statements of operations as we have determined we are the primary obligor under the arrangements relative to the research and development services we perform as lead technical expert.

Collaboration Revenue

We have entered into strategic collaborations and other similar arrangements with third parties for research and other licenses, development and commercialization of certain products and product candidates. Such arrangements provide for various types of payments to us, including upfront fees, funding of research and development services and preclinical and clinical material, technical, development, regulatory, and commercial milestone payments, licensing fees, option exercise fees, and royalty and earnout payments on product sales. Such payments are often not commensurate with the timing of revenue recognition and therefore result in deferral of revenue recognition. We recognize revenue based on the amount of the transaction price that is allocated to each respective performance obligation when or as the performance obligation is satisfied by transferring a promised good or service to the customer.

Licensing and Royalty Revenue

License revenue is recognized when the license is granted to the licensee, provided no significant performance obligations remain. Royalty revenue is recognized based on sales by licensees when the underlying sales occur, in accordance with the terms of the licensing agreement and when collectibility is reasonably assured.

Stand-Ready Manufacturing Revenue

We have entered into long-term strategic agreements with government entities to establish domestic mRNA manufacturing capabilities and support pandemic readiness. Under these agreements, we are required to build and maintain manufacturing facilities and to supply vaccines in accordance with minimum spend commitments or annual demand forecasts established by the respective governments. A portion of the total consideration under each agreement is allocated to a stand-ready obligation to maintain manufacturing readiness and is recognized as other revenue on a straight-line basis over the contractual period. The remaining consideration is related to our obligation to supply vaccines and will be recognized as product sales when control of the vaccines transfers upon delivery.

Cash and Cash Equivalents

We consider all highly liquid investments with an original maturity of 90 days or less from the date of purchase to be cash equivalents.

Cash, Cash Equivalents and Restricted Cash shown in the Consolidated Statements of Cash Flows

The following table provides a reconciliation of cash, cash equivalents and restricted cash in the consolidated balance sheets that sum to the total of the same such amounts shown in the consolidated statements of cash flows (in millions):

	December 31,		
	2025	2024	2023
Cash and cash equivalents	$ 2,595	$ 1,927	$ 2,907
Restricted cash[(1)]	1	1	17
Restricted cash, non-current[(2)]	1	1	4
Total cash, cash equivalents and restricted cash shown in the consolidated statements of cash flows	$ 2,597	$ 1,929	$ 2,928

[(1)]Included in prepaid expenses and other current assets in the consolidated balance sheets.
[(2)]Included in other non-current assets in the consolidated balance sheets.

Investments

We invest our excess cash balances in marketable debt securities. We classify our investments in marketable debt securities as available-for-sale. We report our available-for-sale securities at fair value at each balance sheet date, and include any unrealized holding gains and losses (the adjustment to fair value) in accumulated other comprehensive income (loss), a component of stockholders' equity. Realized gains and losses are determined using the specific-identification method, and are included in other income (expense), net in our consolidated statements of operations. We classify our available-for-sale securities as current or non-current based on each instrument's underlying effective maturity date and for which we have the intent and ability to hold the investment for a period of greater than 12 months. Available-for-sale securities with maturities of less than 12 months are classified as current and are included in investments in the consolidated balance sheets. Available-for-sale securities with maturities greater than 12 months for which we have the intent and ability to hold the investment for greater than 12 months are classified as non-current and are included in investments, non-current in the consolidated balance sheets.

We evaluate securities for impairment at the end of each reporting period. Impairment is evaluated considering numerous factors, and their relative significance varies depending on the situation. Factors considered include whether a decline in fair value below the amortized cost basis is due to credit-related factors or non-credit-related factors, the financial condition and near-term prospects of the issuer, and our intent and ability to hold the investment to allow for an anticipated recovery in fair value. A credit-related impairment is recognized as an allowance on the balance sheet with a corresponding adjustment to earnings. Any impairment that is not credit- related is recognized in other comprehensive income (loss), net of applicable taxes.

Investments in publicly traded equity securities with readily determinable fair values are recorded at quoted market prices for identical securities, with changes in fair value recorded in other income (expense), net, in our consolidated statements of operations. Investments in equity securities without readily determinable fair values are recorded at cost minus impairment, if any, adjusted for changes resulting from observable price changes in orderly transactions for identical or similar securities. Such adjustments are recorded in other income (expense), net, in our consolidated statements of operations.

Accounts Receivable, net

Accounts receivable, net represent amounts due from customers less wholesalers chargebacks, discounts and fees (see our "Revenue Recognition" policy within Note 2 for product sales provision) and allowance for expected credit losses. Amounts payable to us are recorded as accounts receivable when our right to consideration is unconditional. To estimate the allowance for credit losses, we determine the allowance based on ongoing credit evaluation, historical experience and the aging of such receivables, among other factors. The allowance for doubtful accounts was immaterial at December 31, 2025 and 2024. Additionally, bad debt expenses were immaterial for the years ended December 31, 2025, 2024 and 2023.

Concentrations of Credit Risk

Financial instruments that subject us to significant concentrations of credit risk consist primarily of cash, cash equivalents, marketable debt securities, and accounts receivable, net. Our investment portfolio comprises money market funds and marketable debt securities, including U.S. Treasury securities, debt securities of U.S. government agencies and corporate entities and commercial paper. Our cash management and investment policy limits investment instruments to investment-grade securities with the objective to preserve capital

and to maintain liquidity until the funds can be used in business operations. We invest in a variety of financial instruments and limit the amount of credit exposure with any individual financial institution.

We are also subject to credit risk from our accounts receivable, net related to our net product sales and other revenue. We sell our products primarily to wholesalers and distributors and other governments and organizations. We do not require collateral or other security to support accounts receivable. To date, we have not experienced material losses with respect to the collection of our accounts receivable.

Significant Customers

Our accounts receivable, net are generally unsecured and are from customers in different countries. We generated revenue from product sales to wholesalers and distributors and other governments and organizations, and to a lesser extent, stand-ready manufacturing revenue from international government entities, grants made by government-sponsored and private organizations, and collaboration revenue from our strategic alliances.

A significant portion of our revenue to date has been generated from the following entities that accounted for more than 10% of total revenue and accounts receivable for the periods presented:

	Percentage of Revenue Years Ended December 31,			Percentage of Accounts Receivable December 31,	
	2025	2024	2023	2025	2024
FFF Enterprises	15 %	13 %	*	*	*
Public Works and Government Services Canada	13 %	*	*	33 %	*
Cardinal Health	11 %	*	*	*	*
United Kingdom Health Security Agency	*	16 %	*	*	65 %
Mexico Medistik	*	*	*	29 %	*
Boryung Biopharma Korea	*	*	*	17 %	*
Taiwan Food and Drug Administration	*	*	*	*	10 %
Ministry of Health, Labor, and Welfare of Japan	*	*	21 %	*	*

** - Represents an amount of less than 10%*

Fair Value Measurements

Fair value is defined as the price that would be received from selling an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date. When determining the fair value measurements for assets and liabilities, which are required to be recorded at fair value, we consider the principal or most advantageous market in which we would transact and the market-based risk measurements or assumptions that market participants would use in pricing the asset or liability, such as risks inherent in valuation techniques, transfer restrictions and credit risk. ASC 820 *(Fair Value Measurement*) establishes a fair value hierarchy for instruments measured at fair value that distinguishes between assumptions based on market data (observable inputs) and our assumptions (unobservable inputs). Observable inputs are inputs that market participants would use in pricing the asset or liability based on market data obtained from our independent sources. Unobservable inputs are inputs that reflect our assumptions about the inputs that market participants would use in pricing the asset or liability, and are developed based on the best information available in the circumstances. The following fair value hierarchy is used to classify assets and liabilities based on the observable inputs and unobservable inputs used to value the assets and liabilities:

- Level 1: Unadjusted quoted prices in active markets that are accessible at the measurement date for identical, unrestricted assets or liabilities;
- Level 2: Quoted prices for similar assets and liabilities in active markets, quoted prices in markets that are not active, or inputs which are observable, either directly or indirectly, for substantially the full term of the asset or liability; or
- Level 3: Prices or valuation techniques that require inputs that are both significant to the fair value measurement and unobservable (i.e., supported by little or no market activity).

To the extent that the valuation is based on models or inputs that are less observable or unobservable in the market, the determination of fair value requires more judgment. A financial instrument's level within the fair value hierarchy is based on the lowest level of any input that is significant to the fair value measurement.

Our cash equivalents and marketable debt securities are reported at fair value determined using Level 1 and Level 2 inputs (Note 6). The fair value of our foreign currency forward contracts is calculated using Level 2 inputs, which include currency spot rates, forward rates, interest rate curve and credit or non-performance risk.

Inventory

Inventory is recorded at the lower of cost or net realizable value, with cost determined principally using first-in, first-out method. We periodically review the composition of inventory in order to identify excess, obsolete, slow-moving or otherwise unsaleable items. If unsaleable items are observed and there are no alternate uses for the inventory, we will record a write-down to net realizable value in the period that the decline in value is first recognized through a charge to cost of sales. The determination of whether inventory costs will be realizable requires estimates by management. If actual market conditions are less favorable than projected by management, additional write-downs of inventory may be required. We also assess whether we have any excess firm, non-cancelable, purchase commitment liabilities, resulting from our supply agreements with third-party vendors. The determination of net realizable value and firm purchase commitment liabilities requires judgment, including consideration of many factors, such as estimates of future product demand, product net selling prices, current and future market conditions, potential product obsolescence, expiration and utilization of raw materials under firm purchase commitments and contractual minimums, among others. We hold raw materials beyond our one year forecasted production plan, which were classified as non-current and included in other non-current assets in our consolidated balance sheets.

Pre-launch Inventory

Costs relating to raw materials and production of inventory in preparation for product launch prior to regulatory approval are capitalized when future commercialization is considered probable, the future economic benefit is expected to be realized, and we believe that material uncertainties related to the ultimate regulatory approval have been significantly reduced. For pre-launch inventory that is capitalized, we consider a number of factors based on the information available at the time, including the product candidate's current status in the drug development and regulatory approval process, results from the related clinical trials, results from meetings with relevant regulatory agencies prior to the filing of regulatory applications, potential impediments to the approval process such as product safety or efficacy, historical experience, viability of commercialization and market trends. As of December 31, 2025 and 2024, we did not have any capitalized pre-launch inventory on our consolidated balance sheets.

Property, Plant and Equipment

Property, plant and equipment are stated at cost, net of accumulated depreciation. Depreciation is calculated using the straight-line method over the estimated useful lives of the assets. The estimated useful lives of property, plant and equipment are described below:

	Estimated Useful Life
Land and permanent land improvements	Not depreciated
Limited-life land improvements	Determined on a project-by-project basis
Buildings and building improvements	Up to 40 years
Manufacturing and laboratory equipment	5-12 years
Leasehold improvements	Lesser of estimated useful life of improvement or remaining life of related lease
Computer equipment and software	3 to 5 years
Furniture, fixtures and other	5 years
Right-of-use asset, financing	Lease term

Construction in progress includes direct costs related to the construction of various property, plant and equipment, and is stated at original cost. Once the asset is placed into service, these capitalized costs will be allocated to certain property, plant and equipment categories and will be depreciated over the estimated useful life of the underlying assets.

Goodwill and Intangible Assets

Goodwill represents the excess of purchase price over the fair value of net assets acquired and is carried at cost. Goodwill is tested at least annually for impairment by assessing qualitative factors in determining whether it is more likely than not that the fair value of net assets is below their carrying amounts. To date, an impairment of goodwill has not been recorded.

Finite-lived intangible assets primarily consist of acquired intellectual property and regulatory approval and commercial sales milestone payments. These assets are recorded at cost, net of accumulated amortization, and amortized on a straight-line basis over their estimated useful lives.

Impairment of Long-Lived Assets, including Intangibles and Lease Right-of-Use Assets

We evaluate our long-lived assets, which consist of property, plant and equipment, finite-lived intangibles and right-of-use assets to determine if facts and circumstances indicate that the carrying amount of assets may not be recoverable. If such facts and circumstances exist, we assess the recoverability of the long-lived assets by comparing the projected future undiscounted net cash flows associated with the related asset or group of assets over their remaining lives against their respective carrying amounts. If such review indicates that such cash flows are not expected to be sufficient to recover the recorded value of the assets, the assets are written down to their estimated fair values based on the expected discounted future cash flows attributable to the assets or based on appraisals. Impairment expenses for the years ended December 31, 2025, 2024 and 2023 were immaterial.

Leases

Leases are classified at their commencement date, which is defined as the date on which the lessor makes the underlying asset available for use by the lessee, as either operating or finance leases based on the economic substance of the agreement. We recognize lease right-of-use assets and related liabilities in our consolidated balance sheets for both operating and finance leases. Lease liabilities are measured at the lease commencement date as the present value of the future lease payments using the interest rate implicit in the lease. If the rate implicit is not readily determinable, we will utilize our incremental borrowing rate as of the lease commencement date. Lease right-of-use assets are initially measured as the lease liability plus direct costs and prepaid lease payments less lease incentives. The lease term is the non-cancelable period of the lease and includes options to extend or terminate the lease when it is reasonably certain that an option will be exercised.

We recognize operating lease cost in operating expenses in our consolidated statements of operations, inclusive of rent escalation provisions and rent holidays, on a straight-line basis over the respective lease term. For our finance leases, we recognize depreciation expense associated with the leased asset acquired and recognize interest expense related to the portion of the financing in our consolidated statements of operations.

We do not separate non-lease components from lease components for all classes of underlying assets. We do not recognize right-of-use assets and lease liabilities for leases with a lease term of 12 months or less. Instead, these lease payments are recognized in the statements of operations on a straight-line basis over the lease term.

Collaboration Arrangements

We analyze our collaboration arrangements to assess whether they are within the scope of ASC 808 (*Collaborative Arrangements*) to determine whether such arrangements involve joint operating activities performed by parties that are both active participants in the activities and exposed to significant risks and rewards that are dependent on the commercial success of such activities. To the extent the arrangement is within the scope of ASC 808, we assess whether aspects of the arrangement between us and our collaborator are within the scope of other accounting literature. If we conclude that some or all aspects of the arrangement represent a transaction with a customer, we account for those aspects of the arrangement within the scope of ASC 606 *(Revenue from Contracts with Customers*). See our "Revenue Recognition" policy within Note 2 for additional discussion of revenue recognition under these types of arrangements. If we conclude that some or all aspects of the arrangement are within the scope of ASC 808 and do not represent a transaction with a customer, we recognize our allocation of the shared costs incurred with respect to the jointly conducted activities as a component of the related expense in the period incurred. Additionally, for any payments related to capital expenditures that are partially reimbursed by a collaborator, to the extent the underlying capital costs qualify for capitalization, any reimbursements received will offset the capitalized cost of the asset.

Research and Development Funding Arrangements

We have entered into research and development funding arrangements to support a targeted programs, under which a third party has committed to fund us up to a specified amount. Contingent upon the regulatory approval of the program, we are obligated to pay certain milestones and royalties as outlined in the respective agreements. We account for funding received under research and development arrangements as a reduction to expenses when substantive financial risk is transferred to the funding party. These arrangements are accounted for as an obligation to conduct research and development activities. The funding is recognized proportionally as the related costs are incurred, using an input method. Contingent payments, such as milestones or royalties, will be recognized separately upon achievement of the related conditions.

Research and Development Costs

Research and development costs are expensed as incurred. Research and development expenses consist of costs incurred in performing research and development activities, including salaries and benefits, facilities costs, overhead costs, contract services, and other outside costs. The value of goods and services received from contract research organizations and contract manufacturing organizations in the reporting period are estimated based on the level of services performed, and progress in the period in cases when we have not received an invoice from the supplier. Research and development costs also include costs and shared cost associated with third-party collaboration arrangements, including upfront fees and milestones paid to third-parties in connection with technologies that had not reached technological feasibility and did not have an alternative future use.

Assets that are acquired or constructed for research and development activities and that have alternative future uses, in research and development projects or otherwise, are capitalized and depreciated over their useful lives. However, the costs of equipment or facilities that are acquired or constructed and intangibles that are purchased from others for a particular research and development project, and that have no alternative future uses and therefore no separate economic values, are considered research and development costs and expensed when incurred.

Advertising Costs

Costs associated with advertising are expensed as incurred and are included in selling, general and administrative expense in the consolidated statements of operations. Advertising expenses were $133 million in 2025, $146 million in 2024, and $204 million in 2023.

Stock-Based Compensation

We issue stock-based awards to employees and non-employees, generally in the form of stock options, restricted stock units (RSUs), and performance stock units (PSUs). We account for our stock-based compensation awards in accordance with ASC 718 *(Compensation—Stock Compensation)*. Most of our stock-based awards have been made to employees. We measure compensation cost for equity awards at their grant-date fair value and recognize compensation expense over the requisite service period, which is generally the vesting period, on a straight-line basis. The grant date fair value of stock options is estimated using the Black-Scholes option pricing model, which requires management to make assumptions with respect to the fair value of our common stock on the grant date, including the expected term of the award, the expected volatility of our stock, calculated based on a period of time generally commensurate with the expected term of the award, risk-free interest rates and expected dividend yields of our stock.

Prior to 2025, we estimated the expected term of our stock options using the simplified method, whereby, the expected term equals the average of the vesting term and the original contractual term of the option. During 2025, we used historical exercise data in estimating the expected term of stock options, utilizing a methodology that combines historical settlement data with a hypothetical settlement pattern for outstanding options, based on estimated time to arrival at-the-money and adjusted for employee departure rates. The change in methodology used during 2025 did not have a material impact on stock-based compensation and our financial statements. However, during the fourth quarter of 2025, in connection with our stock option exchange program (see Note 13), which resulted in changes to our stock option population, we determined that we no longer have sufficient historical exercise data to provide a reasonable basis upon which to estimate the expected term, as such data is no longer indicative of future exercise patterns. Accordingly, beginning in 2026, we will estimate the expected term of stock options using the simplified method. We will continue to apply this process until a sufficient amount of historical information regarding the expected term becomes available and we believe that historical experience is relevant to our expectations for current grants. The expected volatility is based on our stock price volatility and incorporates both historical volatility of our stock price and implied volatility derived from the price of exchange traded options on our stock.

The grant date fair value of RSUs is estimated based on the fair value of our underlying common stock. For performance-based stock awards, we recognize stock-based compensation expense over the requisite service period using the accelerated attribution method when achievement is probable. We classify stock-based compensation expense in our consolidated statements of operations in the same manner in which the award recipient's salary and related costs are classified or in which the award recipient's service payments are classified. We made an accounting policy election to recognize forfeitures of stock-based awards as they occur.

Income Taxes

We account for income taxes based on an asset and liability approach. We recognize deferred tax assets and liabilities for the expected future tax consequences of temporary differences between the financial reporting and tax bases of assets and liabilities. These

differences are measured using the enacted statutory tax rates and laws that will be in effect when the differences are expected to reverse. Valuation allowances are provided when the expected realization of deferred tax assets does not meet a “more likely than not” criterion. We periodically reassess the need for valuation allowances on our deferred tax assets, considering both positive and negative

evidence to evaluate whether it is more likely than not that all or a portion of such assets will not be realized. We make estimates and judgments about our future taxable income that are based on assumptions that are consistent with our plans and estimates. Should the actual amounts differ from our estimates, the amount of our valuation allowance could be materially impacted. Changes in these estimates may result in significant increases or decreases to our tax provision in a period in which such estimates are changed, which in turn would affect net income or loss. We recognize tax benefits from uncertain tax positions if we believe the position is more likely than not to be sustained on examination by the taxing authorities based on the technical merits of the position. We make adjustments to these tax reserves when facts and circumstances change, such as the closing of a tax audit or the refinement of an estimate. The provision for income taxes includes the effects of any reserves for uncertain tax positions, as well as the related net interest and penalties.

Earnings (Loss) per Share

We calculate diluted net earnings (loss) per share attributable to common stockholders by dividing net earnings (loss) by the weighted average number of common shares outstanding after giving consideration to the dilutive effect of restricted stock units, performance stock units, stock options, and shares under the employee stock purchase plan that are outstanding during the period. For periods in which we have generated a net loss, the basic and diluted net loss per share attributable to common stockholders are the same, as the inclusion of the potentially dilutive securities would be anti-dilutive.

Comprehensive Income (Loss)

Comprehensive income (loss) includes net income (loss) and other comprehensive income (loss) for the period. Other comprehensive income (loss) consists of unrealized gains and losses on our investments, derivatives designated as hedging instruments, and foreign currency translation, as well as, pension and postretirement obligation adjustments. Total comprehensive income (loss) for all periods presented has been disclosed in the consolidated statements of comprehensive income (loss).

The components of accumulated other comprehensive loss for the years ended December 31, 2025 and 2024 were as follows (in millions):

	Unrealized Gains on Available-for-Sale Securities	Pension and Postretirement Obligation Adjustments	Losses (gains) on Foreign Currency Translation	Total
Accumulated other comprehensive loss, balance at December 31, 2023	$ (114)	$ (9)	$ —	$ (123)
Other comprehensive income	124	(3)	(8)	113
Accumulated other comprehensive loss, balance at December 31, 2024	10	(12)	(8)	(10)
Other comprehensive income	28	16	11	55
Accumulated other comprehensive income, balance at December 31, 2025	$ 38	$ 4	$ 3	$ 45

Share Repurchases

Shares of our common stock repurchased pursuant to our repurchase programs are retired. The purchase price of such repurchased shares of common stock is recorded as a reduction to additional paid-in-capital. If the balance in additional paid-in-capital is exhausted, the excess is recorded as a reduction to retained earnings.

Recently Issued Accounting Standards

From time to time, new accounting pronouncements are issued by the FASB or other standard setting bodies and adopted by us as of the specified effective date. Except as noted below, we believe that the impact of recently issued standards that are not yet effective will not have a material impact on our consolidated financial statements and disclosures.

In December 2023, the FASB issued ASU No. 2023-09, *Income Taxes (Topic 740): Improvements to Income Tax Disclosures*. This ASU requires disaggregated information about a reporting entity's effective tax rate reconciliation as well as additional information on income taxes paid. The standard requires entities to disclose federal, state, and foreign income taxes in their rate reconciliation tables and elaborate on reconciling items that exceed a quantitative threshold. Additionally, it requires an annual disclosure of income taxes paid, net of refunds, categorized by jurisdiction based on a quantitative threshold. The ASU is effective on a prospective basis for

annual periods beginning after December 15, 2024, and early adoption is permitted. We adopted this ASU prospectively in the fourth quarter of 2025, and the required disclosures are included in Note 14, Income Taxes.

In November 2024, the FASB issued ASU 2024-03, *Income Statement - Reporting Comprehensive Income - Expense Disaggregation Disclosures (Subtopic 220-40): Disaggregation of Income Statement Expenses*. This ASU requires entities to disclose, on an annual and interim basis, disaggregated information in the footnotes related to certain expense categories included in income statement line items. Specifically, entities are expected to provide tabular disclosures for prescribed categories such as inventory purchases, employee compensation, depreciation, and intangible asset amortization for each relevant expense caption. The standard also requires disclosure of total selling expenses and a definition of those expenses in annual filings. Any remaining amounts not quantitatively disclosed are expected to be described qualitatively. This ASU is effective for fiscal years beginning after December 15, 2026 and interim periods beginning after December 15, 2027. Early adoption permitted, and the standard may be applied on a prospective or retrospective basis. We are currently assessing the impact that this new accounting standard will have on our consolidated financial statement disclosures.

In September 2025, the FASB issued ASU No. 2025-06, *Intangibles - Goodwill and Other - Internal-Use Software (Subtopic 350-40): Targeted Improvements to the Accounting for Internal-Use Software.* This ASU makes targeted amendments to the accounting for and disclosure of software costs under ASC 350-40. The amendments modernize the guidance to reflect current software development practices, including nonlinear development approaches, and remove references to "development stages." Under the ASU, the following two criteria must be met for entities to begin capitalizing software costs: (1) management, with the relevant authority, implicitly or explicitly authorizes and commits to funding a computer software project, and (2) it is probable that the project will be completed and the software will be used to perform the function intended (referred to as the "probable-to-complete recognition threshold"). The ASU clarifies that this threshold would not be met when there is "significant uncertainty associated with the development activities of the software (referred to as 'significant development uncertainty')." The ASU is effective for all entities for annual reporting periods beginning after December 15, 2027, and for interim reporting periods within those fiscal years. Early adoption is permitted, and entities may apply the amendments prospectively, retrospectively, or using a modified prospective transition approach. We are currently evaluating the impact that this new accounting standard will have on our consolidated financial statements and disclosures.

In September 2025, the FASB issued ASU No. 2025-07, *Derivatives and Hedging (Topic 815) and Revenue from Contracts with Customers (Topic 606): Derivatives Scope Refinements and Scope Clarification for Share-Based Noncash Consideration From a Customer in a Revenue Contract*. This ASU expands the scope exceptions in the derivatives guidance to exclude certain non-exchange-traded contracts with underlyings based on the operations or activities of one of the parties to the contract, including the occurrence or nonoccurrence of an event specific to those operations or activities. The ASU also clarifies that share-based noncash consideration received from a customer in exchange for goods or services should be accounted for as noncash consideration under ASC 606 unless and until the entity's right to receive or retain such consideration becomes unconditional. The ASU is effective for annual reporting periods beginning after December 15, 2026, including interim periods within those annual reporting periods. Early adoption is permitted. We are currently evaluating the impact that adoption of this new accounting standard will have on our consolidated financial statements and disclosures.

In December 2025, the FASB issued ASU No. 2025-10, *Government Grants (Topic 832): Accounting for Government Grants Received by Business Entities*. This ASU establishes guidance on the recognition, measurement, presentation, and disclosure of government grants received by business entities. The guidance defines a government grant as a transfer of a monetary asset or a tangible nonmonetary asset from a government to a business entity other than in an exchange transaction and excludes transactions within the scope of other U.S. GAAP. Under the ASU, government grants are classified as either grants related to an asset or grants related to income, and recognition is permitted only when it is probable that the entity will comply with the conditions attached to the grant and that the grant will be received. The ASU permits alternative presentation approaches depending on the nature of the grant and requires disclosures regarding the nature of the grant, affected financial statement line items, and significant terms and conditions. The ASU is effective for public business entities for annual reporting periods beginning after December 15, 2028, including interim periods within those annual reporting periods. Early adoption is permitted, and the standard may be applied on a modified prospective, modified retrospective, or full retrospective basis. We are currently evaluating the impact of this new accounting standard and do not expect its adoption to have a material impact on our consolidated financial statements and disclosures.

3. Net Product Sales

Net product sales by customer geographic location were as follows for the periods presented (in millions):

	Years Ended December 31,		
	2025	2024	2023
United States	$ 1,165	$ 1,726	$ 1,720
Europe	50	573	1,353
Rest of world	603	810	3,598
Total	$ 1,818	$ 3,109	$ 6,671

Net product sales by product were as follows (in millions):

	Years Ended December 31,		
	2025	2024	2023
COVID[(1)]	$ 1,810	$ 3,084	$ 6,671
RSV	8	25	—
Total	$ 1,818	$ 3,109	$ 6,671

[(1)]Includes sales of Spikevax and mNEXSPIKE.

As of December 31, 2025, we have three commercial products, our COVID vaccines, Spikevax and mNEXSPIKE, and our RSV vaccine, mRESVIA. mRESVIA was approved by the FDA in May 2024 for adults aged 60 years and older, and in June 2025, the approved use was expanded to include adults aged 18 through 59 years who are at increased risk for lower respiratory tract disease (LRTD) caused by RSV. We launched commercial sales of mRESVIA in the third quarter of 2024. In May 2025, mNEXSPIKE was approved for use in adults aged 65 years and older, as well as individuals aged 12 through 64 years with at least one underlying risk factor. We launched commercial sales of mNEXSPIKE in the third quarter of 2025.

We sell our COVID and RSV vaccines to the commercial market as well as to foreign governments and international organizations. In the U.S., our COVID and RSV vaccines are sold primarily to wholesalers and distributors, and to a lesser extent, directly to retailers and healthcare providers. Wholesalers and distributors typically do not make upfront payments to us. Net product sales are recognized net of estimated wholesaler chargebacks, invoice discounts for prompt payments and pre-orders, provisions for sales returns and government rebates, and other related deductions.

The following table summarizes product sales provision for the periods presented (in millions):

	Years Ended December 31,		
	2025	2024	2023
Gross product sales	$ 3,304	$ 4,517	$ 8,203
Product sales provision:			
Wholesaler chargebacks, discounts and fees	(1,037)	(1,141)	(976)
Returns, rebates and other fees [1]	(449)	(267)	(556)
Total product sales provision	$ (1,486)	$ (1,408)	$ (1,532)
Net product sales	$ 1,818	$ 3,109	$ 6,671

[1] Includes an adjustment of approximately $216 million in 2024, reflecting a reduction in prior year provision estimates, primarily related to returns and chargebacks for the previous COVID vaccine season. Adjustments recorded in 2025 related to prior year provision estimates were not material.

The following table summarizes the activities related to product sales provision recorded as accrued liabilities for the year ended December 31, 2025 (in millions):

	Returns, rebates and other fees
Balance at December 31, 2024	$ (370)
Provision related to sales made in current period	(436)
Provision related to sales made in prior periods	(13)
Payments and returns related to sales made in current period	28
Payments and returns related to sales made in prior periods	282
Balance at December 31, 2025	$ (509)

4. Other Revenue

The following table summarizes other revenue for the periods presented (in millions):

	Years Ended December 31,		
	2025	2024	2023
Grant revenue	$ 22	$ 37	$ 94
Collaboration revenue (Note 5)	13	48	83
Licensing and royalty revenue	11	42	—
Stand-ready manufacturing revenue	80	$ —	$ —
Total other revenue	$ 126	$ 127	$ 177

Grant Revenue

In April 2020, we entered into an agreement with the Biomedical Advanced Research and Development Authority (BARDA), a division of the Administration for Strategic Preparedness and Response (ASPR) within the U.S. Department of Health and Human Services (HHS), for an award of up to $483 million to accelerate development of our original COVID vaccine, mRNA-1273. The agreement has been subsequently amended to provide for additional commitments to support various late-stage clinical development efforts of our original COVID vaccine, including a 30,000 participant Phase 3 study, pediatric clinical trials, adolescent clinical trials and pharmacovigilance studies. The maximum award from BARDA, inclusive of all amendments, was approximately $1.8 billion. All contract options have been exercised. The BARDA contract concluded on June 15, 2025, upon completion of all contractual deliverables. We accrued and recognized revenue through that date for eligible costs incurred in accordance with the agreement. As of December 31, 2025, the remaining available funding, net of revenue earned, was approximately $62 million. While this funding remains available, we do not expect to recognize material additional revenue, as the contract has concluded and all obligations have been fulfilled. Final billing and closeout activities are ongoing and may result in immaterial adjustments to the recognized revenue amount.

The following table summarizes grant revenue for the periods presented (in millions):

	Years Ended December 31,		
	2025	2024	2023
BARDA	$ 9	$ 34	$ 88
Other grant revenue	13	3	6
Total grant revenue	$ 22	$ 37	$ 94

Collaboration Revenue

We have entered into collaboration agreements with strategic collaborators to accelerate the discovery and advancement of potential mRNA medicines across therapeutic areas. As of December 31, 2025, 2024 and 2023, we had collaboration agreements with Merck & Co., Inc (Merck), Vertex Pharmaceuticals Incorporated and Vertex Pharmaceuticals (Europe) Limited (together, Vertex), and others. See Note 5 to for further description of these collaboration agreements.

The following table summarizes our total consolidated net revenue from our strategic collaborators for the periods presented (in millions):

	Years Ended December 31,		
Collaboration Revenue by Strategic Collaborator:	2025	2024	2023
Vertex	$ 13	$ 23	$ 82
Other	—	25	1
Total collaboration revenue	$ 13	$ 48	$ 83

Licensing and Royalty Revenue

In April 2024, we entered a non-exclusive out-licensing agreement with a pharmaceutical company based in Japan for mRNA COVID-related intellectual property for the territory of Japan. Under the terms of the agreement, we received an upfront payment of $50 million, which included a $20 million prepayment creditable against future royalties. Additionally, we are entitled to receive low double-digit royalties on the net sales of the company's COVID product. Upon execution of the agreement, we recognized $30 million of the upfront payment as other revenue in our consolidated statements of operations. The remaining $20 million was recorded as deferred revenue in our consolidated balance sheets and recognized as royalty revenue as the underlying sales occurred. In accordance with the terms of the agreement, the $20 million was fully recognized by the end of the first quarter of 2025.

Stand-Ready Manufacturing Revenue

Stand-ready manufacturing revenue relates to amounts recognized under long-term strategic agreements with government entities for maintaining mRNA manufacturing capacities and support pandemic readiness.

5. Collaboration Agreements and Research and Development Funding Arrangement

Merck – Personalized mRNA Cancer Vaccines (Intismeran Autogene)

In June 2016, we entered into a Collaboration and License Agreement, which was subsequently amended in 2018, with Merck for the development and commercialization of personalized mRNA cancer vaccines (PCV), also known as Individualized Neoantigen Therapy (INT), which has been assigned the generic name intismeran autogene. This agreement was subsequently amended in 2018. Under the PCV Agreement, we received an upfront payment of $200 million from Merck and we were responsible for designing and researching intismeran, providing manufacturing capacity and manufacturing intismeran and conducting Phase 1 and Phase 2 clinical trials for intismeran, alone and in combination with KEYTRUDA (pembrolizumab), Merck's anti-PD-1 therapy, all in accordance with an agreed upon development plan and budget. We concluded that the collaboration arrangement was governed by the revenue recognition standard ASC 606.

In September 2022, Merck exercised its option for intismeran, including mRNA-4157, under the terms of the agreement and in October 2022 paid us an option exercise fee of $250 million. Following this exercise, the Merck Participation Term commenced, under which we and Merck collaborate on development and potential commercialization of PCV, with costs and any profits or losses generally

shared equally on a worldwide basis, subject to certain exceptions as outlined in the agreement. During the development phase, we are primarily responsible for process development and the manufacture of intismeran materials, while Merck generally leads clinical trials. We concluded that the collaboration arrangement under the Merck Participation Term is within the scope of ASC 808. For the years ended December 31, 2025, 2024 and 2023, we recognized expenses, net of Merck's reimbursements, of $407 million,

$390 million, and $184 million, respectively, related to the PCV collaboration under the Merck Participation Term. Additionally, the net cost recovery for capital expenditures during the same periods were $24 million, $109 million, and $102 million, respectively, which were applied to reduce the capitalized cost of the assets.

Development and Commercialization Funding Arrangement with Blackstone Life Sciences (Blackstone)

In March 2024, we entered into a development and commercialization funding arrangement with Blackstone, under which Blackstone has committed to providing up to $750 million in funding to us. This funding supports the development of our investigational mRNA-based influenza vaccine. Contingent upon regulatory approval in the U.S. and only if the approval is dependent on data from the funded activities, Blackstone will be entitled to receive low single-digit percentage royalties and up to $750 million in sales milestone payments. These payments are based on net sales of our future influenza and combination vaccines, with sales milestone payments contingent upon achieving specified cumulative net sales targets.

Given the substantive transfer of financial risk to Blackstone, we account for this arrangement as an obligation to conduct research and development activities. The funding is recognized as a reduction to the expenses of our mRNA-based influenza program. This reduction is recognized proportionally as the related costs are incurred, based on an input method. For the year ended December 31, 2025 and 2024, we recorded research and development expense reductions of $340 million and $267 million, respectively. As of December 31, 2025 and 2024, we had a research and development funding liability of $43 million and $58 million, respectively, related to the advance funding received from Blackstone.

Vertex – Strategic Alliance in Cystic Fibrosis

In July 2016, we entered into a Strategic Collaboration and License Agreement (Vertex Agreement), with Vertex Pharmaceuticals Incorporated, and Vertex Pharmaceuticals (Europe) Limited, together, Vertex. The Vertex Agreement, which was amended in July 2019 (2019 Vertex Amendment), is aimed at the discovery and development of potential mRNA medicines for the treatment of cystic fibrosis (CF) by enabling cells in the lungs of people with CF to produce functional cystic fibrosis transmembrane conductance regulator (CFTR) proteins. Pursuant to the Vertex Agreement, we lead discovery efforts during an initial research period, leveraging our platform technology and mRNA delivery expertise along with Vertex's scientific experience in CF biology and the functional understanding of CFTR. Vertex is responsible for conducting development and commercialization activities for candidates and products that arise from the strategic alliance, including the costs associated with such activities. Subject to customary "back-up" supply rights granted to Vertex, we exclusively manufacture (or have manufactured) mRNA for preclinical, clinical and commercialization purposes. This collaboration arrangement is accounted for under ASC 606 and currently ongoing.

Immatics – Strategic Multi-Platform Collaboration to Develop Oncology Therapeutics

In September 2023, we entered into a strategic collaboration with Immatics to jointly discover and develop T cell-redirecting cancer immunotherapies. Upon effectiveness of the agreement in October 2023, we made an upfront payment of $120 million, recognized as research and development expense. Immatics is also eligible for research funding, milestone payments, tiered royalties on global net sales of TCER® products and certain vaccine products that are commercialized under the agreement. Additionally, Immatics has an option to enter into a global profit and loss share arrangement for the most advanced TCER®.

In addition to the collaborative arrangements mentioned above, we have other collaborative and licensing arrangements that we do not consider to be individually significant to our business at this time. Pursuant to these agreements, we may be required to make upfront payments and payments upon achievement of various development, regulatory and commercial milestones, which in the aggregate could be significant. Future milestone payments, if any, will be reflected in our consolidated financial statements when the corresponding events become probable. In addition, we may be required to pay significant royalties on future sales if products related to these arrangements are commercialized.

6. Financial Instruments and Fair Value Measurements

Cash and Cash Equivalents and Investments

The following tables summarize our cash, cash equivalents, and available-for-sale securities by significant investment category at December 31, 2025 and 2024 (in millions):

	December 31, 2025						
	Amortized Cost	Unrealized Gains	Unrealized Losses	Fair Value	Cash and Cash Equivalents	Current Marketable Securities	Non-Current Marketable Securities
Cash and cash equivalents	$ 2,595	$ —	$ —	$ 2,595	$ 2,595	$ —	$ —
Available-for-sale:							
Certificates of deposit	91	—	—	91	—	86	5
U.S. treasury bills	653	—	—	653	—	653	—
U.S. treasury notes	2,188	5	(3)	2,190	—	1,185	1,005
Corporate debt securities	2,535	6	(1)	2,540	—	1,250	1,290
Government debt securities	66	—	—	66	—	30	36
Total	$ 8,128	$ 11	$ (4)	$ 8,135	$ 2,595	$ 3,204	$ 2,336

	December 31, 2024						
	Amortized Cost	Unrealized Gains	Unrealized Losses	Fair Value	Cash and Cash Equivalents	Current Marketable Securities	Non-Current Marketable Securities
Cash and cash equivalents	$ 1,927	$ —	$ —	$ 1,927	$ 1,927	$ —	$ —
Available-for-sale:							
Certificates of deposit	52	—	—	52	—	52	—
U.S. treasury bills	786	—	—	786	—	786	—
U.S. treasury notes	3,048	3	(15)	3,036	—	1,958	1,078
Corporate debt securities	3,590	3	(13)	3,580	—	2,172	1,408
Government debt securities	138	—	—	138	—	130	8
Total	$ 9,541	$ 6	$ (28)	$ 9,519	$ 1,927	$ 5,098	$ 2,494

The amortized cost and estimated fair value of available-for-sale securities, by contractual maturity at December 31, 2025 and 2024 were as follows (in millions):

	December 31, 2025	
	Amortized Cost	Estimated Fair Value
Due in one year or less	$ 3,199	$ 3,204
Due after one year through five years	2,334	2,336
Total	$ 5,533	$ 5,540

	December 31, 2024	
	Amortized Cost	Estimated Fair Value
Due in one year or less	$ 5,106	$ 5,098
Due after one year through five years	2,508	2,494
Total	$ 7,614	$ 7,592

In accordance with our investment policy, we place investments in investment grade securities with high credit quality issuers, and generally limit the amount of credit exposure to any one issuer. We evaluate securities for impairment at the end of each reporting period. We did not record any impairment charges related to our available-for-sale securities during the years ended December 31, 2025, 2024, and 2023. We did not recognize any credit-related allowance to available-for-sale securities as of December 31, 2025 and 2024.

The following table summarizes the amount of gross unrealized losses and the estimated fair value for our available-for-sale securities in an unrealized loss position by length of time the securities have been in an unrealized loss position at December 31, 2025 and 2024 (in millions):

	Less than 12 Months		12 Months or More		Total	
	Gross Unrealized Losses	Estimated Fair Value	Gross Unrealized Losses	Estimated Fair Value	Gross Unrealized Losses	Estimated Fair Value
As of December 31, 2025:						
U.S. treasury bills	$ —	$ 111	$ —	$ —	$ —	$ 111
U.S. treasury notes	(1)	176	(2)	235	(3)	411
Corporate debt securities	—	608	(1)	40	(1)	648
Government debt securities	—	36	—	8	—	44
Total	$ (1)	$ 931	$ (3)	$ 283	$ (4)	$ 1,214
As of December 31, 2024:						
U.S. treasury bills	$ —	$ 101	$ —	$ —	$ —	$ 101
U.S. treasury notes	(2)	729	(13)	960	(15)	1,689
Corporate debt securities	(4)	843	(9)	1,646	(13)	2,489
Government debt securities	—	—	—	37	—	37
Total	$ (6)	$ 1,673	$ (22)	$ 2,643	$ (28)	$ 4,316

At December 31, 2025 and 2024, we held 108 and 252 available-for-sale securities, respectively, out of our total investment portfolio that were in a continuous unrealized loss position. We neither intend to sell these investments nor conclude that we are more-likely-than-not that we will have to sell them before recovery of their carrying values. We also believe that we will be able to collect both principal and interest amounts due to us at maturity.

Assets and Liabilities Measured at Fair Value on a Recurring Basis

The following tables summarize our financial assets and liabilities measured at fair value on a recurring basis as of December 31, 2025 and 2024 (in millions):

	Fair value at December 31, 2025	Fair Value Measurement Using	
		Level 1	Level 2
Assets:			
Money market funds	$ 963	$ 963	$ —
Certificates of deposit	91	—	91
U.S. treasury bills	1,445	—	1,445
U.S. treasury notes	2,190	—	2,190
Corporate debt securities	3,163	—	3,163
Government debt securities	66	—	66
Equity investments[1]	6	6	—
Derivative instruments	1	—	1
Total	$ 7,925	$ 969	$ 6,956
Liabilities:			
Derivative instruments	$ 4	$ —	$ 4

	Fair value at December 31, 2024	Fair Value Measurement Using	
		Level 1	Level 2
Assets:			
Money market funds	$ 1,195	$ 1,195	$ —
Certificates of deposit	52	—	52
U.S. treasury bills	1,016	—	1,016
U.S. treasury notes	3,036	—	3,036
Corporate debt securities	3,763	—	3,763
Government debt securities	138	—	138
Equity investments[1]	14	14	—
Derivative instruments	10	—	10
Total	$ 9,224	$ 1,209	$ 8,015
Liabilities:			
Derivative instruments	$ 2	$ —	$ 2

[1]Investments in publicly traded equity securities with readily determinable fair values are recorded at quoted market prices for identical securities, with changes in fair value recorded in other expense net, in our consolidated statements of operations.

As of December 31, 2025 and 2024, we did not have non-financial assets or liabilities measured at fair value on a recurring basis.

For the years ended December 31, 2025, 2024 and 2023, we recognized net losses of $8 million, $52 million and $35 million, respectively, on equity investments from changes in fair value of the securities.

Fair Value of Other Financial Instruments

We estimate the fair value of our term loan using Level 2 inputs. The fair value of the term loan approximates its carrying value as of December 31, 2025, because the instrument bears interest at a variable rate that reflects current market rates. See Note 11 for additional information.

7. Inventory

Inventory, as of December 31, 2025 and 2024 consisted of the following (in millions):

	December 31,	
	2025	2024
Raw materials	$ 91	$ 63
Work in progress	29	26
Finished goods	33	28
Total inventory	$ 153	$ 117
Inventory, non-current[(1)]	$ 114	$ 150

[(1)] Consisted of raw materials with an anticipated consumption beyond one year. Inventory, non-current is included in other non-current assets in the consolidated balance sheets.

Inventory write-downs as a result of excess, obsolescence, scrap or other reasons, and losses on firm purchase commitments are recorded as a component of cost of sales in our consolidated statements of operations. For the years ended December 31, 2025, 2024, and 2023, inventory write-downs were $291 million, $495 million, and $2.2 billion, respectively. For the years ended December 31, 2025, 2024, and 2023, losses on firm purchase commitments were $24 million, $60 million, and $141 million, respectively. Inventory write-downs were mainly related to inventory in excess of expected demand and shelf-life expiration. Losses on firm purchase commitments were primarily related to excess raw material purchase commitments that will expire before the anticipated consumption of those raw materials.

These charges in 2025 and 2024 were primarily driven by the continued operation of the COVID vaccine market as a seasonal market, along with a decline in overall customer demand and excess inventory. Forecasting demand for COVID vaccines requires advance production planning ahead of each season, including production of multiple COVID vaccine products, which resulted in excess inventory and capacity as actual demand differed from forecasted demand. The higher charges in 2023 primarily reflect our strategic initiative to resize its manufacturing cost structure in response to the transition to an endemic seasonal COVID vaccine market. In the third quarter of 2023, we completed our long-range financial planning process, incorporating revised forecasts of vaccination rates. This resulted in the reassessment of future demand for our COVID vaccine, leading to a strategic initiative to resize our manufacturing cost structure. This initiative, launched in the same quarter, involved reassessing our inventory levels and renegotiating with our suppliers to reduce our purchase commitments related to raw materials which were not expected to be consumed before expiration. This initiative resulted in a raw materials write-down of $903 million, included in the total inventory write-down amount for the quarter.

As of December 31, 2025 and December 31, 2024, the accrued liability for losses on firm future purchase commitments in our consolidated balance sheets was $5 million and $60 million, respectively. As of December 31, 2025 and December 31, 2024, we had inventory on hand of $267 million and $267 million, respectively, inclusive of inventory for our COVID and RSV vaccines. Our raw materials and work-in-progress inventory have variable shelf lives. We expect that the majority of this inventory will be consumed over the next three years. The shelf life of Spikevax, our original COVID vaccine, is nine to twelve months. mNEXSPIKE has a shelf life of twelve months. The shelf life of mRESVIA, our RSV vaccine, is eighteen months.

8. Property, Plant and Equipment, Net

Property, plant and equipment, net as of December 31, 2025 and 2024 consisted of the following (in millions):

	December 31,	
	2025	2024
Land and land improvements	$ 78	$ 59
Building and building improvements	1,183	743
Manufacturing and laboratory equipment	542	344
Leasehold improvements	403	207
Furniture, fixtures and other	39	31
Computer equipment and software	196	150
Construction in progress	298	1,057
Right-of-use assets, financing (Note 10)	132	132
Total	2,871	2,723
Less: Accumulated depreciation	(737)	(527)
Property, plant and equipment, net	$ 2,134	$ 2,196

Depreciation and amortization expense for the years ended December 31, 2025, 2024, and 2023 was $211 million, $185 million, and $617 million, respectively.

9. Other Balance Sheet Components

Accounts Receivable, Net

Accounts receivable, net, as of December 31, 2025 and 2024 consisted of the following (in millions):

	December 31,	
	2025	2024
Accounts receivable	$ 368	$ 698
Less: Wholesalers chargebacks, discounts and fees	(181)	(340)
Less: Allowance for doubtful accounts	(3)	—
Accounts receivable, net	$ 184	$ 358

Prepaid Expenses and Other Current Assets

Prepaid expenses and other current assets, as of December 31, 2025 and 2024 consisted of the following (in millions):

	December 31,	
	2025	2024
Prepaid services	$ 169	$ 173
Down payments and prepayments related to manufacturing and materials	61	106
Interest receivable	42	59
Value added tax receivable	37	45
Prepaid income tax	33	28
Collaboration receivable	13	58
Income tax receivable	8	72
Other current assets	45	58
Prepaid expenses and other current assets	$ 408	$ 599

Other Non-Current Assets

Other non-current assets, as of December 31, 2025 and 2024 consisted of the following (in millions):

	December 31,	
	2025	2024
Income tax receivable, non-current	$ 161	$ 97
Inventory, non-current[(1)]	114	150
Down payments and prepayments, non-current	100	139
Deferred tax assets	81	81
Goodwill	52	52
Finite-lived intangible asset	45	40
Equity investments	6	14
Other	46	21
Other non-current assets	$ 605	$ 594

[(1)]Consisted of raw materials with an anticipated consumption beyond one year.

Accrued Liabilities

Accrued liabilities, as of December 31, 2025 and 2024 consisted of the following (in millions):

	December 31,	
	2025	2024
Provisions related to product sales (Note 3)	$ 509	$ 370
Compensation-related	420	312
Manufacturing	140	109
Development operations	106	120
Other external goods and services	57	131
Property, plant and equipment	45	99
Royalties	31	46
Raw materials	30	41
Commercial	23	45
Clinical trials	20	94
Loss on future firm purchase commitments[(1)]	5	60
Accrued liabilities	$ 1,386	$ 1,427

[(1)]Related to losses that are expected to arise from firm, non-cancellable, commitments for future raw material purchases (Note 7).

Other Current Liabilities

Other current liabilities, as of December 31, 2025 and 2024 consisted of the following (in millions):

	December 31,	
	2025	2024
Estimated reimbursements to wholesalers and distributors	$ 84	$ 103
Research and development funding liability (Note 5)	43	58
Lease liabilities - financing (Note 10)	25	23
Lease liabilities - operating (Note 10)	17	14
Other	16	23
Other current liabilities	$ 185	$ 221

Other Non-Current Liabilities

Other non-current liabilities, as of December 31, 2025 and 2024 consisted of the following (in millions):

	December 31,	
	2025	2024
Tax liabilities	$ 249	$ 231
Other	36	36
Other non-current liabilities	$ 285	$ 267

Deferred Revenue

The following table summarizes the activities in deferred revenue during the year ended December 31, 2025 (in millions):

	December 31, 2024	Additions	Deductions	December 31, 2025
Net product sales	$ 188	$ 108	$ (189)	$ 107
Other revenue	23	212	(90)	145
Total deferred revenue	$ 211	$ 320	$ (279)	$ 252

10. Leases

We have entered into various long-term, non-cancelable lease arrangements for our facilities and equipment, expiring at various times through 2039. Certain of these arrangements have free rent periods or escalating rent payment provisions. We recognize lease costs under such arrangements on a straight-line basis over the life of the lease. We have two main campuses in Massachusetts, our Moderna Science Center (MSC), located in Cambridge, which serves as our headquarters, and our Moderna Technology Center (MTC), located in Norwood. In addition, we own and lease facilities, including office, laboratory, and manufacturing sites, in various locations globally to support our research, development, manufacturing, and business operations.

Moderna Science Center

Our Cambridge campus previously included multiple leased properties at Technology Square and the MSC, a facility comprising approximately 462,000 square feet, that serves as our principal executive office, along with additional office and laboratory spaces. The MSC lease commenced during the third quarter of 2023 and has a term of 15 years, with options for two additional seven-year extensions. During the fourth quarter of 2023, we amended the expiration dates of our Technology Square leases to conclude in January 2025, as we transitioned operations to the MSC. As of December 31, 2024, we substantially exited our leased spaces at Technology Square, completing the consolidation of our Cambridge operations into the MSC.

Moderna Technology Center

The MTC is a multiple-building campus spanning approximately 722,000 square feet. that previously operated under long-term finance leases expiring in 2042, with options for three five-year extensions. The MTC has been a critical facility for our manufacturing, laboratory, and office operations. In December 2024, we completed the acquisition of the MTC campus, including the underlying land and buildings, for a total purchase price of $385 million. Upon acquisition, we derecognized the right-of-use assets and lease liabilities associated with the Norwood leases. The purchase price, after adjustments related to lease terminations, was allocated to land and buildings, with approximately $231 million recorded as property, plant, and equipment on our consolidated balance sheets as of December 31, 2024.

Operating and financing lease right-of-use assets and lease liabilities as of December 31, 2025 and 2024 were as follows (in millions):

	December 31,	
	2025	2024
Assets:		
Right-of-use assets, operating, net[1] [2]	$ 719	$ 759
Right-of-use assets, financing, net[3] [4]	42	65
Total	$ 761	$ 824
Liabilities:		
Current:		
Operating lease liabilities[5]	$ 17	$ 14
Financing lease liabilities[5]	25	23
Total current lease liabilities	42	37
Non-current:		
Operating lease liabilities, non-current	653	671
Financing lease liabilities, non-current	20	39
Total non-current lease liabilities	673	710
Total	$ 715	$ 747

[1] These assets are real estate related assets, which include land, office, manufacturing, and laboratory spaces.
[2] Net of accumulated amortization.
[3] These assets are related to contract manufacturing service agreements.
[4] Included in property, plant and equipment in the consolidated balance sheets, net of accumulated depreciation.
[5] Included in other current liabilities in the consolidated balance sheets.

The components of the lease costs were as follows for the periods presented (in millions):

	Years ended December 31,		
	2025	2024	2023
Operating lease costs	$ 89	$ 103	$ 88
Financing lease costs:			
Amortization of right-of-use assets, financing leases	24	22	500
Interest expense for financing lease liabilities	3	24	38
Total financing lease costs	$ 27	$ 46	$ 538
Short term lease costs	$ —	$ 13	$ 2
Variable lease costs	$ 22	$ 42	$ 113

Supplemental cash flow information relating to our leases was as follows for the periods presented (in millions):

	December 31,		
	2025	2024	2023
Cash paid for amounts included in measurement of lease liabilities:			
Operating cash flows used in operating leases	$ (71)	$ (78)	$ (93)
Operating cash flows used in financing leases	(3)	(19)	(39)
Financing cash flows used in financing leases	(23)	(12)	(292)
Operating lease non-cash items:			
Decrease in right-of-use assets related to lease modifications and reassessments	$ (10)	$ (4)	$ (67)
Right-of-use assets obtained in exchange for operating lease liabilities	—	100	714
Finance lease non-cash items:			
Decrease in right-of-use assets related to lease modifications and reassessments	$ —	$ (425)	$ (213)
Right-of-use assets obtained in exchange for financing lease liabilities	—	75	—
Changes in financing lease liabilities	(5)	2	3
Lease liability derecognized upon purchase of underlying leased asset	—	579	—

Weighted average remaining lease terms and discount rates as of December 31, 2025 and 2024 were as follows:

	December 31,	
	2025	2024
Remaining lease term:		
Operating leases	12 years	13 years
Finance leases	2 years	3 years
Discount rate:		
Operating leases	7.6 %	7.6 %
Finance leases	5.9 %	5.9 %

Future minimum lease payments under non-cancelable lease agreements as of December 31, 2025, were as follows (in millions):

Fiscal Year	Operating Leases	Financing Leases
2026	$ 65	$ 27
2027	78	20
2028	81	—
2029	82	—
2030	80	—
Thereafter	680	—
Total minimum lease payments	1,066	47
Less amounts representing interest	(396)	(2)
Present value of lease liabilities	$ 670	$ 45

11. Credit Agreement

In November 2025, we entered into a Credit and Guaranty Agreement (the Credit Agreement) with lenders led by Ares Capital Corporation, as administrative agent. The Credit Agreement provides for a senior secured term loan facility with aggregate term loan commitments of $1.5 billion, consisting of a $600 million initial term loan, which was funded at closing, and $900 million of delayed draw term loan commitments. The initial term loan matures on November 24, 2030. The delayed draw term loan commitments consist of (1) a $400 million delayed draw term loan facility (DDTL-1), which is available, subject to customary conditions, through November 24, 2027, and (2) a $500 million delayed draw term loan facility (DDTL-2), which is available, subject to customary conditions and the achievement of specified regulatory approval milestones for certain product candidates, through November 24, 2028.

Borrowings under the Credit Agreement bear interest at a variable rate equal to, at our option, (i) Term SOFR plus a margin of 5.50% or (ii) a base rate plus a margin of 4.50%. The base rate is calculated as the highest of (a) the Wall Street Journal prime rate, (b) the federal funds rate plus one half of one percent and (c) Term SOFR plus one percent. We are also required to pay commitment fees on the undrawn portions of DDTL-1 and DDTL-2. The interest rate applicable to the initial term loan was approximately 9.38% as of December 31, 2025.

The obligations under the Credit Agreement are guaranteed by certain of our subsidiaries and are secured by a first-priority lien on substantially all of our assets, in each case subject to customary exceptions and limitations. The Credit Agreement is subject to compliance with customary representations and warranties, affirmative covenants, restrictive covenants and events of default. The restrictive covenants, subject to specified limitations and exceptions, limit, among other things, our ability to incur additional indebtedness and liens, make certain investments, engage in certain fundamental changes, dispose of assets and make restricted payments. Events of default under the Credit Agreement include, among others, nonpayment of principal, interest or other amounts when due, failure to comply with covenants (subject to applicable notice and cure periods), breaches of certain representations and warranties, the occurrence of certain significant adverse events and certain insolvency-related events. The Credit Agreement also includes a financial covenant requiring us to maintain minimum cash and cash equivalents (as defined in the Credit Agreement, which primarily consist of our cash, cash equivalents, and available-for-sale securities) as of the last business day of each week of at least $500 million, increasing to $750 million if more than $1.0 billion is drawn under the Credit Agreement. The financial covenant is not required to be tested at any time that the trailing 30-day average market capitalization of the Company exceeds $5.0 billion and is subject to a customary equity cure. As of December 31, 2025, we were in compliance with the applicable terms and covenants under the Credit Agreement.

As of December 31, 2025, the initial term loan had an outstanding principal balance of $600 million and a carrying amount of $590 million, net of unamortized original issue discount and debt issuance costs, which was classified as long-term debt in our consolidated balance sheet. No amounts had been drawn under DDTL-1 or DDTL-2. The principal amount of $600 million is due in full at maturity, and no principal payments are required prior to that date.

12. Commitments and Contingencies

Legal Proceedings

We are a party to various legal proceedings and claims. Accruals are recognized for legal matters when a loss is both probable and reasonably estimable. As of December 31, 2025, no material contingent liabilities have been recognized. If a material loss is reasonably possible and we can estimate the amount or range of the loss, we disclose such information. Unless otherwise noted, either the outcome of these matters is not expected to be material, or the potential loss cannot be reasonably estimated.

From time to time, we may be a party to litigation, arbitration, or other legal proceedings in the course of our business. The outcome of such matters is inherently uncertain and often involves significant judgment in assessing risk and estimating potential exposure. While we do not currently expect any pending proceedings to have a material adverse effect on our financial position, results of operations, or cash flows, there can be no assurance that future developments will not have a material impact.

The following summarizes our significant legal proceedings and matters outstanding as of December 31, 2025.

We have brought patent-infringement actions against Pfizer Inc. (Pfizer), BioNTech SE (BioNTech) and related entities in the U.S. District Court for the District of Massachusetts, Germany, the Netherlands, the UK, Ireland and Belgium concerning our mRNA platform technology and disease-specific vaccine designs. Pfizer and BioNTech have commenced actions or asserted defenses seeking to revoke our patents in these jurisdictions.

Arbutus Biopharma Corporation (Arbutus) and Genevant Sciences GmbH (Genevant) have brought patent-infringement actions against us in the U.S. District Court for the District of Delaware, as well as in Canada, Japan, Switzerland and the Unified Patent Court (UPC) asserting patents concerning lipid nanoparticles. Arbutus and Genevant are seeking judgment of infringement of the asserted patents and monetary damages. The U.S. case has set trial to begin March 9, 2026.

GlaxoSmithKline Biologicals SA (GSK) has filed two complaints against us in the U.S. District Court for the District of Delaware asserting certain patents owned by GSK. GSK has filed two patent-infringement lawsuits against us in the UPC and two complaints in Spain concerning liposomes and modified liposomes for RNA delivery.

Northwestern University has filed a complaint against us in the U.S. District Court for the District of Delaware asserting U.S. patents concerning lipid nanoparticle technology.

Bayer CropSciences LLC, Monsanto Company, and Monsanto Technology, LLC have filed a complaint in U.S. District Court for the District of Delaware asserting a U.S. patent directed to methods of modifying gene sequences.

mNG Bio, LLC has filed a complaint against us in U.S District Court for the District of Massachusetts asserting a U.S. patent directed to a yellow-green fluorescent protein.

BioNTech SE has filed a complaint against us in the U.S. District Court for the District of Delaware asserting a U.S. patent directed to modified mRNA compositions encoding a spike protein fragment.

We are subject to shareholder class action and shareholder derivative litigation pending in the U.S. District Court for the District of Massachusetts related to statements about our RSV vaccine (mRNA-1345).

Indemnification Obligations

As permitted under Delaware law, we indemnify our officers, directors, and employees for certain events, occurrences while the officer, or director is, or was, serving at our request in such capacity. The term of the indemnification is for the officer's or director's lifetime.

We have standard indemnification arrangements in our leases for laboratory and office space that require us to indemnify the landlord against any liability for injury, loss, accident, or damage from any claims, actions, proceedings, or costs resulting from certain acts, breaches, violations, or non-performance under our leases.

We enter into indemnification provisions under our agreements with counterparties in the ordinary course of business, typically with business collaborators, contractors, clinical sites and customers. Under these provisions, we generally indemnify and hold harmless the indemnified party for losses suffered or incurred by the indemnified party as a result of our activities. These indemnification provisions generally survive termination of the underlying agreement. The maximum potential amount of future payments we could be required to make under these indemnification provisions is unlimited.

Through December 31, 2025 and 2024, we had not experienced any significant losses related to these indemnification obligations, and no material claims were outstanding. We do not expect significant claims related to these indemnification obligations and, consequently, concluded that the fair value of these obligations is negligible, and no related reserves were established.

Purchase Commitments and Purchase Orders

We enter into agreements in the normal course of business with vendors and contract manufacturing organizations (CMOs) for raw materials and manufacturing services and with vendors for preclinical research studies, clinical trials and other goods or services. As of December 31, 2025, we had $411 million of non-cancelable purchase commitments related to raw materials and manufacturing agreements, which are expected to be paid through 2029. This amount includes $5 million of the purchase commitments related to raw materials that was recorded as an accrued liability for loss on future firm purchase commitments. As of December 31, 2025, we had $1.0 billion of non-cancelable purchase commitments related to research and development and other goods and services which are expected to be paid through 2032. These amounts represent our minimum contractual obligations, including termination fees.

In addition to purchase commitments, we have agreements with third parties for various services, including services related to clinical operations and support and contract manufacturing, for which we are not contractually able to terminate for convenience and avoid any and all future obligations to the vendors. Certain agreements provide for termination rights subject to termination fees or wind-down costs. Under such agreements, we are contractually obligated to make certain payments to vendors, mainly, to reimburse them for their unrecoverable outlays incurred prior to cancellation. At December 31, 2025, we had cancelable open purchase orders of approximately $1.8 billion, reflecting amounts associated with our significant vendor arrangements, under such agreements for our clinical operations and support and contract manufacturing. These amounts represent only our estimate of those items for which we had a contractual commitment to pay at December 31, 2025, assuming we would not cancel these agreements. The actual amounts we pay in the future to the vendors under such agreements may differ from the purchase order amounts.

Licenses to Patented Technology

In 2017, we entered into sublicense agreements with Cellscript, LLC and its affiliate, mRNA RiboTherapeutics, Inc. to sublicense certain patent rights. Pursuant to each agreement, we are required to pay certain license fees, annual maintenance fees, minimum royalties on future net sales and milestone payments contingent on achievement of certain development, regulatory and commercial

milestones for specified products, on a product-by-product basis. Commercial milestone payments and royalties based on annual net sales of licensed products for therapeutic and prophylactic products are accounted for as additional expense of the related net product sales in the period in which the corresponding sales occur.

In December 2022, we entered into a non-exclusive patent license agreement with the National Institute of Allergy and Infectious Diseases (NIAID), an Institute or Center of the National Institutes of Health (NIH) to license certain patent rights concerning stabilizing prefusion coronavirus spike proteins and the resulting stabilized proteins for use in COVID vaccine products. Pursuant to the agreement, we have agreed to pay low single-digit royalties on future net sales, a minimum annual royalty payment, and certain contingent development, regulatory and commercial milestone payments on a licensed product-by-licensed product basis.

In January 2025, we entered into a non-exclusive patent license agreement with NIAID to license certain patent rights related to the development of mRNA-based vaccines for the prevention or treatment of RSV infection. Upon execution of the agreement, we made a total payment of $10 million, which was capitalized as an intangible asset and is amortized to cost of sales on a straight-line basis over the estimated useful life of the licensed patents. In addition, we are obligated to pay low single-digit royalties on future net sales of licensed products.

For the years ended December 31, 2025, 2024, and 2023 we recognized $88 million, $155 million, and $301 million, respectively, of royalties and commercial milestone payments associated with our net product sales, which was recorded to cost of sales in our consolidated statements of operations.

Additionally, we have other in-license agreements with third parties which require us to make future development, regulatory and commercial milestone payments for specified products associated with the agreements. The achievement of these milestones have not yet occurred as of December 31, 2025.

13. Stock-Based Compensation and Share Repurchase Programs

Equity Plans

In connection with our initial public offering (IPO), we adopted the 2018 Stock Option and Incentive Plan (the 2018 Equity Plan) in November 2018. The 2018 Equity Plan became effective on the date immediately prior to the effective date of the IPO and replaced our 2016 Stock Option and Incentive Plan (the 2016 Equity Plan). The 2018 Equity Plan provides flexibility to our compensation committee to use various equity-based incentive awards as compensation tools to motivate our workforce. The shares of common stock underlying any awards that are forfeited, canceled, held back upon exercise or settlement of an award to satisfy the exercise price or tax withholding, reacquired by us prior to vesting, satisfied without any issuance of stock, expire or are otherwise terminated (other than by exercise) under the 2018 Equity Plan and the 2016 Equity Plan will be added back to the shares of common stock available for issuance under the 2018 Equity Plan.

The Board of Directors may grant to employees, non-employee directors, consultants and independent advisors equity-based awards during their period of service, generally in the form of stock options, restricted stock units, and performance stock units. The terms and conditions of stock-based awards are defined at the sole discretion of our Board of Directors. We issue service-based awards, vesting over a defined period of service, and performance-based awards, vesting upon achievement of defined conditions. Service based awards generally vest over a four-year period, with the first 25% of such awards vesting following twelve months of continued employment or service. The remaining awards vest in twelve quarterly installments over the following twelve quarters. Stock options granted under the 2018 Equity Plan and the 2016 Equity Plan expire ten years from the date of grant and the exercise price must be at least equal to the fair market value of common stock on the grant date.

As of December 31, 2025, we had a total of 62 million shares reserved for future issuance under our Equity Plans, of which 38 million shares were reserved for equity awards previously granted, and 24 million shares were available for future grants under the 2018 Equity Plan. No additional awards will be granted under the 2016 Equity Plan as it was replaced by the 2018 Equity Plan.

Stock Option Exchange Program

On November 13, 2025, we commenced an offer to exchange certain eligible stock options held by eligible employees of the Company for new stock options (the “Exchange Offer”). Executive officers and members of our Board of Directors were not eligible to participate. The Exchange Offer expired on December 12, 2025.

Under the Exchange Offer, 2,864 eligible employees elected to exchange, and we accepted for cancellation, eligible stock options to purchase an aggregate of 4,289,694 shares of our common stock, representing approximately 80% of the total shares of common stock underlying eligible options. Immediately following the expiration of the Exchange Offer, we granted replacement stock options to purchase an aggregate of 1,653,121 shares of our common stock. The exercise price of the replacement stock options was $29.46 per

share, which was the closing price of our common stock on December 12, 2025. The replacement stock options are subject to new vesting schedules based on continued service.

The exchange of stock options resulted in an immaterial incremental stock-based compensation expense, calculated using a lattice option pricing model, which will be recognized together with any unrecognized compensation cost remaining on the exchanged options over the remaining requisite service period of the modified awards.

Options

We have granted options generally through the 2018 Equity Plan and 2016 Equity Plan. The following table summarizes our option activity during the year ended December 31, 2025:

	Number of Options (in millions)	Weighted Average Exercise Price per Share	Weighted-Average Remaining Contractual Term	Aggregate Intrinsic Value[(1)] (in millions)
Outstanding at December 31, 2024	26.20	$ 59.64	5.1 years	$ 359
Granted	8.91	30.47		
Exercised	(1.22)	14.06		
Canceled/forfeited	(6.71)	114.24		
Outstanding at December 31, 2025	27.18	38.63	4.8 years	153
Exercisable at December 31, 2025	19.54	38.22	3.3 years	151
Expected to vest at December 31, 2025	7.64	$ 39.67	8.7 years	$ 2

[(1)]Aggregate intrinsic value is calculated as the difference between the exercise price of the underlying options and the fair value of common stock for those options in the money as of December 31, 2025.

The total intrinsic value of options exercised was $19 million, $122 million, and $413 million for the years ended December 31, 2025, 2024, and 2023, respectively. The aggregate intrinsic value represents the difference between the exercise price and the selling price received by option holders upon the exercise of stock options during the period. The excess tax benefits realized from tax deductions from option exercises were $2 million, $24 million, and $84 million during the years ended December 31, 2025, 2024, and 2023, respectively. The total consideration recorded as a result of stock option exercises was approximately $17 million, $42 million, and $25 million, respectively, for the years ended December 31, 2025, 2024, and 2023.

Restricted Common Stock Units (RSUs) and Performance Stock Units (PSUs)

We have granted RSUs and PSUs generally through the 2018 Equity Plan. The following table summarizes our RSU and PSU activity during the year ended December 31, 2025:

	Number of Units (in millions)	Weighted Average Grant Date Fair Value per Unit
Outstanding, non-vested at December 31, 2024	7.86	$ 91.60
Issued	12.67	30.77
Vested	(6.81)	63.70
Canceled/forfeited	(2.52)	60.43
Outstanding, non-vested at December 31, 2025	11.20	46.11

The total grant date fair value of RSUs and PSUs vested during the years ended December 31, 2025, 2024, and 2023, was $434 million, $228 million, and $99 million, respectively. The total intrinsic value of RSUs and PSUs vested during the years ended December 31, 2025, 2024, and 2023, was $186 million, $161 million and $120 million, respectively.

During 2025, 2024 and 2023, we granted an immaterial amount of PSUs, respectively, primarily to certain senior executives with vesting that is contingent upon the achievement of specified preestablished goals over the performance period, generally three years. The actual number of common shares ultimately issued is calculated by multiplying the number of PSUs by a payout percentage ranging from 0% to 200%. The estimated fair value of PSUs is based on the grant date fair value.

Valuation and Stock-Based Compensation Expense

Stock-based compensation for options granted under our Equity Plans (excluding options issued in connection with the Exchange Offer) is determined using the Black-Scholes option pricing model. The weighted-average assumptions used to estimate the fair value of options granted for the years ended December 31, 2025, 2024, and 2023 were as follows:

	Weighted Average Years Ended December 31,		
	2025	2024	2023
Options:			
Risk-free interest rate	4.04 %	4.28 %	4.13 %
Expected term	5.54 years	6.10 years	6.07 years
Expected volatility	66 %	50 %	48 %
Expected dividends	— %	— %	— %
Weighted average fair value per share	$ 18.64	$ 49.76	$ 57.87

Stock-Based Compensation Expense

The following table presents the components and classification of stock-based compensation expense for the years ended December 31, 2025, 2024, and 2023 (in millions):

	Years Ended December 31,		
	2025	2024	2023
Options	$ 152	$ 163	$ 136
RSUs	314	254	144
PSUs	8	5	17
Employee stock purchase plan	9	7	8
Total	$ 483	$ 429	$ 305
Cost of sales	$ 26	$ 25	$ 37
Research and development	291	264	157
Selling, general and administrative	166	140	111
Total	$ 483	$ 429	$ 305

Stock-based compensation expenses related to non-employee awards were immaterial for the years ended December 31, 2025, 2024, and 2023.

As of December 31, 2025, there were $633 million of total unrecognized compensation cost related to non-vested stock-based compensation with respect to options, RSUs and PSUs granted. That cost is expected to be recognized over a weighted-average period of 2.1 years at December 31, 2025.

Share Repurchase Programs

On February 22, 2022, our Board of Directors authorized a share repurchase program of our common stock for up to $3.0 billion, with no expiration date. On August 1, 2022, our Board of Directors authorized an additional $3.0 billion under the repurchase program, with no expiration date (collectively with the February 22, 2022 authorization, the 2022 Repurchase Programs). As of December 31, 2025, $1.7 billion of our Board of Directors' authorization for repurchases of our common stock remains outstanding under the 2022 Repurchase Programs, with no expiration date. The timing and actual number of shares repurchased under the 2022 Repurchase Programs will depend on a variety of factors, including price, general business and market conditions, and other investment opportunities, and shares may be repurchased through open market purchases through the use of trading plans intended to qualify under Rule 10b5-1 under the Securities Exchange Act of 1934, as amended.

The following table summarizes activity related to our share repurchase programs (in millions, except per share data):

	Years Ended December 31,		
	2025	2024	2023
Number of shares repurchased	—	—	8
Average price per share[(1)]	$ —	$ —	$ 143.26
Aggregate purchase price	$ —	$ —	$ 1,153
Remaining authorization at end of period	$ 1,667	$ 1,667	$ 1,667

[(1)]Average price paid per share includes related expenses and excise tax, applicable beginning January 1, 2023.

14. Income Taxes

Loss before income taxes for the years ended December 31, 2025, 2024, and 2023 consisted of the following (in millions):

	Years Ended December 31,		
	2025	2024	2023
United States	$ (2,655)	$ (3,697)	$ (4,056)
Foreign	(113)	90	114
Loss before income taxes	$ (2,768)	$ (3,607)	$ (3,942)

The provision for income taxes for the years ended December 31, 2025, 2024, and 2023 consisted of the following components (in millions):

	Years Ended December 31,		
	2025	2024	2023
Current:			
Federal	$ 11	$ (57)	$ (225)
State	4	(1)	72
Foreign	29	13	24
Total current	$ 44	$ (45)	$ (129)
Deferred:			
Federal	$ —	$ —	$ 888
State	—	—	8
Foreign	10	(1)	5
Total deferred	10	(1)	901
Total provision for (benefit from) income taxes	$ 54	$ (46)	$ 772

The reconciliation of the federal statutory income tax amount and rate to our effective tax rate for the year ended December 31, 2025 was as follows (in millions, except percentages):

	Year Ended December 31, 2025	
Federal statutory tax rate	$ (581)	21.0 %
State and local income tax, net of federal income tax effect	3	(0.2)%
Foreign tax effects	62	(2.1)%
Tax credits	(11)	0.4 %
Changes in valuation allowances	509	(18.4)%
Nontaxable or nondeductible items	2	(0.1)%
Stock-based compensation windfall/shortfall	43	(1.6)%
Changes in unrecognized tax benefits	10	(0.4)%
Other	17	(0.6)%
Effective tax rate	$ 54	(2.0)%

The effective tax rate reconciliation for the year ended December 31, 2025 is presented in accordance with ASU 2023-09, while the effective tax rate reconciliations for the years ended December 31, 2024 and 2023 have not been recast and continue to be presented under the prior guidance. As a result of the adoption of ASU 2023-09, certain items in the effective tax rate reconciliation may have been reclassified between categories compared to prior periods; however, such reclassifications did not have a material impact on any individual line items or the overall effective tax rate.

The reconciliation of the federal statutory income tax rate to our effective tax rate for the years ended December 31, 2024 and 2023 was as follows:

	Years Ended December 31,	
	2024	2023
Federal statutory tax rate	21.0 %	21.0 %
Change in valuation allowance	(23.5)%	(52.6)%
Foreign-derived intangible income	— %	0.2 %
Stock-based compensation windfall/shortfall	0.2 %	2.4 %
Federal research and development credits	5.5 %	4.6 %
State taxes, net of federal benefits	(0.7)%	5.7 %
Non-deductible items	(0.4)%	(0.4)%
Other	(0.8)%	(0.5)%
Effective tax rate	1.3 %	(19.6)%

Our effective tax rate for the year ended December 31, 2025 was (2.0)% and was higher than the federal statutory tax rate, primarily due to our global valuation allowance, which limits our ability to recognize tax benefits from the loss. The higher effective tax rate was also impacted by certain of our foreign subsidiaries that have taxable income, while we incurred a net loss before income taxes in other jurisdictions. For the year ended December 31, 2024, the effective tax rate was lower than the federal statutory tax rate as a result of an increase in valuation allowance against deferred tax assets, which limited the recognition of tax benefits on our pre-tax loss. The tax benefits from research and development credits provided a partial offset. For the year ended December 31, 2023, despite being in a pre-tax loss position, our effective tax rate exceeded the federal statutory tax rate, primarily due to the establishment of a valuation allowance against the majority of the deferred tax assets, which resulted in a net tax expense rather than a benefit. This was partially offset by tax benefits from research and development credits and stock-based compensation.

For the year ended December 31, 2025, income taxes paid, net of refunds, by jurisdiction were immaterial both individually and in the aggregate and, accordingly, have not been separately disclosed.

Deferred income taxes reflect the tax effect of temporary differences between the carrying amount of assets and liabilities for financial reporting and the amounts used for income tax purposes, tax credit carryforwards and the tax effect of net operating loss carryforwards. Significant components of our deferred tax assets and tax liabilities as of December 31, 2025 and 2024 were as follows (in millions):

	December 31,	
	2025	2024
Deferred tax assets:		
Net operating loss carryforwards	$ 1,156	$ 424
Stock-based compensation	129	131
Capitalized licenses, research and development and start-up costs	1,693	1,794
Tax credit carryforwards	308	251
Operating lease liabilities	134	138
Other comprehensive income	3	4
Inventory reserve and capitalization	152	205
Wholesaler chargebacks, discounts and fees	26	43
Returns and other fees	116	86
Outside basis difference	84	125
Other	213	160
Total deferred tax assets	4,014	3,361
Less: valuation allowance	(3,738)	(3,084)
Net deferred tax assets	$ 276	$ 277
Deferred tax liabilities:		
Right-of-use assets, operating	$ (135)	$ (139)
Property, plant and equipment	(58)	(55)
Other	(23)	(16)
Total deferred tax liabilities	(216)	(210)
Net deferred tax assets	$ 60	$ 67

The table below summarizes changes in the valuation allowance for deferred tax assets for the periods presented (in millions):

	Years Ended December 31,		
	2025	2024	2023
Valuation allowance at beginning of the period	$ 3,084	$ 2,224	$ 155
Increases to valuation allowance	654	860	2,069
Valuation allowance at December 31	$ 3,738	$ 3,084	$ 2,224

We periodically reassess the need for valuation allowances on our deferred tax assets, considering both positive and negative evidence to evaluate whether it is more likely than not that all or a portion of such assets will not be realized. During 2023, following the completion of our long-range financial planning process, we reassessed the evidence and concluded that a valuation allowance was necessary due to the preponderance of negative evidence, including:

- A pre-tax loss for the full year 2023, serving as a significant source of objectively verifiable negative evidence in accordance with ASC 740 (Income Taxes).
- A projected three-year cumulative loss resulting from our long-range financial planning process. This projection was due to a significant decrease in expected sales of our COVID vaccine as we transitioned to a seasonal market. Additionally, we anticipated substantial research and development expenses for our on-going Phase 3 clinical trials and to advance our product candidates into later-stage development. These factors contributed additional negative evidence with respect to the realizability of our deferred tax assets. The projections were based upon revenue from our approved drug product, which we believe can be reasonably estimated. In contrast, future taxable income projections from our investigational medicines are deemed inherently subjective and not objectively verifiable; they are insufficient to override negative evidence, and therefore, they were not assigned any weight in our valuation allowance analysis assessment.

Our evaluation also included whether there were other sources of taxable income that would allow us to realize our deferred tax assets, such as taxable income in carryback years, available tax planning strategies and the future reversals of taxable temporary differences. After assessing these strategies and all evidence, we determined it was more likely than not that we will not realize all of our deferred tax assets and therefore increased the valuation allowance by $2.1 billion during 2023.

In 2025 and 2024, we continued to maintain a global valuation allowance against the majority of our deferred tax assets, consistent with the assessment established in 2023. The valuation allowance reflects the ongoing preponderance of negative evidence, including continued and projected losses.

Significant management judgment is required in assessing the realizability of our deferred tax assets. In the event that actual results differ from our estimates, we adjust our estimates in future periods and we may need to modify our valuation allowance, which could materially impact our financial position and results of operations.

At December 31, 2025, we had $4.1 billion, $3.3 billion, and $89 million of federal, state and foreign net operating loss carryforwards, respectively, of which $4.1 billion, $1.6 billion, and $89 million respectively, will not expire and $1.7 billion of state net operating loss carryforwards will begin to expire in 2032. At December 31, 2025, we also had federal and state research and development tax credit carryforwards of $111 million and $208 million, respectively, the majority of which will begin to expire in 2030.

We recognize, in our financial statements, the effect of a tax position when it is more likely than not, based on the technical merits, that the position will be sustained upon examination. A reconciliation of the beginning and ending amounts of unrecognized tax benefits during the years ended December 31, 2025, 2024, and 2023 were as follows (in millions):

	Years Ended December 31,		
	2025	2024	2023
Unrecognized tax benefits at beginning of the period	$ 240	$ 231	$ 128
Decrease due to prior positions:			
Tax positions for prior years	(3)	(10)	—
Settlements with tax authorities	—	—	(27)
Increase due to current year tax positions:			
Additions based on tax positions for current year	4	19	44
Additions based on tax positions for prior years	—	—	86
Unrecognized tax benefits at end of the period	$ 241	$ 240	$ 231

As of December 31, 2025, we had $241 million of net unrecognized tax benefits, which would affect our tax rate if recognized. Unrecognized tax benefits may change during the next twelve months for items that arise in the ordinary course of business. We do not anticipate a material change to our unrecognized tax benefits over the next twelve months that would have an adverse effect on our consolidated operating results. We recognize interest and penalties, if applicable, related to uncertain tax positions as a component of income tax expense.

We file income tax returns in U.S. and various state, local and foreign jurisdictions. The income tax returns of all material taxing jurisdictions remain open to tax examination for all tax years since our date of incorporation for those jurisdictions. We have federal, state, and foreign carryforward attributes generated in past years which may be adjusted upon examination. As of December 31, 2025, we are under audit in various U.S., and foreign jurisdictions; however, no adjustments to our tax positions have been proposed at this time.

15. Loss per Share

The computation of basic earnings (loss) per share (EPS) is based on the weighted-average number of our common shares outstanding. The computation of diluted EPS is based on the weighted-average number of our common shares outstanding and potential dilutive common shares outstanding during the period as determined by using the treasury stock method.

Basic and diluted EPS for the years ended December 31, 2025, 2024 and 2023 were calculated as follows (in millions, except per share data):

	Years Ended December 31,		
	2025	2024	2023
Numerator:			
Net loss	$ (2,822)	$ (3,561)	$ (4,714)
Denominator:			
Basic and diluted weighted-average common shares outstanding	389	384	382
Basic and diluted EPS	$ (7.26)	$ (9.28)	$ (12.33)

The following common stock equivalents, presented based on amounts outstanding as of December 31, 2025, 2024 and 2023, were excluded from the calculation of diluted EPS attributable to common stockholders for the periods indicated because their inclusion would have been anti-dilutive (in millions):

	December 31,		
	2025	2024	2023
Options	27	26	26
RSUs and PSUs	11	8	5
Total	38	34	31

16. Geographic Information

Geographic Revenue

We operate in one reporting segment that primarily focuses on the discovery, development and commercialization of mRNA medicines. Our chief executive officer manages our operations and evaluates our financial performance on a consolidated basis. Most of our principal operations, other than manufacturing, and our decision-making functions are located at our corporate headquarters in the United States.

Total revenue by geographic area of our customers and collaborators was as follows (in millions):

	Years Ended December 31,		
	2025	2024	2023
United States	$ 1,199	$ 1,785	$ 1,895
Europe	53	598	1,355
Rest of world	692	853	3,598
Total	$ 1,944	$ 3,236	$ 6,848

Our property, plant and equipment, including financing right-of-use assets, by geographic area was as follows (in millions):

	December 31,	
	2025	2024
United States	$ 1,468	$ 1,532
Europe	303	283
Rest of world	363	381
Total	$ 2,134	$ 2,196

Item 9. Changes in and Disagreements with Accountants on Accounting and Financial Disclosure

None.

Item 9A. Controls and Procedures

Evaluation of Disclosure Controls and Procedures

Our management, with the participation of our Chief Executive Officer and our Chief Financial Officer, evaluated the effectiveness of our disclosure controls and procedures as of December 31, 2025. The term "disclosure controls and procedures," as defined in Rules 13a-15(e) and 15d-15(e) under the Securities Exchange Act of 1934, or the Exchange Act, means controls and other procedures of a company that are designed to ensure that information required to be disclosed by a company in the reports that it files or submits under the Exchange Act is recorded, processed, summarized and reported, within the time periods specified in the SEC's rules and forms. Disclosure controls and procedures include, without limitation, controls and procedures designed to ensure that information required to be disclosed by a company in the reports that it files or submits under the Exchange Act is accumulated and communicated to the company's management, including its principal executive and principal financial officers, as appropriate to allow timely decisions regarding required disclosure. Management recognizes that any controls and procedures, no matter how well designed and operated, can provide only reasonable assurance of achieving their objectives and management necessarily applies its judgment in evaluating the cost-benefit relationship of possible controls and procedures. Based on the evaluation of our disclosure controls and procedures as of December 31, 2025, our Chief Executive Officer and Chief Financial Officer concluded that, as of such date, our disclosure controls and procedures were effective at the reasonable assurance level.

Management's Report on Internal Control Over Financial Reporting

Our management is responsible for establishing and maintaining adequate internal control over financial reporting (as such term is defined in Exchange Act Rule 13a-15(f)) to provide reasonable assurance regarding the reliability of our financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. Management assessed our internal control over financial reporting as of December 31, 2025. Management based its assessment on criteria established in Internal Control - Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 Framework). Based on that evaluation, our management concluded that our internal control over financial reporting was effective as of December 31, 2025.

The effectiveness of our internal control over financial reporting as of December 31, 2025 has been audited by Ernst & Young LLP, an independent registered public accounting firm, as stated in their report included in this Annual Report on Form 10-K.

Changes in Internal Controls over Financial Reporting

During the three months ended December 31, 2025, there were no changes in our internal control over financial reporting (as defined in Rules 13a-15(f) and 15d-15(f) under the Exchange Act), which have materially affected, or are reasonably likely to materially affect, our internal control over financial reporting.

Inherent Limitations on the Effectiveness of Controls

Our management, including our Chief Executive Officer and Chief Financial Officer, believe that our disclosure controls and procedures and internal control over financial reporting are designed to provide reasonable assurance of achieving their objectives and are effective at the reasonable assurance level. However, our management does not expect that our disclosure controls and procedures or our internal control over financial reporting will prevent all errors and all fraud. A control system, no matter how well-conceived and operated, can provide only reasonable, not absolute, assurance that the objectives of the control system are met. Further, the design of a control system must reflect the fact that there are resource constraints, and the benefits of controls must be considered relative to their costs. Because of the inherent limitations in all control systems, no evaluation of controls can provide absolute assurance that all control issues and instances of fraud, if any, have been detected. These inherent limitations include the realities that judgments in decision making can be faulty, and that breakdowns can occur because of a simple error or mistake. Additionally, controls can be circumvented by the individual acts of some persons, by the collusion of two or more people or by a management override of the controls. The design of any system of controls also is based in part upon certain assumptions about the likelihood of future events, and there can be no assurance that any design will succeed in achieving its stated goals under all potential future conditions; over time, controls may become inadequate because of changes in conditions, or the degree of compliance with policies or procedures may deteriorate. Because of the inherent limitations in a cost-effective control system, misstatements due to error or fraud may occur and not be detected.

Report of Independent Registered Public Accounting Firm

To the Stockholders and the Board of Directors of Moderna, Inc.

Opinion on Internal Control Over Financial Reporting

We have audited Moderna, Inc.'s internal control over financial reporting as of December 31, 2025, based on criteria established in Internal Control-Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework) (the COSO criteria). In our opinion, Moderna, Inc. (the Company) maintained, in all material respects, effective internal control over financial reporting as of December 31, 2025, based on the COSO criteria.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the consolidated balance sheets of the Company as of December 31, 2025 and 2024, the related consolidated statements of operations, comprehensive income (loss), stockholders' equity, and cash flows for each of the three years in the period ended December 31, 2025, and the related notes and our report dated February 20, 2026 expressed an unqualified opinion thereon.

Basis for Opinion

The Company's management is responsible for maintaining effective internal control over financial reporting and for its assessment of the effectiveness of internal control over financial reporting included in the accompanying Management's Report on Internal Control Over Financial Reporting. Our responsibility is to express an opinion on the Company's internal control over financial reporting based on our audit. We are a public accounting firm registered with the PCAOB and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audit in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether effective internal control over financial reporting was maintained in all material respects.

Our audit included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, testing and evaluating the design and operating effectiveness of internal control based on the assessed risk, and performing such other procedures as we considered necessary in the circumstances. We believe that our audit provides a reasonable basis for our opinion.

Definition and Limitations of Internal Control Over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (1) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

/s/ Ernst & Young LLP

Boston, Massachusetts

February 20, 2026

Item 9B. Other Information

10b5-1 Plans

On November 23, 2025, Stéphane Bancel, our Chief Executive Officer, terminated a trading arrangement that was intended to satisfy the affirmative defense of Rule 10b5-1(c), which had been entered into on September 3, 2025, and was scheduled to commence as early as December 17, 2025, with a termination date of August 10, 2026. The plan provided for the potential sale of up to 1,439,788 shares of common stock. No shares of common stock were sold under the plan prior to its termination.

Mr. Bancel has two different stock option awards that will reach their ten-year expiration on August 10, 2026, including an option to purchase 558,394 shares and another option to purchase 193,321 shares (the Bancel Expiring Options). Mr. Bancel intends to contribute all of the after-tax proceeds from the exercise and sale of the Bancel Expiring Options to charitable causes. On November 28, 2025, Mr. Bancel adopted a trading arrangement that is intended to satisfy the affirmative defense of Rule 10b5-1(c) (the Bancel 10b5-1 Plan). Between May 13,2026 and August 10, 2026, the Bancel 10b5-1 Plan provides for the potential exercise of vested stock options and the associated sale of the Bancel Expiring Options, which represent up to 751,715 shares of the Company's common stock in the aggregate. The Bancel 10b5-1 Plan expires on August 10, 2026, or upon the earlier completion of all authorized transactions under the plan.

Stephen Hoge, our President, has two different stock option awards that will reach their ten-year expiration on August 10, 2026, including an option to purchase 223,357 shares and another option to purchase 96,660 shares (the Hoge Expiring Options). On November 13, 2025, Dr. Hoge adopted a trading arrangement that is intended to satisfy the affirmative defense of Rule 10b5-1(c) (the Hoge 10b5-1 Plan). Between February 23, 2026 and August 10, 2026, the Hoge 10b5-1 Plan provides for the potential exercise of vested stock options and the associated sale of the Hoge Expiring Options, which represent up to 320,017 shares of the Company's common stock in the aggregate. The Hoge 10b5-1 Plan expires on August 10, 2026, or upon the earlier completion of all authorized transactions under the plan.

Item 9C. Disclosure Regarding Foreign Jurisdictions that Prevent Inspections

None Applicable.

PART III

Item 10. Directors, Executive Officers and Corporate Governance

The information required by this Item is incorporated herein by reference to the information that will be contained in our proxy statement related to the 2026 Annual Meeting of Stockholders, which we intend to file with the Securities and Exchange Commission within 120 days of the end of our fiscal year pursuant to General Instruction G(3) of Form 10-K.

Item 11. Executive Compensation

The information required by this Item is incorporated herein by reference to the information that will be contained in our proxy statement related to the 2026 Annual Meeting of Stockholders, which we intend to file with the Securities and Exchange Commission within 120 days of the end of our fiscal year pursuant to General Instruction G(3) of Form 10-K.

Item 12. Security Ownership of Certain Beneficial Owners and Management and Related Stockholder Matters

The information required by this Item is incorporated herein by reference to the information that will be contained in our proxy statement related to the 2026 Annual Meeting of Stockholders, which we intend to file with the Securities and Exchange Commission within 120 days of the end of our fiscal year pursuant to General Instruction G(3) of Form 10-K.

Item 13. Certain Relationships and Related Transactions, and Director Independence

The information required by this Item is incorporated herein by reference to the information that will be contained in our proxy statement related to the 2026 Annual Meeting of Stockholders, which we intend to file with the Securities and Exchange Commission within 120 days of the end of our fiscal year pursuant to General Instruction G(3) of Form 10-K.

Item 14. Principal Accounting Fees and Services

Our independent public accounting firm is Ernst & Young LLP, Boston, Massachusetts, PCAOB Auditor ID 00042.

The information required by this Item is incorporated herein by reference to the information that will be contained in our proxy statement related to the 2026 Annual Meeting of Stockholders, which we intend to file with the Securities and Exchange Commission within 120 days of the end of our fiscal year pursuant to General Instruction G(3) of Form 10-K.

PART IV

Item 15. Exhibits, Financial Statement Schedules

(a) Documents filed as part of this report.

(1) Financial statements.

For a list of the consolidated financial statements included herein, see "Index to Consolidated Financial Statements" under Part II, Item 8 of this Annual Report on Form 10-K.

(2) Schedules.

No financial statement schedules have been submitted because they are not required or are not applicable or because the information required is included in the consolidated financial statements or the notes thereto.

(3) Exhibits.

Exhibit No.	Exhibit Index
3.1	Restated Certificate of Incorporation of the Registrant. (2)
3.2	Second Amended and Restated By-laws of the Registrant. (2)
4.1	Specimen Common Stock Certificate. (1)
4.2	Description of Capital Stock. (9)
10.1#	2016 Stock Option and Grant Plan, as amended, and forms of award agreements thereunder. (1)
10.2#	2018 Stock Option and Incentive Plan. (1)
10.3#	Form of Indemnification Agreement between the Registrant and each of its directors. (1)
10.4†	Master Collaboration and License Agreement, by and between Moderna Therapeutics, Inc. and Merck Sharp & Dohme Corp., dated as of January 12, 2015, as amended by Amendment No. 1 dated as of January 8, 2016, Amendment No. 2 dated as of June 28, 2016, Amendment No. 3 dated as of June 28, 2016 and Amendment No. 4 dated as of June 28, 2016. (1)
10.5†	Amended and Restated mRNA Cancer Vaccine Collaboration and License Agreement, by and between ModernaTX, Inc. and Merck Sharp & Dohme Corp., dated as of April 17, 2018. (1)
10.6†	Patent Sublicense Agreement, by and among ModernaTX, Inc. and Cellscript, LLC and mRNA RiboTherapeutics, Inc. (solely with respect to certain provisions), dated as of June 26, 2017. (1)
10.7	Credit and Guaranty Agreement, dated as of November 19, 2025, by and among Moderna, certain subsidiary guarantors, Ares Capital Corporation and the lenders. (10)
10.8#	Amended and Restated Executive Severance Plan and Form of Participation Letter, as amended on February 23, 2023. (7)
10.9#	Letter Agreement by and between the Company and Stéphane Bancel, dated as of June 13, 2018, as amended by Amendment No. 1 dated as of November 4, 2018. (1)
10.10#	Letter Agreement by and between the Company and Stephen Hoge, dated as of October 17, 2017. (1)
10.11#	Employment Letter Agreement between ModernaTX, Inc. and Shannon Klinger, dated as of March 4, 2021. (5)
10.12#	Offer Letter by and between ModernaTX, Inc. and James Mock, dated as of August 15, 2022. (6)
10.13#	Senior Executive Cash Incentive Bonus Plan. (1)
10.14#	Amended and Restated Non-Employee Director Compensation Policy, effective October 1, 2022. (6)
10.15#	Form of Indemnification Agreement between the Registrant and each of its officers. (1)
10.16#*	2018 Employee Stock Purchase Plan.
10.17#	Form of Employee Restricted Stock Unit Award Agreement. (8)
10.18#	Form of Employee Non-Qualified Stock Option Agreement. (8)
10.19#	Form of Non-Employee Director Restricted Stock Unit Award Agreement. (5)
10.20#	Form of Non-Employee Director Non-Qualified Stock Option Agreement. (5)
10.21#	Form of Performance-Based Restricted Stock Unit Award Agreement under the 2018 Stock Option and Incentive Plan. (3)
19	Moderna, Inc. Insider Trading Policy. (9)

21.1*	Subsidiaries of the Registrant.
23.1*	Consent of Ernst & Young LLP, Independent Registered Public Accounting Firm.
31.1*	Certification of Principal Executive Officer pursuant to Rule 13a-14(a) and Rule 15d-14(a) of the Securities Exchange Act of 1934, as adopted pursuant to Section 302 of the Sarbanes-Oxley Act of 2002
31.2*	Certification of Principal Financial Officer pursuant to Rule 13a-14(a) and Rule 15d-14(a) of the Securities Exchange Act of 1934, as adopted pursuant to Section 302 of the Sarbanes-Oxley Act of 2002
32.1+	Certification pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002
32.2+	Certification pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002
97	Moderna, Inc. Policy for Recoupment of Executive Incentive Compensation. (8)
101.INS*	XBRL Instance Document
101.SCH*	XBRL Taxonomy Extension Schema Document
101.CAL*	XBRL Taxonomy Extension Calculation Document
101.DEF*	XBRL Taxonomy Extension Definition Linkbase Document
101.LAB*	XBRL Taxonomy Extension Labels Linkbase Document
101.PRE*	XBRL Taxonomy Extension Presentation Link Document
104*	Cover Page Interactive Data File (formatted as Inline XBRL with applicable taxonomy extension information contained in Exhibits 101)

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* Filed herewith.

† Pursuant to 17 C.F.R. §§230.406 and 230.83, the confidential portions of this exhibit have been omitted and are marked accordingly.

\# Indicates a management contract or any compensatory plan, contract or arrangement.

\+ The certifications furnished in Exhibits 32.1 and 32.2 hereto are deemed to accompany this Annual Report on Form 10-K and will not be deemed "filed" for purposes of Section 18 of the Securities Exchange Act of 1934, as amended. Such certifications will not be deemed to be incorporated by reference into any filings under the Securities Act of 1933, as amended, or the Securities Exchange Act of 1934, as amended, except to the extent that the Registrant specifically incorporates it by reference.

(1) Incorporated by reference to the Registration Statement on Form S-1 (File No. 333-228300) filed with the Securities and Exchange Commission on November 9, 2018.

(2) Incorporated by reference to the Current Report on Form 8-K (File No. 001-38753) filed with the Securities and Exchange Commission on May 9, 2024.

(3) Incorporated by reference to the Quarterly Report on Form 10-Q (File No. 001-38753) filed with the Securities and Exchange Commission on May 6, 2021.

(4) Incorporated by reference to the Quarterly Report on Form 10-Q (File No. 001-38753) filed with the Securities and Exchange Commission on November 7, 2024.

(5) Incorporated by reference to the Annual Report on Form 10-K (File No. 001-38753) filed with the Securities and Exchange Commission on February 25, 2022.

(6) Incorporated by reference to the Quarterly Report on Form 10-Q (File No. 001-38753) filed with the Securities and Exchange Commission on November 3, 2022.

(7) Incorporated by reference to the Annual Report on Form 10-K (File No. 001-38753) filed with the Securities and Exchange Commission on February 24, 2023.

(8) Incorporated by reference to the Annual Report on Form 10-K (File No. 001-38753) filed with the Securities and Exchange Commission on February 23, 2024.

(9) Incorporated by reference to the Annual Report on Form 10-K (File No. 001-38753) filed with the Securities and Exchange Commission on February 21, 2025.

(10) Incorporated by reference to the Current Report on Form 8-K (File No. 001-38753) filed with the Securities and Exchange Commission on November 24, 2025.

Item 16. Form 10-K Summary

None.

SIGNATURES

Pursuant to the requirements of the Section 13 or 15(d) of the Securities Exchange Act of 1934, the Registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

MODERNA, INC.

Date:
February 20, 2026

By: /s/ Stéphane Bancel

Stéphane Bancel
Chief Executive Officer and Director

POWER OF ATTORNEY AND SIGNATURES

Each individual whose signature appears below hereby constitutes and appoints each of Stéphane Bancel and James M. Mock as such person's true and lawful attorney-in-fact and agent with full power of substitution and resubstitution, for such person in such person's name, place and stead, in any and all capacities, to sign any and all amendments to this Annual Report on Form 10-K, and to file the same, with all exhibits thereto, and all documents in connection therewith, with the Securities and Exchange Commission granting unto each said attorney-in-fact and agent full power and authority to do and perform each and every act and thing requisite and necessary to be done in and about the premises, as fully to all intents and purposes as such person might or could do in person, hereby ratifying and confirming all that any said attorney-in-fact and agent, or any substitute or substitutes of any of them, may lawfully do or cause to be done by virtue hereof.

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed by the following persons on behalf of the Registrant and in the capacities and on the dates indicated.

Signature	Title	Date
/s/ Stéphane Bancel Stéphane Bancel	Chief Executive Officer and Director *(Principal Executive Officer)*	February 20, 2026
/s/ James M. Mock James M. Mock	Chief Financial Officer *(Principal Financial Officer and Principal Accounting Officer)*	February 20, 2026
/s/ Noubar B. Afeyan, Ph.D. Noubar B. Afeyan, Ph.D.	Chairman and Director	February 20, 2026
/s/ Sandra Horning, M.D. Sandra Horning, M.D.	Director	February 20, 2026
/s/ Abbas Hussain Abbas Hussain	Director	February 20, 2026
/s/ Elizabeth Nabel, M.D. Elizabeth Nabel, M.D.	Director	February 20, 2026
/s/ Francois Nader, M.D. Francois Nader M.D.	Director	February 20, 2026
/s/ David M. Rubenstein David M. Rubenstein	Director	February 20, 2026
/s/ Paul Sagan Paul Sagan	Director	February 20, 2026
/s/ Elizabeth Tallett Elizabeth Tallett	Director	February 20, 2026